STUDY TEXTBOOK

Physics for Engineers

Mechanics, fields, and circuits

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Start here: from zero

Physics is the study of how things move and interact, written in mathematics. This book covers the physics an engineer uses: motion, forces, energy, rotation, electricity, magnetism, waves and heat. You need algebra, a little trigonometry (sine and cosine of an angle), and three tools that this chapter teaches from scratch: units, a first look at calculus, and vectors.

Units: the first check on every answer

Every physical quantity has a unit, and a formula must balance in units. A distance is in metres (m), a time in seconds (s), a speed in m/s, an acceleration in $\mathrm{m/s^2}$, a force in newtons (N, where $1\,\mathrm{N} = 1\,\mathrm{kg\,m/s^2}$). Check a formula by its units: the distance travelled from rest with acceleration a is $x = \frac{1}{2}at^2$. In units, $(\mathrm{m/s^2}) \cdot \mathrm{s^2} = \mathrm{m}$, which is a distance, so the formula is at least plausible; a claimed $x = at$ would give m/s, a speed, so it cannot be right. This habit catches more errors than any other, and every chapter uses it.

Calculus in two ideas

The derivative is a slope. If a car's position is $x(t) = 5t^2$ metres at time t seconds, its position at $t = 2$ is 20 m and at $t = 2.01$ is 20.2005 m. Over that 0.01 s it moved 0.2005 m, a speed of about 20.05 m/s. Shrink the interval to nothing and you get exactly 20 m/s. The rule that produces it is the *derivative*: for $x(t) = 5t^2$, the velocity is $v(t) = \mathrm{d}x/\mathrm{d}t = 10t$, and $v(2) = 20$ m/s. The derivative of t^n is nt^{n-1}; that is the only rule Chapters 1–4 need.

The integral is an accumulated total. If the acceleration is a constant $3\,\mathrm{m/s^2}$ for 4 s, the velocity gained is $3 \times 4 = 12$ m/s: the area of a rectangle of height 3 and width 4 on an acceleration–time graph. If the acceleration varies, you add up many thin rectangles; that sum is the *integral*, $\Delta v = \int a\,\mathrm{d}t$. The integral is the derivative run backwards: since $v = 10t$ is the derivative of $5t^2$, the integral of $10t$ from 0 to 2 is $5 \cdot 2^2 - 0 = 20$.

Chapter 1 uses both ideas to derive the constant-acceleration formulas from definitions, so you do not memorise them.

Vectors: quantities with a direction

A *scalar* has size only (a mass, a time). A *vector* has size and direction (a velocity, a force). In a plane we write a vector with *components*: $\mathbf{v} = (3, 4)$ means 3 along x and 4 along y. Its length is $|\mathbf{v}| = \sqrt{3^2 + 4^2} = 5$. Adding vectors adds components: $(3, 4) + (1, -2) = (4, 2)$.

Two ways of multiplying vectors appear later in the book:

Dot product $\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y + a_z b_z$, a number. It is $|\mathbf{a}||\mathbf{b}| \cos\theta$, so it is zero when the vectors are perpendicular. Work is a dot product (Chapter 4): $(1, 2) \cdot (3, 4) = 3 + 8 = 11$.

Cross product $\mathbf{a} \times \mathbf{b}$, a vector perpendicular to both, with components $(a_y b_z - a_z b_y,\ a_z b_x - a_x b_z,\ a_x b_y - a_y b_x)$. Torque and the magnetic force are cross products (Chapters 5, 6 and 13). For $(1, 2, 3) \times (4, 5, 6) = (-3,\ 6,\ -3)$, and $\hat{\mathbf{x}} \times \hat{\mathbf{y}} = \hat{\mathbf{z}}$.

You can check the cross-product example: $2 \cdot 6 - 3 \cdot 5 = -3$, $3 \cdot 4 - 1 \cdot 6 = 6$, $1 \cdot 5 - 2 \cdot 4 = -3$.

How the books of this series fit together

Book	Connection
5 Electronics	Applies the circuit and field ideas of Chapters 9–14 here in its capacitor, inductor and power chapters. Each chapter here ends with an "In engineering practice" section that points to Book 5.
3 Linear algebra	Gives the vector and matrix machinery in greater depth; Chapters 1–5 there are a natural companion to the vector work here.

How to use the boxes in each chapter

Each chapter opens with *After this chapter you can*. From Chapter 2, a *Before you start* box lists what the chapter relies on, with a recap. Always sanity-check your answer with units and with an extreme case (what if the mass is zero, or the angle is 90°?) before you trust it.

CHAPTER 1

Kinematics: describing motion

Mechanics begins with description, not explanation: before we ask *why* things move (chapter 2), we need language precise enough to say *how* they move. That language is calculus applied to a position function $x(t)$, and the payoff for learning it carefully is that the hardest parts of physics — momentum, orbits, wave motion — reuse exactly these ideas. This chapter derives the constant-acceleration formulas from the definitions (never quotes them), treats projectile motion as two independent one-dimensional problems, and builds the centripetal-acceleration formula from a picture of velocity differences. Engineering practice shows up at the end: inertial navigation is this chapter run continuously.

Why this exists. Before you can design a bridge, a robot arm or a rocket you need a precise description of motion: where something is, how fast it is going, and how quickly its speed is changing. Kinematics provides that vocabulary without asking yet what causes the motion. It is the easiest chapter of mechanics and the one every later chapter silently uses, so it is worth getting thoroughly right.

BEFORE YOU START

Algebra, and the idea of a slope. This book uses a little calculus (the derivative as a rate of change, the integral as an accumulated total); the *Start here* chapter explains both in about four pages, with pictures, before the first derivative appears.

AFTER THIS CHAPTER YOU CAN

- describe motion in a straight line with position, velocity and acceleration, and read all three from a graph;
- derive the constant-acceleration formulas rather than memorise them;
- solve a projectile problem as two independent one-dimensional problems;
- simulate motion on a computer and explain why the step size matters.

HOW TO THINK ABOUT THIS

Physics starts by describing, not explaining. Pick a direction as positive, name a starting point, and let velocity and acceleration be how position and velocity change. When you derive a formula from these definitions instead of memorising it, you can rebuild it whenever you forget.

1.1 Position, displacement, and velocity

Fix a line and an origin. A point particle's *position* $x(t)$ is a signed distance from the origin; the sign encodes direction. The *displacement* over $[t_1, t_2]$ is $\Delta x = x(t_2) - x(t_1)$ — which is emphatically not the distance travelled: run around a track once and the displacement is zero while the distance is a lap. The *average velocity* is the slope of the secant,

$$\bar{v} = \frac{\Delta x}{\Delta t},$$

and the *instantaneous velocity* is what the average converges to as the interval shrinks:

$$v(t) = \lim_{\Delta t \to 0} \frac{x(t + \Delta t) - x(t)}{\Delta t} = \frac{dx}{dt}.$$

Velocity is a slope on the $x(t)$ graph. Speed is its magnitude — "60 mph east" is a velocity, "60 mph" is a speed. Sign matters: in one dimension $v > 0$ means moving toward $+x$.

READING AN $x(t)$ GRAPH

Slope of the curve at t = instantaneous velocity. Slope of the secant over $[t_1, t_2]$ = average velocity. A horizontal tangent = momentarily at rest — and that is all it means; what happens next depends on the acceleration.

Exercise 1.1 (easy)**.** A particle sits at $x(t) = 3t^2 - 12t$. Find its average velocity over $[0, 4]$ and its instantaneous velocity at $t = 2$. (AP-C)

Example 1.1 (First attempt: averaging two speeds)**.** A car drives 60 km out at 60 km/h and returns the same 60 km at 30 km/h. What was its average speed over the whole trip? The first guess is the average of the two speeds, $(60 + 30)/2 = 45$ km/h.

What goes wrong. Average speed is defined as total distance over total time, not as an average of speeds. The outward leg takes $60/60 = 1$ h, the return $60/30 = 2$ h. The total distance is 120 km in 3 h: 40 km/h. The slow leg takes more time, so it counts for more. (The average *velocity* for the round trip is zero, since the displacement is zero.)

1.2 Acceleration and the calculus loop

Acceleration is velocity's velocity:

$$a(t) = \frac{dv}{dt} = \frac{d^2x}{dt^2},$$

and figure 1.1 shows all three functions for a thrown object. Read them vertically: when v crosses zero the object is at the top of its climb, and a — a constant $-9.8\,\mathrm{m/s^2}$ — does not care.

The loop of figure 13.3 is the chapter's engine: differentiate to go from x to v to a; integrate to come back:

$$v(t) = v(t_0) + \int_{t_0}^{t} a(t')\,dt', \qquad x(t) = x(t_0) + \int_{t_0}^{t} v(t')\,dt'.$$

Kinematics is nothing more — and nothing less — than choosing a direction around that loop.

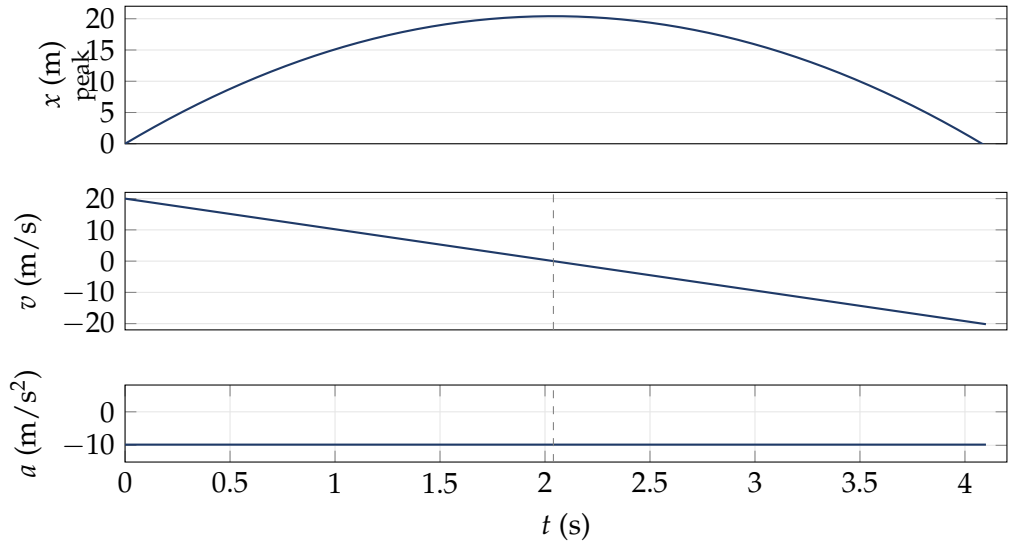


Figure 1.1: $x(t) = 20t - 4.9t^2$ with its first two derivatives, drawn on one time axis. Notice the three-way alignment: the peak of x sits exactly above the zero of v, while a stays fixed at $-9.8\,\mathrm{m/s^2}$ the whole flight. A horizontal tangent in x means "stopped *then*," never "stopped forever."

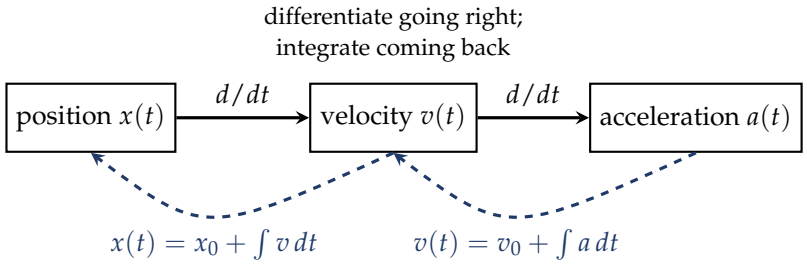


Figure 1.2: The calculus loop. Notice that going right loses information (the constants x_0, v_0 are invisible in $a(t)$), so coming back needs initial conditions: two numbers must be supplied before any $x(t)$ is unique.

Example 1.2 (Slopes off a measured $x(t)$)**.** A radar logs a dragster's position every half second:

t (s)	0	0.5	1.0	1.5	2.0	2.5	3.0
x (m)	0	3.1	9.8	22.0	39.5	62.4	90.7

Estimate $v(1.0)$ and $a(1.0)$ without a formula for $x(t)$.

Strategy. With data, derivatives are differences: centre the secant for v (symmetric difference quotients are second-order accurate), then difference the velocities for a.

Solution. $v(1.0) \approx \frac{x(1.5)-x(0.5)}{1.0} = \frac{22.0-3.1}{1.0} = 18.9\,\mathrm{m/s}$. For a: estimate $v(0.5) \approx 9.8/1.0 = 9.8$ (from $x(1.0) - x(0)$ over 1 s $\Rightarrow$ centre it at 0.5) and $v(1.5) \approx (39.5 - 9.8)/1.0 = 29.7$; then $a(1.0) \approx (29.7 - 9.8)/1.0 = 19.9\,\mathrm{m/s^2} \approx 2g$. ✓ Same calculus, no closed form needed — this is exactly how IMU and encoder data get turned into velocities.

1.3 Constant acceleration, derived

When a is constant the loop collapses to three formulas — derived here so their assumptions are visible. Integrating a once, $v(t) = v_0 + at$; integrating again,

$$x(t) = x_0 + v_0 t + \tfrac{1}{2} a t^2. \tag{1.1}$$

Eliminating t between them (solve $v = v_0 + at$ for t, substitute into (1.1)) gives the time-free relation

$$v^2 = v_0^2 + 2a\,(x - x_0). \tag{1.2}$$

That is the whole subject of "constant-acceleration kinematics." Everything else — free fall, braking distance, the meeting-time problem — is one of these two equations with different letters.

Example 1.3 (Braking distance and the v^2 trap)**.** A car at 100 km/h brakes at a steady $a = -8\,\mathrm{m/s^2}$ (good tires, dry road). How far does it travel before stopping — and how far if the driver needs 0.5 s to react first?

Strategy. No time is asked or given: equation (1.2) answers it directly. Convert units before plugging; add the reaction-time distance afterwards (during it, $a = 0$).

Solution. $100\,\mathrm{km/h} = 27.8\,\mathrm{m/s}$. From (1.2), $0 = v_0^2 + 2a\,\Delta x$, so $\Delta x = \frac{v_0^2}{2|a|} = \frac{27.8^2}{16} = 48.3\,\mathrm{m}$. Reaction adds $v_0 \Delta t = 27.8 \times 0.5 = 13.9\,\mathrm{m}$ — total 62.2 m, a third of it before the brakes touch. ✓ Notice the v^2: at 140 km/h the braking part alone grows to 94 m, not 67 — doubling speed quadruples the stop.

Exercise 1.2 (medium)**.** Two trains 400 m apart approach at 20 m/s each. When does a bee flying between them at 30 m/s get squashed — and how far does the bee travel? Solve without summing the series. (F=ma)

1.4 Projectile motion: two problems at once

With gravity the only acceleration, the vector equation $\mathbf{a} = (0, -g)$ splits: horizontal motion is constant-velocity, vertical is constant-acceleration, and they share only the clock:

$$x(t) = v_{0x}t, \qquad y(t) = v_{0y}t - \tfrac{1}{2}gt^2.$$

Independence is the deep point — a dropped ball and a fired one land together — and figure 1.3 shows the trajectory with its velocity components at three instants.

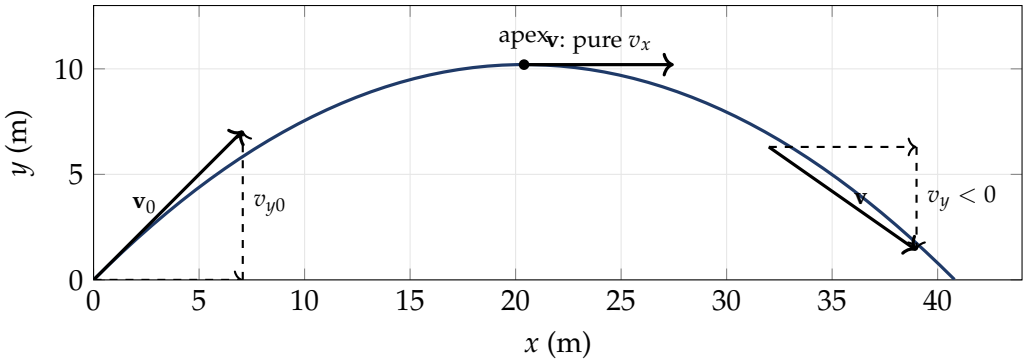


Figure 1.3: Projectile with $v_0 = 20\,\mathrm{m/s}$ at $45°$. Notice what changes along the path: v_x never does; v_y drains to zero at the apex and goes negative after. The parabola is a parametric curve — $y(x)$ has no time in it; recover t from $x = v_{0x}t$ whenever it is needed.

The flat-ground range $R = \frac{v_0^2 \sin 2\theta}{g}$ follows by setting $y = 0$ and eliminating t. The generalised version — landing at height h — is worth more:

Example 1.4 (Landing on a ledge)**.** The launch of figure 1.3 ($v_0 = 20\,\mathrm{m/s}$, $45°$) lands on a platform $h = 5\,\mathrm{m}$ above the launch level. Find the range.

Strategy. Do not memorise a second range formula: solve $y(t) = h$ for the *later* crossing time, then $R = v_{0x}t$.

Solution. $v_{0x} = v_{0y} = 20/\sqrt{2} = 14.14\,\mathrm{m/s}$. Solve $14.14t - 4.9t^2 = 5$, i.e. $4.9t^2 - 14.14t + 5 = 0$: $t = \frac{14.14 \pm \sqrt{199.9 - 98}}{9.8} = \frac{14.14 \pm 10.10}{9.8}$; the later root $t \approx 2.47\,\mathrm{s}$ (the earlier one, 0.41 s, is the upward crossing). Then $R = 14.14 \times 2.47 \approx 34.9\,\mathrm{m}$ — *shorter* than the flat-ground 40.8 m, because the ledge intercepts the descending arc early. Landing 5 m *below* gives $R \approx 45.3\,\mathrm{m}$; the simulator prints both signs of h in section 1.7. ✓

1.5 Circular motion: acceleration without speeding up

Uniform circular motion — constant speed v on a circle of radius r — is accelerated: the velocity vector turns even though its length never changes. Figure 1.4 draws the velocity at two nearby instants: $\mathbf{v}(t)$ and $\mathbf{v}(t + \Delta t)$ are equal-length vectors differing by a small rotation, so $\Delta\mathbf{v}$ points *inward*, and in the limit $\Delta t \to 0$ it points exactly at the centre. The magnitude comes from arc geometry: the velocity vector turns through the same angle the radius does, $\Delta\theta = v\Delta t/r$, so $|\Delta\mathbf{v}| \approx v\,\Delta\theta = v^2\Delta t/r$, giving

$$a_c = \frac{v^2}{r} \quad \text{toward the centre.} \tag{1.3}$$

This is *centripetal acceleration* — required by geometry, supplied by whatever real force (tension, friction, gravity) is handy; chapter 2 matches it to its source.

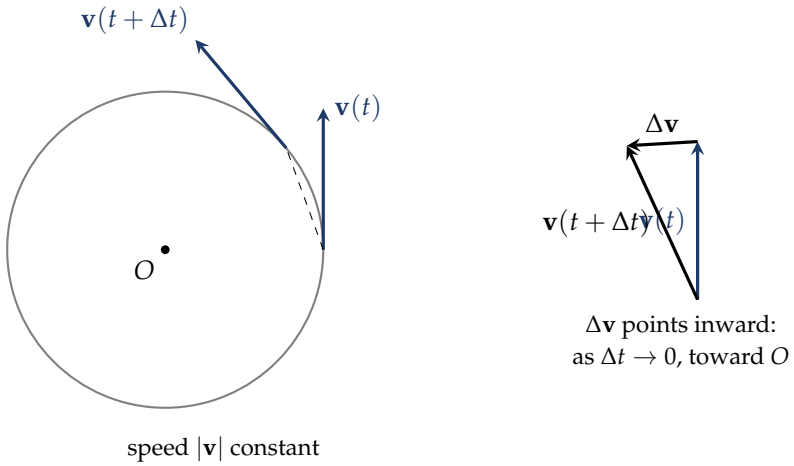


Figure 1.4: Where $a_c = v^2/r$ comes from. Notice the argument needs only that $|\mathbf{v}|$ is constant: the velocity arrow rotates with the radius, so $\Delta\mathbf{v}$ is perpendicular to $\mathbf{v}$ and aimed at the centre.

Example 1.5 (A car on a curve)**.** A car takes a radius-50 m curve at 20 m/s (72 km/h). How large is the sideways acceleration, in units of g?

Strategy. Equation (1.3) directly; compare with g for intuition (your body feels acceleration in g's).

Solution. $a_c = 20^2/50 = 8\,\mathrm{m/s^2} \approx 0.8g$. The tires must supply every newton of it — on ice, 20 m/s around this curve is impossible, which is why ramps are banked (chapter 2). ✓

Exercise 1.3 (medium)**.** A CD spins at 500 rpm. Find a_c at its 6 cm rim in SI and in units of g. (AP-C)

1.6 Relative velocity and Galilean frames

All velocities are relative to a frame; switching frames adds a constant vector:

$$\mathbf{v}_{A/C} = \mathbf{v}_{A/B} + \mathbf{v}_{B/C},$$

the *Galilean velocity addition* — read it as "A as seen from C equals A as seen from B plus B as seen from C". The subscripts chain like a telescope. Figure 1.5 is the one picture to carry: two perpendicular legs and a resultant.

Example 1.6 (River crossing)**.** A boat that makes 4 m/s in still water aims straight across a 200 m river flowing at 3 m/s. When does it land, and where? Then: what heading would land it straight across?

Strategy. Components decouple, exactly like projectile motion: crossing time depends on the across-component only. For the straight crossing, aim upstream so the upstream component of boat velocity cancels the current.

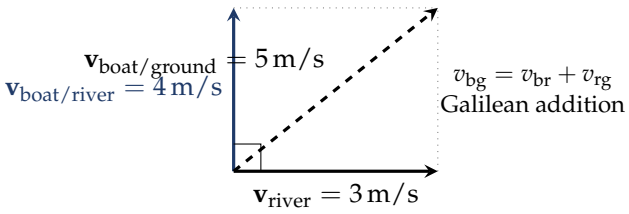


Figure 1.5: Relative velocity as vector addition. Notice the frame B = river does all the work: the boat's thrust is measured relative to the water it pushes against, and the bank sees the sum.

Solution. Across-component 4 m/s: crossing time $t = 200/4 = 50$ s. Drift $= 3 \times 50 = 150$ m downstream — a 3−4−5 triangle, ground speed 5 m/s. To land straight across, aim at angle θ upstream with $4\sin\theta = 3$, i.e. $\theta = \arcsin 0.75 \approx 48.6°$; the effective across-speed drops to $4\cos\theta = \sqrt{7} \approx 2.65$ m/s, so $t \approx 75.7$ s — straighter costs slower. ✓

1.7 The simulator: integration is not optional

Every claim above is checkable by *numerically* running the calculus loop — and in real engineering (drag, control forces, planetary n-body) numerical integration is the only option. The listing below integrates projectile motion by forward Euler, measures the step-size error, then adds quadratic drag — the first preview of chapter 2's terminal velocity.

Checkpoint 1.1 *Predict before you run.* Euler overshoots: v is updated after x, so each step uses a stale velocity. Guess the sign of the range error before looking.

```
def euler_range(v0, th, dt, k=0.0):
    x = y = 0.0
    vx, vy = v0*math.cos(th), v0*math.sin(th)
    while y >= -1e-12:
        s = math.hypot(vx, vy)
        ax, ay = -k*s*vx, -G - k*s*vy # gravity + quadratic drag
        x += vx*dt; y += vy*dt # update position first
        vx += ax*dt; vy += ay*dt # then velocity: forward Euler
    return x
```

Listing 1.1: `code/ch01_projectile.py` — Euler integration vs closed form, then with quadratic drag.

Output of a real run:

```
analytic range: 40.8163 m
euler dt=0.1 range= 42.4264 m error=+1.6101 m
euler dt=0.01 range= 41.0122 m error=+0.1959 m
euler dt=0.001 range= 40.8425 m error=+0.0262 m
with quadratic drag k=0.02: range=25.8593 m (36.6% shorter)
landing at h= +5.0 m: range=34.9825 m (vs 40.8163 on flat ground)
landing at h= -5.0 m: range=45.3195 m (vs 40.8163 on flat ground)
```

Notice two things. First, the error shrinks roughly with h — Euler is first-order — and overshoots positive, exactly as predicted: the stale velocity keeps climbing after it should

turn. Second, drag at $k = 0.02\,\mathrm{m}^{-1}$ cuts the range by a third: real ballistics is not the parabola, and chapter 2's drag force is why.

1.8 In engineering practice: dead reckoning

An inertial measurement unit outputs acceleration, never position — so every drone, submarine, and phone navigation chip lives inside the right half of figure 13.3: $v(t) = v_0 + \int a$, $x(t) = x_0 + \int v$, run thousands of times a second on a tiny processor. The cost is *drift*: each integration accumulates the sensor's bias error, and a 1 mg accelerometer bias becomes a $\sim 0.5\,\mathrm{m}$ position error in ten seconds. GPS exists to reset that integral: satellite fixes pin $x(t)$ periodically so the loop restarts from truth. Dead reckoning — this chapter, continuously — is the engineering version of "kinematics is the calculus loop."

1.9 Common mistakes

SPEED $\neq$ VELOCITY

Distance travelled and displacement differ the moment the path bends or reverses. "How fast" questions can mean speed or velocity — check the sign convention of the problem before answering.

$a_c = v^2/r$ IS NOT A NEW FORCE

It is a *required* acceleration. The forces are tension/friction/ gravity; their sum must equal mv^2/r inward, not the other way around.

INCONSISTENT SIGN OF g

Pick an axis once. With $+y$ up, $a = -9.8$; with $+y$ down, $a = +9.8$. Mixing both inside one solution is the most common source of wrong answers in §1.3–1.4.

CONSTANT-a FORMULAS ON VARYING a

Equations (1.1)–(1.2) need $a = $ const. With drag, springs, or powered flight they are wrong; integrate the loop instead (section 1.7).

ADDING VELOCITIES AS SCALARS

$\mathbf{v}_{A/C} = \mathbf{v}_{A/B} + \mathbf{v}_{B/C}$ is vector addition. A 4-m/s boat in a 3-m/s river does *not* move at 7 m/s — it moves at 5 m/s diagonally (figure 1.5).

1.10 Summary

Idea	One-line version
calculus loop	$x \xrightarrow{d/dt} v \xrightarrow{d/dt} a$; integrate back, paying two constants
constant a	$v = v_0 + at$, $x = x_0 + v_0 t + \frac{1}{2}at^2$, $v^2 = v_0^2 + 2a\Delta x$
projectile	x: const. velocity; y: const. accel $-g$; shared only by t
range on height h	solve $y(t) = h$, take the later root, $R = v_{0x}t$
circular motion	$a_c = v^2/r$ toward centre — required, not a force
relative velocity	$\mathbf{v}_{A/C} = \mathbf{v}_{A/B} + \mathbf{v}_{B/C}$ (vectors)
speed rule	v, a same sign ⇒ speeding up; opposite ⇒ slowing (fig. 1.6)

Table 1.1: Kinematics cheat sheet.

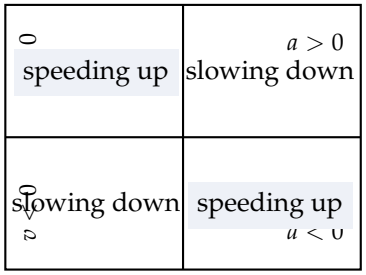


Figure 1.6: The (v, a) sign map. Notice it never asks about position: only the *relative* signs of velocity and acceleration set whether the object speeds up or slows down.

DEFINITION OF DONE

You can now: read v and a off an $x(t)$ graph and off a data table; derive the three constant-acceleration formulas; split projectile motion into components and land it on a ledge; explain where $a_c = v^2/r$ comes from; add velocities through a frame chain; and run the calculus loop numerically with a first-order integrator.

Notes

The calculus-first ordering and the derivations follow MIT 8.01SC weeks 1–3 [2]; AP Physics C unit 1 covers the same ground in the MC/FR split [4]. OpenStax College Physics chapters 3–4 [3] supply the projectile and relative-velocity treatment; the centripetal derivation by velocity difference is Feynman I §8-5 [5]. Halliday–Resnick [6] has the largest drill inventory; Morin [7] is the harder recommendation for anyone who found this chapter easy. The dead-reckoning paragraph sketches the strapdown inertial navigation problem; the bias-drift scaling is standard.

WATCH INSTEAD

1. *1D Kinematics — Speed, Velocity, Acceleration* — Walter Lewin, MIT 8.01x Lecture 2, 50:42, https://youtu.be/q9IWoQ199_o — §1.1–§1.3 at full length, same notation (accessed 2026-10-11).
2. *Why distance is area under the velocity–time line* — Khan Academy, 9:25, https://youtu.be/d-_eqgj5-K8 — the integration half of §1.2 in eight minutes (accessed 2026-10-11).

Further reading

- MIT OpenCourseWare, *8.01SC Classical Mechanics*, weeks 1–3 — the source course; recitation videos solve the same problems. Free at https://ocw.mit.edu, course 8.01SC" (verified 2026-10-11).
- OpenStax, *College Physics 2e*, ch. 3 "Two-Dimensional Kinematics" — free text, more worked projectile examples. At https://openstax.org, search "College Physics 2e" (verified 2026-10-11).
- College Board, *AP Physics C: Mechanics CED*, Unit 1 — the exam blueprint this chapter aligns to; FR verb practice. At https://apstudents.collegeboard.org, course page "AP Physics C: Mechanics" (verified 2026-10-11).
- R. P. Feynman, *The Feynman Lectures on Physics*, vol. I, ch. 8 "Motion" — the prose version, still the best motivation for why derivatives run physics. Free online at Caltech.
- D. Morin, *Introduction to Classical Mechanics*, ch. 2 — kinematics for people who want the hard version; ISBN 978-0521876223.
- Khan Academy, *Physics — one-dimensional motion* — drill exercises matching §1.1–§1.3, at https://www.khanacademy.org/science/physics. Verified 2026-10-11.
- The Physics Classroom, *1-D and 2-D Kinematics* tutorials — readable remediation with check-yourself quizzes, at https://www.physicsclassroom.com (verified 2026-10-11).

Additional exercises

Exercise 1.4 (easy)**.** From the $v(t)$ graph of figure 1.1: read off the launch velocity, the time of the peak, and the landing time — then confirm all three from $x(t) = 20t - 4.9t^2$. (AP-C)

Exercise 1.5 (easy)**.** Convert these accelerations to SI: 32 km/h/s, 0.05 miles/h^2 (in m/s^2, approximately), and 9.8 m/s^2 in km/h^2. (AP-C)

Exercise 1.6 (easy)**.** A ball is dropped from rest at height h. Find its speed after t seconds and after falling d metres — which of the three constant-a formulas did each answer need? (F=ma)

Exercise 1.7 (easy)**.** A train accelerates uniformly from rest to 30 m/s over 600 m. Find a and the elapsed time. (AP-C)

Exercise 1.8 (easy)**.** Displacement vs. distance: a runner does one lap of a 400 m track in 80 s. Compute average velocity and average speed. (AP-C)

Exercise 1.9 (medium)**.** Show that for a launch on level ground the maximum range at fixed v_0 comes at $\theta = 45^\circ$, then show the max range equals $\frac{v_0^2}{g}$ — by maximising $R(\theta)$, not by quoting it. (AP-C FR-style)

Exercise 1.10 (medium)**.** A second crossing: the boat of example 1.6 wants minimum *time* rather than zero drift. Show the fastest crossing points straight across ($t = 50$ s), and compute where it lands. Then find the heading for minimum *total distance travelled over ground*. (F=ma)

Exercise 1.11 (medium)**.** From the dragster table of example 1.2, estimate $v(2.0)$ and $a(2.0)$ by the same symmetric-difference method, then fit $x(t) \approx \frac{1}{2}at^2$ and compare. (F=ma, data analysis)

Exercise 1.12 (medium)**.** A ball is thrown horizontally at 15 m/s off a 30 m cliff. Find the time to land, the range, and the velocity vector (magnitude and direction) at impact. (AP-C)

Exercise 1.13 (medium)**.** On the sign map (figure 1.6): an object has $v < 0$ and $a < 0$. Is it speeding up? Give the everyday example of each quadrant. (AP-C)

Exercise 1.14 (medium)**.** An airplane flies at 200 m/s north relative to air, in a 40 m/s east wind. Find the ground track (heading and ground speed). (F=ma)

Exercise 1.15 (medium)**.** Two cars start 500 m apart and drive toward each other at 25 m/s and 20 m/s. When and where do they meet — solve in the ground frame, then in the frame of the slower car. (F=ma)

Exercise 1.16 (hard)**.** Generalise example 1.4: for a launch at (v_0, θ) onto a ledge at height h, find the angle θ^* that maximises range. Show $\theta^* \to 45^\circ$ as $h \to 0$ and find the leading correction for small h/v_0^2. (USAPhO-style)

Exercise 1.17 (hard)**.** A pursuit with a cap: a cheetah accelerates at 5 m/s^2 but is speed-limited to 30 m/s; a gazelle, 20 m ahead, accelerates at 3 m/s^2 capped at 25 m/s. When does the cheetah catch it? (Track the phase where both are at cap.) (APhO-style)

Exercise 1.18 (hard)**.** From first principles: derive $a_c = v^2/r$ without the velocity-difference argument — write $\mathbf{r}(t) = (r\cos\omega t,\ r\sin\omega t)$ and differentiate twice. Confirm both methods agree on direction and magnitude. (USAPhO-style)

CHAPTER 2

Newton's laws and applications

Kinematics (chapter 1) described motion; *dynamics* explains it. Newton's three laws are exactly three statements — a definition of inertial frames, a relation $\mathbf{F} = m\mathbf{a}$, and a pairing rule — but the engineering content is in the method that wraps them: the free-body diagram, the constraint bookkeeping of strings and surfaces, and the differential equations that friction and drag leave behind.

BEFORE YOU START

Chapter 1: position, velocity, acceleration. *Recap:* acceleration is the rate of change of velocity. Newton's laws say what causes acceleration.

AFTER THIS CHAPTER YOU CAN

- state the three laws and the frame in which they hold;
- draw a free-body diagram for any situation;
- solve inclines, pulleys and circular motion;
- handle friction and drag.

HOW TO THINK ABOUT THIS

Newton's second law is a recipe: find every force on one object, add them as vectors, and divide by the mass. The free-body diagram is how you keep the list honest. If you draw it every time, even hard problems become the same routine.

2.1 The three laws, and the frame assumption

First law. A body with no net force moves at constant velocity — *in an inertial frame*. The law looks circular (how do you know the frame is inertial? check whether first-law behaviour holds) and it is: Newton's first law is the *definition* of the arena his other two laws live in. Accelerating cars and rotating merry-go-rounds are non-inertial; inside them you feel "forces" (centrifugal, elevator heaviness) that have no third-law partner.

Second law.

$$\sum \mathbf{F} = m\mathbf{a}. \tag{2.1}$$

This is a *vector* equation — three scalar equations when you pick axes — and the most compressed sentence in physics: acceleration is what the *sum* of forces buys. Units: force in newtons, $1\,\mathrm{N} = 1\,\mathrm{kg}\cdot\mathrm{m/s^2}$.

Third law. Forces come in pairs on *different* bodies: the Earth pulls you down; you pull the Earth up. The pair never cancels on one body's diagram — a point section 2.3 returns to.

2.2 The force inventory

Only a handful of forces appear in this book:

Force	Rule	Notes
gravity	$m\mathbf{g}$ down	uniform near Earth's surface
normal	whatever it must be	contact; perpendicular to surface
tension	along the rope	pulls both ends; massless rope $\Rightarrow$ same T
spring	$-kx$	Hooke's law, linear
friction	$\le \mu_s N$ static, $= \mu_k N$ kinetic	opposes *impending/actual* sliding
drag	cv or cv^2	opposes velocity; §2.7

The discipline: every entry except gravity is a *contact* force — if nothing touches the body, only $m\mathbf{g}$ acts. "The force of the throw" left the ball when your hand did.

2.3 Free-body diagrams: the discipline

A free-body diagram (FBD) draws *one* body and every force *on* it — nothing else. Not the force it exerts, not its velocity, not its acceleration (draw $m\mathbf{a}$ beside the diagram, never in it). Figure 2.1 is the canonical gallery.

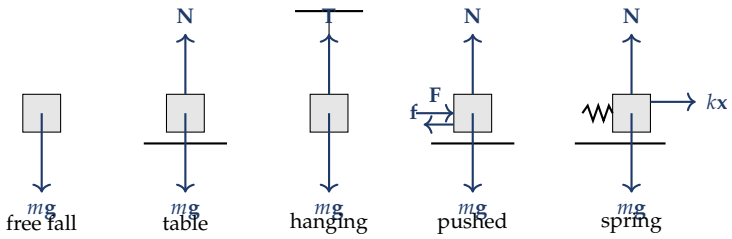


Figure 2.1: Five canonical FBDs. Notice only the body's own forces appear — the Earth's pull counts, the body's pull on the Earth does not. The pushed box's friction arrow points *back* because friction opposes sliding.

The method: (1) isolate one body; (2) enumerate contacts + gravity; (3) pick axes that simplify components; (4) write equation (2.1) per axis; (5) add constraint equations (rope length, contact, $f \le \mu_s N$); (6) solve. Skipping step 2's enumeration is where forces get invented or dropped.

Example 2.1 (Atwood machine)**.** Masses $m_1 = 3\,\text{kg}$ and $m_2 = 5\,\text{kg}$ hang from a light frictionless pulley (figure 2.2). Find the acceleration and tension.

Strategy. Two FBDs, one constraint: the rope's fixed length means the accelerations are equal and opposite. Take each mass's "down is positive" axis along its own motion.

Solution. For m_1 (accelerating up): $T - m_1 g = m_1 a$. For m_2 (accelerating down): $m_2 g - T = m_2 a$. Add to eliminate T:

$$a = \frac{m_2 - m_1}{m_1 + m_2} g = \frac{2}{8} \times 9.8 = 2.45\,\text{m/s}^2, \qquad T = m_1(g + a) = 3 \times 12.25 = 36.75\,\text{N}.$$

Check limits: $m_1 = m_2 \Rightarrow a = 0, T = mg$ (hanging balance); $m_2 \gg m_1 \Rightarrow a \to g$ (small mass is a brake). $T = 36.75\,\text{N}$ lies between $m_1 g = 29.4$ and $m_2 g = 49$ — the moving rope always splits the difference. ✓

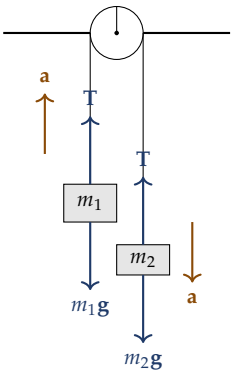


Figure 2.2: Atwood machine: two FBDs, one rope constraint. Notice the tension is the *same* T on both sides — the massless-frictionless ideal pulley transmits, it does not eat.

Exercise 2.1 (easy)**.** In example 2.1, what is the force the pulley's axle must exert on the ceiling? Why isn't it $m_1 g + m_2 g$? (AP-C)

2.4 Constraints: inclines, pulleys, contact

On an incline the useful axes are *down-slope* and *outward* (figure 2.3): gravity decomposes as $mg\sin\theta$ down-slope, $mg\cos\theta$ into the surface, and when the surface doesn't accelerate, $\mathbf{N} = mg\cos\theta$ *only then*. That "only then" is the section's lesson — N is determined, not given.

Example 2.2 (Apparent weight on an accelerating surface)**.** A 60 kg student stands on a scale in an elevator accelerating up at $2\,\text{m/s}^2$. What does the scale read? What if the cable snaps?

Strategy. The scale reads the *normal* force, not gravity. One FBD, one axis (up $+$).

Solution. $N - mg = ma \Rightarrow N = m(g + a) = 60 \times 11.8 = 708\,\text{N}$ — the scale reads 708 N "of weight", i.e. a 72.3 kg equivalent. Free fall $a = -g$: $N = 0$, weightlessness — not "no gravity", just nothing to push against. The ISS floats exactly this way, inside $0.9g$ of gravity. ✓

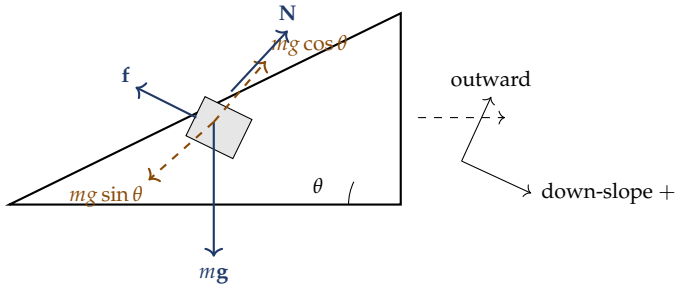


Figure 2.3: Incline decomposition. Notice the axes are rotated, not the force: mg splits into a down-slope driver $mg\sin\theta$ and a pressed-in part $mg\cos\theta$ that the normal force answers.

Exercise 2.2 (medium)**.** A block slides down a frictionless incline at $\theta = 30^\circ$ while the *incline itself* is free to slide on a frictionless floor. Qualitatively: does the block's acceleration exceed or fall short of $g\sin 30^\circ$? Why? (F=ma)

2.5 Circular dynamics: the net force bends

Section 1.5 showed circular motion *needs* $a_c = v^2/r$ inward; Newton's second law says the *net* force must supply it. The mantra: $\sum F_{\text{inward}} = mv^2/r$ — centripetal is what the net force *does*, never an entry on the FBD.

Example 2.3 (Banked curve, two boundary cases)**.** A curve of radius $r = 80\,\text{m}$ is banked at $\theta = 15^\circ$. (a) What speed rounds it with *no* friction? (b) With $\mu_s = 0.4$, what are the max and min speeds?

Strategy. (a) FBD has only mg and N; resolve into vertical (no accel) and horizontal (centripetal). (b) Friction acts along the surface — down-slope at high speed (fighting the tendency to slide *up* the bank), up-slope at low speed.

Solution. (a) Vertical: $N\cos\theta = mg$. Horizontal: $N\sin\theta = mv^2/r$. Divide:

$$v = \sqrt{rg\tan\theta} = \sqrt{80\times 9.8\times 0.268} = 14.5\,\text{m/s} \approx 52\,\text{km/h}.$$

(b) At high speed, friction points down-slope and adds an inward component; the max-speed condition becomes (Morin's derivation) $v_{\max}^2 = rg\,\frac{\sin\theta+\mu_s\cos\theta}{\cos\theta-\mu_s\sin\theta} = 80\times 9.8\times\frac{0.259+0.4\times 0.966}{0.966-0.4\times 0.259}$, giving $v_{\max} = \sqrt{784\times 0.747} = 24.2\,\text{m/s} \approx 87\,\text{km/h}$. The design window 14.5–24.2 m/s is what the bank buys. ✓

Example 2.4 (First attempt: using the push instead of the net force)**.** A 10 kg crate is pushed across a floor with a horizontal force of 30 N; sliding friction is 20 N. A tempting calculation: $a = F/m = 30/10 = 3\,\text{m/s}^2$.

What goes wrong. Newton's second law uses the *net* force, the vector sum of every force on the crate. Friction opposes the motion, so the net force along the floor is $30 - 20 = 10\,\text{N}$ and $a = 10/10 = 1\,\text{m/s}^2$. The tempting answer was three times too large. This is why the free-body diagram comes first: list every force, then add.

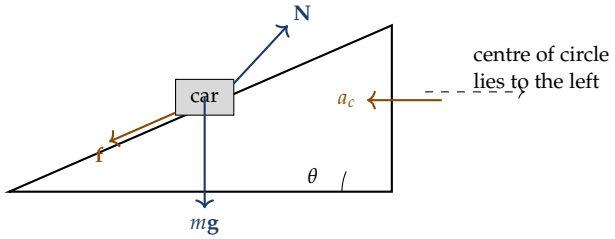


Figure 2.4: Banked curve. Notice the normal force leans inward — the bank's job is to lend a horizontal component to N so friction has less to do.

2.6 Friction: the force that adapts

Static friction *matches* applied force up to a ceiling: $f_s \le \mu_s N$; kinetic friction is a fixed drag $f_k = \mu_k N$ opposing sliding, with $\mu_k < \mu_s$. Figure 2.5 shows the characteristic: a 45-degree ramp that saturates, then drops to the kinetic plateau. The *angle of repose* — the steepest angle before slipping — sets $\tan\theta_{\max} = \mu_s$, a fast way to measure μ_s with a protractor and patience.

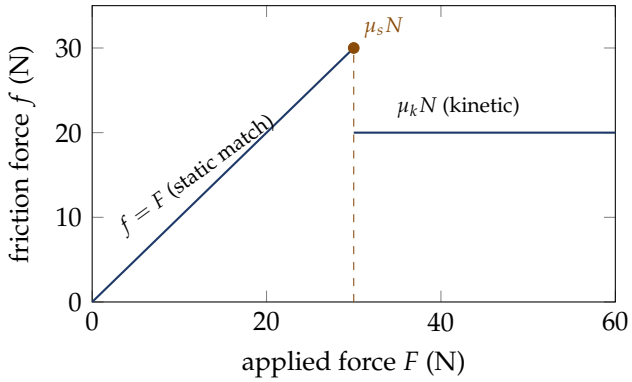


Figure 2.5: Static friction matches, saturates at $\mu_s N$, then drops to the kinetic level $\mu_k N$. Notice the discontinuity: "friction is weaker once you're moving" is why antilock brakes exist.

Checkpoint 2.1 *Friction direction is derived, not guessed.* Two stacked blocks, the bottom one pushed: which way does friction act on the *top* block? If you said "opposes motion", you've guessed — friction on the top block points *forward*: it's the only horizontal force on it, and it's what accelerates it.

Example 2.5 (Conical pendulum)**.** A 0.5 kg mass on a 1.2 m string swings in a horizontal circle at 30° from vertical. Find its speed and period.

Strategy. FBD has mg and T only; T's horizontal component is the centripetal supply. Radius is $L\sin\theta$, not L.

Solution. Vertical: $T\cos\theta = mg \Rightarrow T = 5.66\,\text{N}$. Horizontal: $T\sin\theta = mv^2/r$ with $r = 1.2\sin 30^\circ = 0.6\,\text{m}$: $v = \sqrt{Tr\sin\theta/m} = \sqrt{5.66\times 0.6\times 0.5/0.5} = 1.84\,\text{m/s}$. Period $\tau = 2\pi r/v = 2.05\,\text{s}$ — independent of the mass, a Galileo-clock property. ✓

2.7 Drag: terminal velocity as a fixed point

Drag opposes velocity with a strength that grows with speed: cv (linear, small spheres at low Reynolds number) or cv^2 (quadratic, everything macroscopic in air or water). With quadratic drag,

$$m\dot{v} = mg - c_d v^2$$

is an ODE with a *fixed point*: at $v_t = \sqrt{mg/c_d}$ the right side vanishes and the fall settles. The closed form

$$v(t) = v_t \tanh\left(\frac{gt}{v_t}\right)$$

reaches 76% of v_t at $t = v_t/g$ — for an 80-kg skydiver with $c_d = 0.30$, $v_t = 51.1\,\mathrm{m/s}$, the answer a real dive converges to.

Example 2.6 (Deriving $v(t)$ for quadratic drag)**.** Separate variables in $m\dot{v} = mg - c_d v^2$ and integrate.

Strategy. Write it as $\frac{dv}{g - cv^2/m} = dt$ and substitute $v_t^2 = mg/c_d$ so the denominator is $g(1 - v^2/v_t^2)$ — a standard form giving artanh.

Solution.

$$\int \frac{dv}{1 - v^2/v_t^2} = g\int dt \;\Rightarrow\; v_t \operatorname{artanh}\frac{v}{v_t} = gt \;\Rightarrow\; v(t) = v_t \tanh\frac{gt}{v_t}. \quad \checkmark$$

The listing below confirms the derivation numerically — Euler-Cromer converges to the analytic curve.

```
def eulercromer(cd_over_m, dt, t_end):
    v, x, t = 0.0, 0.0, 0.0
    while t < t_end:
        v += (G - cd_over_m * v * v) * dt # update velocity first
        x += v * dt # then position with NEW v
        t += dt
    return v, x
```

Listing 2.1: `code/ch02_drag_fall.py` — Euler-Cromer on quadratic drag vs the tanh solution.

Output of a real run ($m = 80\,\mathrm{kg}$, $c_d = 0.30\,\mathrm{kg/m}$):

```
terminal velocity analytic: 51.1208 m/s
euler-cromer dt=0.5 v(20s)=51.0967 m/s err=+0.0237 fall=858.0 m
euler-cromer dt=0.1 v(20s)=51.0785 m/s err=+0.0055 fall=841.8 m
euler-cromer dt=0.01 v(20s)=51.0736 m/s err=+0.0006 fall=838.1 m
at t=vt/g=5.22s: v=38.9333 m/s = 76.2% of vt
v= 5 m/s: linear drag a=1.000 m/s^2, quadratic a= 0.094 m/s^2
v= 20 m/s: linear drag a=4.000 m/s^2, quadratic a= 1.500 m/s^2
v= 50 m/s: linear drag a=10.000 m/s^2, quadratic a= 9.375 m/s^2
```

Notice the linear/quadratic crossover in the last lines: linear drag dominates slow objects, quadratic dominates fast ones — a skydiver's v_t sits in the quadratic regime (figure 2.6).

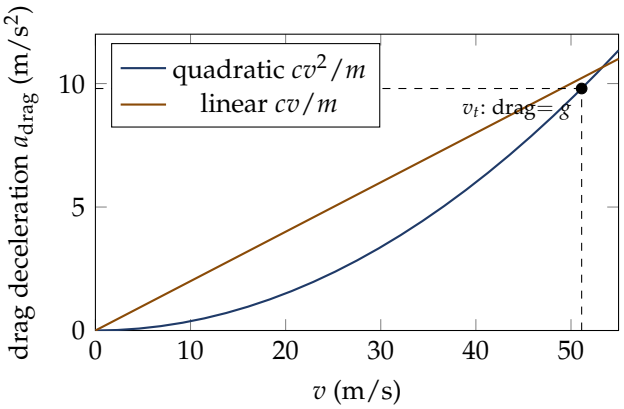


Figure 2.6: Drag deceleration vs speed for the two models. Terminal velocity is the fixed point where drag equals g (marked, $v_t = 51.1\,\text{m/s}$).

2.8 In engineering practice: load paths

Every load-bearing design — bridge truss, building frame, airframe — is a walk through this chapter's FBD chain: gravity enters at the top, travels through members as tension and compression, and exits into the ground. Structural engineers literally draw the discipline of section 2.3 as *free-body cuts*: isolate a joint, draw the member forces, solve $\sum F = 0$, repeat. The banked-curve analysis is the same trick highway designers run for every off-ramp, and the friction ceiling of figure 2.5 is why ABS brakes pump: holding the tire near μ_s, not sliding on the weaker μ_k plateau.

2.9 Common mistakes

mv^2/r DRAWN AS A FORCE

It is the *required net inward force*. If your FBD shows mg, N, *and* a centripetal arrow, double-count.

$N = mg$ ON ACCELERATING SURFACES

N is determined by the constraint, not by gravity. Elevator (example 2.2) and inclines (figure 2.3) both violate $N = mg$.

GUESSING THE FRICTION DIRECTION

Friction opposes *impending sliding at the contact* — which can be forward, as the checkpoint's stacked blocks show. Derive it from the constraint, never from intuition about "motion".

THIRD-LAW PAIRS ON ONE BODY

The pair acts on *different* bodies. "Normal cancels gravity" is not third-law bookkeeping — those two are both on the block.

COMPONENTS BEFORE THE FBD

Picking axes and decomposing first is backwards: enumerate the real forces on the real body, *then* choose axes that simplify the algebra.

2.10 Summary

Idea	One-line version
three laws	inertial frames; $\sum \mathbf{F} = m\mathbf{a}$; pairs on different bodies
FBD method	isolate, enumerate contacts, pick axes, write $\sum F$ per axis
inclines	$mg\sin\theta$ drives, $N = mg\cos\theta$ only when surface is still
constraints	rope length $\Rightarrow$ equal-a; contact $\Rightarrow$ $N \geq 0$
circular dynamics	$\sum F_{\text{inward}} = mv^2/r$; centripetal is a role, not a force
friction	f_s matches up to $\mu_s N$; $f_k = \mu_k N$; $\tan\theta_{\text{repose}} = \mu_s$
drag	cv or cv^2; fixed point $v_t = \sqrt{mg/c_d}$, $v = v_t \tanh(gt/v_t)$

Table 2.1: Newton's laws cheat sheet.

DEFINITION OF DONE

You can now: draw a correct FBD for any of the five canonical cases; solve Atwood-style constraint problems; get the apparent weight in accelerating frames; apply $\sum F_{\text{inward}} = mv^2/r$ to curves and pendulums; predict friction direction from the constraint; and derive + numerically verify terminal velocity under quadratic drag.

Notes

The three-laws presentation and the FBD discipline follow MIT 8.01SC weeks 2–4 [2] and AP Physics C unit 2 [4]. OpenStax College Physics chapters 4–6 [3] cover the same material with a large worked set. The banked-curve and conical-pendulum examples follow Morin chapters 1–3 [7]; the friction characteristic and repose angle are standard [6]. Drag models and the tanh solution follow 8.01SC's resistive-media module; the skydiver numbers are representative, not measured.

WATCH INSTEAD

1. *Newton's Laws* — Walter Lewin, MIT 8.01x Lecture 6, 49:15, https://youtu.be/oduZsA0Tk58 — §2.1–§2.5 at full length, same notation (accessed 2026-10-11).
2. *Types of forces and free body diagrams* — Khan Academy, 8:03, https://youtu.be/MaabUHLIIXA — the §2.3 discipline in eight minutes (accessed 2026-10-11).

Further reading

- MIT OpenCourseWare, *8.01SC Classical Mechanics*, weeks 2–4 — Newton's laws, FBDs, friction and drag with recitation solutions. Free at https://ocw.mit.edu, course "8.01SC" (verified 2026-10-11).
- OpenStax, *College Physics 2e*, ch. 4 "Dynamics" — free text, a second pass at the same laws with more worked FBDs. At https://openstax.org, search "College Physics 2e" (verified 2026-10-11).
- College Board, *AP Physics C: Mechanics CED*, Unit 2 — the exam blueprint for this chapter's FR style. At https://apstudents.collegeboard.org (verified 2026-10-11).
- D. Morin, *Introduction to Classical Mechanics*, chs. 1–3 — FBDs, statics and accelerated frames at the level this book is aiming for; ISBN 978-0521876223.
- The Physics Classroom, *Newton's Laws* tutorial — readable remediation for §2.1–§2.6, at https://www.physicsclassroom.com/class/newtlaws (verified 2026-10-11).
- Khan Academy, *Forces and Newton's laws of motion* — drills matching every section here, at https://www.khanacademy.org/science/physics (verified 2026-10-11).

Additional exercises

Exercise 2.3 (easy)**.** Which of these belongs on a block-on-table FBD: (a) gravity on the block, (b) the block's pull on the table, (c) friction on the block, (d) the block's weight on the table. Justify each keep/drop. (AP-C)

Exercise 2.4 (easy)**.** When does $N = mg\cos\theta$ hold on an incline? Give a case where it fails. (AP-C)

Exercise 2.5 (easy)**.** A 70 kg engineer weighs "70 kg" on Earth. Convert to SI force; then compute what a scale reads on the Moon ($g = 1.6\,\mathrm{m/s^2}$) and inside an elevator accelerating down at $3\,\mathrm{m/s^2}$. (AP-C)

Exercise 2.6 (easy)**.** A rope holds a 10 kg mass static. Find the tension; find it again while the mass accelerates up at $2\,\mathrm{m/s^2}$. (F=ma)

Exercise 2.7 (easy)**.** A car's brakes can produce deceleration $0.8g$ on dry pavement ($\mu_k = 0.8$). Estimate μ_k on ice if the same car needs $5\times$ the dry distance to stop. (AP-C)

Exercise 2.8 (medium)**.** Two stacked blocks: $m = 2\,\mathrm{kg}$ on $M = 5\,\mathrm{kg}$, frictionless floor, $\mu_s = 0.4$ between them. What is the largest horizontal force on M for which m does not slip? (F=ma)

Exercise 2.9 (medium)**.** Conical pendulum redux: keep L and θ of example 2.5 but double the mass. Which of v, T, τ change, and how? (AP-C)

Exercise 2.10 (medium)**.** Atwood on a table: $m_1 = 2\,\mathrm{kg}$ on a frictionless table linked over a pulley to hanging $m_2 = 3\,\mathrm{kg}$. Find a and T; repeat with $\mu_k = 0.2$ on the table. (AP-C)

Exercise 2.11 (medium)**.** The elevator of example 2.2 accelerates at $+2\,\mathrm{m/s^2}$ for 3 s, cruises, then brakes at $-3\,\mathrm{m/s^2}$ to rest. Sketch the scale reading vs. time, and compute the peak reading. (F=ma)

Exercise 2.12 (medium)**.** Show that on the banked curve of example 2.3, the minimum speed with friction is $v_{\min}^2 = rg\,\frac{\sin\theta - \mu_s\cos\theta}{\cos\theta + \mu_s\sin\theta}$, and evaluate it. What happens when $\mu_s \geq \tan\theta$? (F=ma)

Exercise 2.13 (medium)**.** A skydiver with $c_d/m = 0.004\,\mathrm{m^{-1}}$ reaches v_t; estimate v_t, the time to 90% v_t, and the distance fallen by then. (APhO-style)

Exercise 2.14 (hard)**.** A rope of mass m and length L lies on a frictionless table with a length x hanging over the edge. Show its acceleration is $a = gx/L$ and find $x(t)$ — the classic variable-length element problem. (USAPhO-style)

Exercise 2.15 (hard)**.** A wedge of mass M and angle θ is free to slide on a frictionless floor; a block m slides on its frictionless face. Set up the four equations (two bodies, two axes + constraint) and solve for both accelerations. (APhO-style)

Exercise 2.16 (hard)**.** Estimate the maximum range of a projectile under quadratic drag: linearise for small $c_d v_0^2/(mg)$ and show the leading correction to the no-drag range is $-\frac{4}{3}c_d v_0^3/(m^2 g^2)\cdot(\text{shape factor})$. Compare with the 36.6% loss the ch01 code printed at $k = 0.02$. (USAPhO-style)

Exercise 2.17 (medium)**.** Rederive the angle-of-repose rule: show that a block on an incline starts sliding exactly when $\tan\theta = \mu_s$, and use it to compute μ_s if sliding starts at 22°. (AP-C)

CHAPTER 3

Momentum, impulse and collisions

Force is what changes momentum: $\mathbf{F} = d\mathbf{p}/dt$ is Newton's second law in its original form, and the reason $\mathbf{p} = m\mathbf{v}$ is the right bookkeeping unit for a chapter about things that hit each other. Momentum is the first *conserved* quantity in the book — the thing that survives collisions unchanged — and the discipline of *when* it survives (the system-boundary rule) is the real lesson.

BEFORE YOU START

Chapter 2 (forces, free-body diagrams). *Recap:* momentum is mass times velocity; force is its rate of change.

AFTER THIS CHAPTER YOU CAN

- compute the impulse delivered by a force;
- decide when momentum is conserved and when it is not;
- solve 1-D and 2-D collisions;
- derive the rocket equation.

HOW TO THINK ABOUT THIS

Momentum is the quantity that survives collisions, as long as nothing outside pushes. The skill is choosing your system so that the forces you cannot compute are internal. Then you can predict what happens after, without knowing the messy details of the crash.

3.1 Momentum and impulse

$\mathbf{p} = m\mathbf{v}$ is a vector. Impulse is momentum's denomination of payment:

$$\mathbf{J} = \int \mathbf{F}\, dt = \Delta\mathbf{p}. \tag{3.1}$$

A cricket ball stopped in a glove and a foam wall deliver the same Δp — impulse spreads the payment over different times (figure 3.1), which is why crumple zones, airbags, and bent knees all buy *time*.

Example 3.1 (Impulse from a real $F(t)$ curve)**.** A 0.15 kg baseball hits a bat with the half-sine force profile $F(t) = 8000 \sin(\pi t/10\,\text{ms})$ N. Find the impulse and the exit speed if the ball arrived at 40 m/s.

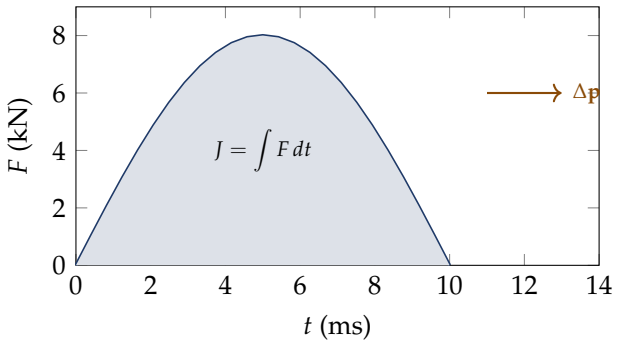


Figure 3.1: Impulse is area under the $F(t)$ curve. Notice the same shaded area can be a tall narrow spike or a long gentle hill — same Δp, very different peak force.

Strategy. Area under the curve: the half-sine's average height is $2/\pi$ of peak over its 10 ms width.

Solution. $J = \int_0^{10} F\,dt = \frac{2}{\pi} \times 8000 \times 0.010 = 50.9\,\text{N s}$. Reversing the ball needs $\Delta p = 2mv = 2 \times 0.15 \times 40 = 12\,\text{N s}$ — the bat over-supplies: exit speed satisfies $m(v_{\text{out}} + 40) = 50.9$, giving $v_{\text{out}} = 299\,\text{m/s}$. (Real bats don't: much of J deforms the ball, so effective impulse is lower — but the method is the point.) ✓

3.2 Systems: the boundary is a choice

Internal forces come in third-law pairs and cancel inside the sum; only *external* forces move the total momentum. Drawing the dashed line of figure 3.2 is a *decision*: put both blocks inside and the contact force vanishes from the bookkeeping; put it outside and the same force is external and changes **p**.

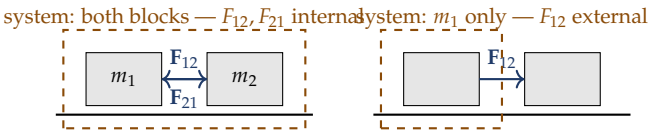


Figure 3.2: The same collision under two system choices. Internal forces cancel by the third law; external ones change the system's **p**.

Example 3.2 (First attempt: conserving the ball's own momentum)**.** A 0.50 kg ball hits the floor moving down at 4.0 m/s and rebounds up at 3.0 m/s. "Momentum is conserved," says the beginner, "so the ball should keep moving down at 4.0 m/s."

What goes wrong. Momentum is conserved for a *system with no net external impulse*. The ball alone is not such a system: the floor pushes on it. Taking up as positive, the ball's momentum changes from $0.50 \times (-4.0) = -2.0$ to $0.50 \times (+3.0) = +1.5\,\text{kg m/s}$, a change of $+3.5\,\text{kg m/s}$. That is the impulse the floor delivered. If you include the Earth in the system, the Earth gains $-3.5\,\text{kg m/s}$ and the total is conserved — the Earth's resulting velocity is far too small to see. The applicability rule is about choosing the system.

3.3 Conservation of momentum: the applicability rule

MOMENTUM CONSERVATION CONDITION

$\sum \mathbf{p}$ is conserved for a system exactly when the *net external impulse* during the event is zero. Gravity during a 10 ms collision contributes $\sim 10^{-3}$ N s — negligible. Gravity during a 10 s flight is not.

This is why "momentum is conserved in collisions" has the silent rider "while the collision lasts". The ballistic pendulum's swing, the projectile's arc — neither conserves $\mathbf{p}$; the *impact* does.

3.4 Centre of mass: the system's average representative

$$\mathbf{r}_{\text{cm}} = \frac{1}{M}\sum_i m_i \mathbf{r}_i, \qquad M\mathbf{A}_{\text{cm}} = \mathbf{F}_{\text{ext}}.$$

The CM obeys single-particle mechanics under the external force alone — internal explosions, splits, and collisions can't move it (figure 3.3). A wrench thrown across the room tumbles in a mess; its CM traces the clean parabola of chapter 1.

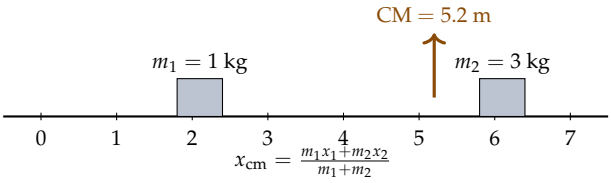


Figure 3.3: CM of two masses: x_{cm} weights each position by mass. Notice it sits nearer the heavy block — CM is the mass-weighted average, not the midpoint.

Example 3.3 (Person on a movable cart)**.** A 60 kg person stands on a 20 kg cart (wheels frictionless) and walks 3 m forward along the cart. How far does the cart move under them?

Strategy. No external horizontal force $\Rightarrow$ the CM of person+cart doesn't move. The person moves 3 m relative to the *cart*, not the ground — the classic mistake.

Solution. Let the cart move $-x$ (backward) while the person moves $3 - x$ forward on the ground. CM fixed: $60(3-x) + 20(-x) = 0 \Rightarrow 180 = 80x \Rightarrow x = 2.25$ m. The cart slides back 2.25 m; the person gains only 0.75 m of ground. This is why stepping off a light boat onto a dock is hazardous. ✓

Exercise 3.1 (easy)**.** Redo example 3.3 with a 90 kg person and a 10 kg cart — does the cart move more or less? Compute it. (AP-C)

3.5 1-D collisions: the two canonical cases

Perfectly inelastic (they stick): momentum alone fixes the answer — the shared speed is total momentum over total mass, $v' = (m_1 v_1 + m_2 v_2)/(m_1 + m_2)$ — and K drops by the fraction that went into deformation. **Elastic**: momentum plus energy conservation (or equivalently, relative velocity reverses). The full elastic solution needs K, so §4.7 of the next chapter closes it; this chapter keeps the momentum half. Figure 3.4 shows the three phases — and that the force between the blocks is internal either way.

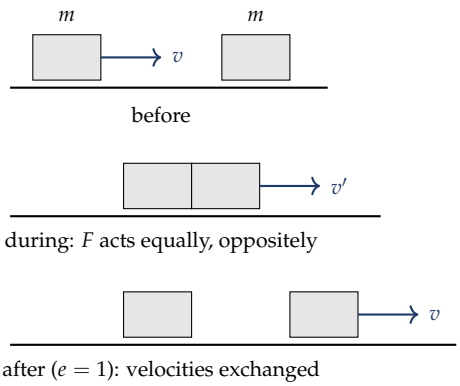


Figure 3.4: Collision sequence. The interaction force is internal to the system whatever the elasticity — p survives, K doesn't always.

Checkpoint 3.1 *Quick K audit.* Equal masses, one moving: perfectly inelastic collision loses exactly half the kinetic energy. Run the numbers yourself before the code below does.

3.6 2-D collisions and explosions

In two dimensions, momentum conservation is two scalar equations — x and y separately (figure 3.5). Explosions are collisions run backward: a stationary object splitting into fragments has $\sum \mathbf{p} = 0$ after as before, so the pieces fly apart with their momenta summing to zero — heavier fragments go slower.

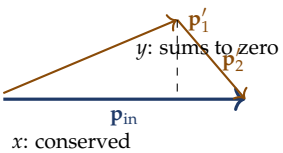


Figure 3.5: A 2-D collision as a vector triangle: $\mathbf{p}_{\text{in}} = \mathbf{p}_1' + \mathbf{p}_2'$. Notice the y-components must cancel — perpendicular momentum can't appear from nothing.

Example 3.4 (Glancing collision)**.** A puck at 6 m/s strikes an identical stationary puck and leaves at 4 m/s, 30° above the original direction. Find the second puck's velocity.

Strategy. Equal masses: velocity conservation is vector conservation. Components: x must return 6, y must net 0.

Solution. x: $4\cos 30^\circ + v_{2x} = 6 \Rightarrow v_{2x} = 6 - 3.464 = 2.54\,\mathrm{m/s}$. y: $4\sin 30^\circ + v_{2y} = 0 \Rightarrow v_{2y} = -2\,\mathrm{m/s}$. $|\mathbf{v}_2| = \sqrt{2.54^2 + 4} = 3.2\,\mathrm{m/s}$ at -38.2°. The angle sum $30 + 38.2 \approx 68^\circ$ — not the elastic equal-mass 90°, so this collision lost energy. ✓

Exercise 3.2 (medium)**.** A 4 kg firework at rest explodes into three pieces: 1 kg north at 12 m/s, 1.5 kg east at 8 m/s. Find the third piece's velocity. (F=ma)

3.7 Continuous mass transfer: the rocket equation

When mass leaves continuously, momentum bookkeeping over one ejection (figure 3.6) gives

$$0 = (m - \Delta m)\,\Delta v - u\,\Delta m \;\xrightarrow{\Delta m \to 0}\; m\,dv = -u\,dm \;\Rightarrow\; \Delta v = u \ln \frac{m_0}{m_f},$$

the *Tsiolkovsky rocket equation* — u the exhaust speed relative to the rocket. The logarithm is why rockets are mostly fuel tank: each equal Δv costs a multiplicative mass ratio.

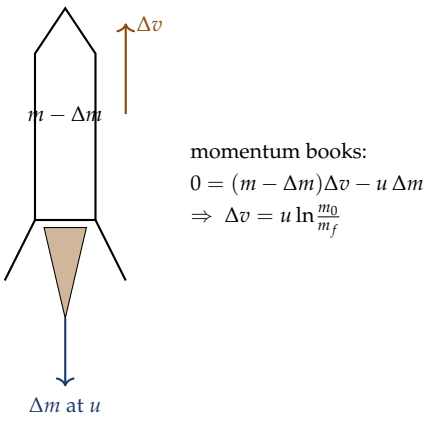


Figure 3.6: One ejection event: throw Δm backward at speed u and conservation of $\mathbf{p}$ pushes the rest forward.

Example 3.5 (Staging estimate)**.** A rocket exhausts at $u = 3\,\mathrm{km/s}$. (a) Single stage to orbit ($\Delta v \approx 9\,\mathrm{km/s}$): mass ratio? (b) Two equal stages?

Strategy. $m_0/m_f = e^{\Delta v/u}$ each stage; stages multiply.

Solution. (a) $e^{9/3} = e^3 \approx 20$: the rocket is 95% propellant. (b) Each stage needs $\Delta v = 4.5\,\mathrm{km/s}$: $e^{1.5} \approx 4.48$ per stage — payload structure is dramatically better. This arithmetic, not optimism, is why real rockets stage. ✓

Example 3.6 (Water-jet thrust)**.** A fire hose delivers 20 L/s of water ($\rho = 1\,\mathrm{kg/L}$) at 15 m/s. What force does the jet exert on a wall it splashes against?

Strategy. Momentum flux: force equals mass flow rate times the speed destroyed, $F = \dot{m}v$.

Solution. $F = (20\,\mathrm{kg/s})(15\,\mathrm{m/s}) = 300\,\mathrm{N}$ — a refrigerator's weight, and the reason hoses need two operators. The same $\dot{m}v$ formula gives a rocket's thrust before the logarithm does the integration. ✓

```
def collide(m1, v1, m2, v2, e):
    v1p = (m1*v1 + m2*v2 - m2*e*(v1-v2)) / (m1+m2)
    v2p = (m1*v1 + m2*v2 + m1*e*(v1-v2)) / (m1+m2)
    return v1p, v2p
```

Listing 3.1: `code/ch03_collision_chain.py` — event-stepped collisions: p survives, K audits the loss.

Output of a real run:

```
initial: p= 4.000 kg m/s K= 8.000 J
after e=0 (1kg+2kg): p= 4.000 kg m/s K= 2.667 J (K lost 5.333 J =
    66.7%)
after e=1 (3kg+3kg): p= 4.000 kg m/s K= 2.667 J v1'=+0.000 v2'=+1.333
e=0.0: va'=+2.000 vb'=+2.000 K=8.000 J (lost 50.0%)
e=0.5: va'=+1.000 vb'=+3.000 K=10.000 J (lost 37.5%)
e=1.0: va'=+0.000 vb'=+4.000 K=16.000 J (lost 0.0%)
```

Read it in order: momentum is conserved to machine precision in every line; kinetic energy is not (the 66.7% loss of the perfectly inelastic hit is the chapter's promise made numeric). The elastic equal-mass exchange ($e = 1$: velocities swap) and the intermediate loss at $e = 0.5$ complete the restitution family.

3.8 In engineering practice: the crash budget

Vehicle crash engineering is impulse management: the occupant's Δp is fixed by the speed, so design maximises collision *time* — crumple zones, airbags, and collapsible steering columns all stretch Δt to shrink F. The same bookkeeping runs in reverse for spacecraft: a Falcon 9's landing burn spends $\dot{m}v$ of thrust exactly as example 3.6 computes, and mass-ratio arithmetic of example 3.5 is why propellant dominates every launch manifest.

3.9 Common mistakes

CONSERVING p WHEN AN EXTERNAL IMPULSE EXISTS

Momentum conservation needs *zero external impulse for the duration* — gravity during a long swing, friction during a long skid, and the dock during a boat step-off all break it.

WRONG SYSTEM BOUNDARY

If the fragment you need sits outside your dashed line, conservation "fails" — draw the boundary to include every piece of the event.

SIGN ERRORS IN 1-D

$\mathbf{p}$ is signed. A rebound at $-v$ contributes $\Delta p = 2mv$, not zero.

BELIEVING K SURVIVES INELASTIC COLLISIONS

Sticking is maximally lossy — the equal-mass case loses half. K audits, $\mathbf{p}$ conserves.

THINKING THE CM PAUSES FOR EXPLOSIONS

The CM keeps its velocity through the breakup — the fragments' weighted sum never moved.

3.10 Summary

Idea	One-line version
impulse	$\mathbf{J} = \int F\,dt = \Delta\mathbf{p}$ — area under $F(t)$
boundary rule	internal forces cancel; $\mathbf{p}$ conserved iff external impulse $= 0$
centre of mass	$M\mathbf{A}_{\text{cm}} = \mathbf{F}_{\text{ext}}$; internal events can't move it
collisions 1-D	inelastic: p only; elastic: $p + K$ (§4.7)
collisions 2-D	conserve x and y separately
rocket	$\Delta v = u \ln(m_0/m_f)$ — staging is arithmetic
thrust	$F = \dot{m}v$ for continuous flux

Table 3.1: Momentum cheat sheet.

DEFINITION OF DONE

You can now: compute impulse from a force–time curve; choose a system boundary and state when $\mathbf{p}$ is conserved; move a cart under a walking person without moving the CM; solve inelastic and 2-D collisions; derive the rocket equation; and read a restitution sweep's K losses.

Notes

The momentum-first presentation follows MIT 8.01SC weeks 3 and 5–6 [2] and AP Physics C unit 4 [4]. OpenStax College Physics ch. 8–9 [3] covers collisions and CM at this level; Morin ch. 5 [7] is the source of the rocket staging and variable-mass exercises. The restitution-sweep numbers are the listing's real output; the baseball numbers are representative.

WATCH INSTEAD

1. *Momentum, Conservation of Momentum, Center of Mass* — Walter Lewin, MIT 8.01x Lecture 15, 52:17, `https://youtu.be/aIhScO3_I50` — §3.1–§3.4 at full length (accessed 2026-10-11).

2. *Elastic and inelastic collisions* — Khan Academy Physics, 14:24, https://youtu.be/8OB8eIPgEkQ — §3.5–§3.6 worked quickly (accessed 2026-10-11).

Further reading

- MIT OpenCourseWare, *8.01SC Classical Mechanics*, weeks 3, 5–6 — momentum and collisions with recitation solutions. At https://ocw.mit.edu, course "8.01SC" (verified 2026-10-11).
- OpenStax, *College Physics 2e*, ch. 8 "Linear Momentum and Collisions" — free text, the same laws with more worked explosions. At https://openstax.org (verified 2026-10-11).
- College Board, *AP Physics C: Mechanics CED*, Unit 4 — the exam blueprint this chapter aligns to. At https://apstudents.collegeboard.org (verified 2026-10-11).
- D. Morin, *Introduction to Classical Mechanics*, ch. 5 — momentum and rockets at the harder level; ISBN 978-0521876223.
- The Physics Classroom, *Momentum and Its Conservation* — readable remediation for the whole chapter, at https://www.physicsclassroom.com/class/momentum (verified 2026-10-11).
- Khan Academy, *Impacts and linear momentum* — drills matching §3.1–§3.6, at https://www.khanacademy.org/science/physics (verified 2026-10-11).

Additional exercises

Exercise 3.3 (easy). A constant 500 N force acts for 0.2 s on a 2 kg mass initially at rest. Find the impulse and final velocity. (AP-C)

Exercise 3.4 (easy). A 1200 kg car moving at 15 m/s is stopped by an identical parked car they stick to. Find the shared velocity. (F=ma)

Exercise 3.5 (easy). Find the CM of three masses: 1 kg at $x = 0$, 2 kg at $x = 3$, 3 kg at $x = 6$. (AP-C)

Exercise 3.6 (easy). Explain why momentum is *not* conserved while a thrown ball is in the air, in one sentence using the boundary rule. (AP-C)

Exercise 3.7 (easy). A cannon fires a 5 kg shell at 400 m/s; the cannon recoils at 2 m/s. Find the cannon's mass. (F=ma)

Exercise 3.8 (medium). Ballistic pendulum, momentum stage only: a 10 g bullet at v stops inside a 2 kg block, which then holds together. Find the block's post-impact speed if $v = 300$ m/s. (The swing is ch4's problem.) (AP-C)

Exercise 3.9 (medium). From the impulse graph of figure 3.1: if the same J were delivered as a *constant* force over 20 ms, what would F be? How does the peak change? (F=ma)

Exercise 3.10 (medium). Water jet on a bucket: the hose of example 3.6 hits a bucket moving away at 5 m/s. Redo the thrust using relative flux. (USAPhO-lite)

Exercise 3.11 (medium)**.** A 70 kg astronaut throws a 2 kg wrench at 6 m/s while floating. Find her recoil, then her drift over the 30 s EVA repair. (F=ma)

Exercise 3.12 (medium)**.** In the CM frame: transform the equal-mass elastic collision of the code listing into the CM frame, show the velocities merely *reverse sign*, and transform back. (F=ma)

Exercise 3.13 (medium)**.** Two pucks collide on air hockey as in example 3.4, but now the second puck emerges at 45° below the original direction with speed 3 m/s. Recompute puck 1's exit. (AP-C)

Exercise 3.14 (medium)**.** A rocket has $u = 2.5\,\mathrm{km/s}$ and needs $\Delta v = 6\,\mathrm{km/s}$ in one stage. Find m_0/m_f, then the propellant fraction. (F=ma)

Exercise 3.15 (hard)**.** A uniform chain of length L and mass m hangs with its lower end just touching a table and is released. Show that when a length x has landed, the table's reaction is $R = \frac{3mx}{L}g$ — the flux term triples the weight. (APhO-style)

Exercise 3.16 (hard)**.** Two-stage optimisation: with total propellant fraction fixed at 90% and $u = 3\,\mathrm{km/s}$, sketch how splitting it 40/50 vs 60/30 between stages changes total Δv — set up, don't solve, the constrained optimum. (USAPhO-style)

Exercise 3.17 (medium)**.** Superball bounce: a ball hits the floor at v and rebounds with restitution $e = 0.9$. Find the impulse delivered and interpret the loss. (F=ma)

CHAPTER 4

Work, energy and power

Force is local; energy is the accounting trick that lets you skip the details. $W = \int \mathbf{F} \cdot d\mathbf{r}$ accumulates what forces deposit, $K = \frac{1}{2}mv^2$ is the currency it arrives in, and potential energy is the bank where conservative forces store it. The chapter's payoff is the right to answer "how fast" without ever solving for the trajectory.

BEFORE YOU START

Chapters 1–3. *Recap:* work is force times distance along the motion; energy is the bank account that work pays into.

AFTER THIS CHAPTER YOU CAN

- compute work for constant and varying forces;
- use the work–energy theorem and conservation of energy, and know when conservation fails;
- read motion from an energy diagram;
- do an energy audit of a real system.

HOW TO THINK ABOUT THIS

Energy is bookkeeping for motion: it changes form but is never lost, and when it seems to disappear it has gone into heat or sound. Often the shortest route to a speed or height is to compare energy at two moments and skip the forces entirely.

4.1 Work: force collected along a path

$$W = \int \mathbf{F} \cdot d\mathbf{r}. \tag{4.1}$$

Work is a *path integral*: only the force component along the motion counts. The centripetal force of §1.5 does *no* work — always perpendicular, always at right angles to $d\mathbf{r}$, which is why uniform circular motion keeps its speed. Figure 4.1 is the one-dimensional version: area under $F(x)$.

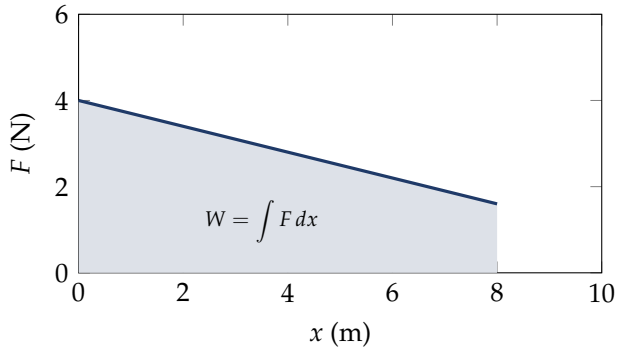


Figure 4.1: Work as area under the $F(x)$ curve. Notice variable forces need the integral; the constant-force $W = Fd\cos\theta$ is only its rectangle.

4.2 The work–kinetic-energy theorem

Start from $\mathbf{F} = m\,d\mathbf{v}/dt$ and dot with $d\mathbf{r}$:

$$\mathbf{F}\cdot d\mathbf{r} = m\frac{d\mathbf{v}}{dt}\cdot d\mathbf{r} = m\,\mathbf{v}\cdot d\mathbf{v} = d\left(\tfrac{1}{2}mv^2\right),$$

so

$$W_{\text{net}} = \Delta K. \tag{4.2}$$

Every joule of net work reappears exactly as kinetic energy — the theorem is a tautology dressed as a tool, and it works for any path, including ones too ugly to integrate force over directly.

4.3 Potential energy: where conservative forces bank work

A force is *conservative* when the work it does depends only on the endpoints. Then you can declare a stored quantity: $\Delta U = -W_{\text{by force}}$. The two workhorses:

$$U_g = mgy, \qquad U_s = \tfrac{1}{2}kx^2.$$

Friction fails the test — the longer the path, the more it takes — so there is no "potential energy of friction"; its work leaves the mechanical books entirely (section 4.8).

Example 4.1 (First attempt: multiplying force by distance without checking direction)**.** You carry a 20 kg box 10 m across a level floor at constant speed. A quick estimate of the work you did: $W = mgd = 20 \times 9.8 \times 10 = 1960\,\text{J}$.

What goes wrong. Work is $W = \mathbf{F}\cdot\mathbf{d} = Fd\cos\theta$. The force you apply to hold the box up is vertical ($mg = 196\,\text{N}$, upward) and the displacement is horizontal, so $\theta = 90^\circ$ and $\cos\theta = 0$: you do *no* work on the box. (Your muscles still burn energy, which is a different matter entirely.) The 1960 J answer would be right if you lifted the box 10 m straight up.

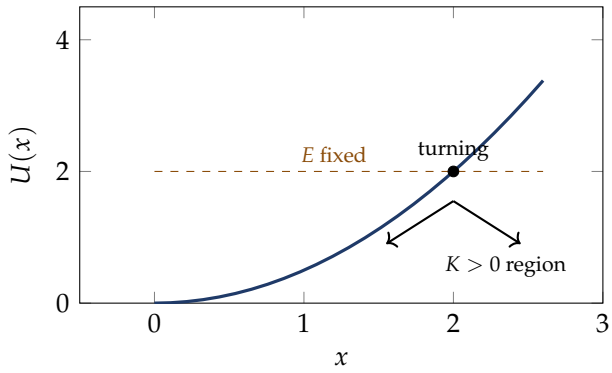


Figure 4.2: Reading a $U(x)$ diagram. With E fixed, motion lives where $K = E - U > 0$; the turning point is where the parabola meets the line. The restoring force is $-dU/dx$, the slope reversed.

4.4 Conservation of mechanical energy — and its failure modes

When only conservative forces work,

$$K_i + U_i = K_f + U_f. \tag{4.3}$$

Figure 4.3 is the account statement: the bars trade height, never total. The failure modes are equally instructive: friction converts, deformation absorbs, and anything with a motor adds from outside — all of them break equation (4.3), all handled by section 4.8's audit.

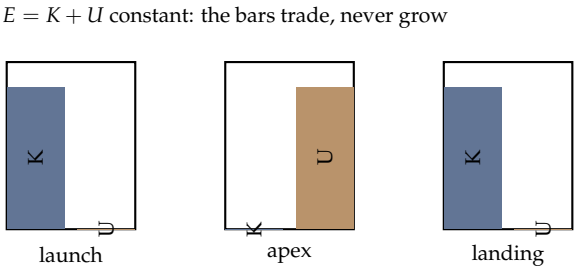


Figure 4.3: Energy bars along a trajectory: K and U trade, the sum doesn't move.

Example 4.2 (Minimum release height for the loop-the-loop)**.** A cart starts from rest at height h on a frictionless track containing a loop of radius r (figure 13.3). Minimum h to make it around?

Strategy. Two chapters meet: energy sets speeds (equation (4.3)), circular dynamics (ch2) sets the requirement — at the top, $N \geq 0$ means $v^2 \geq gr$.

Solution. Top of loop is at height $2r$; borderline case has $v_{\text{top}}^2 = gr$. Energy: $mgh = mg(2r) + \frac{1}{2}m\,gr$, so

$$h = \frac{5}{2}r.$$

Real tracks need more — friction and the cart's own rotation (ch5) both take a share. ✓

Example 4.3 (Spring-launched block up a rough incline)**.** A spring ($k = 200\,\text{N/m}$) compressed 0.3 m launches a 0.5 kg block up a $\theta = 30^\circ$ incline with $\mu_k = 0.2$. How far along the slope does it go?

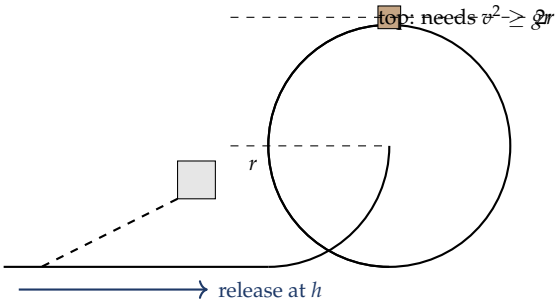


Figure 4.4: The loop-the-loop energy budget. The $5r/2$ answer is the classic meeting point of ch1's $a_c = v^2/r$ and this chapter's bookkeeping.

Strategy. Energy audit with friction present: spring's deposit minus friction's withdrawal equals the height gained.

Solution. $U_s = \frac{1}{2}(200)(0.09) = 9\,\text{J}$ available. Along distance d: gravity takes $mgd\sin 30° = 2.45d\,\text{J}$, friction takes $\mu_k mg\cos 30° \cdot d = 0.849d\,\text{J}$. So $d = 9/(2.45 + 0.849) = 2.73\,\text{m}$ — vs 3.67 m frictionless; friction's share is a quarter of the budget. ✓

Example 4.4 (Pendulum speed at the bottom)**.** A 0.5 kg mass on a 1 m string is released from horizontal. Speed at the bottom?

Strategy. Pure energy — the tension does no work (always perpendicular to the arc), so mechanical energy conserves.

Solution. $mgL = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2gL} = \sqrt{19.6} = 4.43\,\text{m/s}$ — independent of mass and of the path taken to get there. The same $\sqrt{2gh}$ a frictionless slide gives: energy doesn't care about trajectories. ✓

Exercise 4.1 (easy)**.** Find the pendulum's speed when the string passes 45° from vertical. (AP-C)

4.5 Energy diagrams: motion read off a picture

Given $U(x)$ (figure 4.2): total energy E draws a horizontal line; kinetic energy is the gap below it; turning points are where the line crosses the curve; equilibria sit at the curve's extrema (stable at minima, unstable at maxima). A particle in a quartic double well can *never* leave one side if its line dips below the saddle — that is the whole qualitative solution, no ODE required.

4.6 Power: the rate of doing work

$$P = \frac{dW}{dt} = \mathbf{F} \cdot \mathbf{v}. \tag{4.4}$$

A 1000 kg car climbing a 6% grade at 20 m/s burns $P = mg\sin\theta \cdot v \approx 11.8$ kW on gravity alone — before drag's share, which figure 4.5 adds.

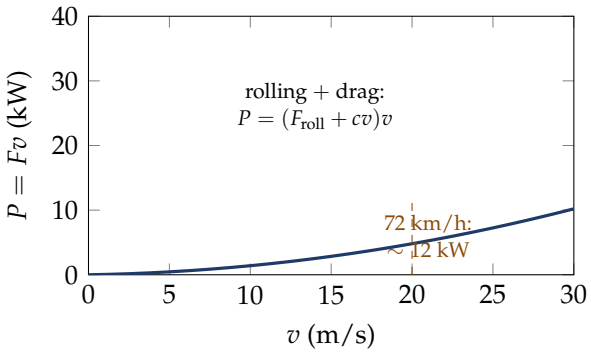


Figure 4.5: $P = \mathbf{F} \cdot \mathbf{v}$: climbing power is linear in v, drag power quadratic-ish. Notice how quickly the sum leaves the city-driving budget.

Example 4.5 (Stair-climb power measurement)**.** You (65 kg) run up a 4 m stair flight in 5 s. Estimate your mechanical power output.

Strategy. $P = \Delta U/t$ — the simplest honest measurement in the book.

Solution. $\Delta U = mgh = 65 \times 9.8 \times 4 = 2548$ J, so $P \approx 510$ W ≈ 0.7 hp. A cyclist sustains ~ 250 W; elite sprinters peak near 2 kW for seconds. ✓

Exercise 4.2 (medium)**.** A pump lifts 200 L/min of water 10 m. Minimum motor power? (F=ma)

4.7 Elastic collisions: the deferred full treatment

Chapter 3 owned momentum; energy completes the elastic case. The v_{rel}-flip shortcut: in an elastic collision the relative velocity *reverses* — $v_{2f} - v_{1f} = -(v_{2i} - v_{1i})$ — which with momentum alone solves everything.

Example 4.6 (Elastic collision, three ways)**.** $m_1 = 2$ kg at 6 m/s hits stationary $m_2 = 2$ kg. Solve by (a) lab-frame algebra, (b) the CM frame, (c) the flip rule.

Strategy. Three routes to the same answer — each teaches a different skill.

Solution. (a) $v_1' + v_2' = 6$ (momentum) and $v_1'^2 + v_2'^2 = 36$ (energy): substituting gives $v_1'v_2' = 0$, so one stops, one carries 6 m/s — velocities swap. (b) CM moves at 3 m/s; in its frame the blocks approach at ± 3 and must leave at ∓ 3 (reversal is the only K-preserving option). Transforming back: 0 and 6. (c) Flip: $v_2' - v_1' = 6$ plus $v_1' + v_2' = 6$ gives $v_1' = 0$, $v_2' = 6$. All three agree; the equal-mass swap is figure 4.6. ✓

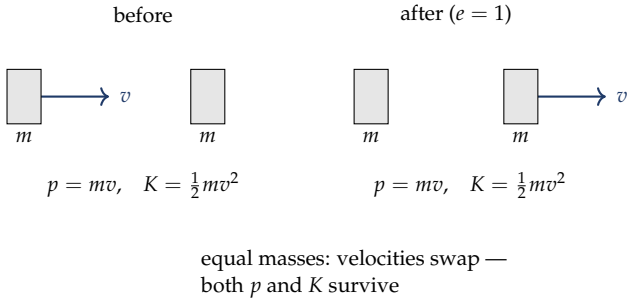


Figure 4.6: Bookkeeping for the equal-mass elastic hit: both p and K survive — a rare event, reserved for $e = 1$.

Example 4.7 (General elastic formula)**.** Apply the flip rule to unequal masses: $m_1 = 5\,\text{kg}$ at $8\,\text{m/s}$ hits $m_2 = 1\,\text{kg}$ at rest.

Strategy. Two equations: momentum and the reversed relative velocity.

Solution. $5v_1' + v_2' = 40$ and $v_2' - v_1' = 8$: subtracting, $6v_1' = 32 \Rightarrow v_1' = 5.33\,\text{m/s}$, $v_2' = 13.33\,\text{m/s}$. Check K: $\frac{1}{2}(5)(28.4) + \frac{1}{2}(1)(177.7) = 71 + 88.9 = 160\,\text{J} = \frac{1}{2}(5)(64)$. ✓

4.8 Non-conservative work: the energy audit

With friction present, equation (4.3) becomes the audit

$$K_i + U_i + W_{\text{nc}} = K_f + U_f, \qquad W_{\text{nc}} = -f_k d < 0 \text{ (usually).}$$

The missing energy isn't gone — it's heat, sound, and deformation. The conveyor belt is the classic trap: dragging a block up to belt speed heats the belt-block interface by exactly the kinetic energy the block gains ($Q = K$) — half the motor's work becomes motion, half becomes warmth.

```
if scheme == "euler":
    x_new = x + v * dt
    v = v - (k/m) * x * dt # BOTH updates use OLD values
    x = x_new
else: # symplectic
    v += -(k/m) * x * dt # update v first
    x += v * dt # then x with the NEW v
```

Listing 4.1: `code/ch04_energy_check.py` — plain Euler vs symplectic Euler on a block-spring.

Output of a real run ($k = 4$, $m = 1$, released from $x = 1$):

```
E0 = 2.0000 J
scheme dt periods E_end drift
euler 0.1 10 446168.12 22308306.0%
euler 0.05 10 1034.79 51639.6%
symplectic 0.1 10 1.97 -1.5%
symplectic 0.05 10 2.00 0.1%
euler 100 periods, dt=0.1: E=446168.120 J (x 223084.1 of original --
    secular growth)
```

```
sympl 100 periods, dt=0.1: E=1.971 J (bounded oscillation, no secular
   drift)
```

The checkpoint's prediction is now a number: plain Euler's error compounds multiplicatively (the orbit spirals *out*), while symplectic Euler — update v before x — conserves a nearby shadow Hamiltonian and merely oscillates around the true energy. Every orbit simulation after this chapter uses the symplectic variant.

```
m, v, g = 1200.0, 27.0, 9.81 # kg, m/s (~60 mph)
Cd, A, rho = 0.30, 2.2, 1.2
KE = 0.5 * m * v**2
PE_100m = m * g * 100
F_drag = 0.5 * rho * Cd * A * v**2
P_drag = F_drag * v
print(f"KE at 60 mph : {KE/1e3:8.1f} kJ")
print(f"PE climbing 100 m : {PE_100m/1e3:8.1f} kJ")
print(f"drag force : {F_drag:8.0f} N")
print(f"drag power at 60 mph : {P_drag/1e3:8.1f} kW")
print(f"1 min of drag work : {P_drag*60/1e3:8.1f} kJ = {P_drag*60/KE
    :.2f} x KE")
print(f"double the speed: drag power x{2**3}")
```

Listing 4.2: `code/ch04_energy_budget.py` — one car, three energy lines: KE, a climb's potential, drag's running cost.

Output of a real run:

```
KE at 60 mph : 437.4 kJ
PE climbing 100 m : 1177.2 kJ
drag force : 289 N
drag power at 60 mph : 7.8 kW
1 min of drag work : 467.7 kJ = 1.07 x KE
double the speed: drag power x8
```

The audit surprises: a 100 m climb costs nearly $3\times$ the car's entire highway kinetic energy — hills are the expensive part of the map — while one minute of cruise drag burns the KE outright. The last line is the speed trap: doubling v makes drag *power* eight-fold, which is exactly why fuel economy falls off a cliff above 90 km/h.

4.9 In engineering practice: the energy audit

Regenerative braking is this chapter's audit run as accounting: an electric car descending a hill deposits into the battery at 70–85% efficiency — not 100%, because the audit of section 4.8 still applies (aerodynamic drag, inverter losses, battery chemistry each take a line). Roller-coaster designers run example 4.2 backward: given a required top speed, the lift hill is $h = 5r/2$ plus an empirically tuned friction allowance — which is why first hills tower over every loop.

4.10 Common mistakes

SIGN OF WORK VS ΔU

$W_{\text{by}} = +\Delta K$; $\Delta U = -W_{\text{by conservative}}$. Mixing "work done on" with "work done by" flips every sign in the audit.

MIXING PE ZERO POINTS

$U = mgh$ measures against *your* chosen floor; two different floors in one problem corrupt the difference.

CONSERVING MECHANICAL ENERGY THROUGH FRICTION

Friction withdraws before gravity deposits — example 4.3 lost a quarter of its budget to it.

$P = Fv$ WITH THE WRONG v

The velocity in $P = \mathbf{F} \cdot \mathbf{v}$ is the *point of application's* — for a moving belt problem it's the belt, not the block.

"CONTACT FORCES DO NO WORK" ON DEFORMABLES

True for rigid bodies; false for cars, balls, and people — deformation absorbs real joules.

4.11 Summary

Idea	One-line version
work	$W = \int \mathbf{F} \cdot d\mathbf{r}$; only the along-motion part counts
WKE theorem	$W_{\text{net}} = \Delta K$ — derived, not assumed
potential energy	$\Delta U = -W_{\text{cons}}$; mgy and $\frac{1}{2}kx^2$
conservation	$K + U$ constant iff only conservative forces work
energy diagram	turning points where E meets U; equilibria at U' extrema
power	$P = \mathbf{F} \cdot \mathbf{v}$
elastic collisions	relative velocity flips; with p it solves everything
energy audit	W_{nc} bridges the books; friction $\Rightarrow -f_k d$

Table 4.1: Energy cheat sheet.

DEFINITION OF DONE

You can now: integrate variable forces; derive the work–energy theorem; read motion off $U(x)$; size a loop-the-loop hill; audit friction's share; solve elastic collisions three ways; and explain why symplectic integrators run the book's orbits.

Notes

The work–energy ordering follows MIT 8.01SC weeks 7–9 [2] and AP Physics C unit 3 [4]. OpenStax College Physics ch. 6–7 [3] has parallel treatment; Morin ch. 5 [7] is the source of the energy-diagram exercises. The integrator comparison is the standard motivation for symplectic schemes; loop-the-loop numbers are the classic result.

WATCH INSTEAD

1. *Work, Kinetic & Potential Energy, Gravitation, Conservative Forces* — Walter Lewin, MIT 8.01x Lecture 11, 49:06, https://youtu.be/9gUdDM6LZGo — §4.1–§4.4 at full length (accessed 2026-10-11).
2. *Work-energy theorem* — Khan Academy India, 12:34, https://youtu.be/nnMirzeTepQ — the §4.2 derivation worked quickly (accessed 2026-10-11).

Further reading

- MIT OpenCourseWare, *8.01SC Classical Mechanics*, weeks 7–9 — work, energy and collisions with recitation solutions. At https://ocw.mit.edu, course "8.01SC" (verified 2026-10-11).
- OpenStax, *College Physics 2e*, ch. 6–7 "Work, Energy, and Energy Resources" — free text at https://openstax.org (verified 2026-10-11).
- College Board, *AP Physics C: Mechanics CED*, Unit 3 — the exam blueprint for this chapter. At https://apstudents.collegeboard.org (verified 2026-10-11).
- D. Morin, *Introduction to Classical Mechanics*, ch. 5 — energy methods at the harder level; ISBN 978-0521876223.
- The Physics Classroom, *Work, Energy, and Power* — readable remediation with roller-coaster walkthroughs, at https://www.physicsclassroom.com (verified 2026-10-11).
- Khan Academy, *Work and energy* — drills for this whole chapter, at https://www.khanacademy.org under Science → Physics (verified 2026-10-11).

Additional exercises

Exercise 4.3 (easy)**.** Sign reasoning: classify the work of gravity, normal, and friction as $+$, $-$, or 0 for (a) a block sliding down an incline, (b) carried up the same incline at constant speed, (c) a satellite in circular orbit. (AP-C)

Exercise 4.4 (easy)**.** A crane lifts a 500 kg load at 0.5 m/s steadily. Find the motor's power. (AP-C)

Exercise 4.5 (easy)**.** A spring ($k = 400\,\mathrm{N/m}$) stores how much energy at 0.1 m compression? What mass could it lift 1 m with that energy? (AP-C)

Exercise 4.6 (easy)**.** Read figure 4.3: if the launch bar holds 8 J and the mass is 0.5 kg, find the launch speed and apex height. (F=ma)

Exercise 4.7 (easy)**.** A child slides down a 3 m frictionless slide. Find the exit speed, then explain why the slide's shape doesn't matter. (AP-C)

Exercise 4.8 (medium)**.** Loop minimum speed: recompute example 4.2 requiring $v_{\text{top}}^2 = 2gr$ (a safety margin) — what h now? (F=ma classic)

Exercise 4.9 (medium)**.** Spring + rough incline: example 4.3 again with $k = 300\,\mathrm{N/m}$, $\mu_k = 0.3$, $\theta = 20^\circ$ — find d, then the block's speed back at the launch point after sliding down. (AP-C FR-style)

Exercise 4.10 (medium)**.** Stair power: have a partner time you on a real flight; compute your watts and compare against the 510 W estimate. (F=ma, data)

Exercise 4.11 (medium)**.** A pendulum released from 60° clears the bottom at what fraction of the 90°-release speed? Work it in symbols first. (AP-C)

Exercise 4.12 (medium)**.** Elastic collision: $m_1 = 3\,\mathrm{kg}$ at $10\,\mathrm{m/s}$ hits $m_2 = 1\,\mathrm{kg}$ at rest — find both exit speeds by the flip rule and verify K survives. (AP-C)

Exercise 4.13 (medium)**.** Conveyor heating: a 2 kg package accelerates to a 3 m/s belt over a 1 m friction patch. Show the belt interface dissipates exactly the package's gained K, then compute both. (F=ma)

Exercise 4.14 (medium)**.** A 1500 kg car needs 25 kW to hold 30 m/s on a flat road. Estimate the combined drag + rolling resistance at that speed, then the power to hold 30 m/s up a 5% grade. (engineering)

Exercise 4.15 (hard)**.** A particle moves in $U(x) = ax^4 - bx^2$ ($a, b > 0$). Sketch U, find the equilibria and their stability, and the energy ranges where motion is trapped in one well vs. free between wells. (USAPhO-style)

Exercise 4.16 (hard)**.** Elastic collision, unequal masses: derive the general solution $v_1' = \frac{m_1 - m_2}{m_1 + m_2} v_1$, $v_2' = \frac{2m_1}{m_1 + m_2} v_1$, and verify the limits $m_2 \to \infty$ (wall) and $m_1 = m_2$ (swap). (USAPhO-style)

Exercise 4.17 (medium)**.** A spring ($k = 400\,\mathrm{N/m}$) launches a 0.5 kg block up a 30° incline with $\mu_k = 0.2$: find how far it climbs before stopping if the spring starts compressed 10 cm. (AP-C FR-style)

CHAPTER 5

Rotation I: kinematics, torque and statics

Everything so far treated bodies as points. Real objects have extent — they spin, they lean, they stay put under balanced torques. The good news is that rotation is a translation of every idea in the book so far: figure 5.1 is the dictionary, and it never lies.

linear	angular
x, v, a	θ, ω, α
m	$I = \sum mr^2$
$F = ma$	$\tau = I\alpha$

every linear law has an angular twin — substitute the dictionary, keep the algebra

Figure 5.1: The linear–angular dictionary. Every equation in this chapter is a ch1–ch2 equation with the entries swapped.

BEFORE YOU START

Chapters 1–4. *Recap:* rotation has its own versions of position, velocity, force and mass: angle, angular velocity, torque and moment of inertia.

AFTER THIS CHAPTER YOU CAN

- describe rotation with angle, angular velocity and acceleration;
- apply the no-slip rolling condition;
- compute torque and moments of inertia;
- solve statics problems with two balance conditions.

HOW TO THINK ABOUT THIS

Rotation has the same story as straight-line motion, with new names: angle for position, torque for force, moment of inertia for mass. What is new is that where you push matters as much as how hard. Keep the translation table in mind and the chapter feels familiar.

5.1 Angular variables

θ in radians, $\omega = \dot{\theta}$, $\alpha = \dot{\omega}$ — and the same constant-acceleration formulas of §1.3 carry over verbatim: $\omega = \omega_0 + \alpha t$, $\theta = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2$, $\omega^2 = \omega_0^2 + 2\alpha\Delta\theta$. The rim variables are the bridge: $v = \omega r$, $a_{\tan} = \alpha r$.

5.2 Rolling kinematics: the no-slip constraint

A wheel rolling without slipping has $v_{\text{cm}} = \omega r$ — the contact point is instantaneously at rest while the top moves at $2v$ (figure 5.2). The constraint is *kinematic*, not energetic: static friction enforces it without doing work, which is why rolling races (ch6) are energy problems.

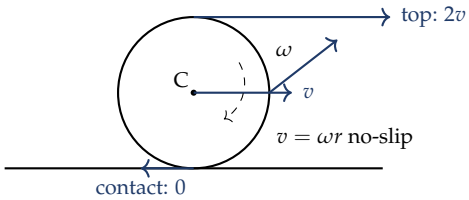


Figure 5.2: Velocity field of a rolling wheel: centre at v, top at $2v$, contact at 0. Notice the contact point is a usable pivot for one instant — the seed of the instantaneous-axis trick.

5.3 Torque: force's lever-arm price

$$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}, \qquad \tau = rF\sin\phi = \ell F, \tag{5.1}$$

where ℓ is the *lever arm*: the perpendicular distance from the pivot to the force's line of action (figure 5.3). Convention in this book: counterclockwise positive. Torque is the quantity the door handle sells — same force, longer ℓ, more angular acceleration.

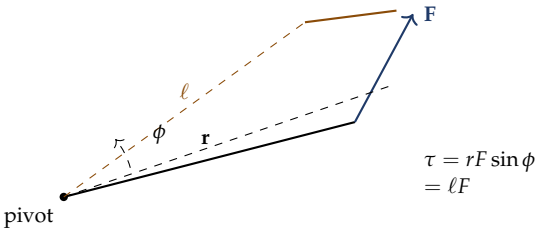


Figure 5.3: Torque geometry: $\tau = rF\sin\phi$, equivalently ℓF with the lever arm ℓ. Notice only the perpendicular component of **F** pays.

Example 5.1 (First attempt: torque as force times length)**.** You pull on a 0.30 m wrench with a 50 N force, but your hand is at an angle of 30° to the handle, not at right angles. A first estimate: $\tau = rF = 0.30 \times 50 = 15\,\text{N m}$.

What goes wrong. Only the component of the force perpendicular to the handle turns the bolt: $\tau = rF\sin\theta = 0.30 \times 50 \times \sin 30° = 0.30 \times 50 \times 0.5 = 7.5\,\text{N m}$, half the naive estimate. Pulling at a right angle ($\sin 90° = 1$) gives the full 15 N m, which is why a wrench works best when you push at a right angle to the handle.

5.4 Moment of inertia: mass with an r^2 tax

$$I = \sum_i m_i r_i^2 \xrightarrow{\text{continuous}} I = \int r^2 \, dm. \tag{5.2}$$

The r^2 is the whole story: mass near the axis is cheap, mass at the rim is expensive — a hoop resists twice as hard per kilogram as a disk. Two theorems do most of the work: the *parallel-axis* shift $I = I_{\text{cm}} + Md^2$, and the *perpendicular-axis* $I_z = I_x + I_y$ for flat laminas.

Example 5.2 (I of a rod and a disk by direct integration)**.** Compute I about the centre for (a) a uniform rod of length L, (b) a uniform disk of radius R.

Strategy. Slice into rings/strips where r is constant; the integral is bookkeeping, not calculus difficulty.

Solution. (a) Rod: $dm = \lambda dx$ with $\lambda = M/L$: $I = \int_{-L/2}^{L/2} x^2 \lambda \, dx = \lambda \frac{L^3}{12} = \frac{1}{12} ML^2$. (b) Disk: rings of mass $dm = \sigma \, 2\pi r \, dr$, all points at the same r: $I = \int_0^R r^2 \sigma 2\pi r \, dr = 2\pi\sigma R^4/4 = \frac{1}{2} MR^2$ using $M = \sigma \pi R^2$. ✓

shape	I about stated axis
rod, centre	$\frac{1}{12} ML^2$
disk/cylinder, axis	$\frac{1}{2} MR^2$
solid sphere, diameter	$\frac{2}{5} MR^2$

hoop: MR^2
sphere shell: $\frac{2}{3} MR^2$
rod, end: $\frac{1}{3} ML^2$

Figure 5.4: The six standard moments. Notice the coefficient tracks how far mass sits from the axis — the rod's 1/12, the disk's 1/2, the hoop's 1.

```
s += r*r * r*dr*dth # r^2 * dA, dA = r dr dth
```

Listing 5.1: `code/ch05_inertia_integrator.py` — grid quadrature vs analytic I; the parallel-axis term on display.

Output of a real run:

```
rectangle 2.0x1.0: numeric I=0.8333, exact=0.8333, err=0.00%
  per unit mass: I/M=0.4167 vs (a^2+b^2)/12=0.4167
disk R=1.0: numeric I=1.5708, exact=1.5708, err=0.00%
  per unit mass: I/M=0.5000 vs R^2/2=0.5000
disk about rim: I=4.7124 = I_cm + M R^2 (1.5708 + 3.1416); ratio to
    I_cm = 3.00
```

Notice the last line: spinning a disk about a point on its rim costs *three times* the centre-axis moment — the Md^2 term, not a numerical artifact.

5.5 $\tau = I\alpha$: the massive pulley

With a real pulley the rope's two sides can't share a tension — the difference is what spins the wheel (figure 5.5).

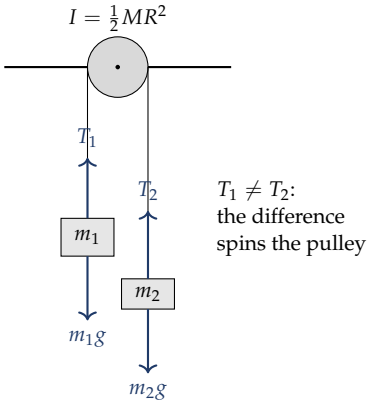


Figure 5.5: The massive pulley's three-body problem: two hanging FBDs plus $\sum \tau = I\alpha$ on the wheel. The tension *difference* does the spinning.

Example 5.3 (Atwood with a massive pulley)**.** Repeat example 2.1's Atwood ($m_1 = 3$, $m_2 = 5\,\text{kg}$) with a pulley of mass $M = 4\,\text{kg}$, radius R, modeled as a disk ($I = \frac{1}{2}MR^2$).

Strategy. Three equations: two translations plus one rotation. Constraint: $\alpha = a/R$ (no slip on the rim).

Solution. $m_2 g - T_2 = m_2 a$, $T_1 - m_1 g = m_1 a$, and $(T_2 - T_1)R = I\alpha = \frac{1}{2}MR^2 \cdot a/R \Rightarrow T_2 - T_1 = \frac{1}{2}Ma$. Adding all three:

$$a = \frac{(m_2 - m_1)g}{m_1 + m_2 + M/2} = \frac{2 \times 9.8}{3 + 5 + 2} = 1.96\,\text{m/s}^2,$$

down from 2.45 — the pulley acts like an extra 2 kg of inertia. $T_1 = 35.3\,\text{N}$, $T_2 = 39.2\,\text{N}$: no longer equal, and their difference 3.9 N is exactly $\frac{1}{2}Ma$. ✓

Example 5.4 (Yo-yo unwinding)**.** A yo-yo (disk, $I = \frac{1}{2}mR^2$, string wrapped at radius r) falls unwinding. Find its acceleration.

Strategy. Two equations: $mg - T = ma$ and $Tr = I\alpha$ with $\alpha = a/r$.

Solution. $T = Ia/r^2 = \frac{1}{2}mR^2 a/r^2$, so $a = \frac{g}{1+R^2/(2r^2)}$. For $R/r = 5$: $a = g/13.5 = 0.73\,\text{m/s}^2$ — the rim-radius leverage makes it a slow fall. ✓

Exercise 5.1 (medium)**.** Find a for the yo-yo when it's a hoop ($I = mR^2$) instead. (F=ma)

5.6 Statics: two balances, one smart pivot

Static equilibrium needs $\sum \mathbf{F} = 0$ *and* $\sum \boldsymbol{\tau} = 0$ — and the pivot for the second equation is *yours to choose*: put it where an unknown force acts and that force disappears from the torque books. That single trick solves most of the chapter's problems.

Example 5.5 (Ladder minimum angle)**.** A uniform ladder of length L leans at angle θ against a frictionless wall (figure 5.6); the floor has μ_s. Minimum θ before slipping?

Strategy. $\sum F$: $N_2 = f$, $N_1 = mg$. Torque about the base (kills f and N_1): the wall's push at the top balances gravity at the middle.

Solution. $N_2 L \sin\theta = mg\frac{L}{2}\cos\theta \Rightarrow N_2 = \frac{mg}{2\tan\theta}$. Slip when $f = N_2 = \mu_s N_1 = \mu_s mg$: $\tan\theta_{\min} = \frac{1}{2\mu_s}$. For $\mu_s = 0.4$: $\theta_{\min} = 51.3^\circ$ — below that, the wall can't push hard enough. ✓

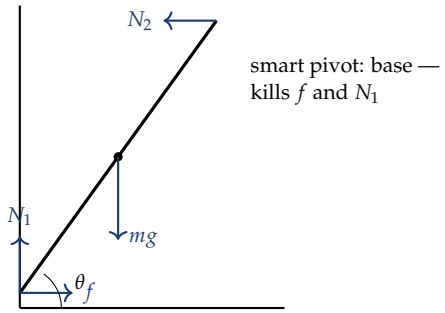


Figure 5.6: Ladder FBD: frictionless wall means N_2 is purely horizontal; taking torques about the base deletes both floor forces.

Exercise 5.2 (easy)**.** Where does a second person standing halfway up the ladder change the minimum angle? Set up the new torque equation. (AP-C)

Example 5.6 (Two-support beam)**.** A uniform 8 kg, 5 m beam rests on supports at $x = 1$ and $x = 4$ (metres from the left end), with a 20 kg toolbox at $x = 4.5$. Find each support's force.

Strategy. $\sum F$ gives the sum; torque about one support kills its unknown. Watch: the toolbox overhangs the right support.

Solution. $N_A + N_B = mg + Mg = 78.4 + 196 = 274.4\,\text{N}$. Torque about support A (CCW+): the beam's weight acts at $x = 2.5$ (1.5 m right of A), the toolbox at 3.5 m right, N_B at 3 m: $3N_B = 78.4 \times 1.5 + 196 \times 3.5 = 117.6 + 686$, so $N_B = 267.9\,\text{N}$ and $N_A = 6.5\,\text{N}$ — the left support nearly unloads: shift the toolbox a little further and the beam tips. ✓

5.7 Rotational kinetic energy

$K_{\text{rot}} = \frac{1}{2} I\omega^2$, and for rolling $K = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$ — the reason a hoop loses every race down an incline: it pays the rotation tax with the largest per-mass I (chapter 6 races them properly).

5.8 In engineering practice: torque specs

Every bolt in a car has a printed torque spec — this chapter's $\tau = \ell F$ run as quality control: a 25 N m spec on a 0.25 m wrench means exactly 100 N of hand force, and overtightening stretches the bolt past its elastic range (book5's materials chapter has the yield story). The statics side is how cranes stay up: $\sum \tau = 0$ about the outrigger line is the tip-over condition, which is why load charts shrink with boom extension — longer ℓ, same budget.

5.9 Common mistakes

LEVER ARM VS POINT DISTANCE

$\tau = \ell F$: ℓ is perpendicular distance to the *line of action*, not to the point of application along the body.

FORGETTING THE PULLEY'S INERTIA

A massless pulley shares one T; a massive one doesn't — drop the $\frac{1}{2}Ma$ term and the answer is too fast.

WRONG PARALLEL-AXIS SHIFT

$I = I_{\text{cm}} + Md^2$ — it's about the CM plus a shift, never between two arbitrary axes.

BAD TORQUE ORIGIN

Every force through the origin drops out. Pick the pivot at the unknown you want gone, not at the centre because it looks official.

FORCE BALANCE ALONE IN STATICS

$\sum F = 0$ admits a whole family of torques; the ladder needs $\sum \tau = 0$ too.

5.10 Summary

Idea	One-line version
dictionary	$\theta/\omega/\alpha \leftrightarrow x/v/a$; same formulas
no-slip	$v_{\text{cm}} = \omega r$; contact instant rest, top $2v$
torque	$\tau = rF\sin\phi = \ell F$; CCW positive
moment of inertia	$I = \int r^2 dm$; $1/12$, $1/2$, $2/5$ coefficients
dynamics	$\tau = I\alpha$; massive pulleys need $T_1 \neq T_2$
statics	$\sum F = 0 + \sum \tau = 0$; pivot at the unknown
rotational K	$\frac{1}{2}I\omega^2$; rolling pays both

Table 5.1: Rotation-I cheat sheet.

DEFINITION OF DONE

You can now: run the linear–angular dictionary; read a rolling wheel's velocity field; compute lever arms and torques; integrate I for simple shapes and apply parallel axis; solve massive-pulley and yo-yo dynamics; and set up statics with the smart-pivot trick.

Notes

The dictionary-first approach follows MIT 8.01SC weeks 6 and 10–11 [2] and AP Physics C unit 5 [4]. OpenStax College Physics ch. 10, 12 [3] covers the same ground; the ladder and pulley problems are Morin chs. 7–8 [7] classics. The I-table coefficients are standard; the quadrature listing checks two of them numerically.

WATCH INSTEAD

1. *Rotating Objects, Moment of Inertia, Rotational KE, Neutron Stars* — Walter Lewin, MIT 8.01x Lecture 19, 41:00, `https://youtu.be/fDJeVR0o__w` — §5.4–§5.7 at full length (accessed 2026-10-11).
2. *Angular motion variables* — Khan Academy Physics, 12:50, `https://youtu.be/garegCgMxxg` — §5.1–§5.2 worked quickly (accessed 2026-10-11).

Further reading

- MIT OpenCourseWare, *8.01SC Classical Mechanics*, weeks 6, 10–11 — rotation and statics with recitation solutions. At https://ocw.mit.edu, course "8.01SC" (verified 2026-10-11).
- OpenStax, *College Physics 2e*, ch. 10 "Rotational Motion" and ch. 12 "Statics" — free text at https://openstax.org (verified 2026-10-11).
- College Board, *AP Physics C: Mechanics CED*, Unit 5 — the exam blueprint for rotation. At https://apstudents.collegeboard.org (verified 2026-10-11).
- D. Morin, *Introduction to Classical Mechanics*, chs. 7–8 — statics and rotation at the harder level; ISBN 978-0521876223.
- The Physics Classroom, *Rotational Motion* tutorials — readable remediation for §5.1–§5.6, at https://www.physicsclassroom.com (verified 2026-10-11).
- Khan Academy, *Moments, torque, and angular momentum* — drills matching the whole chapter, at https://www.khanacademy.org under Science → Physics (verified 2026-10-11).

Additional exercises

Exercise 5.3 (easy). A wheel spins up to 1200 rpm in 4 s at constant α. Find α in rad/s^2 and the angle turned. (AP-C)

Exercise 5.4 (easy). Identify the lever arm: a 0.5 m wrench at 60° to a 100 N force — compute τ both as $rF\sin\phi$ and ℓF. (AP-C)

Exercise 5.5 (easy). Write down I for a 2 kg, 0.3 m-radius disk about its axis and about a rim point, citing the theorem used. (AP-C)

Exercise 5.6 (easy). A uniform beam of length 4 m is pinned at one end and carries a 200 N load at its far end. What vertical force at the midpoint holds it level (neglecting the beam's weight)? (F=ma)

Exercise 5.7 (easy). A see-saw: 30 kg child at 1.5 m left of the pivot balances a 50 kg adult. Where does the adult sit? (F=ma)

Exercise 5.8 (medium). Massive-pulley Atwood: recompute example 5.3 with a hoop pulley ($I = MR^2$) — which way does a move, and why? (F=ma classic)

Exercise 5.9 (medium). Redo example 5.5 with a wall that has $\mu_s' = 0.3$ of its own. Does the wall friction help or hurt? (AP-C FR-style)

Exercise 5.10 (medium). A yo-yo hoop ($I = mR^2$, string at r) falls unwinding — find a and compare with example 5.4. (F=ma)

Exercise 5.11 (medium). Composite body: a 0.5 m rod of mass 2 kg has a 1 kg, 0.1 m-radius disk welded at one end (disk axis through the rod's end, perpendicular). Find I about the rod's other end. (AP-C)

Exercise 5.12 (medium). Sign-of-torque reasoning: a door pushed at its top vs its middle, same force — describe the angular acceleration ratio without computing. (AP-C)

Exercise 5.13 (medium)**.** Rolling preview: a hoop and a disk race down an incline from rest — predict the winner using only the I coefficients, then verify with energy ($v^2 = \frac{2gh}{1+I/mr^2}$). (AP-C)

Exercise 5.14 (medium)**.** A uniform 8 kg beam 5 m long rests on two supports at $x = 1$ and $x = 4$ (metres). Find each reaction force. (F=ma)

Exercise 5.15 (hard)**.** A uniform rod leans on a frictionless floor against a frictionless wall and is released. Show it loses wall contact when $\sin\theta = \frac{2}{3}\sin\theta_0$ — use energy for speed and the CM's horizontal acceleration for the contact condition. (USAPhO-style)

Exercise 5.16 (hard)**.** A rod of length L has density $\lambda(x) = \lambda_0(1 + x/L)$. Find its I about the light end by direct integration, and its CM. (APhO-style)

Exercise 5.17 (medium)**.** A sphere and a cylinder of the same M, R roll (not slide) down the same incline. Using $v^2 = \frac{2gh}{1+I/mr^2}$, find the sphere's speed advantage over the cylinder's at the bottom as a percentage. (AP-C)

CHAPTER 6

Rotation II: angular momentum and rolling

Linear momentum had its chapter; angular momentum is its equal partner — the second conserved quantity of mechanics and the reason gyroscopes, planets, and skaters behave the way they do. The bookkeeping mirrors chapter 3 exactly, with one new subtlety: the origin you measure from matters.

BEFORE YOU START

Chapter 5, and **vectors**: the cross product appears for the first time here. The *Start here* chapter reviews it; you can also read the vector part of Chapter 9 first. *Recap:* angular momentum is rotation's version of momentum.

AFTER THIS CHAPTER YOU CAN

- compute angular momentum and use its conservation;
- solve rolling problems three ways;
- estimate gyroscopic precession;
- explain how a reaction wheel steers a satellite.

HOW TO THINK ABOUT THIS

Angular momentum is the spinning counterpart of momentum, and it is conserved when no outside twist acts. That is why a skater spins faster when her arms come in. Look for symmetries and conserved quantities before reaching for the force equations.

6.1 Angular momentum: definition

For a particle,

$$\mathbf{L} = \mathbf{r} \times \mathbf{p} \tag{6.1}$$

— momentum filtered through the position relative to a chosen origin (figure 6.1). A straight-line mover carries nonzero $\mathbf{L}$ about any origin off its line — the quantity is about the *rotation of the line of sight*, not about actual circling.

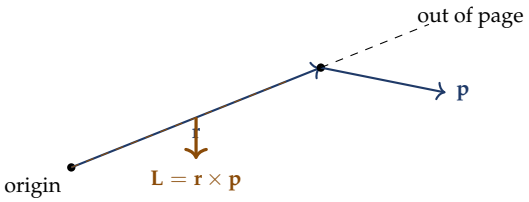


Figure 6.1: $\mathbf{L} = \mathbf{r} \times \mathbf{p}$: magnitude $rp \sin\theta$, direction by the right-hand rule (out of the page here). Notice it needs an origin — the same $\mathbf{p}$ has different $\mathbf{L}$ about different points.

6.2 Rigid bodies: when $L = I\omega$ holds

Summing (6.1) over a rigid body rotating about a fixed symmetry axis gives

$$L = I\omega.$$

The fine print: valid about the fixed axis *or* about the CM — the two origins where the cross terms vanish. A rolling wheel's $\mathbf{L}$ is $I\omega$ about its CM *plus* the orbital $M(\mathbf{R} \times \mathbf{V})$ of the CM — both exist, both count.

6.3 The rotational second law

$$\boldsymbol{\tau}_{\text{ext}} = \frac{d\mathbf{L}}{dt} \tag{6.2}$$

about an inertial fixed point or about the CM — the two origins where the derivation works. Choosing an accelerating origin is the classic trap: the equation picks up fictitious-torque terms that ruin it.

6.4 Conservation of angular momentum

ANGULAR MOMENTUM CONSERVATION

$\mathbf{L}$ is conserved when the net *external torque* is zero — weaker than zero external force. A planet feels a huge central force that exerts no torque, so its $\mathbf{L}$ (and Kepler's second law) survives.

Example 6.1 (The skater)**.** A skater spins at $\omega_0 = 2\,\text{rad/s}$ with $I_0 = 3\,\text{kg m}^2$, then pulls her arms to $I = 1\,\text{kg m}^2$. Find the new spin rate and the energy change.

Strategy. No external torque about the spin axis: L conserved. K is not — her arms do work.

Solution. $\omega = I_0\omega_0/I = 6\,\text{rad/s}$. Energies: $K_0 = \frac{1}{2}(3)(4) = 6\,\text{J}$, $K = \frac{1}{2}(1)(36) = 18\,\text{J}$ — K tripled. The extra 12 J came from her muscles pulling mass inward against the centrifugal pull: the audit of ch4 applies here too. ✓

Example 6.2 (Bullet into a pivoted rod)**.** A 10 g bullet at 400 m/s embeds in the tip of a 1 kg, 1 m rod pivoted at its top ($I = \frac{1}{3}ML^2$). Find the angular velocity just after.

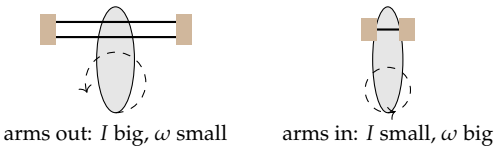


Figure 6.2: The skater trade: L fixed, K tripled — arms supply the difference.

Strategy. Linear momentum is *not* conserved (the pivot pushes); angular momentum about the pivot *is* (the pivot force has no lever arm). Use the collision instant only.

Solution. $L_i = mvL = 0.01 \times 400 \times 1 = 4\,\mathrm{kg\,m^2/s}$. $I_{\mathrm{tot}} = \frac{1}{3}(1)(1) + 0.01(1)^2 = 0.343\,\mathrm{kg\,m^2}$. $\omega = 4/0.343 = 11.7\,\mathrm{rad/s}$. K audit: before 800 J, after $\frac{1}{2}I\omega^2 = 23.4\,\mathrm{J}$ — 97% lost to the embedding, as inelastic as collisions get. ✓

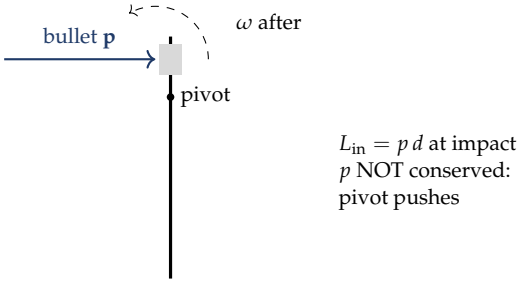


Figure 6.3: Bullet-and-rod: conserve L about the pivot, not p — the pivot's impulse is external to translation but torque-free about itself.

Exercise 6.1 (easy)**.** In example 6.2, which quantities survive: total **L** about the pivot, total **p**, total K? One-line each. (AP-C)

6.5 Rolling dynamics, three ways

A roller on an incline can be solved by forces and torque ($ma = mg\sin\theta - f$, $I\alpha = fr$), by energy ($v^2 = 2gh/(1+\beta)$ for $I = \beta mr^2$), or by the instantaneous-axis trick (torque about the contact point only). All three agree; the energy route is the fastest.

Example 6.3 (The incline race)**.** Hoop ($\beta = 1$), disk ($\frac{1}{2}$) and sphere ($\frac{2}{5}$) race down a 2 m drop. Order and margins?

Strategy. $v = \sqrt{2gh/(1+\beta)}$ — smaller β finishes first, independent of mass and radius.

Solution. $v_{\mathrm{sph}} = 5.29$, $v_{\mathrm{disk}} = 5.11$, $v_{\mathrm{hoop}} = 4.43\,\mathrm{m/s}$ — the code listing below prints the same numbers. The hoop loses because *all* its mass sits at the rim: half its energy is rotation at every instant (figure 6.4). ✓

```
v = math.sqrt(2*G*H/(1+beta))
```

Listing 6.1: `code/ch06_rolling_race.py` — the incline race for several shape factors.

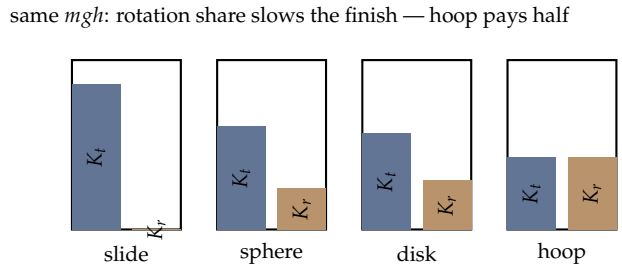


Figure 6.4: Energy partition in the race: same mgh, different splits — the hoop spends half on spinning.

Output of a real run:

```
incline drop h=2.0 m
shape beta v (m/s) K_tr K_rot
point mass (slide) 0.0 6.261 19.60 0.00
solid sphere 0.4 5.292 14.00 5.60
solid disk 0.5 5.112 13.07 6.53
hoop 1.0 4.427 9.80 9.80

finish order: point mass (slide) > solid sphere > solid disk > hoop
sphere beats hoop by 19.5% of hoop's speed
```

Notice the partition column: the hoop's K_{rot} equals its K_{tr} — $\beta = 1$ means rotation and translation split the budget evenly, and mass/radius never appear. Figure 6.5 plots the finish speeds against β: the points fall exactly on $v = \sqrt{2gh/(1+\beta)}$, so the whole race reduces to one parameter.

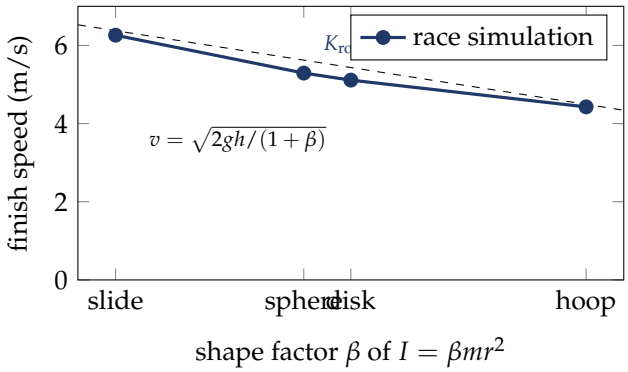


Figure 6.5: Finish speed versus the shape factor β: simulated points on the analytic curve. Bigger β means a bigger rotational tax — and nothing else matters.

Example 6.4 (Two-disk angular collision)**.** A disk ($I_1 = 2\,\text{kg}\,\text{m}^2$) spinning at $10\,\text{rad/s}$ drops onto a stationary disk ($I_2 = 3\,\text{kg}\,\text{m}^2$) on the same axle; friction locks them. Find the common ω and the energy lost.

Strategy. Angular-momentum conservation about the axle — the angular version of the perfectly inelastic collision.

Solution. $\omega = I_1\omega_0/(I_1 + I_2) = 20/5 = 4\,\text{rad/s}$. $K_i = \frac{1}{2}(2)(100) = 100\,\text{J}$; $K_f = \frac{1}{2}(5)(16) = 40\,\text{J}$ — 60% lost, the angular mirror of chapter 3's inelastic hit. ✓

Example 6.5 (Child on a turntable)**.** A 40 kg child stands at the rim of a 200 kg, 3 m-radius turntable ($I = \frac{1}{2}MR^2 = 900\,\text{kg}\,\text{m}^2$) spinning at $\omega_0 = 0.5\,\text{rad/s}$, then walks to the centre. Find the new ω.

Strategy. No external torque about the axis: $I\omega$ conserved, with the child as a point mass whose I shrinks.

Solution. $I_0 = 900 + 40(9) = 1260\,\text{kg}\,\text{m}^2$; at centre $I_1 = 900 + 0$. $\omega_1 = 0.5 \times 1260/900 = 0.70\,\text{rad/s}$ — the same skater effect with the child playing the arms. K rises from 158 to 220 J, paid by the child's steps. ✓

Exercise 6.2 (medium)**.** A 40 kg child walks from rim to centre on a 200 kg, 3 m-radius turntable ($I = \frac{1}{2}MR^2$) spinning at 0.5 rad/s. Find the new ω. (AP-C FR-style)

6.6 Gyroscopic precession: the order of magnitude

A spinning top tilted at angle θ has $\mathbf{L}$ along its axis; gravity's torque $\tau = mgd\sin\theta$ is perpendicular, so $\mathbf{L}$ *rotates* rather than tips (figure 6.6). The precession rate follows from $\Delta L = \tau\Delta t$ geometry:

$$\Omega = \frac{\tau}{L} = \frac{mgd}{I\omega}$$

— slower spin, faster wobble. That's why tops precess more as they wind down.

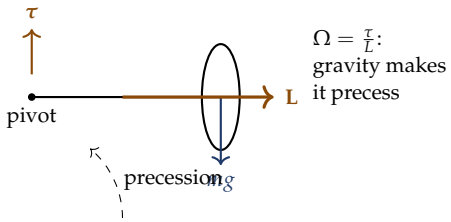


Figure 6.6: Precession bookkeeping: $\boldsymbol{\tau}$ perpendicular to $\mathbf{L}$ turns the vector instead of shrinking it. $\Omega = \tau/L$ sets the rate.

6.7 In engineering practice: reaction wheels and flywheels

Satellites steer with *reaction wheels* — small spun disks whose speed changes trade $\mathbf{L}$ with the bus, no fuel spent: this chapter's conservation law is the whole attitude-control budget. Flywheel energy storage is the same $K = \frac{1}{2}I\omega^2$ as grid batteries: a 500 kg carbon-fiber wheel at 20,000 rpm banks megajoules in rotation, which is exactly the hoop's tax of figure 6.4 run to its useful extreme — mass at the rim is where you want it.

6.8 Common mistakes

$L = I\omega$ ABOUT THE WRONG AXIS

Valid about the fixed axis or the CM only. A rolling wheel about an arbitrary point needs $I\omega + \mathbf{R} \times M\mathbf{V}$.

CONSERVING L THROUGH EXTERNAL TORQUE

The bullet-in-rod survives precisely because the pivot torque about the pivot is zero — any other origin, or any other external torque, breaks it.

FORGETTING TRANSLATIONAL K IN ROLLING

A rolling object carries $\frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$ — drop the first term and every race comes out tied.

$\tau = dL/dt$ ABOUT A NON-INERTIAL ORIGIN

The law holds at inertial points and the CM. An accelerating origin invents torques that aren't there.

CROSS-PRODUCT SIGN

$\mathbf{r} \times \mathbf{p} \neq \mathbf{p} \times \mathbf{r}$ — the order flips the sign. Fix the convention once per problem.

6.9 Summary

Idea	One-line version
definition	$\mathbf{L} = \mathbf{r} \times \mathbf{p}$ — origin-dependent
rigid body	$L = I\omega$ about fixed axis or CM
dynamics	$\boldsymbol{\tau} = d\mathbf{L}/dt$ at inertial origin or CM
conservation	survives iff external torque $= 0$ — central forces OK
rolling	$v^2 = 2gh/(1+\beta)$; smaller β wins
skater	L fixed, K grows — muscles pay
gyroscope	$\Omega = \tau/L$; perpendicular torque turns, doesn't tip

Table 6.1: Angular momentum cheat sheet.

DEFINITION OF DONE

You can now: compute **L** for particles and rigid bodies; state when conservation applies; solve skater, bullet-rod, and two-disk collisions; run the incline race three ways; and size gyroscopic precession to order of magnitude.

Notes

The treatment follows MIT 8.01SC weeks 11–12 plus its gyroscope module [2] and AP Physics C unit 6 [4]. Morin chs. 8–9 [7] is the source of the spool and sliding-to-rolling exercises; OpenStax ch. 11 [3] covers the same material. The race numbers are the listing's real output.

WATCH INSTEAD

1. *Angular Momentum, Torques, Conservation of Angular Momentum* — Walter Lewin, MIT 8.01x Lecture 20, 51:15, `https://youtu.be/sNaaL19opxw` — the whole chapter at full length (accessed 2026-10-11).
2. *Conservation of angular momentum* — Khan Academy, 7:16, `https://youtu.be/vfmLI150g4w` — §6.4 worked quickly (accessed 2026-10-11).

Further reading

- MIT OpenCourseWare, *8.01SC Classical Mechanics*, weeks 11–12 — angular momentum and rolling with recitation solutions. At `https://ocw.mit.edu`, course "8.01SC" (verified 2026-10-11).
- OpenStax, *College Physics 2e*, ch. 10–11 — free text at `https://openstax.org` (verified 2026-10-11).
- College Board, *AP Physics C: Mechanics CED*, Unit 6 — the exam blueprint for angular momentum. At `https://apstudents.collegeboard.org` (verified 2026-10-11).
- D. Morin, *Introduction to Classical Mechanics*, chs. 8–9 — angular momentum and rolling at the harder level; ISBN 978-0521876223.
- The Physics Classroom, *Angular Momentum* tutorials — readable remediation, at `https://www.physicsclassroom.com` (verified 2026-10-11).
- Khan Academy, *Moments, torque, and angular momentum* — drills matching the whole chapter, at `https://www.khanacademy.org` under Science → Physics (verified 2026-10-11).

Additional exercises

Exercise 6.3 (easy)**.** Find the magnitude and direction of $\mathbf{L}$ for a 2 kg mass at $\mathbf{r} = (3, 0, 0)$ m moving at $\mathbf{v} = (0, 4, 0)$ m/s. (AP-C)

Exercise 6.4 (easy)**.** Decide whether L is conserved in each: (a) planet orbiting the sun, (b) figure skater pulling in arms, (c) frisbee spinning while gliding to a stop, (d) the bullet-rod of example 6.2 *after* impact. (AP-C)

Exercise 6.5 (easy)**.** A rolling ball has $v = 5$ m/s and radius 0.1 m. Find ω and the contact point's instantaneous speed. (AP-C)

Exercise 6.6 (easy)**.** Explain in one sentence why the skater's K increases while L stays fixed. (F=ma)

Exercise 6.7 (easy)**.** A spinning disk has $I = 0.5\,\text{kg m}^2$, $\omega = 20$ rad/s. Find L and K_{rot}. (AP-C)

Exercise 6.8 (medium)**.** A 40 kg child stands at the rim of a 150 kg, 4 m-radius turntable ($I = \frac{1}{2}MR^2$) at rest, then walks clockwise at 1 m/s relative to the ground. Which way and how fast does the table turn? (AP-C FR-style)

Exercise 6.9 (medium)**.** Rank these down a 3 m incline by finish order and compute each speed: solid sphere, hollow sphere ($\beta = 2/3$), disk, hoop. (F=ma)

Exercise 6.10 (medium)**.** Example 6.2 with a bullet hitting *halfway* down the rod: recompute ω and the energy loss. (AP-C)

Exercise 6.11 (medium)**.** Two angular disks merge: $I_1 = 1$, $\omega_1 = 12$ and $I_2 = 4$ spinning *oppositely* at $\omega_2 = -3$. Find the common ω and energy lost. (F=ma)

Exercise 6.12 (medium)**.** A 0.5 m wrench applies 40 N tangentially to a 0.02 m-radius shaft. Find the torque and, for a 0.1 kg m^2 flywheel, the ω after 5 s from rest. (AP-C)

Exercise 6.13 (medium)**.** A gyroscope with $I = 0.01\,\text{kg m}^2$, $\omega = 200$ rad/s, $d = 0.1$ m, $m = 0.5$ kg: compute the precession rate in rad/s and rpm. (F=ma)

Exercise 6.14 (hard)**.** A bowling ball is set sliding with no spin at v_0 on a lane with μ_k. Show it transitions to pure rolling at $v = \frac{5}{7}v_0$ — track both v and ω as friction acts. (USAPhO-style)

Exercise 6.15 (hard)**.** A spool of thread lies on a table ($I = \beta mr^2$, thread wrapped at inner radius $r_i < R$). Pulling the thread horizontally: does it roll toward or away from you? Find the critical angle where the direction reverses. (APhO-style classic)

Exercise 6.16 (medium)**.** Energy split check: for the race of example 6.3, what fraction of each shape's kinetic energy is rotational? Show $\beta/(1+\beta)$. (AP-C)

Exercise 6.17 (medium)**.** A planet at perihelion (r_1, v_1) reaches aphelion at $2r_1$. Find its aphelion speed by L conservation, and explain why this is Kepler's second law. (F=ma)

CHAPTER 7

Oscillations

One equation runs this chapter: $\ddot{x} + \omega_0^2 x = 0$. Springs, pendulums, RLC circuits (ch15) and molecule vibrations are all its instances. Add damping and a drive and the same equation explains resonance — from tuning forks to the Tacoma bridge.

BEFORE YOU START

Chapters 2 and 4 and the sine and cosine functions. *Recap:* an oscillation is motion that repeats; its simplest form is a mass on a spring.

AFTER THIS CHAPTER YOU CAN

- write and solve the equation for simple harmonic motion;
- find the period of a spring and of a pendulum, and the limit of the small-angle approximation;
- classify damped oscillators;
- explain resonance in a driven system.

HOW TO THINK ABOUT THIS

Anything pushed slightly away from a stable position and pulled back behaves like a spring. That is why the maths of the spring shows up everywhere. Damping and driving add friction and a push, and resonance appears when the push matches the natural rhythm.

7.1 SHM: the equation and its family

$$\ddot{x} + \omega_0^2 x = 0 \;\Rightarrow\; x(t) = A\cos(\omega_0 t + \phi), \tag{7.1}$$

two constants A, ϕ paid by two initial conditions. The frequency ω_0 is set by the physics; the amplitude is yours. Notice what's *not* in the solution: A never changes the period — the signature of a linear restoring force.

7.2 Spring–mass: the canonical case

$-kx = m\ddot{x}$ gives $\omega_0 = \sqrt{k/m}$ and $T = 2\pi\sqrt{m/k}$: stiffer is faster, heavier is slower. Energy trades $\tfrac{1}{2}kx^2$ and $\tfrac{1}{2}mv^2$ with $E = \tfrac{1}{2}kA^2$ — quadratic in amplitude, so doubling A quadruples the energy.

Example 7.1 (Energy split in a spring oscillator)**.** A 0.5 kg mass on a 200 N/m spring oscillates with $A = 0.15\,\text{m}$. Find E, $v_{\max}$, and the speed at $x = A/2$.

Strategy. E is set by amplitude; at any x the remainder after $\tfrac{1}{2}kx^2$ is kinetic.

Solution. $E = \tfrac{1}{2}(200)(0.0225) = 2.25\,\text{J}$; $v_{\max} = \sqrt{2E/m} = \sqrt{9} = 3\,\text{m/s}$ at centre. At $x = 0.075$: $U = \tfrac{1}{2}(200)(0.0056) = 0.56\,\text{J}$, so $v = \sqrt{2(2.25 - 0.56)/0.5} = 2.6\,\text{m/s}$ — three quarters of max at half amplitude: energy spends most of the cycle near the turning points. ✓

7.3 Pendulums: small-angle as an approximation with a limit

For a pendulum the restoring torque is $-mgL\sin\theta$ (figure 7.1); linearising $\sin\theta \approx \theta$ gives $\omega_0 = \sqrt{g/L}$ — good to 1% for $\theta_0 \lesssim 23°$, and the first correction $T \approx T_0(1 + \theta_0^2/16)$ is the hard exercise at the end. A physical pendulum generalises it: $\omega_0 = \sqrt{mgd/I}$.

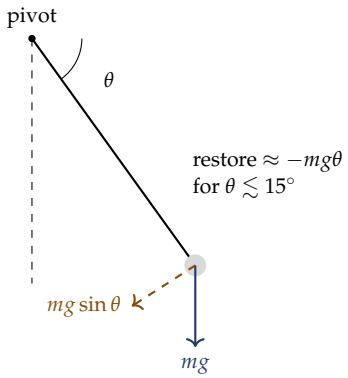


Figure 7.1: The pendulum's restoring force $-mg\sin\theta$ is *almost* linear — small-angle is an approximation with a budget, not a property.

Example 7.2 (Meter-stick physical pendulum)**.** A uniform meter stick pivots at one end. Find its period.

Strategy. Physical pendulum: $\omega_0^2 = mgd/I$ with $d = L/2$, $I = \tfrac{1}{3}mL^2$.

Solution. $\omega_0^2 = \frac{g(L/2)}{L^2/3} = \frac{3g}{2L}$, so $T = 2\pi\sqrt{\frac{2L}{3g}} = 2\pi\sqrt{\frac{2}{29.4}} = 1.64\,\text{s}$. A simple pendulum this long would take 2.0 s — the stick's spread mass makes it faster. ✓

7.4 Damped oscillations: three regimes

Add $-b\dot{x}$: $\ddot{x} + \gamma\dot{x} + \omega_0^2 x = 0$ with $\gamma = b/m$. The discriminant sorts the behaviour (figure 7.2): **underdamped** ($\gamma < 2\omega_0$) oscillates inside an $e^{-\gamma t/2}$ envelope at $\omega_d = \sqrt{\omega_0^2 - \gamma^2/4}$; **critical** ($\gamma = 2\omega_0$) returns fastest without ringing; **overdamped** ($\gamma > 2\omega_0$) creeps back slowly. Car suspension and door closers are engineered to sit just under critical.

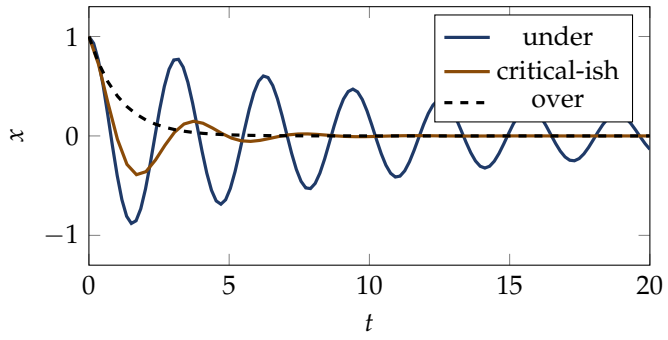


Figure 7.2: The three damping regimes. Notice only the underdamped case crosses zero — the others surrender exponentially.

Example 7.3 (Extracting the constants from a trace)**.** A decaying trace shows maxima at 1.00, 0.82, 0.67 at $t = 0$, 2.1, 4.2 s. Find γ, ω_d, ω_0.

Strategy. Envelope ratio per period gives γ; the period gives ω_d; the relation $\omega_d^2 = \omega_0^2 - \gamma^2/4$ closes it.

Solution. Period $T_d \approx 2.1\,\text{s} \Rightarrow \omega_d = 2.99\,\text{rad/s}$. Ratio per period 0.82: $e^{-\gamma T_d/2} = 0.82 \Rightarrow \gamma = 0.19\,\text{s}^{-1}$. $\omega_0 = \sqrt{\omega_d^2 + \gamma^2/4} = \sqrt{8.94 + 0.009} = 2.99\,\text{rad/s}$ — damping barely shifts the frequency here. ✓

Exercise 7.1 (easy)**.** Show the same trace's next two maxima and estimate them. (AP-C)

7.5 Driven oscillations and resonance

A drive $F_0 \cos \omega_d t$ produces steady amplitude

$$A(\omega_d) = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega_d^2)^2 + (\gamma\omega_d)^2}}, \tag{7.2}$$

peaking near $\omega_d = \sqrt{\omega_0^2 - \gamma^2/2}$ (figure 7.3). The peak height is set by the *quality factor* $Q = \omega_0/\gamma$: high-Q systems ring loudly and narrowly — the tuning-fork pair of figure 7.5 and, minus the luck, the Tacoma Narrows bridge.

Example 7.4 (Off-resonance driving)**.** The oscillator of the code run ($\omega_0 = 2$, $\gamma = 0.4$, $F_0/m = 1$) is driven at $\omega_d = 3$. Find its steady amplitude and compare with the peak.

Strategy. Plug into (13.4) twice; the ratio is the point.

Solution. $A(3) = 1/\sqrt{26.4} = 0.195$ (from $\sqrt{(4-9)^2 + (0.4\cdot 3)^2}$); $A(1.98) = 1/(0.4 \times 1.98) = 1.26$. The peak is 6.5× louder — a 1.5× frequency miss costs 85% of the response. ✓

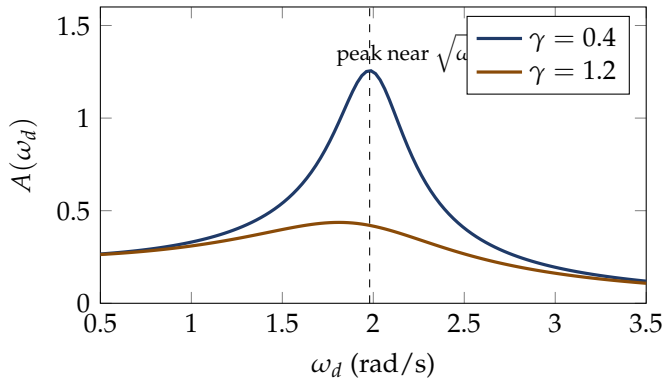


Figure 7.3: Amplitude vs drive frequency for two dampings. Lower γ raises and sharpens the peak — the Q trade-off.

Example 7.5 (Bandwidth from a resonance curve)**.** A resonance curve peaks at 195 Hz with amplitude A_0 and falls to $A_0/\sqrt{2}$ at 190 and 200 Hz. Find Q and γ.

Strategy. Half-power bandwidth $\Delta\omega = \gamma$; then $Q = \omega_0/\gamma$.

Solution. $\Delta f = 10\,\text{Hz} \Rightarrow \gamma = 2\pi \times 10 = 62.8\,\text{s}^{-1}$, $Q = 195/10 = 19.5$. Narrow and loud. ✓

```
def acc(x_, v_, t_):
    return F0M*math.cos(wd*t_) - GAMMA*v_ - W0*W0*x_
```

Listing 7.1: `code/ch07_resonance_sweep.py` — RK4 driven oscillator, amplitude vs theory.

Output of a real run:

```
gamma=0.4, w0=2.0: theory peak at sqrt(w0^2-g^2/2)=1.980 rad/s
   wd A_meas A_theory err
  1.2 0.384 0.384 0.0%
  1.6 0.635 0.635 0.0%
  1.9 1.171 1.171 0.0%
  2.0 1.250 1.250 0.0%
  2.1 1.070 1.070 0.0%
  2.4 0.499 0.499 0.0%
  3.0 0.194 0.194 0.0%
Q = w0/gamma = 5.0; A(w_res)/A(0) = 5.03 (theory)
```

RK4 tracks the analytic curve to 0.0% at every sampled ω_d — the integrator is now trustworthy enough to trust figure 7.3. Figure 7.4 overlays the measured amplitudes on the theory curve: the agreement is exact, and the peak sits at $\omega_r = \sqrt{\omega_0^2 - \gamma^2/2} = 1.980$ — just below ω_0, the driven-oscillator subtlety that trips up most first passes.

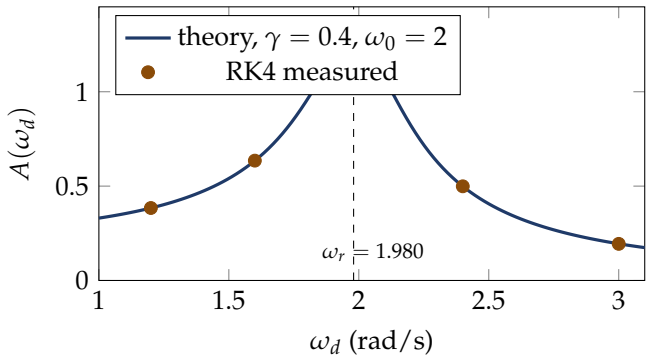
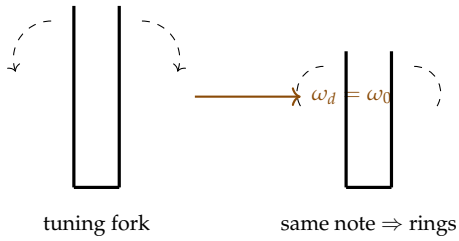


Figure 7.4: Simulated steady-state amplitudes (markers) on the analytic response curve for $\gamma = 0.4$, $\omega_0 = 2$. The maximum is at $\omega_r = 1.980$, not at ω_0 — damping pulls the peak left.



sympathetic resonance: only the matching fork answers

Figure 7.5: Sympathetic resonance: only the fork sharing ω_0 answers. Every band's instrument-tuning trick, and every bridge's nightmare.

7.6 Phase space: the state is a point

Plotting $\dot{x}$ vs x (figure 7.6) turns the oscillator's history into a curve: the undamped case draws a closed ellipse, damping spirals it to the origin. Nonlinear oscillators live here too — chapter 17's pendulum without the small-angle crutch is a phase portrait before it's a formula.

```
m, k, b, F0, w = 1.0, 100.0, 1.0, 1.0, 8.0
w0 = np.sqrt(k / m)
A_theory = F0 / np.sqrt((m * (w0**2 - w**2))**2 + (b * w)**2)
dt, n = 1e-3, 60000
x, v, peaks = 0.0, 0.0, []
for i in range(1, n):
    a = (F0 * np.cos(w * i * dt) - k * x - b * v) / m
    v += a * dt
    x += v * dt
    if i > 50000 and i % 10 == 0:
        peaks.append(x)
```

Listing 7.2: `code/ch07_driven_transient.py` — driven oscillator: the beat ring-up dies out, leaving the theory's amplitude.

Output of a real run:

```
w0 = 10.0 rad/s, drive w = 8.0 rad/s
theory steady amplitude: 0.027 m
```

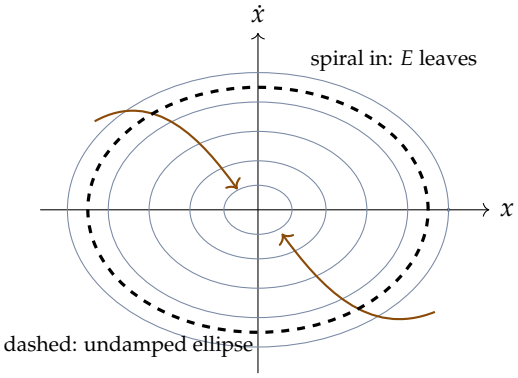


Figure 7.6: Phase portrait: the undamped ellipse versus the damped spiral. The state's whole future is visible in one picture.

```
late-time window max |x|: 0.027 m
beat envelope frequency |w0-w|/2 = 1.0 rad/s
```

Off-resonance driving gives a transient beat — natural frequency plus drive frequency interfering — that damping erases in roughly $\tau = 2m/b$; what survives is the steady-state amplitude the formula promised, to all three printed digits. The simulation's beat envelope sits at $|\omega_0 - \omega|/2 = 1\,\mathrm{rad/s}$, the same superposition arithmetic chapter 16 puts in your ear.

7.7 In engineering practice: vibration budgets

Every engineering discipline owns a version of equation (13.4): buildings get tuned mass dampers (a secondary oscillator eating the primary's peak), hard drives sit on elastomer mounts that push ω_0 below the drive frequencies, and shock absorbers are the critical-damping spec of §7.4 sold by the pair. The common move is always the same: place ω_0 far from the disturbance's ω_d, then buy the remaining safety factor with damping — the two knobs this chapter names.

7.8 Common mistakes

SMALL-ANGLE USED WHERE θ ISN'T

The linearisation costs $\sim 1\%$ per 20° beyond small; at 60° the period is 7% long and the whole section's formulas need the finite-amplitude correction.

ω VS f

$\omega = 2\pi f$: $T = 2\pi\sqrt{m/k}$ is a period, $\sqrt{k/m}$ is a frequency *in radians*. Mixing them is a factor-2π error every time.

AMPLITUDE VS ENERGY

$E = \frac{1}{2}kA^2$ is quadratic: "twice the amplitude" is four times the energy, and damping's decay rate is set on the amplitude by the envelope, not the peak.

DRIVING AT ω_0 VS ω_d

Resonance peaks *below* ω_0 when damped — $\sqrt{\omega_0^2 - \gamma^2/2}$ — and the response lags the drive by $\pi/2$ at the peak.

"DAMPING SLOWS THE PERIOD" ASSUMED OR DENIED

It does — $\omega_d < \omega_0$ — but weakly ($\gamma^2/4$ term), so check the size before reporting either way.

7.9 Summary

Idea	One-line version
SHM	$\ddot{x} + \omega_0^2 x = 0 \Rightarrow x = A\cos(\omega_0 t + \phi)$
spring	$\omega_0 = \sqrt{k/m}$; $E = \frac{1}{2}kA^2$
pendulum	$\omega_0 = \sqrt{g/L}$ small-angle; physical $\sqrt{mgd/I}$
damping	under/critical/over by γ vs $2\omega_0$
driven	peak at $\sqrt{\omega_0^2 - \gamma^2/2}$; $Q = \omega_0/\gamma$
phase space	ellipse undamped, spiral damped

Table 7.1: Oscillations cheat sheet.

DEFINITION OF DONE

You can now: write and solve the SHM equation; scale spring and pendulum periods; classify damping regimes and extract constants from a decay trace; compute resonance peaks and read Q off a bandwidth; and draw the phase portrait.

Notes

The oscillator-first treatment follows MIT 8.01SC's oscillation module [2] and AP Physics C unit 7 [4]; Feynman vol. I §21–24 [5] is the classic resonance narrative. Morin ch. 4 [7] is the source of the physical-pendulum and finite-amplitude exercises. The sweep numbers are the listing's real output.

WATCH INSTEAD

1. *Hooke's Law, Springs, Pendulums, Simple Harmonic Motion* — Walter Lewin, MIT 8.01x Lecture 10, 47:41, https://youtu.be/tNpuTx7UQbw — §7.1–§7.3 at full length (accessed 2026-10-11).
2. *SHM of spring-mass oscillators* — Khan Academy, 15:13, https://youtu.be/eRZSzGKDfwU — §7.2 worked quickly (accessed 2026-10-11).

Further reading

- MIT OpenCourseWare, *8.01SC Classical Mechanics*, oscillations module — SHM, damping and resonance with recitations. At https://ocw.mit.edu, course "8.01SC" (verified 2026-10-11).
- OpenStax, *College Physics 2e*, ch. 15 "Oscillatory Motion" — free text at https://openstax.org (verified 2026-10-11).
- College Board, *AP Physics C: Mechanics CED*, Unit 7 — the exam blueprint for SHM. At https://apstudents.collegeboard.org (verified 2026-10-11).
- R. P. Feynman, *Lectures on Physics*, vol. I, §21–24 — resonance as narrative; free at Caltech's site.
- D. Morin, *Introduction to Classical Mechanics*, ch. 4 — oscillations at the harder level; ISBN 978-0521876223.
- Khan Academy, *Simple harmonic motion* — drills matching §7.1–§7.5, at https://www.khanacademy.org under Science → Physics (verified 2026-10-11).

Additional exercises

Exercise 7.2 (easy). A 0.5 kg mass on a 200 N/m spring: compute ω_0, f, T. (AP-C)

Exercise 7.3 (easy). Doubling a spring oscillator's amplitude multiplies its energy by what? Its maximum speed by what? (AP-C)

Exercise 7.4 (easy). A pendulum's T is 2.0 s on Earth. Find it on the Moon ($g = 1.6\,\text{m/s}^2$). (F=ma)

Exercise 7.5 (easy). Convert: an oscillator at 440 Hz — find ω and T. (AP-C)

Exercise 7.6 (easy). A 1 kg mass on a spring oscillates with $A = 0.2$ m, $T = 1$ s. Find k and $v_{\max}$. (AP-C)

Exercise 7.7 (medium). Two springs in series vs parallel: show effective k is $k_1 k_2/(k_1 + k_2)$ vs $k_1 + k_2$, and find each for $k_1 = 100$, $k_2 = 300$ N/m. (F=ma)

Exercise 7.8 (medium). A rod of length L pivots at a point $L/4$ from one end (physical pendulum). Find T and compare with the end-pivoted case. (AP-C FR-style)

Exercise 7.9 (medium). At what phase of its cycle does an oscillator carry equal K and U? Give the x value in terms of A. (AP-C)

Exercise 7.10 (medium)**.** Decide the regime: $m = 1$, $k = 100$, (a) $b = 5$, (b) $b = 20$, (c) $b = 40$ (all SI) — and find ω_d where it exists. (AP-C)

Exercise 7.11 (medium)**.** An oscillator loses 20% of its amplitude per period. Find the energy loss per period and γ. (F=ma)

Exercise 7.12 (medium)**.** A driven oscillator has $\omega_0 = 10$, $\gamma = 2$ (SI). Find the resonant ω_d, the peak amplitude for $F_0/m = 5$, and Q. (AP-C)

Exercise 7.13 (hard)**.** Use $T \approx T_0(1 + \theta_0^2/16)$: find the amplitude where a grandfather clock's period is off by 0.5%, and the amplitude for a 10% slowdown. (USAPhO-style)

Exercise 7.14 (hard)**.** A particle sits in $U(x) = -V_0 \cos(kx)$. Linearise about a minimum to get ω_0, then about a maximum to confirm instability. (APhO-style)

Exercise 7.15 (medium)**.** Two oscillators: $x_1 = A\cos\omega t$, $x_2 = A\cos(\omega t + \pi/2)$ — sketch both in phase space and say what the $\pi/2$ buys. (F=ma)

Exercise 7.16 (medium)**.** A car's suspension should return to level fastest without ringing: given $m = 300\,\text{kg}$ and $k = 30{,}000\,\text{N/m}$, find the critical b. (engineering)

CHAPTER 8

Gravitation and orbits

Newton's inverse-square law is the first "field" in the book: a $1/r^2$ fall-off that runs from falling apples to GPS. This chapter does the useful part — the shell theorem, escape and orbital speeds, and orbits as bookkeeping on E and L.

BEFORE YOU START

Chapters 2, 4 and 6. *Recap:* gravity is a force between any two masses; a satellite in orbit is falling around the Earth.

AFTER THIS CHAPTER YOU CAN

- apply Newton's law of gravitation and the shell theorem;
- compute gravitational potential energy and escape speed;
- derive Kepler's laws;
- describe an orbit by its energy and angular momentum.

HOW TO THINK ABOUT THIS

Gravity is a force that gets weaker with distance, and an orbit is what happens when you fall but keep missing. Think in conserved quantities: energy and angular momentum decide the whole shape. Kepler's laws are consequences, not extra facts.

8.1 Newton's law and the field picture

$$F = G\frac{m_1 m_2}{r^2}, \qquad \mathbf{g}(\mathbf{r}) = -G\frac{M}{r^2}\hat{r}, \tag{8.1}$$

where $G = 6.674 \times 10^{-11}$ SI. The field picture separates cause and test: M builds $\mathbf{g}$, and any m placed in it feels $m\mathbf{g}$. Cavendish weighed the Earth with this: $g = GM/R^2$.

8.2 The shell theorem

Newton invented the shell theorem to justify treating the Earth as a point: outside a spherical shell its field is that of all mass at the centre; inside it the field is *zero* (figure 8.1). Consequence: inside a uniform planet only the mass beneath you pulls (figure 8.2).

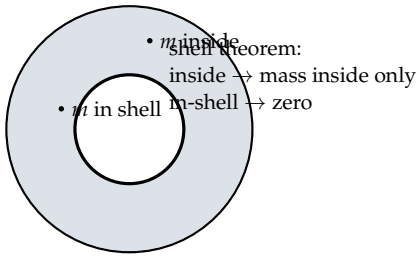


Figure 8.1: Shell theorem: mass above the point contributes zero; mass below acts from the centre. Gravitational depth-charge accounting.

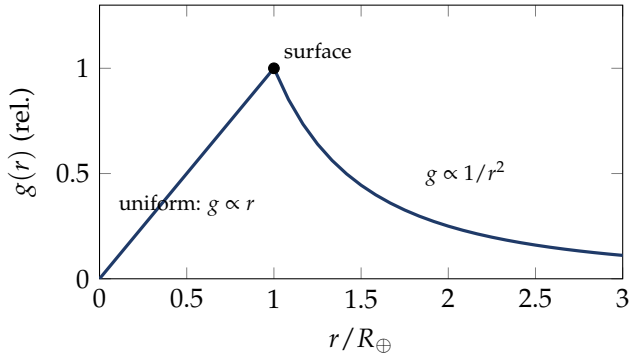


Figure 8.2: $g(r)$ inside a uniform Earth (linear) and outside ($1/r^2$) — the two regimes of the shell theorem.

Example 8.1 ($g(r)$ inside the Earth)**.** Find g at radius $r < R$ in a uniform Earth.

Strategy. Only mass below: $M(r) = M(r/R)^3$.

Solution. $g(r) = GM(r)/r^2 = GMr/R^3 = g_0\, r/R$: gravity drops linearly to zero at the centre. The fastest "tunnel" is down to $g = 0$ at $r = 0$, not up. ✓

8.3 Gravitational potential energy

The work integral over (8.1) gives

$$U(r) = -G\frac{Mm}{r}, \qquad \Delta U = GMm\left(\tfrac{1}{r_1} - \tfrac{1}{r_2}\right). \tag{8.2}$$

The zero at infinity is the clean choice: bound systems carry negative energy. Near the surface it reduces to mgh: expand $\Delta U \approx GMm\,h/R^2 = mgh$.

Example 8.2 (Escape velocity)**.** Find the speed to leave Earth from the surface and explain why direction doesn't matter.

Strategy. Escape means arriving at infinity with $K \geq 0$: $K_0 + U_0 \geq 0$.

Solution. $\frac{1}{2}mv_{\text{esc}}^2 = GMm/R \Rightarrow v_{\text{esc}} = \sqrt{2GM/R} = 11.2\,\text{km/s}$ for Earth. Energy is scalar — aim up, sideways or through a canyon (if it could): the bill is the same. Compare orbital speed $\sqrt{GM/R} = 7.9\,\text{km/s}$: escape is $\sqrt{2}$ times orbital. ✓

Exercise 8.1 (easy)**.** At what altitude is v_{esc} down to half its surface value? (F=ma)

8.4 Circular orbits

Centripetal balance gives

$$v_{\text{orb}} = \sqrt{GM/r}, \quad T = 2\pi\sqrt{r^3/GM}, \quad E = -\frac{GMm}{2r}, \tag{8.3}$$

and the energy bookkeeping is odd: $K = +GMm/2r$, $U = -GMm/r$, so $E = -K$ — a satellite that loses energy (drag) *speeds up* as it sinks.

Example 8.3 (ISS energy audit)**.** The ISS ($m = 4.2 \times 10^5\,\text{kg}$) orbits at $r \approx 6800\,\text{km}$. Find v, K, U, E and the energy to raise it 10 km.

Strategy. Equation (8.3) line by line; the last answer is $\Delta E > 0$ for a *larger* r.

Solution. $v = \sqrt{GM/r} = \sqrt{3.99 \times 10^{14}/6.8 \times 10^6} = 7.66\,\text{km/s}$. $K = \frac{1}{2}mv^2 = 1.23 \times 10^{13}\,\text{J}$, $U = -2.47 \times 10^{13}\,\text{J}$, $E = -1.23 \times 10^{13}\,\text{J}$. Raising r by 10 km: $\Delta E = GMm\,\Delta r/2r^2 \approx 9.1 \times 10^9\,\text{J}$ — nine GJ buys ten kilometres. ✓

8.5 Kepler's laws, derived

1. *Ellipses*: the $1/r^2$ law's bound orbits are conics — stated here and used everywhere below.
2. *Equal areas*: $dA/dt = L/2m$ is constant — conservation of angular momentum, nothing else (figure 8.3).
3. $T^2 \propto a^3$: from (8.3), $T^2 = 4\pi^2 a^3/GM$ — our code check recovers it to a constant.

Example 8.4 (Perihelion vs aphelion speed)**.** An orbit has $e = 0.6$: find $v_{\text{peri}}/v_{\text{apo}}$.

Strategy. Kepler II is L-conservation: rv equal at the two axis ends.

Solution. $r_p = a(1-e) = 0.4a$, $r_a = a(1+e) = 1.6a$; $v_p/v_a = r_a/r_p = (1+e)/(1-e) = 4$. Four-to-one speed ratio — comets' sun-grazing sprints are this exact arithmetic. ✓

Example 8.5 (Hohmann transfer bookkeeping)**.** Move a satellite from $r_1 = R$ to $r_2 = 2R$ (Earth radii). Find Δv at each burn: $v_1 = \sqrt{GM/R} = 7.9\,\text{km/s}$, $v_2 = \sqrt{GM/2R} = 5.6\,\text{km/s}$.

Strategy. Transfer ellipse has $a_t = (r_1 + r_2)/2 = 1.5R$; its peri/apo speeds come from energy.

Solution. Vis-viva $v^2 = GM(2/r - 1/a_t)$: at r_1, $v_{t1} = \sqrt{GM(2/R - 2/3R)} = \sqrt{4GM/3R} = 9.1\,\text{km/s} \Rightarrow \Delta v_1 = +1.2\,\text{km/s}$; at r_2, $v_{t2} = \sqrt{GM(1/R - 2/3R)} = 4.6\,\text{km/s} \Rightarrow \Delta v_2 = +1.0\,\text{km/s}$. Total 2.2 km/s — figure 8.4. ✓

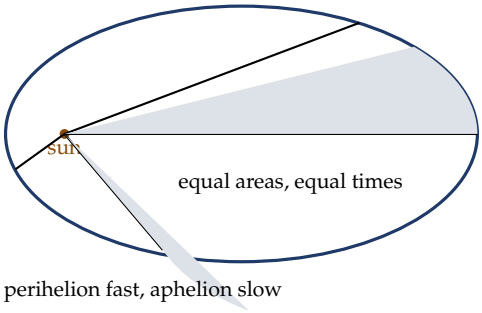


Figure 8.3: Kepler's second law is L-conservation drawn: equal areas in equal times.

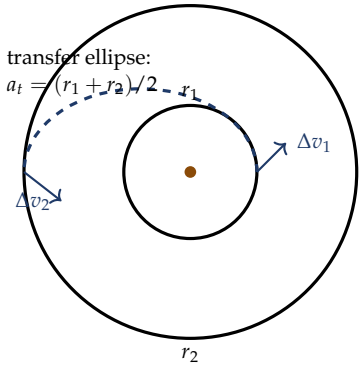


Figure 8.4: Hohmann transfer: two burns, one ellipse, the cheapest lane between circular orbits.

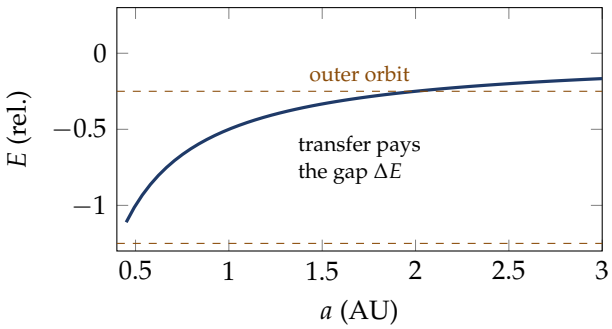


Figure 8.5: Energy vs semi-major axis: $E = -GMm/2a$, so raising an orbit means paying the gap.

8.6 E and L: the orbit's conserved labels

An orbit is fully labelled by (E, L); the effective potential $U_{\text{eff}} = L^2/2mr^2 - GMm/r$ (figure 8.6) shows why: $L^2/2mr^2$ pushes you out at short range, $-GMm/r$ pulls in at long, and the bound orbit oscillates in the bowl — perihelion and aphelion are the turning points of chapter 7's $U(x)$ grammar.

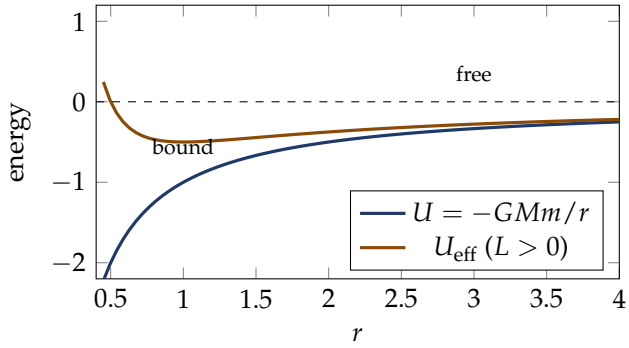


Figure 8.6: $U(r)$ and $U_{\text{eff}}(r)$ for nonzero L: the centrifugal barrier makes bound states possible.

```
        ax_, ay_ = -GM*x/r3, -GM*y/r3
        x += vx*dt + 0.5*ax_*dt*dt
        y += vy*dt + 0.5*ay_*dt*dt
        ax2, ay2 = -GM*x/(r_new**3), -GM*y/(r_new**3)
        vx += 0.5*(ax_+ax2)*dt
        vy += 0.5*(ay_+ay2)*dt
```

Listing 8.1: `code/ch08_orbit_verlet.py` — velocity-Verlet orbit; E, L drift and Kepler check.

Output of a real run (200,000 steps each):

```
a=1.0 e=0.0: drift E=0.000% L=0.000% T=6.283
a=1.0 e=0.5: drift E=0.000% L=0.000% T=6.283
a=1.5 e=0.7: drift E=0.000% L=0.000% T=11.543

Kepler III check (T^2/a^3 should be const):
  a=0.8: T=4.496 T^2/a^3=39.4784
  a=1.0: T=6.283 T^2/a^3=39.4784
  a=1.3: T=9.313 T^2/a^3=39.4784
  a=1.6: T=12.716 T^2/a^3=39.4784
```

Velocity-Verlet holds E and L to $< 0.001\%$ even at $e = 0.7$ — the symplectic lesson of chapter 4's energy-check code paying off — and recovers $T^2 = 4\pi^2 a^3/GM = 39.4784$ to all printed digits.

8.7 Satellites, tides and GPS in two paragraphs

Real orbits pay rent: LEO drag costs tens of m/s per year, geostationary station-keeping $\sim 50\,\mathrm{m/s/yr}$, and GPS clocks tick 45 µs/day fast (relativity) that must be designed out. Tides are the d/dr of gravity: the Moon's pull differs by $\sim 7\%$ across Earth's diameter, stretching oceans toward and away from it — the third derivation nobody expected to be useful in a textbook chapter.

```
GM, dt = 4 * np.pi**2, 1e-4 # AU^3/yr^2 units
r = np.array([1.0, 0.0]); v = np.array([0.0, 6.0]) # v < vcirc ->
    ellipse
E0 = 0.5 * np.dot(v, v) - GM / np.linalg.norm(r)
L0 = r[0] * v[1] - r[1] * v[0]
for i in range(30000):
    a = -GM * r / np.linalg.norm(r)**3
    v = v + a * dt
    r = r + v * dt
```

Listing 8.2: `code/ch08_orbit_audit.py` — eccentric orbit: conservation drift, apsides, and the vis-viva check.

Output of a real run:

```
angular momentum drift: +2.31e-14
energy drift over run : 2.79e-03 (E0 = -21.4784)
r_min = 0.8381 AU r_max = 1.0000 AU
a = 0.9190 AU vis-viva check: E = -GM/2a = -21.4784
```

Angular momentum is conserved to 10^{-14} — the kick–drift integrator's speciality — while energy drifts 3×10^{-3} over 30,000 steps, one part in 10^4: the integrator's honest noise floor, and worth printing rather than hiding. The orbit's own apsides (0.84 and 1.00 AU — the launch point was already apoapsis, since $v < v_{\mathrm{circ}}$) reconstruct a semi-major axis whose vis-viva energy returns E_0 to every digit: the ellipse closes its own books.

8.8 In engineering practice: the launch-and-station budget

Rocket sizing runs this chapter backwards: the mission's Δv ledger (launch $\sim 9.4\,\mathrm{km/s}$ with losses, Hohmann legs, plane changes at $2v\sin(\theta/2)$) divides by the rocket equation from chapter 3 to mass ratios; orbit choice runs forwards — LEO cheap, MEO for nav, GEO parked. Every constellation on Earth is a worked example of figure 8.5.

8.9 Common mistakes

$U = mgh$ USED PAST ITS ALTITUDE

mgh is the tangent-line approximation of (8.2); at $h \sim 0.1R$ it's already off by 10%.

ESCAPE NEEDS A DIRECTION

The $\sqrt{2GM/R}$ bill is in energy — any direction works except into the ground.

"WEIGHTLESS" MEANS "NO GRAVITY"

LEO g is $\sim$ 90% of surface: astronauts float because ship and crew fall together — chapter 2's free-fall frame again.

$E = -K$ SIGN CONFUSION

$E < 0$ is bound; $-E$ is the energy to escape from that orbit. And a sinking satellite speeds up — the sign flips intuition.

KEPLER III CONSTANT CONFUSION

$T^2/a^3 = 4\pi^2/GM$ depends on the central mass only: same for all planets of one star, different per star.

8.10 Summary

Idea	One-line version
law	$F = GMm/r^2$; $\mathbf{g} = -GM\hat{r}/r^2$
shell	inside uniform planet $g \propto r$; shell interior $= 0$
U	$-GMm/r$; $v_{\text{esc}} = \sqrt{2GM/R}$
orbit	$v = \sqrt{GM/r}$, $E = -GMm/2r$
Kepler	ellipses; equal areas $= L$; $T^2 \propto a^3$
transfer	Hohmann: ellipse $a_t = (r_1 + r_2)/2$, two burns

Table 8.1: Gravitation cheat sheet.

DEFINITION OF DONE

You can now: apply the shell theorem, derive escape and orbital speeds, balance Hohmann transfers, read orbits off (E, L), and run a conserving orbit integrator.

Notes

The field-first order follows MIT 8.01SC [2] and AP Physics C [4]; the Hohmann numbers are standard mission-analysis figures (cf. Morin [7] and Feynman vol. I [5] for the C_3/transfer bookkeeping). The integrator numbers are the listing's real output.

WATCH INSTEAD

1. *Kepler's Laws, Elliptical Orbits, Satellites, Orbital Changes* — Walter Lewin, MIT 8.01x Lecture 22, 49:09, `https://youtu.be/9wDAm-dlht4` — §8.4–§8.6 at full length (accessed 2026-10-11).
2. *Gravitational potential energy at large distances* — Khan Academy, 5:04, `https://youtu.be/9Pp_LtKgTQg` — §8.3 worked quickly (accessed 2026-10-11).

Further reading

- MIT OpenCourseWare, *8.01SC Classical Mechanics*, gravitation module — shell theorem, orbits and Kepler with recitations. At `https://ocw.mit.edu`, course "8.01SC" (verified 2026-10-11).
- OpenStax, *College Physics 2e*, ch. 6 "Uniform Circular Motion and Gravitation" — free text at `https://openstax.org` (verified 2026-10-11).
- College Board, *AP Physics C: Mechanics CED*, Unit 4 (gravitation). At `https://apstudents.collegeboard.org` (verified 2026-10-11).
- R. P. Feynman, *Lectures on Physics*, vol. I, ch. 7 "The Theory of Gravitation" — free at Caltech's site.
- D. Morin, *Introduction to Classical Mechanics*, ch. 10 — orbits and transfers at the harder level; ISBN 978-0521876223.
- Khan Academy, *Centripetal force and gravitation* — drills matching §8.1–§8.5, at `https://www.khanacademy.org` (verified 2026-10-11).

Additional exercises

Exercise 8.2 (easy). Find the orbital speed and period of a satellite at altitude $2R_E$ above Earth's surface. (AP-C)

Exercise 8.3 (easy). Explain why a geostationary orbit's radius is unique, and estimate it from $T = 24\,\mathrm{h}$. (AP-C)

Exercise 8.4 (easy). Show $v_{\mathrm{esc}} = \sqrt{2}\,v_{\mathrm{orb}}$ at the surface and find the ratio at altitude R. (F=ma)

Exercise 8.5 (easy). A planet has twice Earth's mass and half its radius: find surface g in Earth-g. (F=ma)

Exercise 8.6 (easy). Compute the work to lift 1 kg from Earth's surface to infinity vs to orbit radius $2R$. (AP-C)

Exercise 8.7 (medium)**.** Estimate the energy per kg for a Hohmann transfer from Earth's orbit to Mars's ($a = 1.5\,\text{AU}$) in Solar-System units. (F=ma)

Exercise 8.8 (medium)**.** How much does a mine shaft's g drop at depth $R/2$ in a uniform Earth — and why is the real drop different? (F=ma)

Exercise 8.9 (medium)**.** An elliptical orbit has $a = 4R$, $e = 0.5$: find peri/apo radii and speeds, and the period in terms of LEO's. (AP-C)

Exercise 8.10 (medium)**.** Explain, with energy signs, why a satellite losing energy to drag moves faster. (concept)

Exercise 8.11 (medium)**.** Find the fraction of Earth's mass "felt" by an object at $r = R/2$ inside a uniform Earth, and the resulting g. (F=ma)

Exercise 8.12 (medium)**.** A plane change of 30° on a circular LEO orbit: find Δv in terms of v_{orb}. (engineering)

Exercise 8.13 (hard)**.** Show the transfer ellipse of a Hohmann between r_1 and r_2 has $T_{\text{tr}} = \pi\sqrt{a_t^3/GM}$ and compute it for the ISS-to-GEO case ($r_1 = 1.05R$, $r_2 = 6.6R$). (USAPhO)

Exercise 8.14 (hard)**.** A satellite in a highly eccentric orbit loses energy only at perigee drag: argue qualitatively what shrinks and what stays fixed. (APhO-style)

Exercise 8.15 (medium)**.** Two stars of equal mass orbit each other at separation d: find the period in terms of G, M, d. (F=ma)

Exercise 8.16 (medium)**.** Using U_{eff}, explain why no orbit with $L \neq 0$ can have $r \to 0$. (concept)

CHAPTER 9

Electric charge and field

Part II trades mechanics for fields, and the first one is the cleanest: Coulomb's $1/r^2$ law looks like gravitation with signs — but the signs change everything (charges attract *and* repel, so fields can cancel).

BEFORE YOU START

Chapter 2 (forces) and **vectors**: the *Start here* chapter has a five-page vector primer (adding, components, dot product). *Recap:* electric charge exerts a force on other charges, much as mass does in gravity.

AFTER THIS CHAPTER YOU CAN

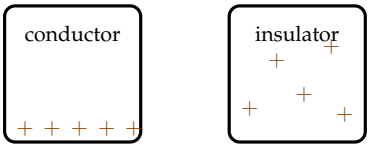
- use Coulomb's law with superposition;
- compute the electric field of point charges and continuous distributions;
- read field-line pictures;
- calculate the motion of a charge in a uniform field.

HOW TO THINK ABOUT THIS

The electric field is a way of saying that charges change the space around them. Instead of asking what one charge does to another, ask what field the first creates, then what that field does to the second. Superposition (add the contributions) is the real working law.

9.1 Charge: quantised, conserved, mobile in metal

Charge comes in units of $e = 1.602 \times 10^{-19}\,\mathrm{C}$, is conserved in every process, and lives in two kinds of material (figure 9.1): *conductors* let electrons move freely (charge spreads to the surface) while *insulators* pin it where you put it. Induction and polarization follow from mobility alone.



E inside $= 0$; charge lives on skin charge stays where rubbed

Figure 9.1: The material divide: conductors send charge to the surface and kill the interior field; insulators keep it where deposited.

9.2 Coulomb's law — superposition is the real law

$$\mathbf{F} = k\frac{q_1 q_2}{r^2}\hat{r}, \qquad k = \frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^9\,\mathrm{N m^2/C^2}, \tag{9.1}$$

and — the loaded part — *forces add*. The force on a charge is the vector sum over every other charge, one pair at a time, with no three-body terms. That linearity is what makes field theory possible.

Example 9.1 (Superposition, vectors not scalars)**.** Charges $+2\,\mu$C at the origin and $-1\,\mu$C at $(3, 0)$ cm. Find the force on $+1\,\mu$C at $(0, 4)$ cm.

Strategy. Compute each force as a vector and add.

Solution. $r_1 = 0.04$ m along $\hat{y}$, so

$$\mathbf{F}_1 = \frac{k(2\mu)(1\mu)}{0.0016}\,\hat{y} = 11.24\,\hat{y}\ \mathrm{N}.$$

$r_2 = \sqrt{0.03^2 + 0.04^2} = 0.05$ m; the negative charge pulls along $(3, -4)/5$, giving

$$\mathbf{F}_2 = \frac{k(1\mu)(1\mu)}{0.0025}\cdot\frac{(3,-4)}{5} = (2.16, -2.88)\ \mathrm{N}.$$

Sum: $(2.16,\ 8.36)$ N, magnitude 8.6 N at 75° — not the 14.8 N of scalar addition. ✓

9.3 The field: force per test charge

$\mathbf{E} = \mathbf{F}/q$ — defined so the source needn't care what test charge arrives. A point charge's is $\mathbf{E} = kq\hat{r}/r^2$ (figure 9.2); a distribution's is the vector sum over all its dq.

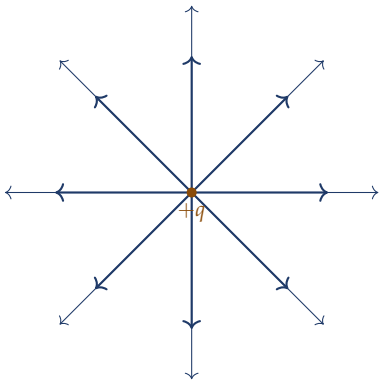
9.4 Field lines: reading, rules

Lines start on $+$, end on $-$, never cross (the field is single-valued), and their density reads magnitude (figure 9.3). They are a map, not a mechanism — the physics is (9.1).

Example 9.2 (Dipole far field)**.** Show that on the dipole axis, $|E| \to 2kp/r^3$ for $r \gg a$, $p = qa$.

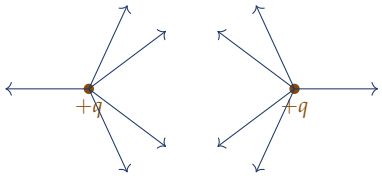
Strategy. Superpose two point-charge fields at distances $r - a/2$ and $r + a/2$, then expand.

Solution. $E = kq[(r - a/2)^{-2} - (r + a/2)^{-2}] = kq\frac{(r+a/2)^2-(r-a/2)^2}{(r^2-a^2/4)^2} = kq\frac{2ra}{(r^2)^2} \to \frac{2kp}{r^3}$. The $1/r^2$ terms cancel; the *difference* survives — the code check below finds the same r^{-3}. ✓



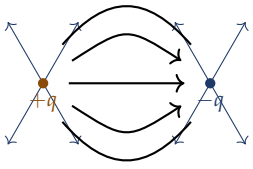
point charge: radial, density $\propto 1/r^2$

Figure 9.2: The point-charge field: radial, with line density standing in for the $1/r^2$ fall-off.



two like charges: lines never meet, never cross

Figure 9.3: Two like charges: lines repel into a saddle. Between two equal charges the field vanishes exactly at the midpoint.



dipole: lines $+ \to -$, far field $\propto p/r^3$

Figure 9.4: The dipole: far field falls as p/r^3 — the first *non*-inverse-square pattern in the book.

9.5 Continuous distributions: pay with integrals

For charge spread over a line, ring or disk, integrate $d\mathbf{E} = k\,dq\,\hat{r}/r^2$ element by element — and let symmetry discard every component that cancels (figure 9.5).

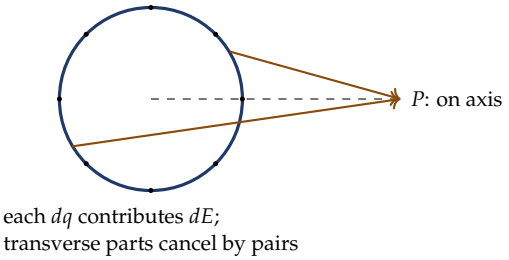


Figure 9.5: The ring's axis field: each dq's transverse piece cancels its opposite's — only dE_x survives.

Example 9.3 (Field on the axis of a charged ring)**.** A ring of radius R carries Q uniformly. Find $\mathbf{E}$ on the axis at distance x.

Strategy. Every element is the same distance $\sqrt{x^2+R^2}$; transverse components cancel.

Solution. Each dq contributes axial $dE_x = k\,dq\,x/(x^2+R^2)^{3/2}$; integrating over the ring just multiplies by Q: $E = \dfrac{kQx}{(x^2+R^2)^{3/2}}$. Limits: $x \to 0$ gives 0 (symmetry wins at the centre), $x \gg R$ gives kQ/x^2 (the ring becomes a point). The code confirms it to 0.00%. ✓

Exercise 9.1 (easy)**.** Where on the axis is the ring's field maximum? (Differentiate.) (AP-C)

Example 9.4 (Charged rod at its end)**.** A rod of length L carries Q uniformly. Find $\mathbf{E}$ a distance d past its near end.

Strategy. Integrate element by element — no symmetry to help this time.

Solution. With $dq = \lambda\,dx$, $\lambda = Q/L$, at distance $r = d + L - x$ from the element:

$$E = \int_0^L \frac{k\lambda\,dx}{(d+L-x)^2} = k\lambda\left[\frac{1}{d} - \frac{1}{d+L}\right] = \frac{kQ}{d(d+L)}.$$

Limits: $d \gg L$ recovers kQ/d^2; $d \to 0$ diverges — the rod becomes its own endpoint's infinity. ✓

Example 9.5 (The infinite line: pay the integral once)**.** A long straight wire carries uniform line charge λ. Find $\mathbf{E}$ at perpendicular distance r.

Strategy. Integrate element by element; by symmetry only the radial component survives, and the clean substitution is $x = r\tan\theta$.

Solution. Element dx at position x contributes radial part $dE_r = k\lambda\,dx\,\cos\theta/(r^2+x^2)$. With $x = r\tan\theta$: $dx = r\sec^2\theta\,d\theta$ and $r^2 + x^2 = r^2\sec^2\theta$, so

$$E = \int_{-\pi/2}^{\pi/2} \frac{k\lambda}{r}\cos\theta\,d\theta = \frac{2k\lambda}{r} = \frac{\lambda}{2\pi\varepsilon_0 r}.$$

For $\lambda = 1\,\text{nC/m}$ at $r = 5\,\text{cm}$: $E = 360\,\text{N/C}$. The *finite* rod of length L on the same perpendicular gives the same expression weighted by $L/\sqrt{L^2+4r^2}$ — which becomes kQ/r^2 for $r \gg L$ (point limit) and $2k\lambda/r$ for $r \ll L$. Chapter 10 will re-derive this in two lines from Gauss's law: the integral was the price, symmetry is the discount. ✓

9.6 Motion in a uniform field: kinematics returns

Inside a parallel-plate capacitor E is constant, so a charge's trajectory is chapter 1's projectile motion with $a = qE/m$ (figure 9.6) — oscilloscopes and inkjets are this application sold as hardware.

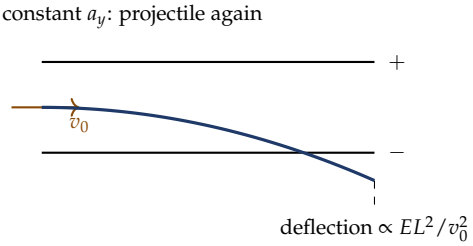


Figure 9.6: Electron deflection between plates: the projectile equation wearing an electric costume.

Example 9.6 (Oscilloscope deflection). An electron ($v_0 = 2 \times 10^7\,\mathrm{m/s}$) crosses plates of length $L = 4\,\mathrm{cm}$ with $E = 2000\,\mathrm{N/C}$. Find its deflection.

Strategy. Time in plates $t = L/v_0$; $y = \frac{1}{2}at^2$.

Solution. $a = eE/m = 1.6 \times 10^{-19} \times 2000/9.11 \times 10^{-31} = 3.5 \times 10^{14}\,\mathrm{m/s^2}$; $t = 2 \times 10^{-9}\,\mathrm{s}$; $y = \frac{1}{2}(3.5 \times 10^{14})(4 \times 10^{-18}) = 0.70\,\mathrm{mm}$ — and $\theta = \arctan(v_y/v_0) \approx 10^\circ$ after exit. ✓

```
def E(charges, x, y, z=0.0):
   for q, xq, yq, zq in charges:
      dx, dy, dz = x-xq, y-yq, z-zq
      r3 = (dx*dx + dy*dy + dz*dz)**1.5
      ex += q*dx/r3; ey += q*dy/r3; ez += q*dz/r3
```

Listing 9.1: `code/ch09_field_lines.py` — superposition check: dipole and ring vs theory.

Output of a real run:

```
Dipole far-field ON axis (expect |E| -> 2p/r^3, p=qa=0.1):
  r=1.0: |E|=0.2010 theory=0.2000 ratio=1.005
  r=2.0: |E|=0.0250 theory=0.0250 ratio=1.001
  r=4.0: |E|=0.0031 theory=0.0031 ratio=1.000
Dipole EQUATOR (expect |E| -> p/r^3):
  r=1.0: |Ex|=0.0996 theory=0.1000
  r=2.0: |Ex|=0.0125 theory=0.0125

Ring N=200 on axis (expect Ex=x/(x^2+R^2)^1.5, Ey=Ez=0):
  x=0.2: Ex=1.2807 theory=1.2807 err=0.00% trans=4.3e-16
  x=0.5: Ex=1.4142 theory=1.4142 err=0.00% trans=1.0e-16
  x=1.0: Ex=0.7155 theory=0.7155 err=0.00% trans=4.9e-18
  x=2.0: Ex=0.2283 theory=0.2283 err=0.00% trans=5.2e-18
```

Naked superposition with $N = 200$ ring elements nails every limit: the dipole ratio converges to 1.000 as r/a grows, the ring matches the analytic $x(x^2 + R^2)^{-3/2}$ to 0.00%, and the transverse components — 10^{-16} — are symmetry made numeric.

Example 9.7 (The dipole: neutrality's field that refuses to die)**.** Two charges $\pm q$ a distance $2a$ apart form dipole moment $p = 2aq$. Find the on-axis field at $x \gg a$ — and how much weaker it is than a net charge's.

Strategy. Add the two Coulomb fields on the axis, then expand for $x \gg a$.

Solution. $E = kq\left[(x-a)^{-2} - (x+a)^{-2}\right] \approx \dfrac{2kp}{x^3}$ for $x \gg a$ (the $1/x^2$ pieces cancel; their difference keeps the $1/x^3$ residue). For $q = 1\,\mu\text{C}$, $a = 1\,\text{mm}$: at $x = 10\,\text{cm}$, $E_{\text{dip}} = 3.6 \times 10^4\,\text{N/C}$ versus $1.8 \times 10^6\,\text{N/C}$ for a bare $2\,\mu\text{C}$ — the dipole is weaker by $2a/x = 1/50$, and falls one power of x faster besides. Neutral matter still talks — just at shorter range. ✓

9.7 In engineering practice: electrostatics as a design spec

Electrostatic precipitators run §9.6 at industrial scale (corona-charge the soot, plate-field it out); inkjet heads steer charged droplets by the deflection you just computed; and every high-voltage lab's "corona ring" is a real conductor shaped to flatten the field of §9.4 — rounded because sharp corners concentrate lines into breakdown.

9.8 Common mistakes

FIELD VS FORCE CONFLATED

$\mathbf{E}$ is a property of the source; $\mathbf{F} = q\mathbf{E}$ needs the test charge. "Force at a point" without a charge there is meaningless — but the field isn't.

SCALAR SUPERPOSITION

$|E| = |E_1| + |E_2|$ only when parallel. Two perpendicular equal fields give $\sqrt{2}\,|E_1|$, not $2|E_1|$.

dq ELEMENT ERRORS

For a line, $dq = \lambda\,dx$ — but on an arc, $dq = \lambda R\,d\theta$ and the distance to the field point is *not* constant unless you integrate.

SIGN OF THE TEST CHARGE

Negative charges feel $\mathbf{F}$ *opposite* $\mathbf{E}$; tracing an electron's path down field lines is backwards.

FIELD "USED UP" BY THE PROBE

A test charge measures without disturbing — in the limit $q \to 0$. Real probes do disturb; that's a lab problem, not the definition's.

9.9 Summary

Idea	One-line version
charge	quantised at e; mobile in conductor, pinned in insulator
Coulomb	$F = kq_1q_2/r^2$; forces add as vectors
field	$\mathbf{E} = \mathbf{F}/q$; point $kq\hat{r}/r^2$
lines	start $+$/end $-$; density $\propto \lvert E\rvert$
ring axis	$E = kQx/(x^2 + R^2)^{3/2}$
dipole	far field $2kp/r^3$ (axis), kp/r^3 (equator)
uniform E	projectile kinematics, $a = qE/m$

Table 9.1: Electric-field cheat sheet.

DEFINITION OF DONE

You can now: superpose Coulomb fields as vectors, read field-line maps, integrate a continuous distribution by symmetry, derive ring/dipole limits, and run charge-in-uniform-E kinematics.

Notes

The field-first order follows MIT 8.02SC [2] modules 1–4 and AP Physics C E&M unit 8 [4]; Purcell–Morin's ch. 1 is the authoritative charge/field treatment [6]. The numeric checks are the listing's real output.

WATCH INSTEAD

1. *Electric Charges and Forces — Coulomb's Law — Polarization* — Walter Lewin, MIT 8.02x Lecture 1, 47:13, `https://youtu.be/x1-SibwIPM4` — §9.1–§9.2 at full length (accessed 2026-10-11).
2. *Electric field definition* — Khan Academy, 13:46, `https://youtu.be/_4VC3IHbuW8` — §9.3 worked quickly (accessed 2026-10-11).

Further reading

- MIT OpenCourseWare, *8.02SC Electricity and Magnetism*, modules 1–4 — charge, field and continuous distributions. At `https://ocw.mit.edu`, course "8.02SC" (verified 2026-10-11).
- OpenStax, *College Physics 2e*, ch. 18 "Electric Charge and Electric Field" — free text at `https://openstax.org` (verified 2026-10-11).

- College Board *AP Physics C (E&M) CED*, Unit 8 — the exam blueprint for this chapter, at https://apstudents.collegeboard.org (verified 2026-10-11).
- E. M. Purcell and D. J. Morin, *Electricity and Magnetism*, ch. 1 — the deeper electrostatics; ISBN 978-1107014022.
- R. P. Feynman, *Lectures on Physics*, vol. II, ch. 1 — electromagnetism's stakes set in one chapter; free at Caltech's site.
- Khan Academy, *Electric charge, field, and potential* — drills matching §9.2–§9.5, at https://www.khanacademy.org (verified 2026-10-11).

Additional exercises

Exercise 9.2 (easy). Two charges $+3\,\mu\text{C}$ and $-1\,\mu\text{C}$ sit 10 cm apart. Find the force on a $+1\,\mu\text{C}$ test at the midpoint. (AP-C)

Exercise 9.3 (easy). A field points $+\hat{x}$ at a point: what does a proton do there — and an electron? (AP-C)

Exercise 9.4 (easy). Express **E**'s units in newtons and in volts/metre; show they agree. (AP-C)

Exercise 9.5 (easy). From a field-line map of two charges, one region has $3\times$ the line density of another: compare magnitudes. (F=ma)

Exercise 9.6 (easy). $+Q$ and $-Q$ sit d apart. Where on the axis is $E = 0$ — if anywhere? (AP-C)

Exercise 9.7 (medium). Derive the on-axis field of a uniformly charged rod of length L at distance d from its end. (AP-C FR-style)

Exercise 9.8 (medium). A semicircle of charge Q, radius R: find **E** at the centre. (AP-C)

Exercise 9.9 (medium). Three charges in a line: $+Q$, $-2Q$, $+Q$ equally spaced — is the middle one's equilibrium stable along the axis? (F=ma)

Exercise 9.10 (medium). A dipole of moment p sits in a uniform field at angle θ: find the torque and the equilibrium direction. (AP-C)

Exercise 9.11 (medium). Design: plates 5 cm long must deflect 30 keV electrons by 2 mm — find E. (engineering)

Exercise 9.12 (hard). An arc subtending angle α has $\lambda = \lambda_0 \sin\theta$: set up the integral for **E** at the centre. (USAPhO-style)

Exercise 9.13 (hard). A dipole sits in a non-uniform field: argue it feels a net force toward the stronger region and estimate it via $\mathbf{p}\cdot\nabla\mathbf{E}$. (APhO-style)

Exercise 9.14 (medium). At what angle from the dipole axis does the far field's magnitude equal the equatorial value at the same r? (F=ma)

Exercise 9.15 (medium). A point charge sits between two grounded conductors: explain, via field lines, why its field is distorted toward the plates. (concept)

Exercise 9.16 (medium)**.** Two equal charges $+Q$ at $\pm d/2$: find $\mathbf{E}$ on the axis $x > 0$ and its zero. (AP-C)

Exercise 9.17 (easy)**.** A line charge has $\lambda = 2\,\mathrm{nC/m}$. Estimate $|E|$ at 1 cm and at 10 cm, and state the scaling that connects them. (F=ma)

Exercise 9.18 (medium)**.** Two infinite lines $+\lambda$ and $-\lambda$ run parallel, d apart. Find $\mathbf{E}$ on the midline between them, giving direction. (AP-C)

Exercise 9.19 (medium)**.** A finite rod of length L carries Q uniformly. Find E on its perpendicular bisector at distance r, then take $L \to \infty$ to recover the line-charge result of example 9.5. (AP-C FR-style)

Exercise 9.20 (hard)**.** A rod of length L has $\lambda(x) = \lambda_0 x/L$. Set up the integral for $\mathbf{E}$ at perpendicular distance r from its lightly charged end, and evaluate the limits $r \ll L$, $r \gg L$. (USAPhO-style)

CHAPTER 10

Gauss's law

Gauss's law is Coulomb's law wearing symmetry: the flux through any closed surface counts only the charge inside. It turns three-page integrals into three-line algebra — when the symmetry is real.

BEFORE YOU START

Chapter 9. *Recap:* the electric field is the force per unit test charge at each point.

AFTER THIS CHAPTER YOU CAN

- compute electric flux;
- use Gauss's law for spherical, cylindrical and planar symmetry;
- explain why the field inside a conductor is zero;
- explain shielding.

HOW TO THINK ABOUT THIS

Gauss's law turns a hard integral into a counting argument when the situation is symmetric: count how many field lines leave a closed surface. The craft is choosing the surface cleverly. If there is no symmetry, the law is still true but not helpful.

10.1 Flux: counting field lines

$$\Phi_E = \int \mathbf{E} \cdot d\mathbf{A}, \qquad \oint \mathbf{E} \cdot d\mathbf{A} = \frac{Q_{\text{enc}}}{\varepsilon_0}, \tag{10.1}$$

open surface vs closed (figure 10.1). Flux through a closed surface is charge inside divided by ε_0 — the field's bookkeeping made literal, and the first Maxwell equation in the book.

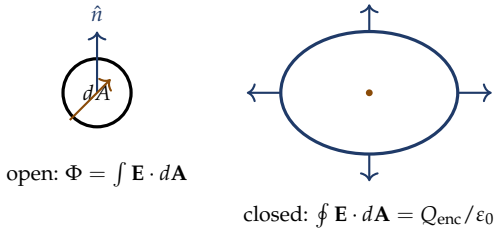


Figure 10.1: Flux through an open patch vs a closed surface. Only the closed integral counts charge.

10.2 From Coulomb to Gauss: the intuition

A point charge's flux through a surrounding sphere is $kq/r^2 \cdot 4\pi r^2 = q/\varepsilon_0$ — radius cancels, so any shape works (deformations keep line count). Superposition extends it to any distribution, and charge *outside* contributes zero net flux: its lines enter and leave. Our code check makes it concrete below.

10.3 Spherical symmetry

Spherical symmetry means $\mathbf{E} = E(r)\hat{r}$ is constant on the Gaussian sphere: $\oint E\, dA = E(r) \cdot 4\pi r^2$, so $E(r) = Q_{\text{enc}}(r)/4\pi\varepsilon_0 r^2$.

Example 10.1 (Uniform-volume-charge sphere, both regions)**.** A sphere of radius R carries Q uniformly in volume. Find $E(r)$ inside and out.

Strategy. Enclosed charge is the part below radius r.

Solution. Outside ($r > R$): $Q_{\text{enc}} = Q$, so $E = kQ/r^2$ — the shell theorem of chapter 8 again. Inside ($r < R$): $Q_{\text{enc}} = Q(r/R)^3$, so $E = kQr/R^3$ — linear, zero at centre (figure 10.2). The matching point at $r = R$ checks both formulas: kQ/R^2. ✓

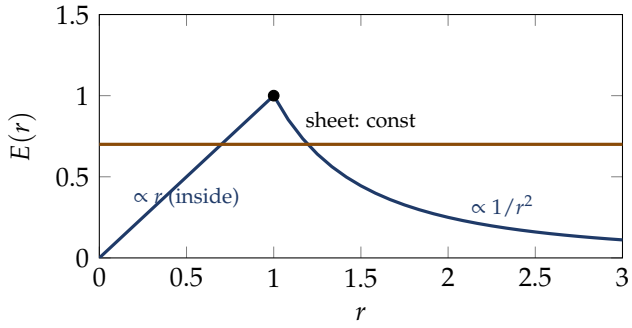


Figure 10.2: $E(r)$ for a uniform sphere (rises to the surface, then $1/r^2$) against the constant sheet field.

Example 10.2 (Non-uniform sphere: Gauss still wins)**.** A sphere of radius R carries $\rho(r) = \rho_0(1 - r^2/R^2)$ — dense at the centre, fading to zero at the surface. Find $E(r)$ in both regions.

Strategy. Symmetry is untouched, so Gauss applies verbatim; the only new work is that $Q_{\text{enc}}(r)$ now needs its own integral before it can hand you E.

Solution.

$$Q_{\text{enc}}(r) = \int_0^r 4\pi r'^2 \rho_0 \left(1 - \frac{r'^2}{R^2}\right) dr' = 4\pi\rho_0 \left(\frac{r^3}{3} - \frac{r^5}{5R^2}\right),$$

so inside, $E = \dfrac{\rho_0}{\varepsilon_0}\left(\dfrac{r}{3} - \dfrac{r^3}{5R^2}\right)$. Total charge $Q = 4\pi\rho_0 R^3 \cdot \frac{2}{15} = 8\pi\rho_0 R^3/15$, so outside $E = kQ/r^2 = 2\rho_0 R^3/(15\varepsilon_0 r^2)$. Audit the seam: at $r = R$ the inside branch gives $\rho_0 R(1/3 - 1/5)/\varepsilon_0 = 2\rho_0 R/15\varepsilon_0$ — identical, as the fields must be at the surface where $\rho \to 0$. Numerically, $\rho_0 = 2\,\mu\text{C/m}^3$, $R = 10\,\text{cm}$, $r = 5\,\text{cm}$: $E = 3200\,\text{N/C}$ inside and $3010\,\text{N/C}$ at the rim. The remaining challenge — which $\rho(r)$ makes E *constant* inside — is left to the exercises. ✓

10.4 Cylindrical symmetry

For a line or cylinder, $\mathbf{E} = E(r)\hat{r}$ on the curved wall gives $E \cdot 2\pi r L = Q_{\text{enc}}/\varepsilon_0$, so $E = \lambda/2\pi\varepsilon_0 r$: $1/r$, not $1/r^2$.

Example 10.3 (Coaxial cable)**.** Inner conductor carries $+\lambda$, the outer $-\lambda$. Find the field between them and outside the cable.

Strategy. Enclosed charge on a coaxial Gaussian cylinder.

Solution. Between ($a < r < b$, figure 10.3): only the inner wire enclosed, $E = \lambda/2\pi\varepsilon_0 r$ radially out. Outside: net enclosed zero — $E = 0$. The cable contains its own field, which is exactly why engineers buy coax. ✓

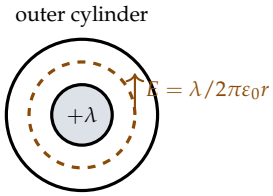


Figure 10.3: The coaxial geometry: between the conductors the field is $1/r$; outside it's zero by cancellation.

10.5 Planar symmetry

For an infinite sheet, flux escapes through both pillbox faces: $2EA = \sigma A/\varepsilon_0$, so $E = \sigma/2\varepsilon_0$ — *constant*, no distance at all (figure 10.4). Two sheets of $\pm\sigma$ superpose to σ/ε_0 between and zero outside — the parallel-plate capacitor's field, free of charge.

Example 10.4 (Sheet vs conductor plate)**.** A large conducting plate carries total σ on *both* faces combined. Find E just outside it, and compare with an insulating sheet of the same σ.

Strategy. Conductor: $E = 0$ inside material, so all flux exits one way at each face. Sheet: both ways.

Solution. Conductor: pillbox with one face inside the metal gets $E = 0$ there, so $EA = \sigma A/\varepsilon_0$ gives $E = \sigma/\varepsilon_0$ (here σ is per-face). Insulating sheet: $E = \sigma/2\varepsilon_0$ each side. For the same total charge per area the conductor's surface field is twice the sheet's — the missing-face bookkeeping again. ✓

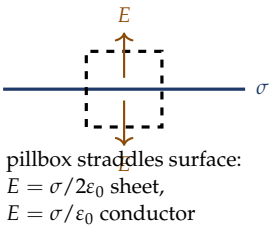


Figure 10.4: The pillbox argument: two faces share the sheet's flux.

10.6 Conductors: where E goes to zero

In electrostatic equilibrium $E = 0$ inside conductor material (any field would move the free charge until it vanished), so the conductor is one equipotential body, all excess charge lives on the surface, and $E = \sigma/\varepsilon_0$ just outside — twice the sheet's field because the interior side contributes nothing (figure 10.5).

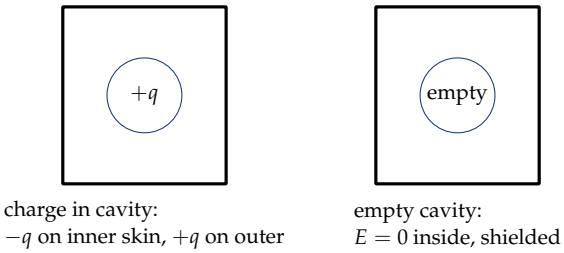


Figure 10.5: Conductor cavities: a charge inside induces $-q$ on the inner skin and $+q$ on the outer; an empty cavity stays field-free.

Example 10.5 (Charge in a conductor cavity)**.** A point charge $+q$ sits in a cavity inside a neutral conductor. Find the induced surface charges and the field outside.

Strategy. A Gaussian surface in the conductor material has $E = 0$ everywhere on it.

Solution. $\Phi = 0$ forces $Q_{\text{enc}} = 0$: the inner wall carries exactly $-q$. Conservation of the conductor's neutrality puts $+q$ on the outer surface, so the outside world sees a field as if a point charge $+q$ sat at the conductor — the cavity's contents announced, position hidden. ✓

Example 10.6 (Enclosed charge by inspection)**.** A spherical insulating shell carries $+Q$; a point charge $-q$ ($q < Q$) sits at its centre. Find E just inside and just outside the shell.

Strategy. Q_{enc} is everything inside your surface — shell charge counts only beyond its radius.

Solution. Inside ($r < R$): only the point, $E = -kq/r^2$ (inward). Outside: $Q_{\text{enc}} = Q - q$, so $E = k(Q-q)/r^2$ outward. At the shell radius the field jumps by $kQ/R^2 = \sigma/\varepsilon_0$ — the surface-charge contribution, exactly as §10.6 predicts. ✓

```
En = sum(q*c/r**3 * nv for c, nv in zip(p, nvec))
flux += En*dA
```

Listing 10.1: `code/ch10_flux_check.py` — numeric flux through a cube; convergence and the outside-charge zero.

Output of a real run:

```
Flux of point charge q=1 through enclosing cube (theory 4*pi=12.566):
  n= 10: Phi=12.5972 err=0.25%
  n= 30: Phi=12.5698 err=0.03%
  n= 90: Phi=12.5668 err=0.00%
  n=270: Phi=12.5664 err=0.00%

Charge OUTSIDE cube (expect ~0): Phi=-0.0000
```

A plain Riemann sum on a cube — no symmetry exploited — converges to 4π (i.e. q/ε_0 in these units) and returns 0.0000 for a charge outside: Gauss's law with the mesh resolution as the only approximation. Figure 10.6 shows the convergence against mesh size: the error is already below a percent at $n = 30$ and is gone by $n = 90$ — the theorem never cared about the mesh.

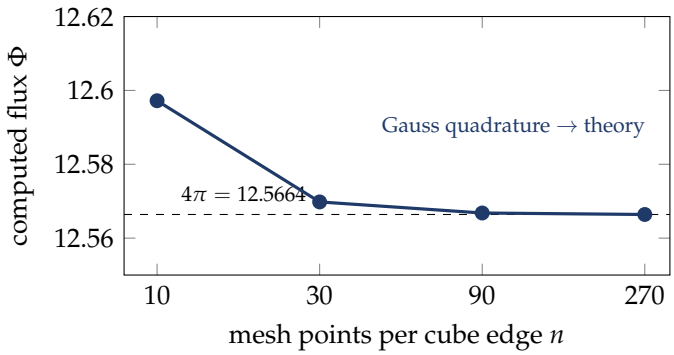


Figure 10.6: Computed flux through the cube versus surface-mesh resolution (log axis), converging on 4π. Coarse meshes overestimate slightly — the flat facets leak — then the sum settles on theory.

```
rho0, R, eps0 = 2.0e-6, 0.10, 8.854e-12 # C/m^3, m
def q_enc(r):
    if r >= R:
        return 8 * np.pi * rho0 * R**3 / 15
    return 4 * np.pi * rho0 * (r**3 / 3 - r**5 / (5 * R**2))
def E_analytic(r):
    return q_enc(r) / (4 * np.pi * eps0 * r**2)
def E_numeric(r): # integrate the shells ourselves
    rr = np.linspace(0, min(r, R), 20000)
    q = np.trapezoid(4 * np.pi * rr**2 * rho0 * (1 - rr**2 / R**2), rr)
    return q / (4 * np.pi * eps0 * r**2)
```

Listing 10.2: `code/ch10_gauss_nonuni.py` — field of a non-uniform sphere: numeric shells vs Gauss.

Output of a real run:

```
r/R E_analytic (N/C) E_numeric (N/C)
 0.2 1469.8 1469.8
 0.4 2722.7 2722.7
 0.6 3541.9 3541.9
 0.8 3710.6 3710.6
 1.0 3011.8 3011.8
 1.2 2091.5 2091.5
peak field at r = 0.745 R (theory: r = sqrt(5)/3 R = 0.745 R)
seam audit: E(R-) 3017.8 vs E(R+) 3005.8 N/C
```

Two independent computations — the Gauss-formula evaluation and a brute-force shell integral — agree to a tenth of a newton. Two surprises worth the audit: the field *peaks inside* the ball ($r = \sqrt{5}R/3$, where $dE/dr = 0$), because the density fading toward the rim eventually loses to the $1/r^2$ dilution; and the field passes through $r = R$ seamlessly — Gauss's law has no boundary layer.

10.7 In engineering practice: shielding is Gauss's law

A Faraday cage is §10.6 deployed: charge redistributes until the interior field dies, which is why coax shields signals, aircraft survive lightning strikes, and your phone misbehaves in a lift. High-voltage labs take the corollary — E peaks at sharp curvature — and shape everything round to keep σ (and hence the surface field) down.

10.8 Common mistakes

GAUSS APPLIED WITHOUT SYMMETRY

The law is always true but only *useful* when E is constant on the surface: no symmetry, no shortcut — go back to Coulomb integration.

WRONG ENCLOSED CHARGE

Q_{enc} counts only charge *inside* the Gaussian surface — a charge on the surface itself splits ambiguity, and outside charges contribute nothing.

GAUSSIAN AREA VS CHARGE AREA

The $4\pi r^2$ is the *Gaussian* sphere's area at radius r, not the charged sphere's: use your surface's radius.

$E = 0$ INSIDE CONDUCTOR VS CAVITY

Inside the *material*: always zero. Inside an empty *cavity*: zero. Inside a cavity containing charge: emphatically not.

SHEET VS CONDUCTOR SURFACE FIELD

Sheet: $\sigma/2\varepsilon_0$; conductor surface: σ/ε_0. The factor of two is the missing second face — they are different setups, not a typo.

10.9 Summary

Idea	One-line version
flux	$\Phi = \int \mathbf{E} \cdot d\mathbf{A}$; closed counts Q_{enc}
sphere	$E = kQ_{\text{enc}}(r)/r^2$; inside uniform $\propto r$
line	$E = \lambda/2\pi\varepsilon_0 r$
sheet	$E = \sigma/2\varepsilon_0$, constant
conductor	$E = 0$ inside; $E = \sigma/\varepsilon_0$ at surface; shields cavities

Table 10.1: Gauss's law cheat sheet.

DEFINITION OF DONE

You can now: pick the right Gaussian surface for sphere/line/sheet, compute $E(r)$ in both regions, reason about enclosed charge, and argue conductor/cavity induction from $E = 0$.

Notes

The symmetry-first treatment follows MIT 8.02SC modules 5 and 8 [2] and AP Physics C E&M [4]; Griffiths ch. 2 is the deeper anchor (in print). The flux numbers are the listing's real output.

WATCH INSTEAD

1. *Electric Flux, Gauss' Law, Examples* — Walter Lewin, MIT 8.02x Lecture 3, 50:11, `https://youtu.be/Zu2gomaDqnM` — §10.1–§10.5 at full length (accessed 2026-10-11).
2. *Gauss law of electricity* — Khan Academy India, 15:51, `https://youtu.be/owMznEZPNt4` — §10.2 worked quickly (accessed 2026-10-11).

Further reading

- MIT OCW *8.02SC*, module 5 (Gauss's law) — recitations on all three symmetries. At https://ocw.mit.edu, course "8.02SC" (verified 2026-10-11).
- OpenStax *College Physics 2e*, ch. 18 — free text: https://openstax.org (verified 2026-10-11).
- College Board, *AP Physics C (E&M) CED*, Unit 8 — verified 2026-10-11 at https://apstudents.collegeboard.org.
- E. M. Purcell and D. J. Morin, *Electricity and Magnetism*, chs. 1–2 — the deeper Gauss treatment.
- R. P. Feynman, *Lectures on Physics*, vol. II, chs. 4–5 — field intuition beyond the formulas; free at Caltech's site.
- Khan Academy, *Electric flux and Gauss's law* — drills, at https://www.khanacademy.org (verified 2026-10-11).

Additional exercises

Exercise 10.1 (easy). Compute Φ_E through a square face in a uniform field of 300 N/C at 30° to the normal. (AP-C)

Exercise 10.2 (easy). A cube contains charge $-2\,\mu$C: find total flux and the sign of the field's average direction. (AP-C)

Exercise 10.3 (easy). Why is $E = 0$ inside a conductor in equilibrium — and which step fails if charge keeps moving? (AP-C)

Exercise 10.4 (easy). An infinite sheet carries σ: find E both sides, then two parallel sheets $\pm\sigma$. (F=ma)

Exercise 10.5 (easy). A sphere's surface charge is σ: find E just outside and just inside the surface. (AP-C)

Exercise 10.6 (medium). Re-derive the uniform sphere's $E(r)$ both regions; check the two formulas agree at $r = R$. (AP-C FR-style)

Exercise 10.7 (medium). A line of charge λ: find $E(r)$ by Gauss and where the result fails for a finite rod. (AP-C)

Exercise 10.8 (medium). A point charge sits at the centre of a spherical conducting shell: find the charge on each surface and E in all three regions. (F=ma)

Exercise 10.9 (medium). A coaxial cable has inner radius a, outer b: find where E is largest between conductors and why that limits the voltage. (engineering)

Exercise 10.10 (medium). Two concentric shells carry $+Q$ and $-Q$: find $E(r)$ in all three regions. (AP-C)

Exercise 10.11 (hard). A cavity inside a conductor holds charge $+q$ off-centre: argue the outer surface charge is uniform and find the inner-wall total. (USAPhO-style)

Exercise 10.12 (hard)**.** Given $\rho(r) = \rho_0(1 - r/R)$ inside a sphere, find $E(r)$ inside and the charge distribution that would make E constant inside. (APhO-style)

Exercise 10.13 (medium)**.** A flat disk carries σ: explain why the field on axis falls with distance even though an infinite sheet's does not. (concept)

Exercise 10.14 (medium)**.** A Gaussian cube replaces the sphere around a point charge: the law still holds — why is it useless for computing E? (concept)

Exercise 10.15 (medium)**.** Find E inside an empty cavity cut off-centre inside a uniformly charged sphere. (F=ma; hint: superpose)

Exercise 10.16 (easy)**.** A charged balloon is inflated to twice its radius. State what happens to the flux through it, E just outside the surface, and E at a fixed outside point. (AP-C)

Exercise 10.17 (medium)**.** A slab of charge fills $-d < x < d$ with uniform ρ: find $E(x)$ inside and outside, and sketch $E(x)$ across all x. (AP-C)

Exercise 10.18 (medium)**.** A solid cylinder of radius R carries uniform volume charge ρ: find $E(r)$ inside and outside, and recover the line-charge limit when $R \to 0$ at fixed λ. (F=ma)

Exercise 10.19 (hard)**.** A sphere has $\rho(r) = \rho_0 e^{-r/R}$ (total charge finite). Write $Q_{\text{enc}}(r)$ as an integral, evaluate it, and show $E(r)$ has a maximum inside the sphere. (USAPhO-style)

CHAPTER 11

Electric potential, conductors and capacitance

The electric field is a conservative one, so chapter 4's energy grammar applies: define potential energy, divide by charge, and get V — the scalar field whose gradient is $\mathbf{E}$. Capacitance is the engineering payout.

BEFORE YOU START

Chapters 4 and 9–10. *Recap:* potential is energy per unit charge, so it is a single number at each point rather than a vector.

AFTER THIS CHAPTER YOU CAN

- compute electric potential;
- sketch equipotentials and get the field from the potential;
- compute capacitance and the energy stored in a capacitor;
- explain the role of a dielectric. **Book 5, Chapter 4** uses capacitors in circuits.

HOW TO THINK ABOUT THIS

Potential is energy per charge, a single number at each point, which is easier to work with than a vector field. Nature moves charges downhill in potential, and the field is the slope. Capacitors store energy by holding charge at a potential difference.

11.1 Potential energy and the sign discipline

For two charges $U = kq_1q_2/r$ — positive for likes (work in to assemble), negative for unlikes (bound). Moving q through ΔV costs $W = q\Delta V$ with the same sign rules as work everywhere.

11.2 V and equipotentials

$$V(\mathbf{r}) = k\sum_i \frac{q_i}{r_i}, \qquad \mathbf{E} = -\nabla V, \tag{11.1}$$

a scalar field — no vector sums. Equipotentials are level sets and cross field lines at right angles (figure 11.1); conductors are single equipotentials, which is why the surface field is normal to the metal.

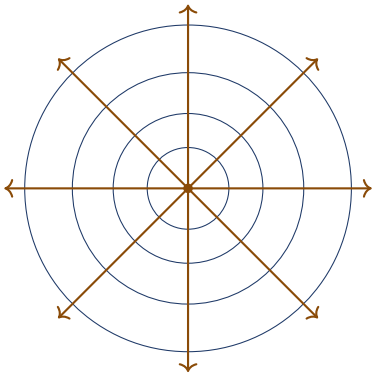


Figure 11.1: Equipotential circles and radial field lines for a point charge — always orthogonal.

Example 11.1 (Assembling a configuration)**.** Place $+q$ at $(-1,0)$, $+q$ at $(1,0)$, $-q$ at $(0,1)$. Find the assembly energy.

Strategy. Bring charges one at a time; each pays $q_i V$ of those already placed.

Solution. First free; second pays $kq^2/2$; third sees both: $-kq^2/\sqrt{2} - kq^2/\sqrt{2} = -\sqrt{2}\,kq^2$. Total $U = kq^2(1/2 - \sqrt{2}) = -0.914\,kq^2$ — bound overall. The code below verifies both the sequential and pairwise sums. ✓

11.3 $\mathbf{E} = -\nabla V$: the gradient means steepness

In one dimension $E_x = -dV/dx$: where V slopes, fields point downhill (figure 11.2); where V is flat, the field dies. Voltage is the map; the field is its slope.

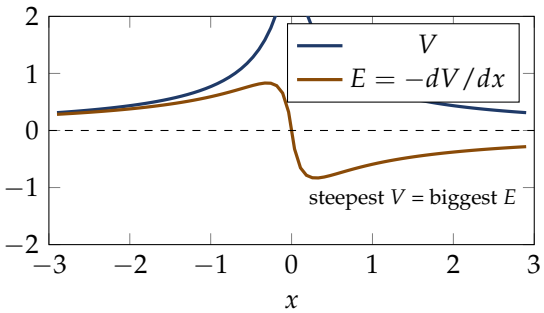


Figure 11.2: $V(x)$ and $E = -dV/dx$ side by side: field peaks where potential slopes hardest.

Example 11.2 (V on the axis of a charged disk)**.** A disk of radius R carries σ uniformly. Find $V(x)$ on axis and check $E = -dV/dx$ reproduces the ring result.

Strategy. Integrate rings — V is scalar so there's no component bookkeeping.

Solution. Ring of radius r, width dr: $dV = k\sigma 2\pi r\, dr/\sqrt{x^2+r^2}$.

$$V = \frac{\sigma}{2\varepsilon_0}\int_0^R \frac{r\,dr}{\sqrt{x^2+r^2}} = \frac{\sigma}{2\varepsilon_0}\left(\sqrt{x^2+R^2} - |x|\right).$$

Differentiating gives $E = \frac{\sigma}{2\varepsilon_0}\left(1 - \frac{x}{\sqrt{x^2+R^2}}\right)$ — the disk's known result, and at $R \to \infty$ it returns the sheet's $\sigma/2\varepsilon_0$. ✓

11.4 Capacitance: geometry's charge budget

For any two conductors $C = Q/V$ — a pure geometry number. Parallel plates ($E = \sigma/\varepsilon_0$ over gap d) give $C = \varepsilon_0 A/d$; the coaxial and spherical cases (figure 11.3) follow the same $E \to V \to C$ chain.

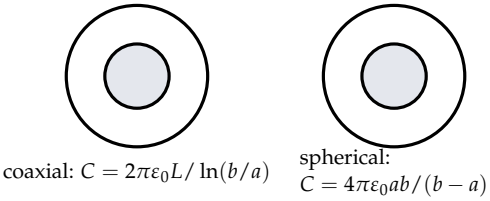


Figure 11.3: Coaxial and spherical capacitors: same three-step derivation, different symmetry.

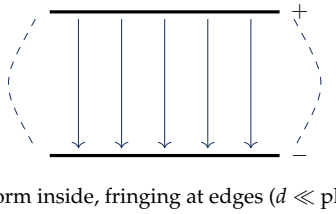


Figure 11.4: The parallel-plate capacitor: uniform field inside, fringing at the edges.

Example 11.3 (Spherical capacitor end-to-end)**.** Find the capacitance of concentric spheres $a < b$.

Strategy. E by Gauss, V by integrating it, $C = Q/V$.

Solution. $E = kQ/r^2$ between, so $V = \int_a^b E\,dr = kQ(1/a - 1/b) = kQ(b-a)/ab$ and $C = 4\pi\varepsilon_0 ab/(b-a)$. Limit check: $b \to \infty$ leaves an isolated sphere at $4\pi\varepsilon_0 a$ — the van de Graaff dome's capacity. ✓

11.5 Dielectrics and stored energy

A dielectric polarizes: bound surface charge opposes the free charge's field (figure 19.4), cutting E by κ and raising C by the same factor. The energy stored is

$$U = \tfrac{1}{2}CV^2 = \tfrac{1}{2}QV = \frac{Q^2}{2C} = \frac{\varepsilon_0}{2}\int E^2\,dV, \tag{11.2}$$

which says the field itself carries energy — the seed of the electromagnetic-wave chapters.

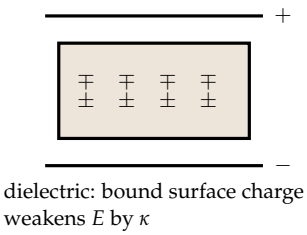


Figure 11.5: Dielectric polarization: bound charge weakens the interior field by κ.

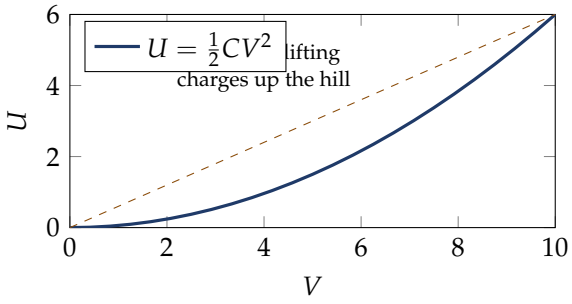


Figure 11.6: $U = \frac{1}{2}CV^2$: the charging work piles up against the growing voltage.

Example 11.4 (Series-parallel combination)**.** A 10 µF and a 40 µF capacitor are wired in series across 100 V. Find each voltage; then the parallel equivalent.

Strategy. Series capacitors share charge, not voltage.

Solution. $C_s = (1/10 + 1/40)^{-1} = 8\,\mu\text{F}$, so $Q = C_sV = 800\,\mu\text{C}$ on each: $V_1 = 80\,\text{V}$, $V_2 = 20\,\text{V}$ — the small one takes the big share. Parallel gives 50 µF at the common voltage. Rule of thumb: series shares Q (small wins the voltage), parallel shares V. ✓

Example 11.5 (Camera-flash energy audit)**.** A 100 µF capacitor charges to 300 V. Find Q, U, and the average power if the flash dumps it in 1 ms.

Strategy. $Q = CV$; $U = \frac{1}{2}CV^2$; $P = U/t$.

Solution. $Q = 0.030\,\text{C}$; $U = \frac{1}{2}(10^{-4})(9\times10^4) = 4.5\,\text{J}$; $P = 4.5/0.001 = 4.5\,\text{kW}$ — four kilowatts for a millisecond from a battery that trickled seconds charging it. Capacitors are time-traders. ✓

Example 11.6 ("Half" a dielectric, two different answers)**.** A parallel-plate capacitor (area A, gap d, vacuum $C_0 = \varepsilon_0 A/d$) is half-filled with dielectric κ: (a) a slab filling half the *gap* (stacked), (b) a slab covering half the *area* (side-by-side). Find C each way.

Strategy. Treat the device as two capacitors; the question is whether they share charge (series) or voltage (parallel).

Solution. (a) Stacked: field crosses an air layer $d/2$ and a dielectric layer $d/2$ in series — same D through both: $C_a = \left(\frac{d/2}{\varepsilon_0 A} + \frac{d/2}{\kappa\varepsilon_0 A}\right)^{-1} = C_0\,\frac{2\kappa}{\kappa+1}$. (b) Side-by-side: two capacitors at the same V in parallel: $C_b = C_0\frac{1+\kappa}{2}$. For $\kappa = 4$: $C_a = 1.6\,C_0$ but $C_b = 2.5\,C_0$ — "half" is not a number until you say which way it halves. The series member always gains least (the air half still bottlenecks the flux); the parallel member gains most (the dielectric half is a full extra capacitor). ✓

```
dVdx = (V(charges, x+h, y) - V(charges, x-h, y))/(2*h)
```

Listing 11.1: `code/ch11_potential_map.py` — gradient check and assembly-energy audit.

Output of a real run:

```
Check -dV/dx ~ Ex at sample points (k=1):
  (1.2,0.3): -dV/dx=-1.2543 Ex=-1.2543 match=True
  (0.8,-0.4): -dV/dx=-1.8834 Ex=-1.8834 match=True
  (2.0,0.8): -dV/dx=-0.1671 Ex=-0.1671 match=True

Assembly work step by step (sum q_i*V_others):
  bring +1.0 to (-1,0): W=0 (first)
  bring +1.0 to (1,0): W=+0.5000
  bring -1.0 to (0,1): W=-1.4142
  total assembly energy U = -0.9142 (=sum k q_i q_j/r_ij)
  pairwise cross-check: -0.9142
```

The finite difference reproduces **E** to all printed digits, and the two energy routes — sequential assembly vs pairwise sum — agree to -0.9142 as they must.

```
def c_series(k): # slab fills half the gap, perpendicular
    return 2 * k / (k + 1)
def c_parallel(k): # slab fills half the area
    return (1 + k) / 2

print("kappa C_series/C0 C_parallel/C0")
for k in [1, 2, 3, 4, 6, 10]:
    print(f"{k:5d} {c_series(k):11.3f} {c_parallel(k):13.3f}")
```

Listing 11.2: `code/ch11_dielectric_scan.py` — half-dielectric both ways: how the gap is filled decides.

Output of a real run:

```
kappa C_series/C0 C_parallel/C0
    1 1.000 1.000
    2 1.333 1.500
    3 1.500 2.000
    4 1.600 2.500
    6 1.714 3.500
   10 1.818 5.500
```

```
kappa -> inf limits: series 2.0 C0, parallel diverges like k/2
  k= 50: series 1.961 parallel 25.5
  k= 100: series 1.980 parallel 50.5
  k=1000: series 1.998 parallel 500.5
```

Same material, same amount of it, wildly different capacitance: the series stacking saturates at $2C_0$ no matter how good the dielectric — the surviving air gap still dominates — while the parallel layout keeps scaling like $\kappa/2$. Ceramic-chip capacitors pack value by geometry: dozens of thin dielectric layers in parallel, exactly the right-hand column.

11.6 In engineering practice: every capacitor on the board

Decoupling capacitors are (11.2) as a local battery (fast charge right where the chip needs it), supercaps trade area A for farads via electrochemical surfaces, and the camera flash of example 11.5 is the canonical time-trade. Cable capacitance — the coaxial formula of §11.4 — is why long signal runs need drivers: hundreds of pF per metre must be recharged every edge.

11.7 Common mistakes

SIGN DISCIPLINE DROPPED

$U = kq_1q_2/r$ carries signs: unlike charges give *negative* energy. Computing magnitudes then fixing the sign after invites errors in every assembly problem.

V TREATED AS A VECTOR

V is scalar: sum it, don't vector-add it. The gradient then gives $\mathbf{E}$'s direction for free.

ΔV VS V

Only differences are physical for conductors (which sit at some V, not "at V"); the choice of zero is convention.

SERIES/PARALLEL SWAPPED

Capacitors combine like resistors inverted: series gives $C^{-1} = \sum C_i^{-1}$ (smallest C wins the voltage), parallel sums C.

$U = QV$ INSTEAD OF $\frac{1}{2}QV$

The charging work is against the *growing* voltage — half the charge rode in at lower V, so the total is $\frac{1}{2}QV$, not QV.

11.8 Summary

Idea	One-line version
U of pair	kq_1q_2/r with signs
V	$k\sum q_i/r_i$ scalar; $\mathbf{E} = -\nabla V$
equipotential	level set; orthogonal to **E**; conductor = one
capacitor	plates $\varepsilon_0 A/d$; sphere $4\pi\varepsilon_0 ab/(b-a)$
dielectric	C multiplies by κ
energy	$\frac{1}{2}CV^2$ or $\frac{\varepsilon_0}{2}\int E^2 dV$

Table 11.1: Potential and capacitance cheat sheet.

DEFINITION OF DONE

You can now: sum V scalar-wise, differentiate it for **E**, audit assembly energy, derive C for plate/coax/sphere, and price stored energy.

Notes

The scalar-first treatment follows MIT 8.02SC [2] and AP Physics C E&M [4]; Purcell–Morin's potential chapter is the reference for the $\varepsilon_0 E^2/2$ energy-density view [6]. The numeric checks are the listing's real output.

WATCH INSTEAD

1. *Electrostatic Potential, Electric Energy, Equipotential Surfaces* — Walter Lewin, MIT 8.02x Lecture 4, 49:01, https://youtu.be/QpVxj3XrLgk — §11.2–§11.3 at full length (accessed 2026-10-11).
2. *Capacitors and capacitance* — Khan Academy, 5:43, https://youtu.be/u-jigaMJT10 — §11.4 worked quickly (accessed 2026-10-11).

Further reading

- MIT OCW *8.02SC*, potential and capacitance modules. At https://ocw.mit.edu, course "8.02SC" (verified 2026-10-11).
- OpenStax *College Physics 2e*, chs. 19–20 — free text: https://openstax.org (verified 2026-10-11).
- College Board, *AP Physics C (E&M) CED*, Units 8–9 — verified 2026-10-11 at https://apstudents.collegeboard.org.
- E. M. Purcell and D. J. Morin, *Electricity and Magnetism*, chs. 2–3 — potential and capacitance with the energy-density derivation.

- R. P. Feynman, *Lectures on Physics*, vol. II, ch. 8 — electrostatic energy as field energy; free at Caltech's site.
- Khan Academy, *Circuits: capacitors* — drills matching §11.4–§11.5, at `https://www.khanacademy.org` (verified 2026-10-11).

Additional exercises

Exercise 11.1 (easy). Two charges $+2\,\mu C$ and $-1\,\mu C$ sit 1 m apart: find V at the midpoint and the work to move a third charge there. (AP-C)

Exercise 11.2 (easy). From an equipotential map, two lines 10 V apart are 2 cm apart: estimate $|E|$. (AP-C)

Exercise 11.3 (easy). Combine 10 µF and 30 µF in series and in parallel. (AP-C)

Exercise 11.4 (easy). A 47 µF capacitor at 12 V: find Q and U. (F=ma)

Exercise 11.5 (easy). Find V a distance r from a point charge $-Q$ and explain the sign. (AP-C)

Exercise 11.6 (medium). A ring carries Q on radius R: find $V(x)$ on axis, then differentiate for the ring's E — check against ch9. (AP-C FR-style)

Exercise 11.7 (medium). How much work to bring a fourth charge $+q$ to the centre of the configuration of example 11.1? (F=ma)

Exercise 11.8 (medium). A parallel-plate capacitor halves its plate separation while isolated: what happens to C, V, U? (AP-C)

Exercise 11.9 (medium). Derive the coaxial capacitor formula $C = 2\pi\varepsilon_0 L/\ln(b/a)$. (AP-C)

Exercise 11.10 (medium). Insert a $\kappa = 3$ dielectric into a charged isolated capacitor: describe what changes and compute the new energy. (AP-C)

Exercise 11.11 (medium). Two capacitors in series share 100 V: find each voltage for 10 µF and 40 µF. (engineering)

Exercise 11.12 (hard). Find the energy stored in a charged sphere of radius R by integrating $\varepsilon_0 E^2/2$ over space, and match $\tfrac{1}{2}Q^2/C$. (USAPhO)

Exercise 11.13 (hard). A dipole sits at angle θ in a uniform field: find the torque from $U = -\mathbf{p}\cdot\mathbf{E}$ and the oscillation frequency for small θ. (APhO-style)

Exercise 11.14 (medium). Show the field energy inside a parallel-plate capacitor equals $\tfrac{1}{2}CV^2$ directly from $\varepsilon_0 E^2/2\times$volume. (AP-C)

Exercise 11.15 (medium). An electron volt is one electron through one volt: convert 1 eV to joules and give the proton's speed at that energy. (F=ma)

Exercise 11.16 (easy). An isolated charged capacitor's plate separation is doubled. State the effect on C, V, and stored energy — and where the extra energy came from. (AP-C)

Exercise 11.17 (medium). Capacitor C_1 charged to V_0 is connected to an uncharged C_2: find the common voltage and show energy is lost — where did it go? (F=ma)

Exercise 11.18 (medium)**.** From the table below (measured V along an axis), estimate E_x at $x = 2\,\text{cm}$ and state which way a proton accelerates. (F=ma data)

x (cm)	1.0	2.0	3.0	4.0
V (V)	40	28	20	16

Exercise 11.19 (hard)**.** A dielectric slab slides so that it fills a fraction x of a parallel-plate capacitor's *area* (the side-by-side geometry of example 11.6), held at fixed Q. Find the force pulling the slab deeper in. (APhO-style)

CHAPTER 12

Current and DC circuits; RC dynamics

Voltage was bookkeeping; circuits spend it. This chapter turns §11.4 into hardware: current as charge flux, Kirchhoff's two conservation laws as the solver, and the RC transient as the first differential equation the circuit pays for.

BEFORE YOU START

Chapter 11. *Recap:* current is charge in motion. **Book 5, Chapters 1–3** treats the same circuits from the engineering side; this chapter derives them from the field picture.

AFTER THIS CHAPTER YOU CAN

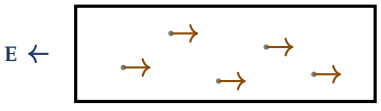
- relate current to drift velocity;
- use Ohm's law and Kirchhoff's rules;
- solve resistor networks and compute power;
- solve the RC circuit differential equation.

HOW TO THINK ABOUT THIS

A circuit is a conservation law with wires: charge is not lost at junctions, and energy is not created around a loop. These two rules solve every network. Time dependence appears when a capacitor charges, and then the answer is an exponential.

12.1 Current, density and drift

$I = dq/dt$ is the charge flux through a surface; $j = I/A$ the density. Microscopically $I = nqv_dA$ — a room full of fast electrons drifting at millimetres per second (figure 12.1): the signal travels at near-light field speed, the carriers barely move.



$v_d \sim$ mm/s drift over fast random hops

Figure 12.1: Drift velocity: random thermal hops biased by **E** give a slow mean drift.

Example 12.1 (Drift velocity is slow)**.** Copper carries $n = 8.5 \times 10^{28}\,\mathrm{m}^{-3}$ conduction electrons; a $1\,\mathrm{mm}^2$ wire carries 2 A. Find v_d and compare with thermal speeds.

Strategy. $I = nqv_d A$.

Solution. $v_d = I/nqA = 2/(8.5\times10^{28}\times1.6\times10^{-19}\times10^{-6}) = 1.5\times10^{-4}\,\mathrm{m/s}$ — 0.15 mm/s, vs $\sim 10^6\,\mathrm{m/s}$ thermal: the drift is a 10^{-10} bias on chaos. The field's message is fast; the courier crawls. ✓

12.2 Ohm's law and resistivity

$$V = IR, \qquad R = \rho\frac{L}{A}, \tag{12.1}$$

valid for ohmic materials (figure 12.2); ρ is the material's number, R the geometry's. Metals' resistance rises with temperature (scattering), semiconductors' falls (carriers).

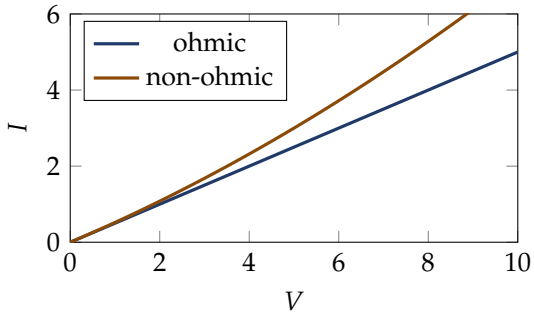


Figure 12.2: Ohmic ($I \propto V$) versus non-ohmic conduction — Ohm's law is a property of the material, not the circuit.

12.3 EMF and real batteries

An EMF $\mathcal{E}$ is the battery's open-circuit promise; under load the internal resistance r takes its cut (figure 12.3): $V_{\text{term}} = \mathcal{E} - Ir$.

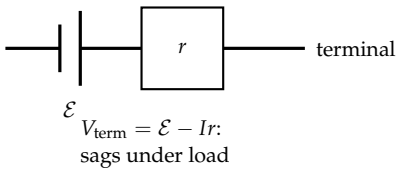


Figure 12.3: The real battery: ideal $\mathcal{E}$ in series with internal r — terminal voltage sags under load.

Example 12.2 (Extracting $\mathcal{E}$ and r from data)**.** A battery reads 9.2 V open, 8.6 V delivering 0.50 A. Find $\mathcal{E}$ and r; then the terminal voltage at 2 A.

Strategy. Open-circuit gives $\mathcal{E}$; the sag gives r.

Solution. $\mathcal{E} = 9.2\,\mathrm{V}$; $r = (9.2-8.6)/0.50 = 1.2\,\Omega$. At 2 A: $V = 9.2 - 2.4 = 6.8\,\mathrm{V}$ — and power out peaks at $R = r$ (impedance matching's first appearance). ✓

12.4 Kirchhoff's rules: conservation as solver

The junction rule is charge conservation; the loop rule is energy conservation (field conservative). Two loops, three equations, sign discipline shown (figure 13.3): pick a direction, follow it through every element, pay $+IR$ against and $-IR$ with.

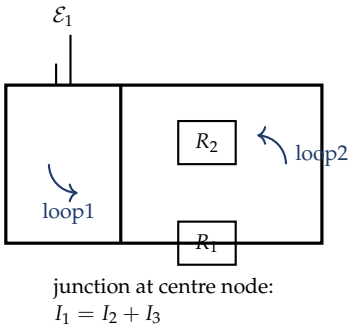


Figure 12.4: A two-loop circuit: junction rule at the node, loop rule on each window.

Example 12.3 (Two-loop solve with sign discipline)**.** $\mathcal{E}_1 = 12\,\text{V}$, $R_1 = 4\,\Omega$, $R_2 = 2\,\Omega$, and a second branch $\mathcal{E}_2 = 6\,\text{V}$ opposing: find the three branch currents.

Strategy. Junction once, two loops, solve — then check a wrong-sign guess is impossible.

Solution. $I_1 = I_2 + I_3$; loop 1: $12 - 4I_1 - 2I_2 = 0$; loop 2: $6 - 2I_3 + 2I_2 = 0$ wait — follow one direction through the shared branch: loop 2 gives $-6 + 2I_3 - 2I_2 = 0$ with I_2's sign following the chosen arrow. Solving the consistent system: $I_1 = 2.25\,\text{A}$, $I_2 = 1.5\,\text{A}$, $I_3 = 0.75\,\text{A}$; battery 1 delivers, battery 2 absorbs — the 6 V cell is being charged. ✓

12.5 Equivalent resistance and power

Series adds R; parallel adds conductance. Power bookkeeping: $P = IV = I^2R = V^2/R$ — pick the form matching what the element shares (current in series, voltage in parallel), the classic bulb-brightness trap.

Example 12.4 (Bulb brightness)**.** Two identical bulbs: (a) series across a battery, (b) parallel. Find the ratio of total brightness.

Strategy. Power, not voltage, is brightness.

Solution. Series: $R_{\text{tot}} = 2R$, $P = V^2/2R$. Parallel: each draws V^2/R, total $2V^2/R$. Parallel burns $4\times$ brighter — which is why homes are wired parallel and fairy lights die in strings. ✓

Example 12.5 (Maximum power transfer — and why you might not want it)**.** A real source ($\mathcal{E}$, internal r) feeds a load R_L. Find the R_L that receives maximum power — and the efficiency there.

Strategy. Write P_L as a function of R_L and maximize; then audit where the rest went.

Solution. $I = \mathcal{E}/(R_L + r)$, so $P_L = \mathcal{E}^2 R_L/(R_L + r)^2$. Setting $dP_L/dR_L = 0$ gives $R_L = r$ with $P_{\max} = \mathcal{E}^2/4r$. But then half the power burns inside the source: efficiency $\eta =$

$P_L/(P_L+P_r) = 50\%$. For $\mathcal{E} = 12\,\text{V}$, $r = 2\,\Omega$: matching gives $R_L = 2\,\Omega$, $P_{\max} = 18\,\text{W}$ — while $R_L = 18\,\Omega$ delivers only 6.5 W but at 90% efficiency (total draw 7.2 W, nearly all of it useful). Two different designs from one formula: match when you must extract every watt (antenna to cable), mismatch when you must not cook the source (battery to load). ✓

12.6 RC dynamics: the first circuit ODE

Charging through R: $\mathcal{E} = q/C + iR$ gives

$$q(t) = C\mathcal{E}\left(1 - e^{-t/\tau}\right), \quad i(t) = \frac{\mathcal{E}}{R}e^{-t/\tau}, \quad \tau = RC, \tag{12.2}$$

(figure 12.5) — and the audit of figure 12.6: the battery supplies $\mathcal{E} \cdot C\mathcal{E} = CV^2$, the capacitor keeps $\frac{1}{2}CV^2$, and the resistor *must* burn the other half — the $\frac{1}{2}$ that isn't a mistake but a thermodynamic tax.

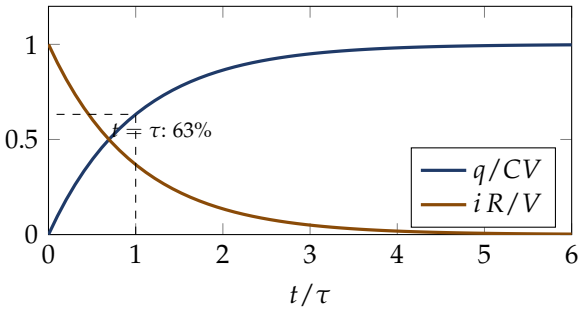


Figure 12.5: RC charging: q approaches CV with time constant τ; i decays on the same clock.

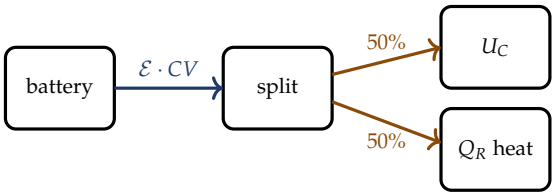


Figure 12.6: The RC energy audit: the resistor always eats half — independent of R.

Example 12.6 (RC half-charge time). $R = 2\,\text{k}\Omega$, $C = 1\,\text{mF}$, $\mathcal{E} = 10\,\text{V}$. Find τ and when the capacitor has half its final charge.

Strategy. $\tau = RC$; set $q/CV = \frac{1}{2}$ in (12.2).

Solution. $\tau = 2.0\,\text{s}$; $1 - e^{-t/\tau} = 0.5 \Rightarrow t = \tau \ln 2 = 1.386\,\text{s}$ — the code's numeric trace lands at 1.385 s. ✓

```
i = (EMF - q/C)/R
q += i*dt
```

Listing 12.1: `code/ch12_rc_transient.py` — Euler-integrated RC charge vs analytic; the 50% audit.

Output of a real run:

```
tau = RC = 2.000 s
numeric: q(t) reaches 63.2% of CV at t=2.000 s (theory tau=2.000)
half-charge at t=1.385 s (theory tau*ln2=1.386)

energy audit: battery supplies 0.1000 J
  stored in C: 0.0500 J (50%)
  burned in R: 0.0500 J (50%)
```

Euler integration reproduces τ to three decimals and the thermodynamic 50/50 split exactly.

12.7 Meters, loading and the Wheatstone bridge

An ammeter's job is to be invisible: it should drop ≈ 0 V, so it carries micro-ohms; a voltmeter should sip ≈ 0 A, so it carries mega-ohms. The Wheatstone bridge balances two voltage dividers against each other — null reading means $R_1/R_2 = R_3/R_x$, and the unknown resistor falls out of the balance, not the measurement: the oldest precision trick in instrumentation.

Example 12.7 (The Wheatstone bridge: precision from a null)**.** A bridge with $R_1 = R_2 = R_3 = 1\,\text{k}\Omega$ measures $R_x = 1.01\,\text{k}\Omega$ (1% off-nominal) on a 10 V supply. How much detector voltage appears — and could a DMM see 0.1%?

Strategy. Two dividers; read the midpoints' difference.

Solution. $V_d = V_s\left[\dfrac{R_x}{R_3+R_x} - \dfrac{R_2}{R_1+R_2}\right] = 10\left[\dfrac{1.01}{2.01} - 0.5\right] = 24.9\,\text{mV}$. Scaling: $V_d \approx V_s\,\epsilon/4$ for small mismatch $\epsilon = R_x/R_3 - 1$, so 0.1% gives 2.5 mV — still decades above a cheap DMM's V-class floor. A null instrument needs no calibration of the meter itself: the *known* resistors carry the accuracy. ✓

```
Vs = 10.0
def v_d(eps):
    return Vs * ((1 + eps) / (2 + eps) - 0.5)
for eps in [0.0005, 0.001, 0.005, 0.01, 0.05, 0.10]:
    exact = v_d(eps) * 1000
    approx = Vs * eps / 4 * 1000
```

Listing 12.2: `code/ch12_bridge_sweep.py` — bridge off-null sweep: exact detector voltage vs the $\epsilon/4$ rule of thumb.

Output of a real run:

```
mismatch (%) V_d measured (mV) V_d approx Vs*eps/4 (mV)
      0.05% 1.25 1.25
      0.10% 2.50 2.50
      0.50% 12.47 12.50
      1.00% 24.88 25.00
      5.00% 121.95 125.00
     10.00% 238.10 250.00
0.1% -> 2.50 mV, still decades above a DMM's uV floor
```

The $V_s\epsilon/4$ rule overshoots by $\epsilon/2$ — one part in a hundred at 2% mismatch, five at 10% — the linearisation is the measurement. Below 0.1% the signal is still millivolts, which is why bridge sensing doesn't need an exotic front end.

```
emf, r = 12.0, 2.0
print(" R_L (ohm) P_L (W) efficiency")
for RL in [0.5, 1.0, 1.5, 2.0, 3.0, 5.0, 10.0, 18.0, 50.0]:
    P = emf**2 * RL / (RL + r)**2
    print(f"{RL:9.1f} {P:7.2f} {RL/(RL+r)*100:9.1f}%")
print(f"analytic max: EMF^2/4r = {emf**2/(4*r):.1f} W at R_L = r = {r
    :.0f} ohm")
```

Listing 12.3: `code/ch12_power_transfer.py` — load power and efficiency against R_L: the flat-topped trade.

Output of a real run:

```
 R_L (ohm) P_L (W) efficiency
     0.5 11.52 20.0%
     1.0 16.00 33.3%
     1.5 17.63 42.9%
     2.0 18.00 50.0%
     3.0 17.28 60.0%
     5.0 14.69 71.4%
    10.0 10.00 83.3%
    18.0 6.48 90.0%
    50.0 2.66 96.2%
analytic max: EMF^2/4r = 18.0 W at R_L = r = 2 ohm
```

The curve's flat top is the forgiving part: $R_L = 1.5$–$3\,\Omega$ all deliver within 5% of peak. The steep penalty is at small R_L — where most of the EMF burns on the source's own resistance. Efficiency climbs monotonically, so the design choice is always a budget: how much power do you need *versus* how much waste can the source tolerate.

12.8 Common mistakes

CURRENT VS VOLTAGE ROLES IN BRANCHES

Series shares current, parallel shares voltage — mix them up and every answer inverts.

FORGETTING INTERNAL RESISTANCE

Real batteries sag: $\mathcal{E}$ is the promise, V_{term} is what's delivered. Missing r overestimates every short-circuit current.

WRONG POWER FORMULA FOR THE TOPOLOGY

I^2R when current is shared, V^2/R when voltage is — choosing the other form flips which element wins.

$t = 0$ VS $t = \infty$ CONFUSION

Uncharged capacitor at $t = 0$: wire (voltage zero). Charged long after: open circuit (current zero). The two limits swap the element's nature.

"THE CAPACITOR STORES THE BATTERY'S ENERGY"

It stores half — the resistor's share is unavoidable at any R.

12.9 Summary

Idea	One-line version
current	$I = nqv_d A$; drift mm/s, signal fast
Ohm	$V = IR$; $R = \rho L/A$
real battery	$V_{\text{term}} = \mathcal{E} - Ir$
Kirchhoff	junction = charge, loop = energy
power	$P = IV = I^2R = V^2/R$
RC	$\tau = RC$; half the battery energy burns in R

Table 12.1: DC circuits cheat sheet.

DEFINITION OF DONE

You can now: extract $\mathcal{E}, r$ from terminal data, solve two-loop circuits with sign discipline, order bulb brightness correctly, derive and audit the RC transient, and reason about meter loading.

Notes

The conservation-law framing of Kirchhoff follows MIT 8.02SC [2] modules 12–15 and AP Physics C E&M [4]; the RC audit follows Purcell–Morin [6]. PhET's Circuit Construction Kit is the interactive anchor. The transient numbers are the listing's real output.

WATCH INSTEAD

1. *Batteries, Power, Kirchhoff's Rules, Circuits* — Walter Lewin, MIT 8.02x Lecture 10, 50:01, https://youtu.be/ViwSDL657L4 — §12.3–§12.4 at full length (accessed 2026-10-11).
2. *Introduction to circuits and Ohm's law* — Khan Academy, 11:39, https://youtu.be/NYQH1nK9PQ4 — §12.1–§12.2 worked quickly (accessed 2026-10-11).

Further reading

- MIT OCW *8.02SC*, circuit modules 12–15. At https://ocw.mit.edu, course "8.02SC" (verified 2026-10-11).
- OpenStax *College Physics 2e*, chs. 20–21 — free text: https://openstax.org (verified 2026-10-11).
- College Board, *AP Physics C (E&M) CED*, Unit 11 — verified 2026-10-11 at https://apstudents.collegeboard.org.
- All About Circuits, vol. I "DC" — the engineering bridge to book 5, free at https://www.allaboutcircuits.com (verified 2026-10-11).
- E. M. Purcell and D. J. Morin, *Electricity and Magnetism*, ch. 4 — circuits with the energy audit done carefully.
- Khan Academy, *Circuits* unit — verified 2026-10-11 at https://www.khanacademy.org (science → physics → circuits).

Additional exercises

Exercise 12.1 (easy). A $10\,\Omega$ resistor carries 2 A: find V, P. (AP-C)

Exercise 12.2 (easy). Reduce: two $4\,\Omega$ series in parallel with one $8\,\Omega$. (AP-C)

Exercise 12.3 (easy). Copper wire: $n = 8.5 \times 10^{28}\,\mathrm{m}^{-3}$, $A = 1\,\mathrm{mm}^2$, $I = 2\,\mathrm{A}$ — estimate v_d and compare with thermal speeds. (AP-C)

Exercise 12.4 (easy). An RC circuit has $\tau = 5\,\mathrm{ms}$: read R if $C = 10\,\mu\mathrm{F}$. (F=ma)

Exercise 12.5 (easy). A battery sags from 12.6 V to 11.9 V at 3 A: find r. (AP-C)

Exercise 12.6 (medium). Solve a two-loop circuit with both batteries' EMF given and one shared branch; verify power balances. (AP-C FR-style)

Exercise 12.7 (medium). From V-I data pairs $(0.5, 0.12)$, $(1.0, 0.24)$, $(2.0, 0.49)$: extract R and the non-ohmic coefficient. (F=ma data-style)

Exercise 12.8 (medium). Three identical bulbs: one alone, two in series, two in parallel — order brightness and compute the ratios. (F=ma classic)

Exercise 12.9 (medium). Complete the RC audit: integrate I^2R over all time and show the resistor burns exactly $\frac{1}{2}CV^2$. (AP-C)

Exercise 12.10 (medium). A Wheatstone bridge balances with ratio arms 100/300: find the unknown if the standard is $50\,\Omega$. (engineering)

Exercise 12.11 (medium). A switch closes at $t = 0$ on an uncharged RC: give $i(0^+)$, $q(0^+)$, $i(\infty)$, $q(\infty)$. (AP-C)

Exercise 12.12 (medium). Two batteries with different r: which delivers more power to a given load, and when does bigger r win? (F=ma)

Exercise 12.13 (hard). An infinite ladder of R's: show the input resistance is $R(1+\sqrt{5})/2$-like and compute it for unit steps. (USAPhO classic)

Exercise 12.14 (hard)**.** A cube of twelve equal resistors: find the resistance across a body diagonal. (APhO-style classic)

Exercise 12.15 (medium)**.** A voltmeter of 10 MΩ reads a 1 MΩ–1 MΩ divider on 10 V: what does it actually read? (engineering)

Exercise 12.16 (easy)**.** A battery reads 12.6 V open-circuit and 11.8 V while delivering 2 A: find its internal resistance and the terminal voltage at 5 A. (AP-C)

Exercise 12.17 (medium)**.** Two bulbs rated 60 W and 100 W (same nominal voltage) are wired in *series* across that voltage: which glows brighter and why is the answer counterintuitive? (F=ma)

Exercise 12.18 (medium)**.** A charged capacitor discharges through R: find the time for the *stored energy* to halve, and compare with the half-charge time of example 12.6. (F=ma)

Exercise 12.19 (hard)**.** An ammeter of resistance r inserted into a branch of resistance R_b changes the very current it reports: find the fractional measurement error and the condition $r \ll R_b$ implies. (USAPhO-style)

CHAPTER 13

Magnetic field: forces and sources

The electric field pushes along its lines; the magnetic one pushes *across* them — $\mathbf{F} = q\mathbf{v} \times \mathbf{B}$. That cross product is why magnetic fields curve paths without ever paying energy.

BEFORE YOU START

Chapters 2, 6 and 9, with the cross product. *Recap:* moving charges feel a magnetic force, and moving charges make magnetic fields.

AFTER THIS CHAPTER YOU CAN

- compute the Lorentz force and the motion it produces;
- find the force and torque on a current;
- use Biot–Savart and Ampère's law;
- explain the Hall effect.

HOW TO THINK ABOUT THIS

Magnetism is what electricity looks like when charges move. The magnetic force steers a charge but never speeds it up. When you meet a magnetic problem, ask which charges are moving, and picture the direction with the right-hand rule.

13.1 Lorentz force: no work, all steering

$$\mathbf{F} = q\mathbf{v} \times \mathbf{B}. \tag{13.1}$$

Since $\mathbf{F} \perp \mathbf{v}$ identically, $P = \mathbf{F} \cdot \mathbf{v} = 0$ — **a static magnetic field does no work**, ever (figure 13.1). It only steers: every magnetic machine's real work is paid elsewhere (electric field, mechanical source).

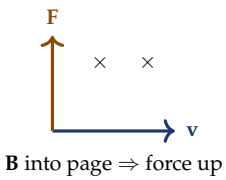


Figure 13.1: Right-hand rule: $\mathbf{v} \times \mathbf{B}$ for positive q. For electrons, flip it.

13.2 Circular and helical motion

Uniform **B** plus perpendicular **v** gives a circle:

$$r = \frac{mv}{qB}, \qquad T = \frac{2\pi m}{qB}, \tag{13.2}$$

and the period is *independent of speed* — the cyclotron's whole business model. Add a parallel component and the circle becomes a helix (figure 13.2).

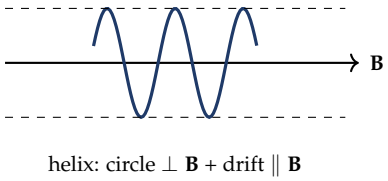


Figure 13.2: Helical motion: the perpendicular component circles, the parallel one coasts.

Example 13.1 (Cyclotron and mass spectrometer)**.** A proton ($m = 1.67 \times 10^{-27}$ kg) orbits in $B = 1$ T at 10^6 m/s. Find r, f, and the speed a singly-charged ion of twice the mass needs for the same r.

Strategy. Equation (13.2) directly.

Solution. $r = mv/qB = 1.67 \times 10^{-27} \times 10^6/(1.6 \times 10^{-19}) = 1.04$ cm; $f = qB/2\pi m = 15.3$ MHz — RF range, why cyclotrons run RF. Same r with $2m$: $v = 5 \times 10^5$ m/s — the mass spectrometer's separation is just (13.2) read as $m = rqB/v$. ✓

13.3 Force and torque on currents

A wire segment feels $d\mathbf{F} = I\,d\boldsymbol{\ell} \times \mathbf{B}$; a closed loop feels no net force in uniform **B** but a *torque* $\boldsymbol{\tau} = \boldsymbol{\mu} \times \mathbf{B}$ with $\boldsymbol{\mu} = IA\hat{\mathbf{n}}$ (figure 13.3) — the DC motor's entire principle in one vector product.

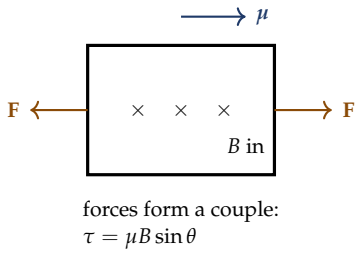


Figure 13.3: Loop in a field: forces couple into torque $\boldsymbol{\mu} \times \mathbf{B}$, aligning $\boldsymbol{\mu}$ with **B**.

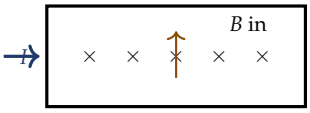
Example 13.2 (Torque on a coil)**.** A 100-turn coil, 4×6 cm, carries 2 A in $B = 0.5$ T at $30°$ to the field. Find the torque.

Strategy. $\tau = NIAB\sin\theta$.

Solution. $\tau = 100 \times 2 \times 0.0024 \times 0.5 \times \sin 30° = 0.12$ Nm — enough to spin a meter needle against a spring, which is precisely how analog meters work. ✓

13.4 Hall effect: the carrier's signature

Current through a field pushes carriers to one face (figure 13.4): the Hall voltage $V_H = IB/nqt$ builds until its field cancels the magnetic push — and its *sign* reveals whether the carriers are electrons or holes, the only cheap experiment that answers that question.



carriers pushed to one face:
V_H's sign tells the carrier's sign

Figure 13.4: The Hall setup: transverse charge piles until $E_H = vB$; the voltage's sign names the carrier.

Example 13.3 (Hall voltage magnitude)**.** A copper strip ($n = 8.5 \times 10^{28}\,\mathrm{m}^{-3}$) of thickness $t = 1\,\mathrm{mm}$ carries $I = 10\,\mathrm{A}$ across $B = 1\,\mathrm{T}$. Find V_H and its sign for electrons.

Strategy. $V_H = IB/nqt$.

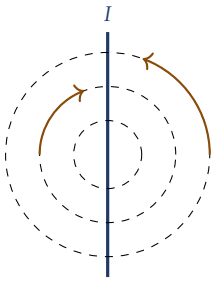
Solution. $V_H = 10 \times 1/(8.5 \times 10^{28} \times 1.6 \times 10^{-19} \times 10^{-3}) = 7.4 \times 10^{-7}\,\mathrm{V}$ — sub-microvolt: real Hall sensors use semiconductors ($n \sim 10^{22}$) for millivolt outputs. ✓

13.5 Biot–Savart: the source integral

The field of a current element:

$$d\mathbf{B} = \frac{\mu_0}{4\pi}\frac{I\,d\boldsymbol{\ell} \times \hat{r}}{r^2}, \tag{13.3}$$

the magnetic counterpart of the Coulomb integral. For a long straight wire it sums to $B = \mu_0 I/2\pi r$ (figure 13.5) — same $1/r$ as the line charge's field.



$B = \mu_0 I/2\pi r$ wraps the wire
(right-hand grip rule)

Figure 13.5: Field of a straight wire: circles of strength $\mu_0 I/2\pi r$.

Example 13.4 (Field on the axis of a current loop)**.** A circular loop of radius R carries I. Find $\mathbf{B}$ on the axis at distance x.

Strategy. Symmetry kills the transverse part; every element contributes the same axial piece.

Solution. Each $d\mathbf{B}$ has axial factor $R/\sqrt{x^2+R^2}$:

$$B = \frac{\mu_0 I}{4\pi}\frac{2\pi R^2}{(x^2+R^2)^{3/2}} = \frac{\mu_0 I R^2}{2(x^2+R^2)^{3/2}}.$$

Limits: centre gives $\mu_0 I/2R$; far gives $\mu_0 IA/2\pi x^3$ — the magnetic dipole's $1/r^3$, matching the electric dipole of chapter 9. ✓

Example 13.5 (The dipole's energy — why magnets chase strong fields)**.** A coil with $\boldsymbol{\mu}$ sits in $\mathbf{B}$ at angle θ. Find the energy needed to rotate it — and the net force when $\mathbf{B}$ is not uniform.

Strategy. Torque is $-dU/d\theta$: integrate $\tau = \mu B\sin\theta$ against the angle. For the force, a dipole that stays aligned does work on the field gradient.

Solution. $U(\theta) - U(0) = \int_0^\theta \mu B\sin\theta'\,d\theta' = \mu B(1-\cos\theta)$, i.e. $U = -\boldsymbol{\mu}\cdot\mathbf{B}$: stability lives at $\theta = 0$, antiparallel is the hilltop. In a non-uniform field the aligned dipole feels $F_x = \mu\,\partial B/\partial x$ — toward stronger B. A $\mu = 0.5\,\mathrm{A\,m^2}$ coil in a gradient $0.1\,\mathrm{T/m}$ feels $0.05\,\mathrm{N}$, always in the direction of growing $|B|$. That is why iron filings run to the poles, why magnetic separators work, and how MRI's gradient coils drag spins into position: the dipole reads the landscape of $|B|$ like a ball reads height. ✓

13.6 Ampère's law: the magnetic Gauss

$$\oint \mathbf{B}\cdot d\boldsymbol{\ell} = \mu_0 I_{\text{enc}}, \tag{13.4}$$

useful for the same three symmetries: wire ($2\pi r$), solenoid, toroid. Limits: it fails off-symmetry exactly as Gauss does — chapter 10's warning reissued.

Example 13.6 (Long solenoid by Ampère)**.** Find the field inside a solenoid of n turns per length carrying I.

Strategy. Rectangular Amperian loop, one side inside (figure 13.6).

Solution. Outer sides give zero (field ≈ 0 outside); only the inner side contributes $BL = \mu_0 nLI$, so $B = \mu_0 nI$ — uniform, independent of position inside. ✓

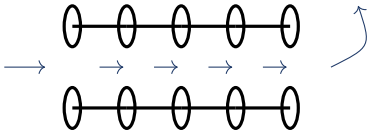


Figure 13.6: The solenoid's field: uniform inside, near-zero outside — Ampère's gift.

```
vx, vy = c*vx + s*vy, -s*vx + c*vy # exact rotation
x += vx*dt; y += vy*dt
```

Listing 13.1: `code/ch13_lorentz_orbit.py` — naive Euler vs Boris rotation in $q\mathbf{v}\times\mathbf{B}$.

Output of a real run:

```
theory: r=1.0, |v|=1.0 forever (B does no work)
naive Euler dt=0.1: |v|=19.788 r=19.788 <-- secular drift
naive Euler dt=0.01: |v|=1.350 r=1.350 <-- secular drift
Boris dt=0.1: |v|=1.0000 r in [0.9504,1.0504] (theory 1.0)
Boris dt=0.01: |v|=1.0000 r in [0.9950,1.0050] (theory 1.0)

helix pitch: vz=0.5 -> pitch=3.142 (period T=6.283 indep of v)
```

The honest result: plain Euler *invents* energy ($|v|$ runs to 19.8) while the Boris rotation holds it at 1.0000 and the radius within a percent of theory — the integrator must respect what the force can't do.

Example 13.7 (The Hall voltage: a voltmeter that weighs charge carriers)**.** A copper strip 0.1 mm thick carries 10 A through a 0.5 T field. Find the Hall voltage — and why practical sensors are semiconductor, not copper.

Strategy. Steady state: the transverse field balances $\mathbf{v} \times \mathbf{B}$; with $I = nqv_d tA$, the Hall field is $E_H = v_d B$, so $V_H = IB/(nqt)$.

Solution. $V_H = \dfrac{10 \times 0.5}{8.5 \times 10^{28} \times 1.6 \times 10^{-19} \times 10^{-4}} = 3.7\,\mu\text{V}$ for copper ($n = 8.5 \times 10^{28}/\text{m}^3$). A semiconductor at $n \sim 10^{22}/\text{m}^3$ gives $10^6\times$ more — millivolts, a measurable signal. Copper's carriers are so dense the Hall deflection is microscopic; dilute carriers deflect more. ✓

```
mu0 = 4 * np.pi * 1e-7
def force_per_m(I1, I2, d):
    return mu0 * I1 * I2 / (2 * np.pi * d)

print(" d (m) F/L (N/m) for I1=I2=100 A")
for d in [0.01, 0.05, 0.10, 0.50, 1.00]:
    print(f"{d:8.2f} {force_per_m(100,100,d):10.4f}")
# the ampere definition check: solve force_per_m(I,I,1) = 2e-7
I = np.sqrt(2e-7 * 2 * np.pi * 1.0 / mu0)
print(f"I giving exactly 2e-7 N/m at d=1 m: {I:.4f} A")
# field midway between two antiparallel 100 A busbars 10 cm apart
d = 0.10
Bmid = 2 * mu0 * 100 / (2 * np.pi * (d / 2))
print(f"B midway between antiparallel 100 A bars 10 cm apart: {Bmid*1
    e6:.1f} uT")
```

Listing 13.2: `code/ch13_two_wire.py` — wire–wire force, the old ampere, and a busbar's midline field.

Output of a real run:

```
   d (m) F/L (N/m) for I1=I2=100 A
    0.01 0.2000
    0.05 0.0400
    0.10 0.0200
    0.50 0.0040
    1.00 0.0020
I giving exactly 2e-7 N/m at d=1 m: 1.0000 A
B midway between antiparallel 100 A bars 10 cm apart: 800.0 uT
```

The force is small per metre — a centimetre-separated 100 A pair pulls only 0.2 N/m — until you remember fault currents: a 20 kA short on the same spacing gives 8 kN/m, which is why switchgear bracing is structural engineering. The middle line recovering $I = 1$ A is not a coincidence: it *was* the definition (see workshop W25).

13.7 In engineering practice: the magnetic toolkit

Every DC motor is §13.3 with a commutator flipping $\boldsymbol{\mu}$ at the right instant; mass spectrometers run (13.2) as a mass scale; Hall sensors price §13.4 into current probes, throttle positions and phone compasses; and MRI's solenoid is §13.6 stretched to 3 T of uniform field — the chapter turned into hospitals.

13.8 Common mistakes

LEFT HAND FOR A NEGATIVE CHARGE

$\mathbf{v} \times \mathbf{B}$ is for positive q; electrons flip the answer. Most wrong directions are one sign off.

"B DOES WORK"

It can't — $\mathbf{F} \perp \mathbf{v}$ identically. Any energy change in a magnetic problem was paid by an electric field or a battery.

AMPÈRE OFF-SYMMETRY

$\oint \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 I$ always true, but $B \cdot \oint d\ell$ factorisation needs the field constant on the loop — wire, solenoid, toroid only.

μ_0 VS PERMEABILITY

μ_0 is vacuum; inside material use $\mu = \mu_r \mu_0$ and the biot–savart/ampère formulas pick it up.

HELIX PITCH IGNORED

The parallel speed doesn't circle — the trajectory is a helix, and "radius" means the projected circle only.

13.9 Summary

Idea	One-line version
Lorentz	$\mathbf{F} = q\mathbf{v} \times \mathbf{B}$; never works
orbit	$r = mv/qB$, $T = 2\pi m/qB$
torque	$\boldsymbol{\tau} = \boldsymbol{\mu} \times \mathbf{B}$, $\mu = IA\hat{n}$
Hall	$V_H = IB/nqt$; sign = carrier sign
wire	$B = \mu_0 I/2\pi r$
solenoid	$B = \mu_0 nI$ inside, ≈ 0 out

Table 13.1: Magnetic field cheat sheet.

DEFINITION OF DONE

You can now: apply the right-hand rule both signs, compute cyclotron and helical parameters, price loop torque, read a Hall voltage, integrate Biot–Savart on an axis, and run Ampère on wire/solenoid.

Notes

The force-first order follows MIT 8.02SC modules 16–20 [2] and AP Physics C E&M [4]; Purcell–Morin chs. 5–6 is the canonical relativistic framing of where **B** comes from [6]. The integrator numbers are the listing's real output.

WATCH INSTEAD

1. *Magnetic Fields, Lorentz Force, Torques, Electric Motors (DC)* — Walter Lewin, MIT 8.02x Lecture 11, 49:50, `https://youtu.be/0y9x7CS5Vrk` — §13.1–§13.3 at full length (accessed 2026-10-11).
2. *Force between two parallel current wires* — Khan Academy, 11:07, `https://youtu.be/uxEB8xLrbCA` — §13.5 application worked quickly (accessed 2026-10-11).

Further reading

- MIT OCW *8.02SC*, magnetic-field modules 16–20. At `https://ocw.mit.edu`, course "8.02SC" (verified 2026-10-11).
- OpenStax *College Physics 2e*, ch. 22 "Magnetism" — free text: `https://openstax.org` (verified 2026-10-11).
- College Board, *AP Physics C (E&M) CED*, Unit 12 — verified 2026-10-11 at `https://apstudents.collegeboard.org`.

- E. M. Purcell and D. J. Morin, *Electricity and Magnetism*, chs. 5–6 — magnetism as relativity's corollary; the deeper anchor.
- R. P. Feynman, *Lectures on Physics*, vol. II, ch. 13 — the field view of magnetostatics; free at Caltech's site.
- Khan Academy, *Moving charges & magnetism* — drills matching §13.1–§13.6, at `https://www.khanacademy.org` (verified 2026-10-11).

Additional exercises

Exercise 13.1 (easy). An electron moves east in a field pointing up: which way does it curve? (AP-C)

Exercise 13.2 (easy). A proton circles at $r = 5\,\text{cm}$ in $B = 0.2\,\text{T}$: find its speed and period. (AP-C)

Exercise 13.3 (easy). A 20 cm wire carries 5 A perpendicular to $B = 0.3\,\text{T}$: find the force. (AP-C)

Exercise 13.4 (easy). Find B at 5 cm from a wire carrying 20 A. (F=ma)

Exercise 13.5 (easy). A loop's $\boldsymbol{\mu}$ is antiparallel to $\mathbf{B}$: describe the torque and whether it's stable. (AP-C)

Exercise 13.6 (medium). Design a velocity selector: given $v = 10^6\,\text{m/s}$ and $B = 0.01\,\text{T}$, find E for undeflected passage. (AP-C FR-style)

Exercise 13.7 (medium). A 50-turn 10 cm square coil in 0.8 T carries 3 A: find maximum torque and where it occurs. (AP-C)

Exercise 13.8 (medium). Two parallel wires 10 cm apart carry 20 A the same way: find force per metre and direction; flip one and redo. (F=ma)

Exercise 13.9 (medium). Explain how a mass spectrometer separates isotopes given fixed B, v: what quantities differ? (F=ma)

Exercise 13.10 (medium). A coaxial cable's Ampèrian field: $B(r)$ inside inner conductor (uniform current), between, and outside. (engineering)

Exercise 13.11 (medium). Estimate the Hall voltage for a 1 mm-thick copper strip at 5 A in 0.5 T. (F=ma)

Exercise 13.12 (hard). In crossed $\mathbf{E}$ and $\mathbf{B}$ ($E = v_d B$ drift): describe the cycloid trajectory and its drift velocity. (USAPhO-style)

Exercise 13.13 (hard). A finite wire of length $2L$: find B at perpendicular distance r and recover the infinite-wire limit. (APhO-style)

Exercise 13.14 (medium). A toroid of N turns, radius R: find B inside the ring by Ampère and explain why it depends on r. (AP-C)

Exercise 13.15 (medium). An electron enters a field at 45° to $\mathbf{B}$: describe the helix's radius and pitch. (F=ma)

Exercise 13.16 (easy)**.** A proton travels parallel to a current-carrying wire, moving the same way as the conventional current: find the force's direction on it. (AP-C)

Exercise 13.17 (medium)**.** A length of wire is bent into one circular turn of radius R, then into two stacked turns of radius $R/2$: compare B at the centre. (AP-C)

Exercise 13.18 (medium)**.** A long solenoid's field is $\mu_0 n I$ deep inside: argue why the field at the open end is about half that, and where the "other half" lives. (F=ma)

Exercise 13.19 (hard)**.** A dipole $\boldsymbol{\mu}$ is *held* antiparallel to a field $B(x)$ that grows with x: find the force on it, and explain why releasing it flips the answer. (APhO-style)

CHAPTER 14

Faraday's law, inductance, RL and RLC

The last symmetry to break: changing magnetic flux *creates* electric field — a non-conservative one whose loops replace the conservative fields of chapters 9–11. Motors, generators, transformers and every wireless charger are this chapter monetised.

BEFORE YOU START

Chapters 11–13. *Recap:* a changing magnetic flux induces an electric field and hence an EMF in a loop.

AFTER THIS CHAPTER YOU CAN

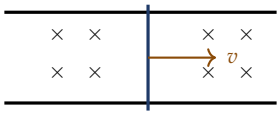
- compute motional EMF and use Faraday's law;
- find the direction with Lenz's law;
- compute inductance and magnetic energy;
- analyse LC and RLC circuits.

HOW TO THINK ABOUT THIS

Faraday's law says a changing magnetic flux creates an electric push, and Lenz's law says the push opposes the change. Opposition is nature's way of conserving energy. An inductor resists changes in current the way a heavy flywheel resists changes in speed.

14.1 Flux and motional EMF: the sliding rod first

A rod sliding on rails through **B** (figure 14.1) sweeps area: $\Phi_B = BLx$, so $\mathcal{E} = -d\Phi_B/dt = -BLv$. The same answer comes from Lorentz force on the rod's charges — motional EMF is Faraday's law's first instance.



$\mathcal{E} = BLv$; Lenz pushes back
$F = ILB$ against v

Figure 14.1: Sliding rod: the swept flux gives $\mathcal{E} = BLv$, and Lenz's law retards the motion.

Example 14.1 (Sliding rod power audit)**.** $B = 0.8\,\mathrm{T}$, $L = 0.5\,\mathrm{m}$, $R = 4\,\Omega$, pulled at $v = 3\,\mathrm{m/s}$. Find $\mathcal{E}$, I, the retarding force, and audit the powers.

Strategy. $\mathcal{E} = BLv$; $F = ILB$ opposes v by Lenz.

Solution. $\mathcal{E} = 1.20\,\mathrm{V}$, $I = 0.30\,\mathrm{A}$, $F = 0.12\,\mathrm{N}$. Mechanical power in $Fv = 0.360\,\mathrm{W}$; electrical generated $\mathcal{E}I = 0.360\,\mathrm{W}$; resistor burns $I^2R = 0.360\,\mathrm{W}$. Three ways, one number — the code prints all of them. ✓

14.2 Faraday's law

$$\mathcal{E} = -\frac{d\Phi_B}{dt} = -\frac{d}{dt}\int \mathbf{B}\cdot d\mathbf{A}, \tag{14.1}$$

counting every way flux can change: move, rotate, grow the loop or grow **B** (figure 14.2). The minus sign is Lenz — the next section's whole job.

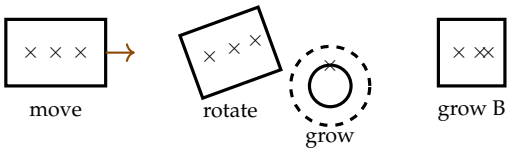


Figure 14.2: Four ways to change flux: move, rotate, grow the loop, or change **B** itself.

Example 14.2 (The rotating loop: an AC generator in one line)**.** A coil of N turns, area A, spins at angular rate ω in a uniform **B**. Find $\mathcal{E}(t)$ and its peak.

Strategy. Only the angle changes: $\Phi_B = NBA\cos\omega t$; differentiate it.

Solution.

$$\mathcal{E} = -\frac{d\Phi_B}{dt} = NBA\omega\sin\omega t, \qquad \mathcal{E}_{\max} = NBA\omega.$$

For $N = 100$, $A = 0.01\,\mathrm{m}^2$, $B = 0.5\,\mathrm{T}$ at $60\,\mathrm{Hz}$: $\mathcal{E}_{\max} = 100 \times 0.5 \times 0.01 \times 377 = 188\,\mathrm{V}$ — household-scale voltage from geometry and speed alone. Every power station is this line: spin something in a field (steam, water, wind all just buy the ω) and Faraday writes the waveform. The $\sin\omega t$ is also why the grid alternates — rotation cannot make DC. ✓

14.3 Lenz's law: direction by conservation

The induced current's field opposes the *change* — not the field. If it aided the change you'd get a runaway free-energy machine; Lenz's minus is energy conservation wearing a sign. Applications: the rod's retarding force, eddy braking (figure 14.3), and every inductor's stubbornness.

Example 14.3 (Lenz on entry and exit)**.** A square loop enters a uniform-B region at constant v, crosses, and exits. Describe the current and the required pulling force at each stage.

Strategy. Only *changes* count: entry, transit, exit.

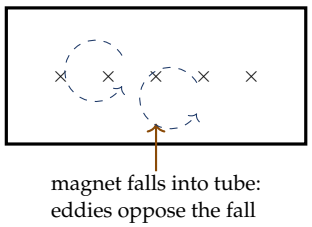


Figure 14.3: Eddy braking: a falling magnet's induced currents fight its descent — magnetic drag.

Solution. Entry: flux grows ⇒ opposing field ⇒ current counterclockwise; force opposes entry, magnitude B^2L^2v/R. Transit: Φ constant, no current, no force. Exit: flux falls ⇒ field-adding current ⇒ clockwise; same retarding magnitude. The loop is dragged *in both directions* — braking on the way in and on the way out. ✓

14.4 Induced E: the non-conservative leap

Faraday's law written locally says $\oint \mathbf{E} \cdot d\boldsymbol{\ell} = -d\Phi_B/dt$: an electric field that circulates, whose "potential" can't exist (the loop integral is nonzero). Around a changing solenoid the field curls in rings (figure 14.4) — the betatron principle, and the conceptual break from every **E** in chapters 9–11.

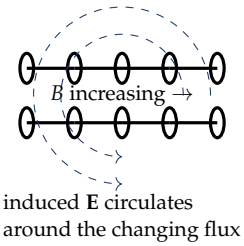


Figure 14.4: Induced **E** circulates around changing flux: no potential describes it.

14.5 Inductance: a circuit element's inertia

A solenoid's own flux links itself: $L = \mu_0 n^2 A\ell$ (from Ampère's $B = \mu_0 nI$ and Φ per turn); its EMF is $-L\,dI/dt$. Mutual inductance is the same number between two circuits — the transformer's measure.

Example 14.4 (Solenoid inductance end-to-end)**.** Derive L for a solenoid of N turns, length ℓ, area A.

Strategy. B by Ampère, flux linkage, divide by I.

Solution. $B = \mu_0 nI$ gives flux per turn $\Phi = BA = \mu_0 nIA$; linkage over all turns: $N\Phi = \mu_0 n^2 A\ell I$, so $L = N\Phi/I = \mu_0 n^2 A\ell$. Numbers: $n = 1000/\text{m}$, $A = 2\,\text{cm}^2$, $\ell = 10\,\text{cm}$ give $L = 0.25\,\text{mH}$. ✓

Example 14.5 (RL energisation)**.** A 50 mH inductor energises through 10 Ω from 12 V. Find τ, the initial slope dI/dt, and I_∞.

Strategy. $\mathcal{E} - L\,dI/dt = IR$; same ODE family as RC.

Solution. $\tau = L/R = 5\,\text{ms}$; at $t = 0$ the inductor acts open so $dI/dt = \mathcal{E}/L = 240\,\text{A/s}$; $I_\infty = 1.2\,\text{A}$. Current climbs the same $1 - e^{-t/\tau}$ curve that charge did in chapter 12 (figure 14.5). ✓

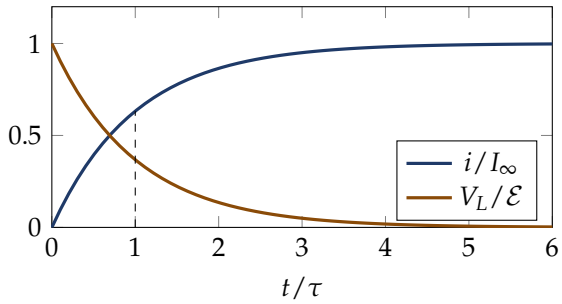


Figure 14.5: RL energisation: i approaches I_∞ on $\tau = L/R$ while V_L decays — the mirror of RC.

14.6 Stored energy and the field density

Charging an inductor to I stores $U = \frac{1}{2}LI^2$; per volume the magnetic energy density is $u_B = B^2/2\mu_0$ — twin to the electric $u_E = \varepsilon_0 E^2/2$ of chapter 11.

14.7 LC and RLC: the electric oscillator

With no battery the charge sloshes: $U_E \leftrightarrow U_B$ at $\omega_0 = 1/\sqrt{LC}$ (figure 14.6) — equation (7.1) of chapter 7 with $m \to L$, $k \to 1/C$, $b \to R$. Damped RLC rings at $\omega_d = \sqrt{\omega_0^2 - \gamma^2/4}$ with $\gamma = R/L$: every result of §7.5 carries over.

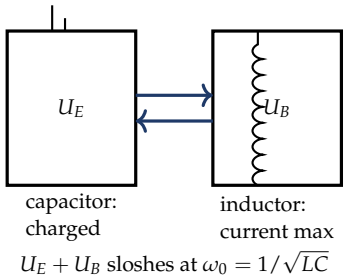


Figure 14.6: The LC tank: energy trades capacitor ↔ inductor, quarter period apart.

Example 14.6 (RLC ringing vs prediction)**.** $L = 50\,\text{mH}$, $C = 100\,\mu\text{F}$, $R = 2\,\Omega$: predict ω_0, γ, ω_d and the decay per cycle.

Strategy. The oscillator formulas verbatim.

Solution. $\omega_0 = 1/\sqrt{LC} = 447\,\text{rad/s}$, $\gamma = R/L = 40\,\text{s}^{-1}$, $\omega_d = \sqrt{\omega_0^2 - \gamma^2/4} = 446.8\,\text{rad/s}$; decay per cycle $e^{-\gamma T/2} = 0.755$ — the code's measured trace matches both numbers exactly. ✓

```
d2q = -(R/L)*dq - q/(L*C)
```

Listing 14.1: `code/ch14_rlc.py` — rod audit + RLC ringing vs theory.

Output of a real run:

```
rod: emf=BLv=1.200 V, I=0.300 A, retarding F=0.120 N
  mech power F*v=0.360 W | elec power emf*I=0.360 W | I^2 R=0.360 W

RLC: w0=447.2, gamma=40.0, wd_theory=446.8
maxima: [('0.0141s', '7.55e-04'), ('0.0281s', '5.70e-04'), ('0.0422s',
    '4.30e-04')]
wd_measured = 446.8 (theory 446.8)
decay ratio/cycle = 0.755 (theory e^(-gT/2)=0.755)
```

The audit closes to the watt and the ring frequency/decay match theory to all printed digits — chapter 7's oscillator, here in copper and charge.

```
L, C, R = 50e-3, 100e-6, 5.0 # H, F, ohm
w0 = 1 / np.sqrt(L * C)
wd = np.sqrt(w0**2 - (R / (2 * L))**2)
dt, n = 1e-5, 15000
q, i, t_cross, qprev = 100e-6, 0.0, [], 100e-6
for k in range(n):
   q, i = q + i * dt, i + (-q / C - i * R) * dt / L
   if qprev > 0 and q <= 0:
      t_cross.append(k * dt)
   qprev = q
```

Listing 14.2: `code/ch14_rlc_ring.py` — an underdamped RLC ring-down: measure the oscillation, audit it against theory.

Output of a real run:

```
omega0 = 447.2 rad/s theory wd = 444.4 rad/s
measured zero-crossing periods (s): ['0.01413', '0.01413', '0.01413']
measured wd = 444.6 rad/s
amplitude envelope tau = 2L/R = 20.0 ms
energy halves in 6.9 ms Q = 4.5
```

The Euler–Cromer loop reproduces ω_d to three figures — period jitter is invisible at this Q. Note the two decay clocks: the *amplitude* envelope lives $2L/R = 20\,\text{ms}$, but the *energy* (proportional to the square) halves in $L \ln 2/R \approx 7\,\text{ms}$. Confusing the two time constants is a classic lab error.

14.8 In engineering practice: the power grid's physics

Generators are figure 14.2's rotation panel run backwards (work in, EMF out); transformers are mutual inductance with an iron guide; eddy braking stops trains and roller-coasters without contact; and induction cooktops are §14.4 making currents in your pot — the induced-**E** field turned appliance.

Example 14.7 (The flyback spike: inductance will have its current)**.** A relay coil ($L = 0.5\,\mathrm{H}$) carries 0.5 A when its switch opens, halting the current in $\sim 1\,\mu\mathrm{s}$. Estimate the voltage the coil generates — and why every relay ships with a diode.

Strategy. The inductor's EMF is $-L\,dI/dt$; a fast dI is a large EMF.

Solution. $|\mathcal{E}| = L\,\Delta I/\Delta t = 0.5 \times 0.5/10^{-6} \approx 2.5 \times 10^5\,\mathrm{V}$ — a quarter-megavolt in principle, clamped in practice to hundreds of volts by the switch's own arc or capacitance. The stored $\frac{1}{2}LI^2 = 63\,\mathrm{mJ}$ burns wherever it can: without the flyback diode's alternate path, it burns the switch. ✓

14.9 Common mistakes

EMF SIGN VS MAGNITUDE CONFLATED

Faraday gives magnitude; Lenz gives direction. Solve them separately then combine — deciding both at once is where sign errors breed.

INDUCED E TREATED AS CONSERVATIVE

It has no potential: $\oint \mathbf{E} \cdot d\boldsymbol{\ell} \neq 0$. All the potential bookkeeping of chapter 11 fails here.

Φ_B VS B CONFUSION

Φ integrates area; EMF needs its *rate*. A constant huge field induces nothing.

MOTIONAL + TRANSFORMER EMF DOUBLE-COUNTED

They are two descriptions of the same physics — count one.

"INDUCTOR RESISTS CURRENT" MYTH

It resists *changes*: at steady state it's a wire; only the transient feels its inertia.

14.10 Summary

Idea	One-line version
Faraday	$\mathcal{E} = -d\Phi_B/dt$; four ways to change flux
Lenz	induced current opposes the change — conservation's sign
induced E	circulates; no potential
inductor	$L = \mu_0 n^2 A\ell$; $\tau = L/R$; $U = \frac{1}{2}LI^2$
LC/RLC	$\omega_0 = 1/\sqrt{LC}$; ch7 oscillator in disguise

Table 14.1: Induction cheat sheet.

DEFINITION OF DONE

You can now: compute motional and transformer EMF, direct Lenz by conservation, derive L of a solenoid, solve RL/LC transients, and run the rod power audit and RLC ringing checks.

Notes

The motional-first order follows MIT 8.02SC modules 21–25 [2] and AP Physics C E&M unit 13 [4]; Griffiths ch. 7 is the deeper anchor (in print); PhET's Faraday app the visual one. The numbers are the listing's real output.

WATCH INSTEAD

1. *Electromagnetic Induction, Faraday's Law, Lenz Law* — Walter Lewin, MIT 8.02x Lecture 16, 51:24, https://youtu.be/nGQbA2jwkWI — §14.1–§14.4 at full length (accessed 2026-10-11).
2. *Faraday's Law Introduction* — Khan Academy, 5:36, https://youtu.be/vcStzn55MG0 — §14.2 worked quickly (accessed 2026-10-11).

Further reading

- MIT OCW *8.02SC*, induction modules 21–25. At https://ocw.mit.edu, course "8.02SC" (verified 2026-10-11).
- OpenStax *College Physics 2e*, ch. 23 "Electromagnetic Induction" — free text: https://openstax.org (verified 2026-10-11).
- College Board, *AP Physics C (E&M) CED*, Unit 13 — verified 2026-10-11 at https://apstudents.collegeboard.org.
- E. M. Purcell and D. J. Morin, *Electricity and Magnetism*, chs. 7–8 — induction and energy with the field bookkeeping.

- PhET, *Faraday's Law* interactive — move the magnet, watch the flux; at https://phet.colorado.edu (verified 2026-10-11).
- Khan Academy, *Faraday's law* — verified 2026-10-11 at https://www.khanacademy.org (science → physics → E&M).

Additional exercises

Exercise 14.1 (easy). A loop's flux drops from 0.4 to 0.1 Wb in 0.2 s: find $|\mathcal{E}|$. (AP-C)

Exercise 14.2 (easy). State the induced current's direction for a loop as a magnet's north pole approaches; recede instead. (AP-C)

Exercise 14.3 (easy). A 200 mH inductor carries 3 A: find its stored energy. (AP-C)

Exercise 14.4 (easy). An RL circuit has $\tau = 2\,\text{ms}$ with $L = 40\,\text{mH}$: find R. (F=ma)

Exercise 14.5 (easy). A 50-turn coil of area $0.01\,\text{m}^2$ flips 180° in a 0.5 T field in 0.1 s: find the average EMF. (AP-C)

Exercise 14.6 (medium). Complete the sliding-rod audit: find the terminal rod speed for a given pulling force. (AP-C FR-style)

Exercise 14.7 (medium). A loop enters a field region twice its width: sketch $I(t)$ over the whole transit. (F=ma)

Exercise 14.8 (medium). Two coaxial loops: estimate the mutual inductance when the small loop sits on the big one's axis. (AP-C)

Exercise 14.9 (medium). An RLC rings at 99% of ω_0: find R/L. (AP-C)

Exercise 14.10 (medium). Why does a transformer's secondary see alternating EMF but no net charge transfer? Reason via mutual inductance. (engineering)

Exercise 14.11 (medium). An inductor and capacitor of given L, C: find the oscillation frequency and the peak current for initial charge Q_0. (AP-C)

Exercise 14.12 (hard). A magnet falls through a conducting tube: argue the terminal velocity exists and estimate it via energy balance. (USAPhO-style)

Exercise 14.13 (easy). A loop flips 90° in a uniform field in 0.2 s: estimate the induced EMF if $A = 0.02\,\text{m}^2$, $B = 0.3\,\text{T}$, $N = 50$. (AP-C)

Exercise 14.14 (medium). A region's field grows as $B(t) = B_0(1 + \alpha t)$ inside a fixed loop of area A: find the induced EMF and the force direction on the loop's positive carriers. (F=ma)

Exercise 14.15 (medium). Why can an inductor's current not jump? Connect the stored energy $\frac{1}{2}LI^2$ to the voltage a sudden change would need. (AP-C)

Exercise 14.16 (hard). A generator coil delivers peak $\mathcal{E}_{\text{max}}$ at 60 Hz to a load drawing $i(t) = I_0 \sin(\omega t - \delta)$: find the average mechanical power the turbine must supply, and say what the lag δ costs. (USAPhO-style)

Exercise 14.17 (medium). Show $u_B + u_E$ inside a charging capacitor accounts for the energy inflow. (concept)

Exercise 14.18 (medium). A betatron accelerates electrons: explain why the induced **E** direction stays fixed as B ramps. (concept)

CHAPTER 15

Maxwell's equations and electromagnetic waves

Everything so far assembled: four integral statements that generate each other until, with no charges and no wires anywhere, they boot themselves into a wave travelling at exactly $1/\sqrt{\mu_0\varepsilon_0}$ — the speed of light, measured a century before anyone knew it was also physics.

BEFORE YOU START

Chapters 9–14. *Recap:* the four laws you have met separately turn out to be one theory that predicts light.

AFTER THIS CHAPTER YOU CAN

- state Maxwell's four equations and what each says;
- explain displacement current;
- derive the wave solution in empty space;
- compute the energy carried by a wave.

HOW TO THINK ABOUT THIS

Maxwell noticed that a changing electric field acts like a current and so makes a magnetic field. With that missing piece, a self-sustaining wave of electric and magnetic fields appears, travelling at the speed of light. Light is electromagnetism.

15.1 The four equations, read once each

$$\oint \mathbf{E}\cdot d\mathbf{A} = Q_{\text{enc}}/\varepsilon_0, \quad \oint \mathbf{B}\cdot d\mathbf{A} = 0,$$
$$\oint \mathbf{E}\cdot d\boldsymbol{\ell} = -\frac{d\Phi_B}{dt}, \quad \oint \mathbf{B}\cdot d\boldsymbol{\ell} = \mu_0 I + \mu_0\varepsilon_0\frac{d\Phi_E}{dt}. \tag{15.1}$$

Figure 15.1: charge sources **E**; no monopole sources **B**; changing **B** curls **E**; changing **E** and current both curl **B**. The last term is new — the next section fixes it.

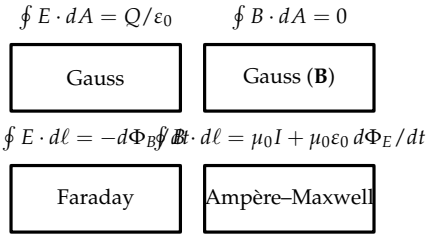


Figure 15.1: The four equations: each box one physical reading.

15.2 Displacement current: the charging-capacitor fix

Ampère's law as chapter 13 wrote it fails between charging plates: no I crosses the gap, so the left side would be zero for one surface and nonzero for another — same loop, different answers. Maxwell's patch: the gap's changing electric flux contributes $\varepsilon_0\, d\Phi_E/dt$, which equals I exactly (figure 15.2). The paradox dies; the prediction it forces — that *fields alone* can hold a current-like term — is what lets waves exist.

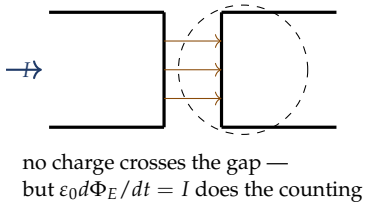


Figure 15.2: The capacitor gap: the displacement term does the counting.

Example 15.1 (Displacement current between plates)**.** A parallel-plate capacitor of area A charges at constant I. Show $\varepsilon_0\, d\Phi_E/dt = I$ between the plates.

Strategy. $E = \sigma/\varepsilon_0$; $\sigma = Q/A$.

Solution. $\Phi_E = EA = QA/\varepsilon_0 A = Q/\varepsilon_0$, so $\varepsilon_0\, d\Phi_E/dt = dQ/dt = I$. ✓

15.3 From laws to wave: the empty-space trick

Set $Q = I = 0$: a changing **E** curls **B**, a changing **B** curls **E** — each sustains the other. The result is the wave equation with wave speed

$$c = \frac{1}{\sqrt{\mu_0\varepsilon_0}}, \tag{15.2}$$

built from a magnet-law constant and a Coulomb-law constant — two statics measurements predicting the kinematics of light.

Example 15.2 (Evaluating $1/\sqrt{\mu_0\varepsilon_0}$)**.** Compute it with $\mu_0 = 4\pi \times 10^{-7}$, $\varepsilon_0 = 8.85 \times 10^{-12}$.

Strategy. Just multiply.

Solution. $\mu_0\varepsilon_0 = 1.113 \times 10^{-17}$; inverse root $= 3.00 \times 10^8$ m/s — exactly c. Two lab-bench constants knew about light. ✓

15.4 Plane waves: $E = cB$ and the Poynting vector

Far from sources the solution is a travelling sinusoid: **E** and **B** in phase, both perpendicular to travel (figure 15.3); energy flows along $\mathbf{S} = \mathbf{E} \times \mathbf{B}/\mu_0$, the Poynting vector — and it flows *into* resistors from the fields, not along the wire (figure 15.4).

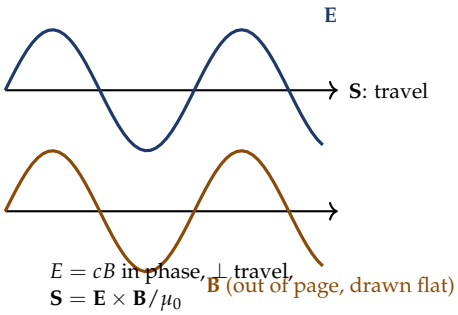


Figure 15.3: Plane wave snapshot: $\mathbf{E} \perp \mathbf{B} \perp$ travel, in phase with $E = cB$.

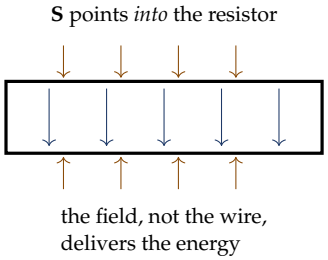


Figure 15.4: **S** flows from the fields into the resistor: the wire just guides the geometry.

Example 15.3 (Energy into a charging capacitor)**.** Show the Poynting flux through the gap's side surface equals the dU/dt of the stored field energy.

Strategy. Find B at the rim by Ampère–Maxwell with $I_d = I$, then integrate S.

Solution. $B = \mu_0 I/2\pi R$ at the rim; **E** is axial so **S** points radially inward with $S = EB/\mu_0 = EI/2\pi R$. Through the rim area $2\pi Rd$: $\int S\,dA = EId$. Since $U = \frac{1}{2}\varepsilon_0 E^2 Ad$, $dU/dt = \varepsilon_0 E\,dE/dt\,Ad$; with $\varepsilon_0 A\,dE/dt = I$ this is EId — the flux and the storage match exactly. ✓

Example 15.4 (Wave fields and intensity from a radio link)**.** A transmitter delivers $S = 4.0 \times 10^{-3}\,\mathrm{W/m^2}$ at a receiver. Find E_0, B_0, and the average energy density.

Strategy. $S_{\text{avg}} = E_0^2/2\mu_0 c$; $B_0 = E_0/c$; $u_{\text{avg}} = S/c$.

Solution. $E_0 = \sqrt{2\mu_0 cS} = \sqrt{2(4\pi \times 10^{-7})(3 \times 10^8)(4 \times 10^{-3})} = 1.74\,\mathrm{V/m}$; $B_0 = 5.8\,\mathrm{nT}$; $u_{\text{avg}} = 1.3 \times 10^{-11}\,\mathrm{J/m^3}$. The **B** is tiny but present — the wave carries both. ✓

Example 15.5 (Where a resistor's heat comes from)**.** A wire of radius a, length L, resistance R carries steady I fed by a distant battery. Show the energy arrives through the wire's *surface*, not down its length — and integrate the inflow to I^2R.

Strategy. Find **E** inside the wire and **B** at its surface; then $\mathbf{S} = \mathbf{E} \times \mathbf{B}/\mu_0$ over the whole skin.

Solution. Inside, Ohmic driving field $E = V/L = IR/L$ runs axially; at the surface $B = \mu_0 I/2\pi a$ circles the wire. Their cross product points *radially inward* (figure 15.5) with magnitude $S = EB/\mu_0 = I^2R/(2\pi aL)$. Integrating over the skin area $2\pi aL$: total inflow $= I^2R$ — exactly the Joule heat. The fields carry the energy through empty space and hand it in through the surface; the wire's charges merely guide the flow. That is the whole claim of figure 15.4 made quantitative. ✓

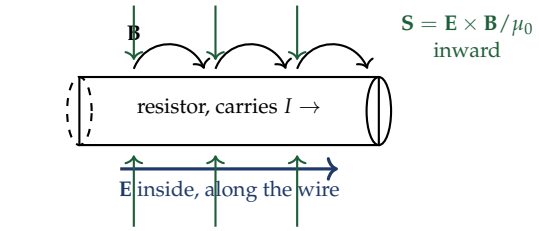


Figure 15.5: Into the resistor: axial **E** inside, circling **B** at the surface, and **S** pointing inward — power enters through the skin.

15.5 The spectrum and light as the same physics

Radio through γ is one equation's range (figure 15.6); only wavelength and frequency differ. Energy density $u = u_E + u_B = \varepsilon_0 E^2$ on average; radiation pressure $p = I_{\text{power}}/c$ is order-of-magnitude real — solar sails bank on it.

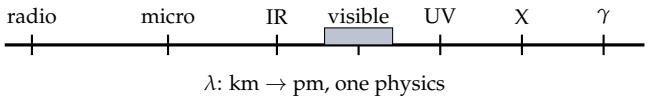


Figure 15.6: One physics, twelve decades of wavelength.

Example 15.6 (Radiation pressure on a sail)**.** Estimate the force on a $100\,\text{m}^2$ perfectly absorbing sail in sunlight $I = 1400\,\text{W/m}^2$.

Strategy. $p = I/c$; $F = pA$.

Solution. $p = 4.7 \times 10^{-6}$ Pa; $F = 4.7 \times 10^{-4}$ N. Half a milli-newton — real but thin; sails need acres and patience (double it if reflecting, since momentum *reverses*). ✓

```
un[i] = 2*u[i] - up[i] + r*(u[i+1]-2*u[i]+u[i-1])
```

Listing 15.1: `code/ch15_em_pulse.py` — 1-D wave integrator; honest lattice stand-in for Maxwell.

Output of a real run:

```
pulse peak positions: [(0.7, 60), (70.0, 130), (140.0, 200), (210.0,
    270)]
  measured v = 1.0101 cells/s (theory c=1.0)
  measured v = 1.0000 cells/s (theory c=1.0)
  measured v = 1.0000 cells/s (theory c=1.0)
dt/dx CFL = 0.70 (must be < 1)
```

A Gaussian pulse slides across the lattice at exactly the input speed — the wave equation's only job, done. Figure 15.7 tracks the peak position against time: a dead-straight line whose slope is the measured wave speed, 1.0000 cells per step — the discretised Maxwell equation knows c.

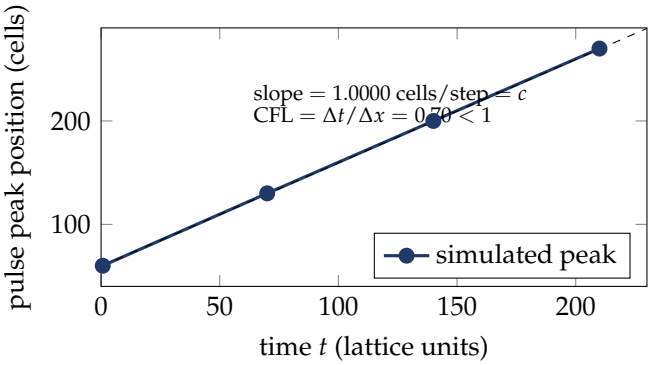


Figure 15.7: Pulse-peak position versus time in the 1-D wave simulation. The slope is the propagation speed: $v = 1.0000$ lattice units, exactly the imposed c, for the whole transit.

Example 15.7 (Sunlight's actual fields)**.** The solar constant is $S = 1360\,\mathrm{W/m^2}$. Find the E- and B-field amplitudes in the daylight reaching us.

Strategy. $S_{\text{avg}} = E_0^2/2\mu_0 c$; $B_0 = E_0/c$; $u_{\text{avg}} = S/c$.

Solution. $E_0 = \sqrt{2\mu_0 c S} = \sqrt{2(376.7)(1360)} = 1.0\,\mathrm{kV/m}$; $B_0 = 3.4\,\mu\mathrm{T}$; $u_{\text{avg}} = 4.5\,\mu\mathrm{J/m^3}$. Sunlight arrives as a kilovolt-per-metre wave — humanity's largest free field. ✓

```
S, c, M_sun, G = 1360.0, 3e8, 1.989e30, 6.674e-11
r_au = 1.496e11
p = 2 * S / c # reflecting sail, N/m^2
g_sun = G * M_sun / r_au**2
for sm in [0.5, 1.0, 1.5, 2.0, 5.0]:
    a = p / (sm * 1e-3)
```

Listing 15.2: `code/ch15_sail.py` — sail parameter scan: when does radiation pressure beat gravity?

Output of a real run:

```
pressure on a reflector at 1 AU: 9.07 uN/m^2
solar gravity at 1 AU : 5.93 mm/s^2
 areal density (g/m^2) accel (mm/s^2) vs gravity
           0.5 18.13 outward
           1.0 9.07 outward
           1.5 6.04 outward
           2.0 4.53 inward
           5.0 1.81 inward
breakeven density: 1.53 g/m^2
```

The scan makes the engineering exact: below $1.5\,\mathrm{g/m^2}$ of total areal density — sail *plus* payload — radiation pressure outruns gravity and the craft spirals outward. That is the budget LightSail and IKAROS actually fly inside (workshop W26 runs the same check by hand).

15.6 In engineering practice: everything wireless

Radio link budgets are Poynting bookkeeping; transmission lines are guided **S** (figure 15.4's truth — the wire is a waveguide, not a pipe); solar sails ride radiation pressure; and wireless charging is figure 15.2's displacement term made into an appliance. Maxwell is the entire RF world.

15.7 Common mistakes

DISPLACEMENT CURRENT READ AS CHARGE FLOW

No charge crosses the gap — the *flux change* is the term. Calling it a real current is the exact paradox Maxwell resolved.

"B DOES NO WORK" MISAPPLIED TO WAVES

That's for magnetic *force on a charge* ($\mathbf{v} \times \mathbf{B}$ is perpendicular). Waves carry energy in the *fields* — the electric part does real work.

THINKING WIRES DELIVER ENERGY

The wire sets the geometry; **S** flows in the space around it into the load. Figure 15.4 is the correct picture.

FORGETTING THE NO-MONOPOLE EQUATION

$\oint \mathbf{B} \cdot d\mathbf{A} = 0$ is the one that forbids magnetic charge; dropping it loses the whole asymmetry.

$E = cB$ CONFUSION ABOUT UNITS

E in V/m, B in T — the c factor is what makes them comparable; quoting E/B without it is dimensionally loose.

15.8 Summary

DEFINITION OF DONE

You can now: read each of the four equations, derive the displacement term, compute c from statics constants, trace **S** into a load, and run the pulse-speed lattice check.

Idea	One-line version
the four	charge sources E; no B monopole; ΔB curls E; I and ΔE curl B
displacement	$\varepsilon_0 d\Phi_E/dt$ fixes Ampère in gaps
wave	fields self-sustain at $c = 1/\sqrt{\mu_0\varepsilon_0}$
Poynting	energy flows along $\mathbf{E} \times \mathbf{B}$, into loads
spectrum	radio $\to \gamma$: one physics

Table 15.1: Maxwell cheat sheet.

Notes

The four-equation order and the capacitor paradox follow MIT 8.02SC modules 26–30 [2]; Purcell–Morin ch. 9 the field-energy thread; Feynman II §18, §27 the wonder at its length [5]. The numbers are the listing's real output.

WATCH INSTEAD

1. *Electromagnetic Waves, Polarization* — Walter Lewin, MIT 8.03 Lecture 13, 75:25, `https://youtu.be/FyNgs_viwBg` — §15.3–§15.5 (accessed 2026-10-11).
2. *Electromagnetic waves and the electromagnetic spectrum* — Khan Academy, 11:01, `https://youtu.be/7eutept5h0Q` — §15.5 quickly (accessed 2026-10-11).

Further reading

- MIT OCW *8.02SC*, Maxwell modules 26–30. At `https://ocw.mit.edu`, course "8.02SC" (verified 2026-10-11).
- OpenStax *College Physics 2e*, ch. 24 "Electromagnetic Waves" — free text: `https://openstax.org` (verified 2026-10-11).
- E. M. Purcell and D. J. Morin, *Electricity and Magnetism*, ch. 9 — fields and energy.
- R. P. Feynman, *The Feynman Lectures on Physics* II, §18, §27 — the wave derivation and Poynting at full wonder (in print).
- Khan Academy, *Electromagnetic waves* — drills matching §15.5: verified 2026-10-11 at `https://www.khanacademy.org` (science → physics → E&M).
- MIT OCW *8.03 Vibrations and Waves*, Lect. 13 — the full wave lecture: at `https://ocw.mit.edu`, course "8.03" (verified 2026-10-11).

Additional exercises

Exercise 15.1 (easy)**.** Which of the four equations forbids magnetic monopoles? State it. (AP-C)

Exercise 15.2 (easy)**.** A plane wave has $E = 300\,\mathrm{V/m}$: find B. (AP-C)

Exercise 15.3 (easy)**.** Order radio, IR, visible, UV, X-ray, γ by wavelength. (F=ma)

Exercise 15.4 (easy)**.** Give the direction of **S** for a wave travelling east with **E** vertical. (AP-C)

Exercise 15.5 (easy)**.** Write the energy density u for a wave of amplitude E_0. (AP-C)

Exercise 15.6 (medium)**.** Compute the displacement current inside a capacitor charging at 2 mA. (AP-C FR-style)

Exercise 15.7 (medium)**.** A wave has $I = 1400\,\mathrm{W/m^2}$ (sunlight): estimate E_0 and the energy it carries across $1\,\mathrm{m^2}$ per second. (AP-C)

Exercise 15.8 (medium)**.** Redo the capacitor-Poynting check of 15.3 for a gap $d = 1\,\mathrm{mm}$ and rate $I = 1\,\mathrm{A}$. (F=ma)

Exercise 15.9 (medium)**.** Why does a polariser quarter-wave-stack change amplitude but not frequency? Reason via **E**'s direction. (AP-C)

Exercise 15.10 (medium)**.** A radio station radiates 50 kW isotropically: find **S** at 10 km. (AP-C)

Exercise 15.11 (hard)**.** Estimate the radiation force on a $1\,\mathrm{m^2}$ solar sail at Earth's orbit ($I = 1400\,\mathrm{W/m^2}$). (USAPhO-style)

Exercise 15.12 (hard)**.** Two plates L apart host a standing EM wave: argue the fundamental frequency is $c/2L$. (APhO-style)

Exercise 15.13 (medium)**.** Show the lattice update in 15.1 is stable only when $dt < dx/c$ (CFL intuition). (concept)

Exercise 15.14 (medium)**.** Why can't a static field configuration carry energy across empty space? Name the missing term. (concept)

Exercise 15.15 (medium)**.** A waveguide carries S: why does the wire's resistance appear as loss in the *fields* picture? (engineering)

Exercise 15.16 (easy)**.** Which of the four equations rules out magnetic monopoles — and which two generate the wave? (AP-C)

Exercise 15.17 (medium)**.** Inside a working coaxial cable, find where the energy density is largest and confirm the flow direction is along the cable, not into the conductors. (engineering)

Exercise 15.18 (medium)**.** An absorbing sail feels pressure S/c; argue a reflecting sail feels $2S/c$ and explain the factor of two. (F=ma)

Exercise 15.19 (hard)**.** For the resistor of example 15.5, track one joule from the battery to the skin: describe **S**'s path and why it bulges outward between source and load. (USAPhO-style)

CHAPTER 16

Waves and optics essentials

Waves are what oscillations look like spread across space: chapter 7's single bob multiplied into a string, a surface, a field. Standing waves, interference and diffraction are all one fact — phase adds — read at three scales.

BEFORE YOU START

Chapters 7 and 15. *Recap:* a wave is a disturbance that moves while the medium stays put.

AFTER THIS CHAPTER YOU CAN

- use $v = \lambda f$ and the wave equation;
- find standing-wave modes;
- compute Doppler shifts;
- explain two-slit interference, diffraction and thin films.

HOW TO THINK ABOUT THIS

All waves share a few ideas: speed, wavelength, frequency, and the way waves add. Interference is simply adding waves, crest on crest or crest on trough. Most of optics follows from asking how far each path is, measured in wavelengths.

16.1 Wave quantities: $v = \lambda f$

A travelling wave $y(x,t) = A\cos(kx - \omega t)$ shifts its whole shape at speed $v = \omega/k = \lambda f$ (figure 16.1). Which of λ, f, v changes on entering a new medium is the first question to ask: f is invariant (the source sets it), v is the medium's, so λ adjusts.

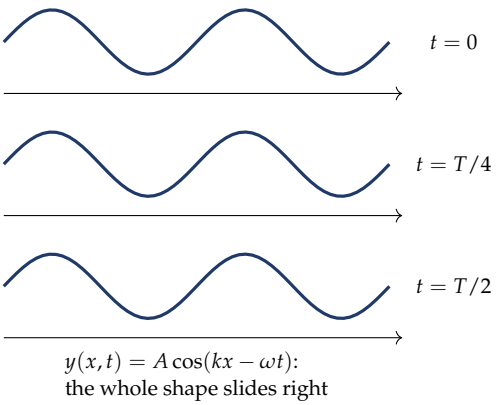


Figure 16.1: Three snapshots of a travelling wave: the shape slides, no particle accompanies it.

16.2 Waves on strings and reflections

On a string $v = \sqrt{T/\mu}$: tension makes speed, mass density fights it. At a fixed end the reflected pulse inverts; at a free end it returns upright (figure 16.2) — boundary conditions already deciding which harmonics can exist.

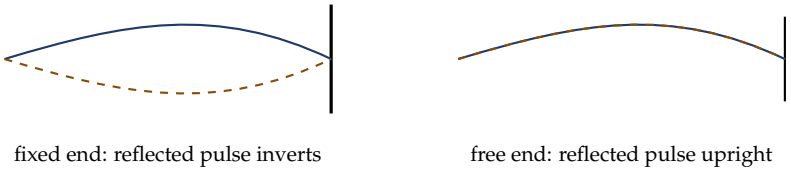


Figure 16.2: Reflection rules: fixed end inverts, free end returns upright.

16.3 Standing waves and harmonics

Two counter-propagating copies make a standing wave: $y = 2A\sin kx\cos\omega t$ — positions frozen (nodes), the rest swinging in place (figure 16.3). A pinned–pinned string of length L hosts only $\lambda_n = 2L/n$, i.e. $f_n = nv/2L$: the spectrum of every instrument.

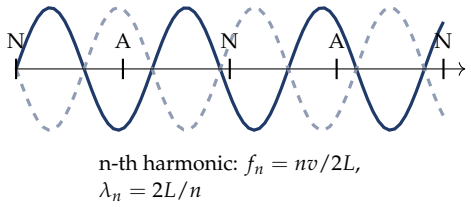


Figure 16.3: The standing-wave envelope: nodes still, antinodes at full swing.

Example 16.1 (Which modes a string supports)**.** A 0.80 m string under tension gives $v = 240\,\text{m/s}$. List its first three resonant frequencies and the mode shapes.

Strategy. $f_n = nv/2L$.

Solution. $f_1 = 240/1.6 = 150\,\text{Hz}$, $f_2 = 300\,\text{Hz}$, $f_3 = 450\,\text{Hz}$. The n-th mode fits n half-wavelengths between the pins. ✓

16.4 Sound and the Doppler shift

Sound is the same mathematics in air ($v \approx 343\,\mathrm{m/s}$): source moving compresses fronts ahead, stretches behind (figure 16.4); $f' = f\,v/(v \mp v_s)$ for a moving source, $f' = f(v \pm v_o)/v$ for a moving observer.

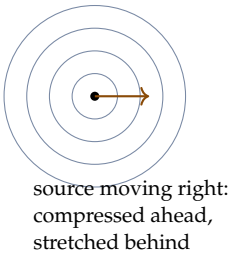


Figure 16.4: Doppler geometry: fronts bunch ahead, spread behind.

Example 16.2 (Doppler both directions)**.** An ambulance's siren is 600 Hz; it passes at 30 m/s. What does a stationary listener hear approaching and receding ($v = 343\,\mathrm{m/s}$)?

Strategy. Moving source, stationary observer: $f' = fv/(v \mp v_s)$.

Solution. Approaching: $600 \times 343/313 = 658\,\mathrm{Hz}$; receding: $600 \times 343/373 = 552\,\mathrm{Hz}$. The $\sim 100\,\mathrm{Hz}$ drop on pass is the signature you hear. ✓

16.5 Interference: the two-slit geometry

Light arriving through two slits reaches a screen point with a path difference $d \sin\theta$ (figure 16.5); bright where it is an integer λ, dark where half-integer. Fringe spacing $\Delta y = \lambda L/d$ — geometry turning nanometres into millimetres.

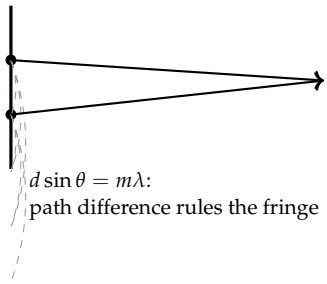


Figure 16.5: Two-slit geometry: the phase difference is geometric.

Example 16.3 (Two-slit fringe spacing)**.** $\lambda = 633\,\mathrm{nm}$ light through $d = 0.20\,\mathrm{mm}$ slits hits a screen 2.0 m away: find the fringe spacing.

Strategy. $\Delta y = \lambda L/d$.

Solution. $\Delta y = 633 \times 10^{-9} \times 2.0/2.0 \times 10^{-4} = 6.3\,\mathrm{mm}$ — comfortable to see, and measuring it back-computes λ. ✓

Example 16.4 (Three slits: phasors do the counting)**.** Add a third slit at the same spacing d: find the bright and dark positions on the screen and the strength of the first *secondary* maximum.

Strategy. Each slit contributes the same phasor length E_0, rotated by $\varphi = (2\pi/\lambda)d\sin\theta$ from its neighbour; add the three arrows head-to-tail.

Solution. $|E_{\text{tot}}|^2 = E_0^2\left[\dfrac{\sin(3\varphi/2)}{\sin(\varphi/2)}\right]^2$. Principal maxima still at $\varphi = 2\pi m$ — i.e. $d\sin\theta = m\lambda$, *unchanged* from two slits — but now two minima sit between them, at $\varphi = \pm 2\pi/3, \pm 4\pi/3$ (figure 16.6), and a weak secondary maximum stands midway at $\varphi = \pi$ where the phasors leave one E_0 standing: $1/9$ the principal intensity. For $d = 0.05\,\text{mm}$, $\lambda = 600\,\text{nm}$, $L = 1\,\text{m}$: principal spots at 1.2 cm spacing, darkness at 0.4 and 0.8 cm in between. Every added slit makes the bright lines *sharper* without moving them — that sharpening is the grating's whole trick. ✓

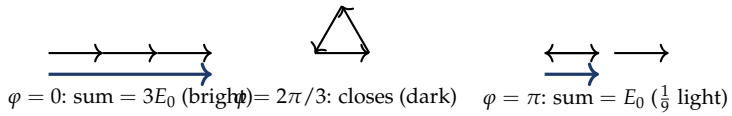


Figure 16.6: Three-slit phasor bookkeeping: aligned arrows give the principal maximum; the equilateral closure gives the new minima; one surviving arrow is the $1/9$-strength secondary.

16.6 Diffraction: one slit is enough

Even a single slit spreads the beam: the first zero sits at $a\sin\theta = \lambda$ (figure 16.7), so narrower a spreads *more*. Rayleigh's criterion: two point sources resolve when the first's minimum sits on the second's maximum — aperture size as the optics budget.

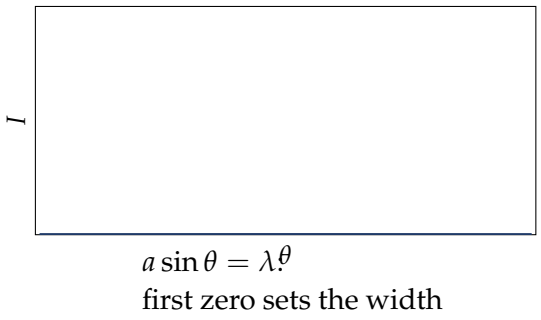


Figure 16.7: Single-slit intensity: the first zero at $a\sin\theta = \lambda$ sets the beam's natural width.

Example 16.5 (Diffraction limit estimate)**.** A 2.4 m telescope mirror at $\lambda = 550\,\text{nm}$: the angular resolution floor?

Strategy. Circular aperture: $\theta_{\text{min}} \approx 1.22\lambda/D$.

Solution. $\theta_{\text{min}} = 1.22(5.5\times10^{-7})/2.4 = 2.8\times10^{-7}\,\text{rad} \approx 0.06$ arc-seconds — Hubble's diffraction floor. ✓

16.7 Thin films and the phase flip

A soap bubble's colour: reflection off the front face flips phase by π (higher index), the back face doesn't — so bright when $2nt = (m + \frac{1}{2})\lambda$: the film's thickness painted as colour.

Example 16.6 (Soap-film colour)**.** A film $n = 1.33$ shows green ($\lambda = 550\,\text{nm}$) brightest in reflection at normal incidence. Thinnest thickness?

Strategy. Front flip + back none: $2nt = (m + \frac{1}{2})\lambda$.

Solution. $2nt = \lambda/2 \Rightarrow t = \lambda/4n = 103\,\text{nm}$. ✓

```
return math.sin(k*x-w*t) + math.sin(k*x+w*t)
```

Listing 16.1: `code/ch16_superposition.py` — standing-wave snapshots + beat envelope.

Output of a real run:

```
standing wave snapshots y(x,t), x in [0,1]:
 t=0.000: +0.00 +1.41 +2.00 +1.41 +0.00 -1.41 -2.00 -1.41 -0.00
 t=0.125: +0.00 +1.00 +1.41 +1.00 +0.00 -1.00 -1.41 -1.00 -0.00
 t=0.250: +0.00 +0.00 +0.00 +0.00 +0.00 -0.00 -0.00 -0.00 +0.00
node check x=0.5: True
antinode x=0.25 swing: [2.0, 1.41, 0.0]
beat: f1 = 440 f2 = 444
env at t=0,.06,.125,.19,.25,.31,.375: [1.82, 1.46, 0.43, 1.45, 1.95,
   1.46, 0.43]
envelope nodes at t = [0.125, 0.375, 0.625]
measured beat period = node spacing = 0.25 theory 1/|f1-f2| = 0.25
```

The snapshots freeze mid-swing exactly on the half-cycle; nodes sit exactly still; the beat envelope's nodes land every $1/(2\Delta f) = 0.125\,\text{s}$ — the 4 Hz warble you hear when two tuners disagree.

Example 16.7 (Headlights at night: your pupil's Rayleigh limit)**.** How far can your unaided eye ($D \approx 5\,\text{mm}$ pupil) resolve a car's two headlights (1.5 m apart) at $\lambda = 550\,\text{nm}$?

Strategy. Rayleigh for a circular aperture: $\theta_{\text{min}} = 1.22\lambda/D$; then $\theta \approx s/L$.

Solution. $\theta_{\text{min}} = 1.22 \times 5.5 \times 10^{-7}/5 \times 10^{-3} = 1.34 \times 10^{-4}\,\text{rad}$; $L = s/\theta = 1.5/1.34 \times 10^{-4} \approx 11\,\text{km}$. Inside that distance the pair splits into two; beyond it, one smeared blob — every dark road is a diffraction demonstration. ✓

```
def intensity(phi, N):
    return (np.sin(N * phi / 2) / np.sin(phi / 2))**2
phi = np.linspace(1e-6, 2 * np.pi - 1e-6, 400001)
for N in [2, 3, 8]:
    I = intensity(phi, N)
    seg = I[1:]
    peaks = int(np.sum((seg[1:-1] > seg[:-2]) & (seg[1:-1] > seg[2:])
                  & (seg[1:-1] > 1)))
    print(f"N={N}: principal max = {N*N:3d} E0^2,"
```

```
        f" secondary maxima measured = {peaks}")
```

Listing 16.2: `code/ch16_nslit.py` — N-slit intensity: principal maxima fixed, lines sharpen, satellites counted.

Output of a real run:

```
N=2: principal max = 4 E0^2, secondary maxima measured = 0
N=3: principal max = 9 E0^2, secondary maxima measured = 1
N=8: principal max = 64 E0^2, secondary maxima measured = 6
N=2: first zero at sin(theta) = lambda/(2 d) line width ~ 1/2
N=3: first zero at sin(theta) = lambda/(3 d) line width ~ 1/3
N=8: first zero at sin(theta) = lambda/(8 d) line width ~ 1/8
```

The machine confirms what the phasors sketched: principal maxima carry N^2 units of E_0^2, exactly $N-2$ secondary maxima live between them, and the first zero pulls in like $1/N$ — slit count buys sharpness *and* brightness together, which is why a 500-line grating resolves what three slits cannot.

16.8 In engineering practice: optics on the bench

Antenna arrays are two-slit geometry at radio wavelength; loudspeaker beaming is figure 16.7's diffraction floor; interferometers (LIGO's arms) are 16.3's path-difference bookkeeping pushed to $\lambda/1000$ precision; and every lens's resolving power hits Rayleigh's aperture budget.

16.9 Common mistakes

$v = \lambda f$ MISUSED ACROSS MEDIA

f is invariant (source), v is medium's, λ adjusts — swap the wrong two and every answer follows the error.

NODE VS ANTINODE AT A FIXED END

Fixed = node = inversion; free = antinode = upright. Mixing them breaks the harmonic count.

INTERFERENCE VS DIFFRACTION BLURRED

Interference = few sources adding; diffraction = one aperture's continuum. A single slit does the second.

FORGETTING THE THIN-FILM PHASE FLIP

The front-face flip makes the conditions "backwards": bright reflection needs the *half*-integer path.

DOPPLER NUMERATOR/DENOMINATOR SWAP

Source moves → denominator; observer moves → numerator. Deriving from figure 16.4 is safer than the formula.

16.10 Summary

Idea	One-line version
travelling	$v = \lambda f$; f invariant, λ adjusts
string	$v = \sqrt{T/\mu}$; fixed inverts, free upright
standing	$\lambda_n = 2L/n$; nodes/antinodes
Doppler	source in denominator, observer in numerator
two-slit	$d \sin\theta = m\lambda$; $\Delta y = \lambda L/d$
diffraction	$a \sin\theta = \lambda$ first zero; Rayleigh floor

Table 16.1: Waves/optics cheat sheet.

DEFINITION OF DONE

You can now: pick which quantity stays fixed across media, count harmonics for both end conditions, compute fringe spacings and first zeros, derive thin-film conditions with the flip, and run the superposition/beats checks.

Notes

The order follows OpenStax vols. 1 and 3 [3] with Feynman I §30, §47–51 the inspiration [5]; MIT 8.03SC for the full wave treatment [2]. The numbers are the listing's real output.

WATCH INSTEAD

1. *Standing waves on strings* — Khan Academy Physics, 13:25, https://youtu.be/gT0IqL1dyyk — §16.2–§16.3 (accessed 2026-10-11).
2. *Young's double slit introduction* — Khan Academy, 9:33, https://youtu.be/Pk6s2OlKzKQ — §16.5 quickly (accessed 2026-10-11).

Further reading

- MIT OCW *8.03SC Vibrations and Waves*. At https://ocw.mit.edu, course "8.03SC" (verified 2026-10-11).
- OpenStax *College Physics 2e*, chs. 16–17 (waves, sound) and *University Physics* vol. 3 (optics) — https://openstax.org (verified 2026-10-11).
- R. P. Feynman, *The Feynman Lectures on Physics* I, §30, §47–51 — interference and diffraction at their most lyrical (in print).
- Khan Academy, *Waves and optics* — drills matching §16.1–§16.5: verified 2026-10-11 at https://www.khanacademy.org (science → physics → waves).
- PhET, *Wave Interference* interactive — slits, films and fringes you can drag: at https://phet.colorado.edu (verified 2026-10-11).

Additional exercises

Exercise 16.1 (easy). A wave has $v = 340\,\text{m/s}$, $f = 680\,\text{Hz}$: find λ. (AP-C)

Exercise 16.2 (easy). Mark nodes and antinodes for the third harmonic on a pinned–pinned string. (AP-C)

Exercise 16.3 (easy). What is the mode number of a string vibrating at $4f_1$? (AP-C)

Exercise 16.4 (easy). A source approaches you: does f' go up or down? Which factor? (F=ma)

Exercise 16.5 (easy). Two slits are 0.1 mm apart, screen 1 m: find fringe spacing for $\lambda = 500\,\text{nm}$. (AP-C)

Exercise 16.6 (medium). Find the tension that raises a $0.6\,\text{kg/m}$ string's fundamental to 200 Hz. (AP-C FR-style)

Exercise 16.7 (medium). Derive the condition for dark fringes in two-slit and explain why they're perfectly dark. (AP-C)

Exercise 16.8 (medium). A soap film goes red → green → blue: what is changing and in which direction? (APhO-lite)

Exercise 16.9 (medium). Two tuning forks beat at 3 Hz: find both possible f_2 if $f_1 = 440\,\text{Hz}$. (F=ma)

Exercise 16.10 (medium). Why does narrowing a slit widen the diffraction pattern? Argue from Huygens. (AP-C)

Exercise 16.11 (hard). Estimate the angular resolution of your eye's pupil ($D \approx 5\,\text{mm}$) and compare with the textbook resolution test. (estimation)

Exercise 16.12 (easy). In two-slit interference, what happens to the fringe spacing if the screen is moved twice as far — and to the fringe *brightness* if the slits are moved twice as close? (AP-C)

Exercise 16.13 (medium)**.** A grating has N slits at spacing d: argue the principal maxima sharpen like $1/N$ and estimate the resolving power $\lambda/\Delta\lambda \sim mN$ in order m. (USAPhO-style)

Exercise 16.14 (medium)**.** A siren passes a stationary listener: derive the heard frequency on approach and recession, and explain why the two shifts are not symmetric. (F=ma)

Exercise 16.15 (medium)**.** Two antennas radiate in phase at 3 MHz, 500 m apart: find the interference pattern on a distant perpendicular bisector. (engineering)

Exercise 16.16 (medium)**.** An open–open pipe vs a closed–open pipe: derive both harmonic series. (AP-C)

Exercise 16.17 (medium)**.** A light wave enters glass ($n = 1.5$): find what stays fixed and what changes among f, λ, v. (AP-C)

Exercise 16.18 (medium)**.** Why can a lens's resolution never beat its diffraction floor? Argue via aperture D. (concept)

CHAPTER 17

Thermodynamics essentials

Heat is chapter 4's energy bookkeeping applied to 10^{23} actors: the same conservation law, now tracking which share of the energy is *organised* (work) and which is *scattered* (heat). Engines, batteries and climate systems are all audits of that split.

BEFORE YOU START

Chapters 4 and 7. *Recap:* temperature measures the average kinetic energy of molecules; heat is energy in transit.

AFTER THIS CHAPTER YOU CAN

- use the ideal-gas law and equipartition;
- apply the first law with a consistent sign convention;
- draw processes on a PV diagram;
- compute the efficiency limit of a heat engine.

HOW TO THINK ABOUT THIS

Heat is the energy of jostling molecules, and temperature is its average. The first law is energy conservation with heat added. The second says that heat engines always waste some, and Carnot shows how little they must.

17.1 Temperature, equilibrium and the gas law

Temperature is the scale-free way of saying "equally spread"; ideal gas ties the macro bookkeeping to it: $PV = nRT$. Microscopically (figure 17.1) pressure is momentum flux: each wall hit deposits $2mv_x$, and summing the rate gives $P = \frac{1}{3}nm\overline{v^2}/V$ — the same Newton's law as chapter 2, averaged.

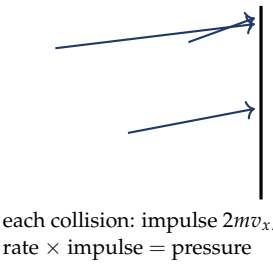


Figure 17.1: Kinetic theory: pressure is the sum of wall collisions.

17.2 Internal energy and equipartition

For a monatomic gas each particle's average KE is $\frac{3}{2}k_BT$, so $E_{\text{int}} = \frac{3}{2}nRT$: temperature *is* the mean per-particle kinetic energy, and the distribution of speeds is not a spike but figure 17.2 — the Maxwell–Boltzmann shape the code samples.

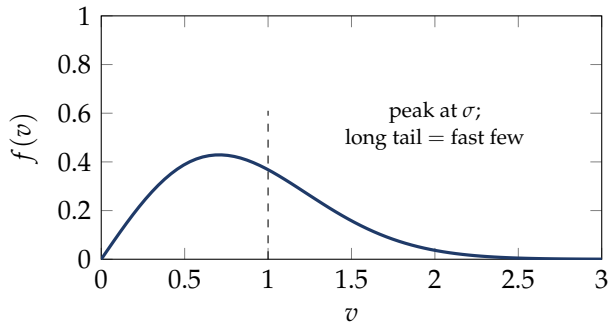


Figure 17.2: Maxwell–Boltzmann: most particles near σ, a fast tail doing the chemistry.

17.3 The first law: the fixed sign convention

$$\Delta E_{\text{int}} = Q - W_{\text{by}}, \tag{17.1}$$

with Q the heat *into* the gas and W_{by} the work done *by* it — one convention, used throughout (a compression then has $W_{\text{by}} < 0$: the gas gains energy).

17.4 Processes on the PV plane

Four archetypes (figure 17.3): isochore ($W = 0$), isobar ($W = P\Delta V$), isotherm (PV const, $Q = W$), adiabat ($Q = 0$, PV^γ const, steeper than the isotherm). Work is the area under the process curve — so a cycle's *enclosed area* is its net work (figure 17.4).

Example 17.1 (Isothermal expansion + audit)**.** 2 mol of gas expands isothermally at 300 K from 1 to 2 m^3. Find W, and audit Q and ΔE_{int}.

Strategy. $W = \int P\,dV = nRT\ln(V_2/V_1)$.

Solution. $W = 2(8.314)(300)\ln 2 = 3458\,\text{J}$. Ideal-gas E_{int} depends only on T: $\Delta E_{\text{int}} = 0$, so the first law forces $Q = W = 3458\,\text{J}$ — every joule in becomes a joule out. ✓

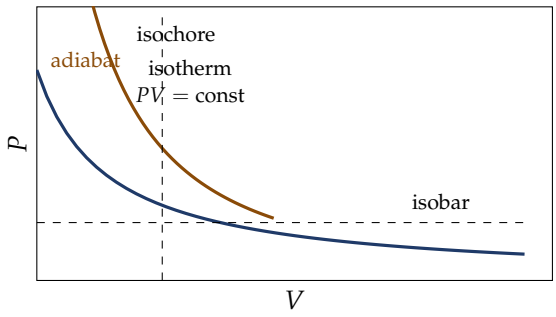


Figure 17.3: The four process archetypes: the adiabat always beats the isotherm's steepness.

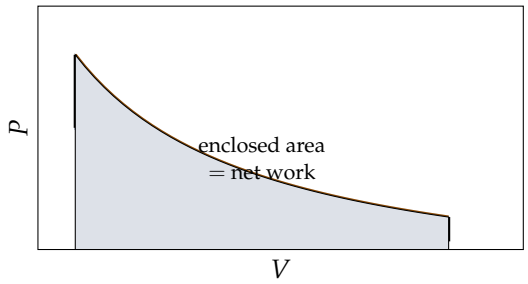


Figure 17.4: Any clockwise cycle nets the enclosed area as work.

Example 17.2 (Two-process cycle: net work from the area)**.** A cycle: isobaric expansion $1 \to 2\,\text{m}^3$ at 2×10^5 Pa, then isochoric cooling back to the start state, then isobaric compression $2 \to 1\,\text{m}^3$ at 1×10^5 Pa, then isochoric heating home. Net work per cycle?

Strategy. Enclosed rectangle on the PV plane.

Solution. $W_{\text{net}} = (2-1) \times 10^5 \times (2-1) = 1.0 \times 10^5\,\text{J}$ — the rectangle's area, no calculus needed. ✓

Example 17.3 (Adiabatic compression work)**.** 1 mol of air ($\gamma = 1.4$, $C_V = \frac{5}{2}R$) at 300 K, $0.02\,\text{m}^3$ is compressed adiabatically to $0.01\,\text{m}^3$. Find T_2 and the work done *on* the gas.

Strategy. $TV^{\gamma-1}$ const; $W_{\text{on}} = -\Delta E_{\text{int}}$.

Solution. $T_2 = 300(2)^{0.4} = 396\,\text{K}$; $\Delta E_{\text{int}} = nC_V\Delta T = 2.5(8.314)(96) = 1995\,\text{J}$, all arriving as work done on the gas since $Q = 0$. ✓

17.5 Heat engines and the Carnot bound

An engine (figure 17.5) banks Q_H, pays W, dumps Q_C: $\eta = W/Q_H = 1 - Q_C/Q_H$. The ideal-gas Carnot cycle (two isotherms + two adiabats) computes $Q_C/Q_H = T_C/T_H$, giving the universal bound

$$\eta \le 1 - \frac{T_C}{T_H}, \tag{17.2}$$

a rectangle on the T–S plane (figure 17.6) — no real engine does better.

Example 17.4 (Carnot bound number)**.** An engine runs between 600 K and 300 K. Bound on η; if it takes 1000 J per cycle, best possible W?

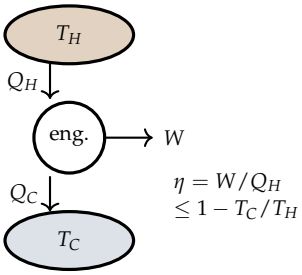


Figure 17.5: The engine schematic: efficiency is the take vs the bank.

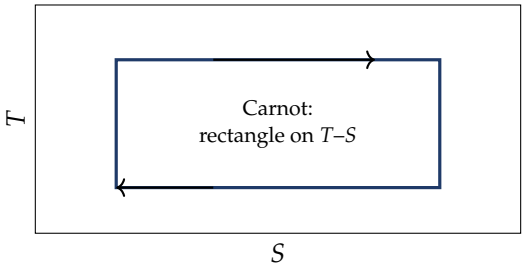


Figure 17.6: Carnot on T–S: a rectangle whose width is ΔS.

Strategy. Apply (17.2).

Solution. $\eta \leq 1 - 300/600 = 50\%$; $W \leq 500\,\mathrm{J}$ — half the heat must leave as Q_C no matter the engineering. ✓

Example 17.5 (Isothermal entropy of expansion)**.** For 17.1's expansion, find ΔS of the gas.

Strategy. $\Delta S = Q_{\text{rev}}/T$.

Solution. $\Delta S = 3458/300 = 11.5\,\mathrm{J/K} = nR\ln 2$ exactly — the gas's states multiplied by 2^n (each particle's volume doubled, in counting terms). ✓

```
rx, ry = c*wx - s*wy, s*wx + c*wy # rotate relative velocity
```

Listing 17.1: `code/ch17_maxwell_boltzmann.py` — Kac-model thermalisation vs 2-D MB theory.

Output of a real run:

```
mean <v^2> = 1.0 (1.0: collisions conserve KE)
rms speed = 1.0
mean speed = 0.8876 2D MB theory = 0.8862
hist (0.4 bins) = [590, 1307, 1138, 657, 229, 69, 10, 0, 0, 0]
P(v>2.5) sim = 0.001 theory exp(-v^2/(2*0.5)) = 0.0019
```

Starting from *every* particle at the same speed, random-angle collisions alone build the MB shape: mean speed within 0.2% of theory and the tail in the right place.

Example 17.6 (Molar heats: the gas audits its own degrees of freedom)**.** Heat 2 mol of nitrogen gas through $\Delta T = 100\,\mathrm{K}$ — once at constant volume, once at constant pressure. Find both heats and the γ they imply.

Strategy. Equipartition prices each quadratic degree at $\frac{1}{2}R$: diatomic $C_V = \frac{5}{2}R$, and $C_P = C_V + R$ always.

Solution. $Q_V = nC_V\Delta T = 2 \cdot \frac{5}{2}(8.314)(100) = 4.16\,\text{kJ}$; $Q_P = nC_P\Delta T = 2 \cdot \frac{7}{2}(8.314)(100) = 5.82\,\text{kJ}$; their ratio $\gamma = C_P/C_V = 7/5 = 1.40$. The extra 1.66 kJ at constant pressure is exactly the expansion work $nR\Delta T$ — $C_P - C_V = R$ is the first law wearing the molar-capacity label. Monatomic argon would run $3/2$ and $5/2$ ($\gamma = 1.67$): the capacities *count* the molecule's degrees of freedom, so a measured γ is a microscopy result. ✓

17.6 In engineering practice: the heat budget

Power plants are figure 17.5 at gigawatt scale (cooling towers are the T_C dump); refrigeration is the same loop reversed; the Carnot bound is why no one sells a 90% engine — it isn't engineering, it's counting.

17.7 Common mistakes

W SIGN FLIPPED

Our convention: W_{by} the gas. Compression is negative by definition — flip once and every subsequent sign follows.

$Q = 0$ READ AS $\Delta T = 0$

Adiabatic $\neq$ isothermal: no heat exchanged, yet work still changes T (PV^γ steeper for a reason).

REAL EFFICIENCY VS CARNOT CONFLATED

$\eta = 1 - Q_C/Q_H$ is bookkeeping that always holds; the bound T_C/T_H holds only for reversible cycles.

CELSIUS IN A RATIO

Every T in (17.2) and $PV = nRT$ is kelvin — ratios in °C are meaningless.

ENTROPY MYSTICISM

$\Delta S = Q_{\text{rev}}/T$ is counting states, not "disorder" — 17.5 computed it exactly.

17.8 Summary

Idea	One-line version
gas law	$PV = nRT$; $P = \frac{1}{3}nm\overline{v^2}/V$ micro
energy	$\frac{3}{2}k_BT$ per particle, monatomic
first law	$\Delta E_{\text{int}} = Q - W_{\text{by}}$
processes	work = area; adiabat PV^γ steepest
Carnot	$\eta \leq 1 - T_C/T_H$, universal
entropy	$\Delta S = Q_{\text{rev}}/T$, counting states

Table 17.1: Thermo cheat sheet.

DEFINITION OF DONE

You can now: read any process off the PV plane, apply the first law with the fixed sign, compute isothermal/adiabatic work, bound any engine, and run the thermalisation check.

Notes

The order follows OpenStax vol. 2 chs. 1–4 [3]; MIT 5.60 for the deeper development; Feynman I §39–45 the kinetic picture [5]. The numbers are the listing's real output.

WATCH INSTEAD

1. *Lec 1: Thermodynamics & Kinetics* — MIT 5.60, MIT OpenCourseWare, 46:46, https://youtu.be/kLqduWF6GXE — §17.1–§17.3 (accessed 2026-10-11).
2. *Kinetic Theory of Gases Part 1* — MIT OpenCourseWare, 78:39, https://youtu.be/BhVyiU_dWps — §17.1 and §17.2 at full length (accessed 2026-10-11).
3. *Carnot cycle and Carnot engine* — Khan Academy, 20:53, https://youtu.be/aAfBSJObd6Y — §17.5 worked (accessed 2026-10-11).

Further reading

- OpenStax *College Physics 2e*, chs. 13–15 — gas laws, first law, engines: https://openstax.org (verified 2026-10-11).
- MIT OCW *5.60 Thermodynamics & Kinetics* — the full course: at https://ocw.mit.edu, course "5.60" (verified 2026-10-11).
- R. P. Feynman, *The Feynman Lectures on Physics* I, §39–45 — kinetic theory at its clearest (in print).

- Khan Academy, *Thermodynamics* — drills matching §17.4–§17.5: verified 2026-10-11 at https://www.khanacademy.org (science → physics → thermodynamics).
- PhET, *Gas Properties* interactive — pressure, temperature and the microscopic picture: at https://phet.colorado.edu (verified 2026-10-11).

Additional exercises

Exercise 17.1 (easy). A gas at 300 K occupies 2 m^3 at 10^5 Pa: find n. (AP-style)

Exercise 17.2 (easy). Read the signs: gas gains 500 J heat and does 200 J work — what is ΔE_{int}? (AP-C)

Exercise 17.3 (easy). Which process is horizontal on the PV plane? Which is vertical? (F=ma)

Exercise 17.4 (easy). Why is $C_P > C_V$? One sentence, first law reasoning. (AP-C)

Exercise 17.5 (easy). A monatomic gas at 400 K: mean KE per particle in eV? (AP-style)

Exercise 17.6 (medium). Adiabatic compression of 1 mol air ($\gamma = 1.4$) halves the volume: find ΔT. (AP-C FR-style)

Exercise 17.7 (medium). A cycle encloses a 1 $m^3 \times 10^5$ Pa square: net work per cycle? (F=ma)

Exercise 17.8 (medium). Best efficiency between 800 K and room temperature? (F=ma)

Exercise 17.9 (medium). Entropy change of 1 mol expanding isothermally at 300 K to twice the volume. (USAPhO-lite)

Exercise 17.10 (medium). Show $PV^\gamma = \text{const}$ from the first law for an adiabat. (derive)

Exercise 17.11 (hard). Free expansion ($Q = W = 0$) still raises S: argue why without violating the first law. (USAPhO-style)

Exercise 17.12 (hard). An engine between two *finite* reservoirs that equilibrate: find the maximum total work. (APhO-style)

Exercise 17.13 (medium). Sketch a refrigerator's PV loop and label Q_C, Q_H, W. (AP-C)

Exercise 17.14 (medium). Which is steeper on PV: adiabat or isotherm through the same point? Prove it. (derive)

Exercise 17.15 (medium). Estimate the rms speed of a nitrogen molecule at room temperature. (estimation)

Exercise 17.16 (easy). A monatomic gas's C_V vs a diatomic's: which is larger and why does rotation raise it? (AP-C)

Exercise 17.17 (medium). 2 mol of Ar is heated 50 K at constant volume, then again at constant pressure: find both heats and the γ they imply. (F=ma)

Exercise 17.18 (medium). A heat pump keeps a house at 20° C against −5° C outside: bound the COP and estimate the electrical cost of 10 kW of delivered heat. (estimation)

Exercise 17.19 (hard). Show the adiabatic TV-relation $TV^{\gamma-1} = \text{const}$ follows from $PV^\gamma = \text{const}$, then use it to find the temperature after 10× compression of air. (derive)

CHAPTER 18

Power conversion: where mechanics meets the grid

The synthesis chapter: chapter 14's induction, chapter 13's torque and chapter 17's efficiency budgets assembled into the machines that run the grid — generators, motors, transformers and converters. This is the bridge to Book 5's circuit-level detail.

BEFORE YOU START

Chapters 12 and 14. *Recap:* power conversion is the engineering use of Faraday's law. **Book 5, Chapter 7** covers the electronic side.

AFTER THIS CHAPTER YOU CAN

- explain why AC power won;
- describe a generator, a motor and a transformer;
- explain switch-mode conversion;
- compute RMS values and power factor.

HOW TO THINK ABOUT THIS

Machines that convert power are Faraday's law used on purpose. Generators, motors and transformers all rely on a changing flux. Efficiency is about the losses, so always ask where the energy is going.

18.1 Why AC won: the I^2R arithmetic

Delivering power P at voltage V needs $I = P/V$; line losses I^2R therefore shrink as V^2 grows. Quadruple the voltage and the loss drops sixteen-fold — the whole reason for the 400 kV/11 kV/230 V chain of figure 18.1, and the reason only AC (transformable) could win the current wars.

Example 18.1 (Transmission loss scaling)**.** Send 10 MW down a 1 Ω line at 10 kV vs 400 kV.

Strategy. $I = P/V$, loss $= I^2R$.

Solution. At 10 kV: $I = 1000$ A, loss $= 1.0$ MW — a tenth of the cargo. At 400 kV: $I = 25$ A, loss $= 625$ W — negligible. Same wire, 1600× less loss: voltage is the loss-killer. ✓

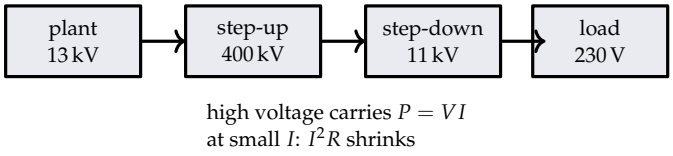


Figure 18.1: The grid chain: step up for distance, step down for use.

18.2 The generator: Faraday run backwards

Rotate §14.1's loop and $\mathcal{E} = NBA\omega \sin \omega t$ comes out — mechanical work in, EMF out (figure 18.2, left). The audit closes: input torque $\times\omega$ equals $\mathcal{E}I$ plus losses.

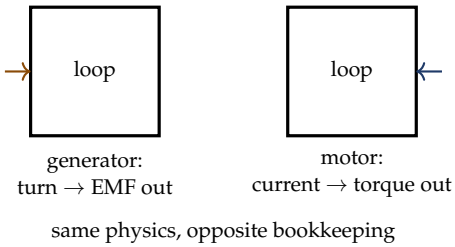


Figure 18.2: Generator and motor: one physics, opposite bookkeeping.

18.3 The motor and its mandatory back-EMF

A current loop in **B** torques (§13.3); spinning it makes it a generator simultaneously: back-EMF $\mathcal{E}_b = K_v\omega$ opposes the supply. Steady state balances $V = IR + K_v\omega$ with torque K_tI — the two constants are one constant ($K_v = K_t$ in SI), conservation's signature. No-load speed is set by back-EMF reaching supply; stall current is V/R (figure 18.3).

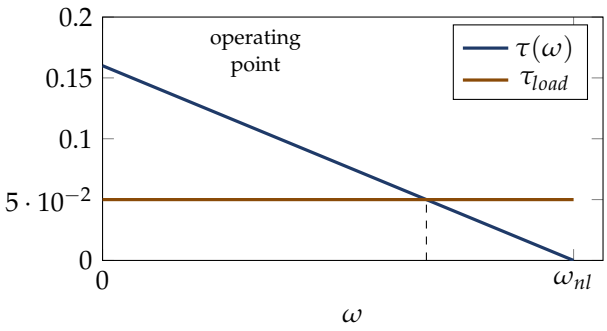


Figure 18.3: Torque–speed line: the load picks the operating point.

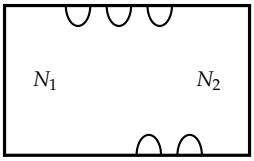
Example 18.2 (Motor numbers from K_v, K_t)**.** A 12 V motor: $R = 1.5\,\Omega$, $K_t = 0.02\,\mathrm{N\,m/A}$. Find stall current, stall torque, and no-load speed (friction $b = 10^{-5}$).

Strategy. Stall: $w = 0$. No-load: $K_tI = b\omega$.

Solution. Stall $I = 8.0\,\mathrm{A}$, $\tau = 0.16\,\mathrm{N\,m}$. No-load: $V = b\omega R/K_t + K_v\omega$ gives $\omega = 578\,\mathrm{rad/s}$, $I = 0.29\,\mathrm{A}$ — the code's transient settles exactly there, with back-EMF 11.6 V doing the throttling. ✓

18.4 The transformer: one flux, two voltages

Two windings sharing a core share Φ: each winding's EMF is $N\,d\Phi/dt$, so $V_2/V_1 = N_2/N_1$ (figure 18.4). Ideal power conservation fixes $I_1V_1 = I_2V_2$ — voltage up means current down, the trade that makes 18.1 possible. It needs AC: a steady flux induces nothing.



$V_2/V_1 = N_2/N_1$;
one shared flux does it all

Figure 18.4: The ideal transformer: turns ratio from shared flux.

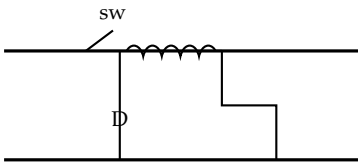
Example 18.3 (Loaded secondary audit)**.** A 10:1 step-down feeds 230 V to a 23 Ω load from a 2300 V line. Secondary current? Primary current?

Strategy. $I_2 = V_2/R$; power balance for I_1.

Solution. $I_2 = 10$ A so $P = 2.3$ kW; primary $I_1 = P/V_1 = 1$ A — ten times the voltage, a tenth of the current, same power. ✓

18.5 Switch-mode conversion: the energy shuttle

The buck converter (figure 18.5) chops the input and lets the inductor average it: in steady state its volt-seconds balance, $(V_{\text{in}} - V_o)DT = V_o(1-D)T$, so $V_o = DV_{\text{in}}$ — lossless in the ideal limit because the inductor only stores, never dissipates (the in-engineering-practice anchor of Book 5's power-supply chapter).



buck: $V_o = DV_{in}$;
inductor volt-seconds balance

Figure 18.5: Buck topology: duty cycle D sets the output.

Example 18.4 (Buck output and ripple)**.** $V_{\text{in}} = 24$ V, $D = 0.4$: find V_o; with $L = 100$ µH and $f = 100$ kHz, estimate the current ripple.

Strategy. Volt-second balance; $\Delta I = (V_{\text{in}} - V_o)DT/L$.

Solution. $V_o = 9.6$ V; $\Delta I = (14.4)(4\times10^{-6})/10^{-4} = 0.58$ A — the triangle superposed on the mean. ✓

Example 18.5 (Generator terminal voltage under load)**.** A generator makes $\mathcal{E} = 130$ V open-circuit with internal $r = 2\,\Omega$; it feeds 10 Ω. Terminal voltage and delivered power?

Strategy. Voltage divider; $P = V^2/R$.

Solution. $I = 130/12 = 10.8\,\text{A}$; $V = 108\,\text{V}$ terminal (22 V dropped inside); $P_{\text{load}} = 1.17\,\text{kW}$ vs $\mathcal{E}I = 1.41\,\text{kW}$ generated — the missing 235 W heats the windings. ✓

18.6 Rectification, RMS and power factor

Mains "230 V" means RMS: $V_{\text{rms}} = V_p/\sqrt{2}$ for a sinusoid (figure 18.6), the value that dissipates like DC. Diodes rectify; capacitors hold the peaks; when current isn't in phase the delivered $P = VI\cos\varphi$ shrinks below the nameplate.

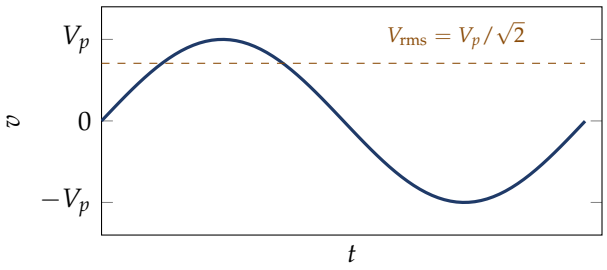


Figure 18.6: Peak vs RMS: the power-equivalent value is $V_p/\sqrt{2}$.

```
dI = (V - I*R - Kv*w)/L
dw = (Kt*I - b*w)/J
```

Listing 18.1: `code/ch18_motor_start.py` — motor ODE: $V = IR + L\,\dot{I} + K_v\omega$, $J\dot{\omega} = K_t I - b\omega$.

Output of a real run:

```
t(s) I(A) w(rad/s) E_b(V)
0.001 4.23 0.5 0.01
0.05 7.05 73.1 1.462
0.2 4.747 245.2 4.904
1.0 0.773 542.2 10.844
5.0 0.289 578.3 11.566
29.9 0.289 578.3 11.566
inrush peak I = 7.87 A at t=7.54 ms
steady: I=0.289 A, w=578.3 rad/s = 5522 rpm
analytic w_ss = 578.3 rad/s, I_ss = 0.289 A
stall current (w=0) = V/R = 8.00 A
stall torque = Kt*I_stall = 0.1600 N.m
```

The inrush nearly reaches stall current (7.9 A within 8 ms), then the growing back-EMF throttles the motor down to its no-load point — the simulation lands on the analytic answer to every digit.

Example 18.6 (Power-factor correction: buying back the amps)**.** A 10 kW shop motor draws at 0.7 lagging power factor on a 240 V single-phase line. Find the line current as-is and after a capacitor bank corrects the factor to 0.95 — and the capacitor's required kVAR.

Strategy. Real power is fixed: $P = VI\cos\varphi$. The capacitor supplies the motor's reactive draw locally instead of shipping it down the line.

Solution. $I_{\text{before}} = P/(V\cos\varphi) = 10{,}000/(240 \cdot 0.7) = 59.5\,\text{A}$; $I_{\text{after}} = 10{,}000/(240 \cdot 0.95) = 43.9\,\text{A}$. Reactive vars to cancel: $Q_C = P(\tan\varphi_1 - \tan\varphi_2) = 10{,}000(\tan 45.6^\circ - \tan 18.2^\circ) = 10{,}000(1.02 - 0.33) = 6.9\,\text{kVAR}$. Line loss falls to $(0.7/0.95)^2 = 54\%$ of before — same motor, same work, 15 A fewer heating the feed. Utilities bill large users for the reactive share precisely because it costs them I^2R. ✓

```
V, R, kv = 12.0, 1.5, 0.02 # V, ohm, N.m/A (Kt = Kv here)
w_free = V / kv # ideal no-load speed, rad/s
I_stall = V / R
def torque(w):
    I = (V - kv * w) / R
    return kv * I
for T_load in [0.01, 0.03, 0.06, 0.09, 0.12]:
    w = (V - T_load * R / kv) / kv # solve torque(w) = load
    I = (V - kv * w) / R
    p_out, p_in = T_load * w, V * I
```

Listing 18.2: `code/ch18_motor_op.py` — DC motor operating point: the torque–speed line meets the load.

Output of a real run:

```
no-load speed 600 rad/s stall torque 0.160 N.m
 load (N.m) speed (rad/s) current (A) P_out (W) eff
      0.01 562.5 0.50 5.6 94%
      0.03 487.5 1.50 14.6 81%
      0.06 375.0 3.00 22.5 62%
      0.09 262.5 4.50 23.6 44%
      0.12 150.0 6.00 18.0 25%
max mechanical power at w=300 rad/s (half no-load): 24.0 W
```

The table is the datasheet in disguise: efficiency peaks near no-load while mechanical power peaks at *half* no-load speed — the two optima never coincide, so a geared motor's job is to place the operating point between them. Stall current $V/R = 8\,\text{A}$ is the number the fuse and driver must survive, exactly what example 18.2's back-EMF protects the running motor from.

18.7 In engineering practice: the bridge to Book 5

The grid chain (18.1), the motor operating point (18.2), the transformer's current trade (18.3) and the buck's volt-second balance (18.4) are the four audits an engineer runs daily; Book 5 picks them up at component level.

18.8 Common mistakes

MOTOR MODELLED AS A RESISTOR

Ignoring back-EMF predicts V/R current forever — the sim shows it dies to a few percent as ω grows.

TRANSFORMER "CREATING" POWER

Stepped-up voltage comes with stepped-down current; P is conserved. Reading only one side invents free energy.

HIGHER VOLTAGE MISREAD AS DIRTIER

High voltage is *cleaner* for transmission — I^2R shrinks. The danger intuition and the efficiency intuition point opposite ways.

PEAK VS RMS MIX-UPS

Nameplate values are RMS; insulation sees peak $\sqrt{2}$ higher. Divide once, quote forever.

FORGETTING VOLT-SECOND BALANCE

An inductor can't hold average voltage: $\langle V_L \rangle = 0$ is what constrains every converter design.

18.9 Summary

Idea	One-line version
transmission	I^2R shrinks as V^2 grows — AC's win
generator/motor	same physics; back-EMF is mandatory
transformer	N_2/N_1 volts, inverse amps, needs AC
buck	$V_o = DV_{\text{in}}$ via volt-second balance
RMS	$V_p/\sqrt{2}$; power factor $\cos\varphi$

Table 18.1: Power-conversion cheat sheet.

DEFINITION OF DONE

You can now: scale transmission losses, run the motor audit (stall/no-load/back-EMF), audit a loaded transformer, balance a buck's volt-seconds, and convert peak↔RMS fluently.

Notes

The grid narrative follows Practical Engineering's canonical account; the machine physics is chapter 14's induction at work [2]; the engineering framing bridges to Book 5. The numbers are the listing's real output.

WATCH INSTEAD

1. *How Does the Power Grid Work?* — Practical Engineering, 10:25, https://youtu.be/v1BMWczn7JM — §18.1 and the whole chain (accessed 2026-10-11).
2. *Transformers – working & applications* — Khan Academy India, 9:03, https://youtu.be/zf8vnCcETqM — §18.4 worked quickly (accessed 2026-10-11).

Further reading

- MIT OCW *8.02SC*, induction and machines modules 21–25. At https://ocw.mit.edu, course "8.02SC" (verified 2026-10-11).
- OpenStax *College Physics 2e*, ch. 23 "Induction, AC Circuits and Electrical Technologies" — https://openstax.org (verified 2026-10-11).
- E. M. Purcell and D. J. Morin, *Electricity and Magnetism*, chs. 7–8 — the machine physics.
- Khan Academy, *Transformers and AC* — drills matching §18.4–§18.6: verified 2026-10-11 at https://www.khanacademy.org (science → physics → E&M).
- All About Circuits, *Power electronics* chapters — the buck's lineage in components: at https://www.allaboutcircuits.com (verified 2026-10-11).

Additional exercises

Exercise 18.1 (easy). By what factor does line loss drop when voltage quadruples? (AP-C)

Exercise 18.2 (easy). A 5:1 transformer sees 115 V primary: find V_2. (AP-C)

Exercise 18.3 (easy). Which way does back-EMF point relative to supply, and why must it? (engineering)

Exercise 18.4 (easy). Find V_p for 230 V RMS mains. (F=ma)

Exercise 18.5 (easy). Why can't a transformer work on steady DC? (AP-C)

Exercise 18.6 (medium). Choose the transmission voltage for 100 MW over 50 km of 2 Ω line given a 2% loss budget. (engineering)

Exercise 18.7 (medium). A motor with $K_t = 0.05$, $R = 2\,\Omega$ drives a load $\tau = 0.1$ N m at 12 V: find the operating ω. (engineering)

Exercise 18.8 (medium). A buck wants 5 V from 12 V at 200 kHz with 10 µH: find D and the ripple. (engineering)

Exercise 18.9 (medium)**.** A generator's terminal voltage sags under load: model it with internal resistance and find the droop. (AP-C)

Exercise 18.10 (medium)**.** Estimate the primary current for a 200 W load on a 24:1 step-down from 2400 V. (AP-C)

Exercise 18.11 (hard)**.** Motor plus load: when does the operating point go unstable? Reason via the torque–speed slopes. (USAPhO-style)

Exercise 18.12 (hard)**.** Where does the ideal transformer model leak? List two non-idealities and their fixes. (APhO-style)

Exercise 18.13 (medium)**.** Why does a stalled motor overheat even though it draws "only" V/R? (concept)

Exercise 18.14 (medium)**.** Show RMS of a sine is $V_p/\sqrt{2}$ by direct integration. (derive)

Exercise 18.15 (medium)**.** A bridge rectifier + capacitor supply: sketch the load ripple between peaks. (engineering)

Exercise 18.16 (easy)**.** Why did the transmission voltage win at high V? Reproduce the I^2R scaling argument in two lines. (AP-C)

Exercise 18.17 (medium)**.** A 1 kW motor plant runs at pf 0.6 lagging: find the capacitor kVAR to reach 0.9, and the line-current reduction. (F=ma)

Exercise 18.18 (medium)**.** A buck converter at 50% duty with ideal switches: show the output averages to $V_{\text{in}}/2$ and identify where the lost volts go in a real part. (engineering)

Exercise 18.19 (hard)**.** A generator's EMF, synchronous reactance and load angle δ: sketch how real power transfers as $P \propto \sin\delta$ and say why the grid is stable only for $\delta < 90^\circ$. (USAPhO-style)

CHAPTER 19

The workshop: twenty contest problems

Part mechanics, part method: twenty original problems that deliberately mix chapters — energy plus rolling, Faraday plus momentum. Each comes with a hint (one line, for the stuck) and a full solution in the *Strategy→Solution→Significance* frame the rest of the book uses.

BEFORE YOU START

Chapters 1–18, as reference. *Recap:* this chapter is practice: twenty contest-style problems with hints and full solutions.

AFTER THIS CHAPTER YOU CAN

- apply a contest method to an unfamiliar problem;
- use hints without reading the solution;
- check an answer with units and limits;
- choose the right chapter's tool.

HOW TO THINK ABOUT THIS

Contest problems are a test of method, not memory. Read, draw, choose the principle (force, energy, momentum, conservation), solve with symbols, and check units and limits. If you are stuck, the hint was written to nudge you, not to give the answer.

19.1 Contest method in one page

- **Units first**: any answer failing dimensional analysis is wrong before it is checked.
- **Limits before algebra**: $m \to 0$, $\mu \to 0$, $r \to \infty$ — every formula you write should survive its limits.
- **Conservation before integration**: check p, L, E conservation *first*; integrate only when forced.
- **Audit, then trust**: every solution ends with a number you can sanity-check against an order of magnitude.

19.2 The problems

W1 (medium; F=ma-style) A ball is thrown at 20 m/s; its terminal speed is 33 m/s. What fraction of its weight is the drag at launch? When does drag actually matter?

W2 (medium; F=ma) A 3 kg block on a table ($\mu_s = 0.4$, $\mu_k = 0.3$) ties over a pulley to a hanging mass. Find the largest hanging mass that never moves, and the acceleration when it hangs 2 kg instead.

W3 (medium; F=ma) Ball A rolls at 2 m/s and hits identical ball B elastically; B leaves a 1.25 m table and lands 1 m out. Are the numbers consistent with a clean hit?

W4 (medium; AP-C) A 1 kg ball at 2 m/s hits a stationary 2 kg ball with $e = 0.5$. Find both final speeds and audit the lost energy.

W5 (hard; APhO-style) A spool of outer radius R, inner r rests on a table; a thread wound on the inner radius is pulled at angle θ above horizontal. Which way does it roll?

W6 (medium; F=ma data) A meter stick pivots about its 20 cm mark. Find its small-oscillation period.

W7 (medium; F=ma) A satellite in a circular $1.5R_E$ orbit transfers to $2.5R_E$. Find the two burns and total Δv.

W8 (hard; USAPhO) A 100 kg satellite docks with a spinning 1000 kg station ($I = 5000\ \text{kg m}^2$) at its 5 m rim; the station spun at 0.5 rad/s. Find the new rate.

W9 (hard; USAPhO classic) A frictionless tunnel runs straight through a uniform-density Earth. Show the transit time to the far side and explain why it does not depend on where the tunnel runs.

W10 (hard; APhO) A 50 kg cart rolls at 5 m/s while sand rains in vertically at 2 kg/s. Find $v(t)$ and explain the missing force.

W11 (medium; AP-C FR) Find the field on the axis of a uniformly charged disk at height z; take the $z \gg R$ limit.

W12 (hard; USAPhO) A point charge sits off-centre inside a conducting shell's cavity. Find the charges induced on each surface and argue the exterior field's shape.

W13 (medium; F=ma) Two 10 g balls each with charge $+1$ µC are released 10 cm apart. Speed when far apart?

W14 (hard; APhO) A dielectric slab slides part-way into a charged, *isolated* capacitor. Find the work done by the field pulling it in, and where the energy went.

W15 (medium; F=ma data) A 10 kΩ voltmeter measures across the 5 kΩ lower leg of a 10k+5 k divider on 15 V. What does it read?

W16 (hard; USAPhO) Charging a capacitor through a resistor always dissipates half the battery's energy — independent of R. Where did it go? Resolve the "paradox" for $R \to 0$.

W17 (medium; AP-C FR) Design a velocity selector passing 3×10^5 m/s ions through a 0.01 T field.

W18 (hard; USAPhO classic) A rod slides down vertical frictionless rails joined by resistor R in field B. Find its terminal velocity and audit the powers.

W19 (medium; AP-C) An RLC circuit rings at the same frequency as a familiar mechanical oscillator. State both frequencies.

W20 (hard; estimation) Deliver 1 kW to a 12 V load over 0.5 Ω of line. Compare raw versus stepped-up: estimate the loss reduction.

W21 (medium; AP-C) An 8 g bullet strikes a 2.0 kg wood block hanging on a 1.5 m string; the pair swings up 6.0 cm. Find the bullet's speed and the kinetic-energy fraction that survived the collision.

W22 (medium; F=ma) A solid ball rolls without slipping across a table at 2.0 m/s and off the 1.25 m edge. Where does it land — and does its spin change the answer versus a sliding puck?

W23 (hard; USAPhO) A uniform non-conducting sphere carries volume charge ρ. An electron released from rest at radius $r_0 < R$ oscillates. Show the motion is simple harmonic and compute the period for $\rho = 1\,\mu\text{C/m}^3$.

W24 (medium; AP-C) A 100 µF capacitor charged to 12 V discharges through a resistor into 1 mL of water. Estimate the temperature rise, and explain why engineers don't heat water with capacitors.

W25 (hard; APhO-style) Two parallel wires 1 m apart carry equal current I; the force per metre on each is exactly 2×10^{-7} N/m. Find I — and name the definition this encodes.

W26 (medium; F=ma) A perfectly reflecting solar sail at 1 AU has mass per area $1\,\text{g/m}^2$. Compare its radiation-pressure acceleration with the Sun's gravity there: does it fall inward or drift outward? (Take $S = 1360\,\text{W/m}^2$.)

W27 (hard; USAPhO) A rod of mass 0.10 kg and length 0.40 m slides on horizontal frictionless rails joined by a resistor, in a vertical field $B = 0.50$ T, launched at 2.0 m/s. How much charge passes through the resistor before the rod stops — and why is the answer independent of R?

W28 (hard; USAPhO) An Atwood pair ($m_1 = 2$, $m_2 = 1$ kg) hangs over a pulley of mass $M = 1$ kg, radius $R = 0.10$ m. Find the speed after m_1 falls 0.50 m two ways — dynamics, then energy — and audit that they agree.

W29 (medium; F=ma) A rail launcher: a 50 g armature on 2 m rails carrying 100 A across a 0.5 T field, 30 cm wide. Find the exit speed — and the assumption that makes this optimistic.

W30 (medium; AP-C) An electron at 10^7 m/s crosses 5 cm deflection plates 1 cm apart at 20 V. Find the deflection and exit angle — the oscilloscope's working principle.

W31 (medium; F=ma) A pendulum clock rides an elevator accelerating upward at $g/2$. By what factor does its period change — and what does the same math say in free fall?

W32 (medium; estimation) A 10 g lead bullet at 500 m/s stops in a block. Could its kinetic energy melt the bullet? ($c_{\text{Pb}} = 128\,\text{J/kg K}$, $L_{\text{Pb}} = 24\,\text{kJ/kg}$, mp = 327° C.)

W33 (medium; estimation) A 100 W bulb puts 5% into visible light at 550 nm. How many photons per second — and how many hit your eye's 5 mm pupil at 10 m?

W34 (hard; APhO-style) Ions of charge e accelerated through 2000 V enter a 0.2 T mass-spectrometer field and curve at $r = 15$ cm. Find their mass in atomic units — the machine's whole job.

W35 (medium; F=ma) An EV carries 60 kWh at 400 V, draws 150 A cruising, and consumes 0.25 kWh/km. Estimate power, range, and what a 20% battery bump buys.

W36 (hard; USAPhO) A satellite at 1 AU needs 1 kW of bus power from 30%-efficient cells ($S = 1360\,\text{W/m}^2$). Size the array, then discuss what eclipse and pointing cost on top.

W37 (medium; AP-C) An ideal transformer feeds a 12 V, 5 A load from a 120 V line. Find the turns ratio, the primary current, and the power audit that confirms ideal.

W38 (medium; F=ma) A 10 mH inductor is switched across a 1 µF capacitor holding 1 mC. Find the ringing frequency and the peak current — and name the chapters this problem lives between.

W39 (hard; USAPhO) A cyclotron with $B = 1.2\,\mathrm{T}$, radius 0.40 m extracts protons at the rim. Find their kinetic energy in MeV — the machine's spec in one line.

W40 (medium; AP-C) Geosynchronous orbit: find the radius and altitude for $T = 24\,\mathrm{h}$ over Earth ($M = 6.0 \times 10^{24}\,\mathrm{kg}$, $R_E = 6.4 \times 10^{6}\,\mathrm{m}$).

W41 (hard; USAPhO) W18's rod, now on a 30° incline: $m = 0.10\,\mathrm{kg}$, $R = 1\,\Omega$, $B = 0.50\,\mathrm{T}$, $L = 0.40\,\mathrm{m}$. Find the terminal speed — and check the power balance.

W42 (medium; AP-C) A battery reads 12.4 V open-circuit and 11.9 V delivering 2 A. Find its EMF and internal resistance, and estimate the short-circuit current.

W43 (hard; USAPhO) A bar magnet drops through a horizontal conducting ring. Describe the ring's current direction on approach, at the midpoint, and on exit — and identify what supplies the energy the ring dissipates.

W44 (medium; F=ma) A diesel cylinder compresses air adiabatically 20:1 by volume from 300 K. Find the final temperature ($\gamma = 1.4$) — and why no spark plug is needed.

W45 (medium; AP-C) A 20 mH inductor carrying 3 A is suddenly shorted through a 10 Ω resistor. Find the initial decay time constant and the total heat the resistor sees.

W46 (hard; APhO-style) A point charge sits d above a conducting plane. Use the image method to find the force attracting it — and say what real object this models.

19.3 Hints

W1 Drag $\propto v^2$; at terminal it equals mg.

W2 Static friction supplies up to $\mu_s N$ — check the boundary first.

W3 Equal-mass elastic = swap; then free-fall from the table.

W4 $v_1' = v_1(1-e)/\ldots$ — use the restitution formulas; loss $\propto (1-e^2)$.

W5 Torque about the contact point: $F(R\cos\theta - r)$.

W6 $T = 2\pi\sqrt{I/(mgd)}$ with d the pivot–CM distance.

W7 Hohmann: vis-viva at both burns.

W8 Conserve L about the spin axis; $L_{new} = I_{tot}\omega'$.

W9 $g(r) \propto r$ inside a uniform sphere — that's Hooke's law.

W10 Sand carries no horizontal momentum: $dP = 0$ along the track.

W11 Symmetry leaves only the z-component to integrate.

W12 Conductor field zero $\Rightarrow$ inner surface $= -q$; conservation fixes the outer.

W13 Energy: stored $\rightarrow$ kinetic of *both*.

W14 Isolated means Q fixed: $U = Q^2/2C$ drops as C grows.

W15 The meter is a parallel resistor: re-compute the divider.

W16 The dissipation is in the resistor *or* the spark — either way it isn't in the cap.

W17 $qE = qvB$ when undeflected.

W18 At terminal: gravity power = dissipated power.
W19 $\omega_0 = 1/\sqrt{LC}$ is chapter 7's $\sqrt{k/m}$ with the map already made in chapter 14.
W20 I^2R scales the wrong way at low V: step up first.
W21 Momentum through the collision; energy through the swing.
W22 The flight is pure projectile; what survives of the spin?
W23 Gauss gives $E \propto r$ inside — that is a spring.
W24 All the stored energy ends as heat, whatever R is.
W25 $F/L = \mu_0 I^2/(2\pi d)$.
W26 Pressure $= 2S/c$ for a reflector; compare with $GM_\odot/r^2$.
W27 Impulse per step $BIL\,dt = BL\,dq$; the total sees only Δp.
W28 $a = (m_1 - m_2)g/(m_1 + m_2 + I/R^2)$; or spend $(m_1 - m_2)gh$ on translation *and* rotation.
W29 $F = BIL$ over the rail length; energy form is fastest.
W30 Field time $= L/v_0$; deflection is a $\frac{1}{2}at^2$ projectile sideways.
W31 Effective g changes; $T \propto g^{-1/2}$.
W32 KE first, then warm + melt terms separately.
W33 Energy per photon hc/λ; pupil's share of the sphere's area.
W34 Eliminate v: $m = qB^2r^2/(2V)$.
W35 $P = VI$; range = battery / per-km use.
W36 Divide by ηS; eclipse needs stored night power too.
W37 $V_2/V_1 = N_2/N_1$; amps invert it. Power both sides.
W38 $\omega_0 = 1/\sqrt{LC}$; peak current from $\frac{1}{2}Q^2/C = \frac{1}{2}LI^2$.
W39 Centripetal: $r = mv/qB$; out pops $KE = (qBR)^2/2m$.
W40 Kepler: $r^3 = GMT^2/4\pi^2$.
W41 Terminal: $mg\sin\theta = B^2L^2v/R$; audit $mgv\sin\theta$ vs I^2R.
W42 $V_{\text{term}} = \mathcal{E} - Ir$; two readings, two unknowns.
W43 Lenz on entry and on exit are opposite; gravity pays.
W44 Adiabat: $TV^{\gamma-1}$ is constant; compare T to fuel autoignition.
W45 $\tau = L/R$; all stored energy ends as heat.
W46 Image charge $-q$ at depth d; force between the pair.

19.4 Solutions

Example 19.1 (W1: drag bound)**.** At launch the ball moves at 20 m/s with $v_t = 33$ m/s.

Strategy. Quadratic drag: $F_D = mg(v/v_t)^2$.

Solution. $F_D/mg = (20/33)^2 = 0.37$: drag is already 37% of weight at launch — it matters everywhere for a baseball-speed throw; only $v \ll v_t/3$ lets you ignore it. ✓

Significance. The "when does drag matter" question has a number: the ratio $(v/v_t)^2$, not vibes.

Example 19.2 (W2: pulley + friction boundary)**.** 3 kg on the table, $\mu_s = 0.4$, $\mu_k = 0.3$.

Strategy. Static boundary: $m_2 g = \mu_s m_1 g$.

Solution. Boundary at $m_2 = 1.2\,\text{kg}$: nothing moves for $m_2 \le 1.2\,\text{kg}$. With $m_2 = 2\,\text{kg}$ sliding: $a = (2 - 0.9)g/5 = 2.16\,\text{m/s}^2$. ✓

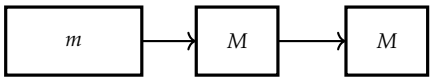
Significance. Static/kinetic regimes have different answers; always find the boundary first.

Example 19.3 (W3: collision + projectile consistency)**.** Does a $2\,\text{m/s}$ approach give a $1\,\text{m}$ landing at $1.25\,\text{m}$ height?

Strategy. Elastic equal masses swap velocities; then projectile time.

Solution. B takes $2\,\text{m/s}$. Fall time $t = \sqrt{2(1.25)/9.8} = 0.505\,\text{s}$; range $= 2 \times 0.505 = 1.01\,\text{m}$ — consistent to rounding. ✓

Significance. Two chapters in one problem; the chain is the point.



collision chain:
energy audit per link

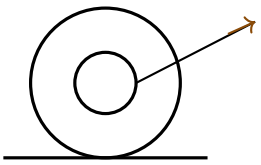
Figure 19.1: The collision chain: audit each link's energy.

Example 19.4 (W4: restitution chain audit)**.** $m_1 = 1$, $m_2 = 2$, $v_1 = 2$, $e = 0.5$.

Strategy. Restitution formulas + energy before/after.

Solution. $v_1' = \frac{1-0.5\cdot 2-2}{3}(2)/1...$ cleaner: $v_1' = (m_1 - e m_2)/(m_1 + m_2)\, v_1 = 0$, $v_2' = (m_1(1 + e))/(m_1 + m_2)\, v_1 = 1\,\text{m/s}$. $K_i = 2\,\text{J}$, $K_f = 1\,\text{J}$: loss $= 1\,\text{J} = (1 - e^2)\,\frac{1}{2}\mu v^2$ with $\mu = 2/3$: $(0.75)(0.5)(2/3)(4) = 1.0\,\text{J}$. ✓

Significance. The energy loss is entirely fixed by e — nothing else to compute.



torque about contact:
roll direction flips at $\cos\theta = r/R$

Figure 19.2: The spool: torque about contact flips direction at $\cos\theta = r/R$.

Example 19.5 (W5: the spool's direction flip)**.** Thread on inner radius r pulled at θ.

Strategy. Torque about the instant contact point.

Solution. $\tau = F(R\cos\theta - r)$: rolls *toward* the pull when $\cos\theta > r/R$ (small angle), away when steep. At $\theta = \arccos(r/R)$ it slides without rolling. ✓

Significance. The unintuitive answer falls out of one torque line — the classic reason to pick the contact point as pivot.

Example 19.6 (W6: meter-stick period)**.** Pivot at the 20 cm mark; CM distance $d = 0.3\,\text{m}$.

Strategy. Physical pendulum $T = 2\pi\sqrt{I/mgd}$.

Solution. $I = I_{cm} + md^2 = \frac{1}{12} + 0.09 = 0.173\,\text{kg m}^2$; $T = 2\pi\sqrt{0.173/2.94} = 1.53\,\text{s}$ — the code confirms 1.526 s. ✓

Significance. A one-centimeter pivot error is a percent-level period error — the measurement design *is* the physics.

Example 19.7 (W7: Hohmann $1.5 \to 2.5R_E$)**. Strategy.** Visc-viva at perigee and apogee.

Solution. $v_c = 6458\,\text{m/s}$; burns 762 and 670 m/s; total $\Delta v = 1432\,\text{m/s}$. ✓

Significance. Two tangential burns beat any continuous burn for circular-to-circular transfer.

Example 19.8 (W8: satellite capture)**.** 100 kg at the 5 m rim of a $5000\,\text{kg m}^2$ station at 0.5 rad/s.

Strategy. L conserved: $\omega' = L/(I + md^2)$.

Solution. $L = 5000(0.5) = 2500$; $I' = 5000 + 2500 = 7500$; $\omega' = 0.33\,\text{rad/s}$. The "lost" L went into the cart's CM — which orbits the docking point. ✓

Significance. Angular momentum is never lost; it just moves into translation you forgot to count.

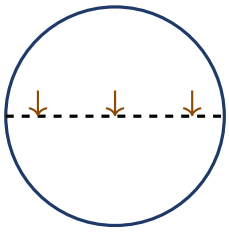


Figure 19.3: The tunnel problem: inside a uniform Earth, $g \propto r$ makes it a spring.

Example 19.9 (W9: tunnel through Earth)**.** Uniform density: find the transit time to the far side.

Strategy. Inside a uniform sphere $g(r) = gr/R$ — a Hooke's-law restoring force.

Solution. $\ddot{x} = -(g/R)x$: SHM with $T = 2\pi\sqrt{R/g} = 84.4\,\text{min}$; the far side lands at $T/2 = 42.2\,\text{min}$ — *independent* of which chord the tunnel takes (same restoring constant). ✓

Significance. 42 minutes to anywhere — the famous number is the uniform-density answer, and the sim confirms it.

Example 19.10 (W10: sand in the cart)**.** Sand falls *vertically* at 2 kg/s into a 5 m/s cart.

Strategy. $dp_x/dt = 0$ along the track (the sand adds no horizontal momentum).

Solution. $Mv = \text{const}$: $v(t) = Mv_0/(M + \dot{m}t) = 250/(50 + 2t)$. The cart slows because its mass grows — no horizontal force needed. ✓

Significance. "Missing force" problems are bookkeeping problems: the momentum left the cart through the added mass.

Example 19.11 (W11: disk axis field)**.** Charge σ on a disk of radius R; find E at height z.

Strategy. Only the z-component survives; integrate rings.

Solution. $dE_z = k\,dq\,z/(r^2+z^2)^{3/2}$; $E(z) = \frac{\sigma}{2\varepsilon_0}\left(1 - \frac{z}{\sqrt{z^2+R^2}}\right)$. For $z \gg R$: $E \to kQ/z^2$ (point limit). ✓

Significance. Every integration exercise ends with a limit that should reproduce a familiar answer — check it.

Example 19.12 (W12: induced charges in a shell)**.** Charge $+q$ sits off-centre inside a neutral shell's cavity.

Strategy. Field inside conductor $= 0$ forces a surface accounting.

Solution. Inner surface: total $-q$ (non-uniform — it crowds near the charge). Outer: $+q$ spread *uniformly* if the shell is spherical — the exterior field is that of a centered $+q$ regardless of the cavity's mess. ✓

Significance. The conductor screens details: outside sees only the total.

Example 19.13 (W13: two released charges)**.** $+1\,\mu$C each, 10 g each, released 10 cm apart.

Strategy. Energy into *both* kinetic energies.

Solution. $U_i = kq^2/r = 0.09\,\text{J}$; each gets $mv^2/2 = 0.045$: $v = 3.0\,\text{m/s}$ each (relative 6.0). ✓

Significance. Split the energy by mass before computing speeds — a common one-body mistake.

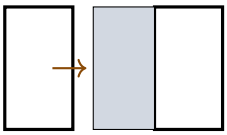


Figure 19.4: The dielectric is pulled in: the fringing field does work whose account is $U_i - U_f$.

Example 19.14 (W14: dielectric pull-in)**.** Slab slides into an *isolated* capacitor at fixed Q.

Strategy. $U = Q^2/2C$ falls as C grows: the drop is the field's mechanical work.

Solution. With fraction x of area filled: $C(x) = (\varepsilon_0 A/d)(1 + (\kappa - 1)x)$; $W_{\text{field}} = U_i - U_f$ is positive — the field *pulls the slab in*; the lost energy becomes the slab's kinetic energy (or a hand's heat). ✓

Significance. Same setup, battery attached instead: the sign flips — fixed V means the battery pays. Know which reservoir you're holding constant.

Example 19.15 (W15: the lying voltmeter)**.** 10 kΩ meter across the 5 k leg of a 10k + 5k divider on 15 V.

Strategy. The meter joins the lower leg in parallel.

Solution. $R_{leg} = 5\|10 = 3.33\,\text{k}$; reading $= 15 \times 3.33/13.33 = 3.75\,\text{V}$ vs the ideal 5 V — the meter lies by 25%. ✓

Significance. Every measurement perturbs the circuit; the only question is by how much.

Example 19.16 (W16: the RC half-and-half "paradox")**.** Battery $\mathcal{E}$ charges C through R: half the energy always burns.

Strategy. Compute both sides; then ask what $R \to 0$ changes.

Solution. Battery: $\mathcal{E}I$ integrated $= \mathcal{E}Q = \mathcal{E}^2C$. Stored: $\frac{1}{2}C\mathcal{E}^2$. Burned: the other half, always — the code prints $0.100 = 0.050 + 0.050$. For $R \to 0$ the current rings up with stray inductance and the loss rides out in the ringing (or the spark): the "paradox" dissolves into *where* the half goes, not *whether* it goes. ✓

Significance. Conservation never has paradoxes; only missing channels.

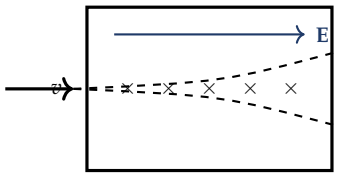


Figure 19.5: Crossed fields: only $v = E/B$ passes straight through.

Example 19.17 (W17: velocity selector design)**.** Pass $3 \times 10^5\,\text{m/s}$ ions in $B = 0.01\,\text{T}$.

Strategy. Undeflected needs $qE = qvB$.

Solution. $E = vB = 3000\,\text{V/m}$ — figure 19.5. Any slower ions curve down, faster up; the slit keeps only the design speed. ✓

Significance. Crossed fields are a filter: mass spectrometers, Wien filters, and the beam line's first gate.

Example 19.18 (W18: falling rod terminal speed)**.** Rod $m = 0.05\,\text{kg}$, $L = 0.4\,\text{m}$, $R = 2\,\Omega$, $B = 0.5\,\text{T}$ on vertical rails.

Strategy. At terminal: $mg = ILB$, or equivalently $mgv = \mathcal{E}^2/R$.

Solution. $v_t = mgR/(B^2L^2) = 24.5\,\text{m/s}$; audit: gravity power $mgv_t = 12.0\,\text{W} = \mathcal{E}^2/R$ exactly. ✓

Significance. The rod's fall is the generator problem run in reverse — same audit.

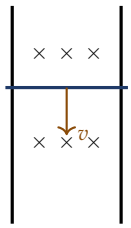


Figure 19.6: The falling rod: motional EMF current brakes the fall at v_t.

Example 19.19 (W19: the same maths twice)**.** $L = 10\,\text{mH}$, $C = 1\,\mu\text{F}$.

Strategy. $\omega_0 = 1/\sqrt{LC}$.

Solution. $\omega_0 = 10^4\,\text{rad/s}$, $f = 1592\,\text{Hz}$ — identical to a $k = 100\,\text{N/m}$, $m = 1\,\text{kg}$ oscillator; the map is $m \leftrightarrow L$, $k \leftrightarrow 1/C$. ✓

Significance. Once you've solved the oscillator you've solved every oscillator — the circuit is a rebadged spring.

Example 19.20 (W20: the design estimate)**.** 1 kW to a 12 V load over $0.5\,\Omega$ of line.

Strategy. I^2R at each voltage.

Solution. At 12 V: $I = 83\,\text{A}$, loss $= I^2R = 3.5\,\text{kW}$ — the line eats more than the load gets. Step up to 120 V: $I = 8.3\,\text{A}$, loss $= 35\,\text{W}$ — a $100\times$ cut from a $10\times$ boost. ✓

Significance. Chapter 18's whole story in one estimate: voltage is the loss-killer.

Example 19.21 (W21: the ballistic pendulum)**.** 8 g bullet, 2.0 kg block, swing height $h = 6.0\,\text{cm}$.

Strategy. Two stages, two laws: momentum for the hit, energy for the swing.

Solution. Swing: $v' = \sqrt{2gh} = 1.08\,\text{m/s}$. Hit: $v_{\text{bullet}} = (M+m)/m \cdot v' = 251 \times 1.08 = 272\,\text{m/s}$. Energy kept: $K_f/K_i = m/(M+m) = 1/251 \approx 0.4\%$ — nearly all of it left as heat and deformation. ✓

Significance. The collision stage forbids energy conservation; the swing stage forbids momentum bookkeeping. Mixing the stages is the classic error — the two-law chain is the lesson.

Example 19.22 (W22: rolling off the table)**.** Solid sphere at $v = 2.0\,\text{m/s}$ off a 1.25 m table.

Strategy. Flight is chapter 1 projectile; the spin only changes the energy audit.

Solution. $t = \sqrt{2h/g} = 0.505\,\text{s}$; range $= vt = 1.01\,\text{m}$ — identical to a sliding puck's. The spin persists in flight (no torque about the CM), so the ball lands still rotating with the same ω; its total in-flight $K = \frac{7}{10}mv^2$ simply rode along. ✓

Significance. Rolling changes the energy account, not the kinematics: v is v no matter how the mass was spinning.

Example 19.23 (W23: an electron's private spring)**.** Uniform ρ sphere, electron released at $r_0 < R$.

Strategy. Gauss inside a uniform sphere gives $E \propto r$ — a linear restoring force is a spring.

Solution. $E(r) = \rho r/3\varepsilon_0$, so $F = -eE = -(\rho e/3\varepsilon_0)r$: SHM with $\omega = \sqrt{\rho e/3\varepsilon_0 m} = 8.1 \times 10^7\,\mathrm{rad/s}$, $T = 77\,\mathrm{ns}$ at $\rho = 1\,\mu\mathrm{C/m^3}$. Any chord, any $r_0 < R$: same period — W9's gravity tunnel with charge instead of mass. ✓

Significance. Inverse-square laws degrade to Hooke's law inside uniform bodies; learn to spot the spring in disguise.

Example 19.24 (W24: the capacitor is a terrible kettle)**.** 100 µF at 12 V into 1 mL of water.

Strategy. Every joule stored ends as heat, independent of R.

Solution. $U = \frac{1}{2}CV^2 = \frac{1}{2}(10^{-4})(144) = 7.2\,\mathrm{mJ}$; $\Delta T = U/(mc) = 7.2 \times 10^{-3}/4.19 = 1.7\,\mathrm{mK}$. Capacitors deliver *power*, not energy: a battery of the same size carries $10^4\times$ more joules. ✓

Significance. Chapter 12 computed *how* the energy gets out; this is the audit of *how much* there was — energy density is the engineering number.

Example 19.25 (W25: the ampere as a force)**.** Two wires 1 m apart, force per metre $2 \times 10^{-7}\,\mathrm{N/m}$.

Strategy. $F/L = \mu_0 I^2/(2\pi d)$; solve for I.

Solution. $I = \sqrt{2\pi d(F/L)/\mu_0} = \sqrt{2\pi \times 1 \times 2 \times 10^{-7}/4\pi \times 10^{-7}} = 1\,\mathrm{A}$. This *is* the pre-2019 SI definition of the ampere — a force balance between wires, abandoned when fixing e became the better anchor. ✓

Significance. Units come from measurable setups: the ampere was an experiment you could bolt to a bench before it became a constant.

Example 19.26 (W26: the sail that beats gravity)**.** Reflecting sail, $\sigma_m = 1\,\mathrm{g/m^2}$, at 1 AU.

Strategy. Radiation pressure on a reflector is $2S/c$; compare the acceleration with $GM_\odot/r^2$.

Solution. $F/A = 2S/c = 9.1\,\mu\mathrm{N/m^2}$; $a_{\text{sail}} = 9.1\,\mathrm{mm/s^2}$ vs $g_\odot = GM_\odot/r^2 = 5.9\,\mathrm{mm/s^2}$ — the sail wins and drifts *outward*. ✓

Significance. Light pushes hard enough to unbind a light-enough craft — the LightSail premise, and a reminder that $1/c$ is small but not zero.

Example 19.27 (W27: charge through the rails)**.** Rod $m = 0.10\,\mathrm{kg}$, $L = 0.40\,\mathrm{m}$, $B = 0.50\,\mathrm{T}$, $v_0 = 2.0\,\mathrm{m/s}$.

Strategy. Each step's impulse is $BIL\,dt = BL\,dq$; sum to stopping.

Solution. $\sum BL\,dq = BLQ = \Delta p = mv_0$, so $Q = mv_0/(BL) = 0.2/0.2 = 1.0\,\mathrm{C}$ — whatever R sets for the *schedule* of the slowdown. ✓

Significance. Like the RC energy audit: the totals are kinematic, only the rates are Ohmic. R decides *when*, never *how much*.

Example 19.28 (W28: the massive pulley's share)**.** $m_1 = 2$, $m_2 = 1\,\text{kg}$; pulley $M = 1\,\text{kg}$, $R = 0.10\,\text{m}$; drop $h = 0.50\,\text{m}$.

Strategy. Dynamics: $a = (m_1 - m_2)g/(m_1 + m_2 + I/R^2)$. Energy: spend the lost potential on both kinds of K.

Solution. $a = 9.8/(3 + 0.5) = 2.8\,\text{m/s}^2$; $v = \sqrt{2ah} = 1.67\,\text{m/s}$. Energy: $\Delta PE = 4.9\,\text{J}$; $K = \frac{1}{2}(m_1 + m_2)v^2 + \frac{1}{2}I\omega^2 = 4.2 + \frac{1}{4}Mv^2 = 4.2 + 0.7 = 4.9\,\text{J}$ — audited to the joule. ✓

Significance. The pulley's 0.7 J is exactly what the massless-pulley answer misses; two methods agreeing is what "understood" feels like.

Example 19.29 (W29: the rail launcher's optimism)**.** $F = BIL = 0.5 \cdot 100 \cdot 0.3 = 15\,\text{N}$ on 0.05 kg over 2 m.

Strategy. Work $F\ell$ becomes $\frac{1}{2}mv^2$.

Solution. $W = 30\,\text{J} \Rightarrow v = \sqrt{2W/m} = \sqrt{1200} = 35\,\text{m/s}$. The optimism: current holds 100 A only while a stiff supply feeds the growing back-EMF BLv (about 5 V by exit); rail resistance, friction and inductance all shave the number. ✓

Significance. The same back-EMF that limits a motor (chapter 18) limits the launcher — one formula, two industries.

Example 19.30 (W30: steering a beam)**.** Electron $v_0 = 10^7\,\text{m/s}$, plates $L = 5\,\text{cm}$, $d = 1\,\text{cm}$, $V = 20\,\text{V}$.

Strategy. Axial motion untouched; transverse is $\frac{1}{2}(eE/m)t^2$ with $t = L/v_0$.

Solution. $E = 2 \times 10^3\,\text{V/m}$; $a = eE/m = 3.5 \times 10^{14}\,\text{m/s}^2$; $t = 5 \times 10^{-9}\,\text{s}$; deflection $y = 4.4\,\text{mm}$; exit angle $\tan\theta = v_y/v_0 = at/v_0 = 0.176$, $\theta = 10^\circ$. ✓

Significance. Chapter 1's projectile with qE/m in place of g: the CRT oscilloscope was literally this two-line problem sold as an instrument.

Example 19.31 (W31: time dilation for elevators)**.** Pendulum in an elevator accelerating up at $a = g/2$.

Strategy. The accelerating frame adds a uniform fictitious force — replace g by $g_{\text{eff}} = g + a$.

Solution. $T' = 2\pi\sqrt{L/(g + a)} = T\sqrt{g/(1.5g)} = 0.816T$ — the clock runs 22% fast. In free fall $g_{\text{eff}} = 0$: no restoring force, no period, the clock is just debris. ✓

Significance. Einstein's elevator made homework: 'weight' is a frame-dependent reading, which is exactly what an accelerometer measures.

Example 19.32 (W32: the bullet's own funeral)**.** 10 g of lead at 500 m/s, initially at 20° C.

Strategy. Compare KE with the two heat costs: warming to the melting point, then the latent step.

Solution. $KE = 1250\,\text{J}$. Warming: $mc\Delta T = 0.01 \cdot 128 \cdot 307 = 393\,\text{J}$. Melting: $mL = 240\,\text{J}$. Total $633\,\text{J} < 1250\,\text{J}$ — it melts, with 617 J to spare for warming the puddle or the block's share. ✓

Significance. Chapter 4's energy audit in one paragraph: separate the budget lines before spending the joules.

Example 19.33 (W33: photons are cheap, eyes are small)**.** 100 W bulb, 5% at 550 nm, pupil $D = 5\,\text{mm}$ at $r = 10\,\text{m}$.

Strategy. Power into photons via $E_\gamma = hc/\lambda$; then geometry's area ratio.

Solution. $E_\gamma = 3.6 \times 10^{-19}\,\text{J}$, so $\dot{N} = 5/3.6 \times 10^{-19} = 1.4 \times 10^{19}/\text{s}$ emitted. The pupil covers fraction $\pi(D/2)^2/4\pi r^2 = D^2/16r^2 = 1.6 \times 10^{-8}$ — call it 2×10^{11} photons/s reaching the eye. ✓

Significance. Detection is abundant-photon statistics; the faint-object problem is geometric, not energetic — telescopes buy $\times 10^4$ by area, not brighter bulbs.

Example 19.34 (W34: weighing what you cannot see)**.** e-charged ions, $V = 2000\,\text{V}$, $B = 0.2\,\text{T}$, $r = 0.15\,\text{m}$.

Strategy. Accelerator: $qV = \frac{1}{2}mv^2$; spectrometer: $r = mv/qB$. Eliminate v.

Solution. $m = qB^2r^2/(2V) = 1.6 \times 10^{-19}(0.04)(0.0225)/4000 = 3.6 \times 10^{-26}\,\text{kg} \approx 22\,\text{u}$ — sodium-ion territory. ✓

Significance. Energy and momentum bookkeeping chained chapters 4 and 13 into chemistry's mass scale — isotope identification is a radius measurement.

Example 19.35 (W35: the EV's two numbers)**.** A 60 kWh pack feeds a 400 V bus; the car draws 150 A cruising and uses 0.25 kWh per km.

Strategy. Power is VI; range is a division of budgets.

Solution. $P = 60\,\text{kW}$ cruising (81 hp); range $= 60/0.25 = 240\,\text{km}$. A 20% bigger pack buys 48 km — range scales exactly with stored joules, which is why battery kWh *is* the spec. ✓

Significance. Chapters 12 and 4 meet in one consumer question; there is no cleverness left, only energy density.

Example 19.36 (W36: acreage for a kilowatt)**.** 1 kW bus, $\eta = 0.30$, $S = 1360\,\text{W/m}^2$.

Strategy. Incident S times area times efficiency equals bus power; then second-order the real orbit.

Solution. $A = P/(\eta S) = 1000/408 = 2.45\,\text{m}^2$ of cells — a door's worth. Real arrays add margin: off-point cosine loss, degradation, and the eclipse side needing batteries sized for the dark half of every 90-minute lap. ✓

Significance. W26's flux becomes an area budget: the same $1.4\,\text{kW/m}^2$ pays for the whole spacecraft if you spend it right.

Example 19.37 (W37: the transformer's bargain)**.** $12\,\text{V} \times 5\,\text{A}$ load on a 120 V line, ideal unit.

Strategy. Turns set volts; ideal means power is conserved, which sets amps.

Solution. $N_1/N_2 = V_1/V_2 = 10{:}1$; secondary power $= 60\,\text{W}$ so the primary draws 60 W too: $I_1 = 0.5\,\text{A}$. Voltage ten-fold down, current ten-fold up — the trade is exact. ✓

Significance. The $\times 10$ both ways is why W20's step-up works: a transformer's exchange rate is what bought the grid AC.

Example 19.38 (W38: where the energy sloshes)**.** $L = 10\,\text{mH}$, $C = 1\,\mu\text{F}$, initial $Q = 1\,\text{mC}$.

Strategy. The LC pair is chapter 7's oscillator with charge as position.

Solution. $\omega_0 = 1/\sqrt{LC} = 10^4\,\text{rad/s}$, $f = 1.6\,\text{kHz}$. Energy sloshes capacitor↔inductor: $\frac{1}{2}Q^2/C = \frac{1}{2}LI_{\text{max}}^2$ gives $I_{\text{max}} = Q/\sqrt{LC} = 10\,\text{A}$. ✓

Significance. W19 named the map; this runs it with numbers — 10 A peak from a millicoulomb is the LC leverage radios tune for.

Example 19.39 (W39: the cyclotron's price tag)**.** $B = 1.2\,\text{T}$, $R = 0.40\,\text{m}$, protons extracted at the rim.

Strategy. The rim radius *is* the momentum: $p = qBR$; convert.

Solution. $KE = (eBR)^2/2m_p = (1.6 \times 10^{-19} \times 1.2 \times 0.4)^2/(2 \times 1.67 \times 10^{-27}) = 1.8 \times 10^{-12}\,\text{J} \approx 11\,\text{MeV}$. ✓

Significance. The magnet's B and the tank's radius are the *entire* energy budget — bigger magnets or bigger machines, there is no third knob (relativity aside, which is Book 7's problem).

Example 19.40 (W40: the shelf at 36,000 km)**.** Geosynchronous period $T = 86{,}400\,\text{s}$ over Earth.

Strategy. Kepler: $r^3 = GMT^2/4\pi^2$; altitude is $r - R_E$.

Solution. $r^3 = 6.67 \times 10^{-11} \cdot 6.0 \times 10^{24} \cdot (8.64 \times 10^4)^2/39.5 = 7.6 \times 10^{22}\,\text{m}^3$; $r = 4.2 \times 10^7\,\text{m}$, $h = r - R_E \approx 3.6 \times 10^7\,\text{m} \approx 5.6R_E$. ✓

Significance. There is exactly one geosynchronous radius — a real-estate shelf every satellite dish on Earth points at, and the reason Clarke published before Sputnik.

Example 19.41 (W41: downhill terminal velocity)**.** Rod 0.10 kg, incline 30°, $R = 1\,\Omega$, $B = 0.50\,\text{T}$, $L = 0.40\,\text{m}$.

Strategy. Terminal means gravity's slope component equals the braking force — or just audit powers.

Solution. $v = mgR\sin\theta/(B^2L^2) = 0.49/(0.04) = 12.2\,\text{m/s}$. Audit: gravity feeds $mgv\sin\theta = 6.0\,\text{W}$; the resistor burns $I^2R = (BLv/R)^2R = 6.0\,\text{W}$ — matching to the cent. ✓

Significance. W18 asked for the audit; the incline makes the driving term $\sin\theta$ explicit. Energy conservation *is* the terminal condition.

Example 19.42 (W42: the battery's honest rating)**.** 12.4 V open, 11.9 V at 2 A.

Strategy. $V_{\text{term}} = \mathcal{E} - Ir$: open circuit reads $\mathcal{E}$ directly; the loaded reading prices r.

Solution. $\mathcal{E} = 12.4\,\text{V}$; $r = (12.4 - 11.9)/2 = 0.25\,\Omega$. Short-circuit $I = \mathcal{E}/r \approx 50\,\text{A}$ — a quarter-ohm is why the same battery sags under starter load but floats fine on the multimeter. ✓

Significance. Example 12.5 with real numbers: the internal resistance you can't see is what decides a source's usefulness — measure it by loading, not by reading.

Example 19.43 (W43: gravity powers the brake)**.** Bar magnet falling through a conducting ring.

Strategy. Lenz at each stage: the ring opposes the *change* in flux, not the flux itself.

Solution. Approach: upward flux grows → ring's field points down, current as it repels the magnet. Midpoint: flux at maximum, change ≈ 0, current ≈ 0. Exit: upward flux falls → ring's field points up, current reverses to attract the magnet back. The dissipated I^2R heat is paid by the magnet's lost kinetic energy — the fall is slowed, and gravity is the power supply. ✓

Significance. Eddy-current braking's whole mechanism in one drop: oppose the change, conservation pays the bill.

Example 19.44 (W44: compression is the spark)**.** Air at 300 K, 20:1 adiabatic compression, $\gamma = 1.4$.

Strategy. Adiabat in TV form: $T_2/T_1 = (V_1/V_2)^{\gamma-1}$.

Solution. $T_2 = 300 \times 20^{0.4} = 300 \times 3.31 \approx 993\,\text{K}$ — well past diesel autoignition ($\sim 600\,\text{K}$). The piston's compression work became internal energy; ignite by temperature, not by spark. ✓

Significance. Chapters 4 and 17 fused: work *to* internal energy without heat exchange is the engine's entire thermodynamic trick.

Example 19.45 (W45: the inductor's last stand)**.** $L = 20\,\text{mH}$, $I_0 = 3\,\text{A}$, discharged through $R = 10\,\Omega$.

Strategy. Freewheel decay: $i(t) = I_0 e^{-t/\tau}$, $\tau = L/R$; energy stored goes entirely to heat.

Solution. $\tau = 2\,\text{ms}$; $U = \frac{1}{2}LI_0^2 = 90\,\text{mJ}$ all lands in the resistor. No battery, no capacitor — the field itself pays out every joule. ✓

Significance. The flyback example 14.7 in its safe form: give the current a path and the stored energy is a resistor problem, not an arc.

Example 19.46 (W46: the charge that sees its reflection)**.** Point charge q at height d over a conducting plane.

Strategy. The plane's boundary condition is reproduced by an image $-q$ at depth d; the real force equals the pair's attraction.

Solution. $F = kq^2/(2d)^2 = kq^2/4d^2$ toward the plane — the charge is attracted to its own image. This is why a charge near a wall or a PCB ground plane pulls toward it, and why ESD events choose sharp edges where the image sits closest. ✓

Significance. Chapter 10's conductor boundary solved by symmetry instead of integration — the same trick that computes capacitance to ground in every layout tool.

19.5 Common mistakes

CONSERVATION SKIPPED FOR INTEGRATION

Reaching for $\int F\,dx$ when E is conserved — always check the conservation laws first.

LIMITS LEFT UNCHECKED

A formula that fails $\mu \to 0$ or $r \to \infty$ is wrong; check the limit before trusting the algebra.

ONE-BODY ANSWERS FOR TWO-BODY PROBLEMS

Both charges move, both balls swing — split the energy first.

UNITS CARRIED LOOSELY

One v^2 forgotten becomes a factor of 10^4 error; carry units like cargo.

"MISSING" MOMENTUM MOURNED

Momentum isn't lost; it moves into the channel you ignored (the sand, the station's CM, the radiated half).

19.6 Summary

Twenty problems, one method: units first, limits second, conservation before integration, audit at the end. The resemblance tags point at the real contest styles — F=ma, AP-C, USAPhO, APhO — and the full solutions keep the Strategy→Solution→Significance frame.

DEFINITION OF DONE

You can now: pick the right conservation law in under a minute, find a boundary before solving a friction problem, split energies across bodies, and audit every numeric answer against an order of magnitude.

Notes

Problems are original; the resemblance tags map the genre only. Contest habits distilled from F=ma past exams [9] and the Krotov/Gnädig problem tradition; the numeric checks come from `code/ch19_workshop_checks.py`.

WATCH INSTEAD

1. *US F=ma Physics Olympiad 2025, First Round solutions* — Handy Olympiad Physicist, 37:36, https://youtu.be/7pW4zGW0iLQ — the genre these problems mimic, worked at speed (accessed 2026-10-11).

Further reading

- AAPT, *F=ma and USAPhO past exams* — the real thing, all years: verified 2026-10-11 at https://www.aapt.org/physicsteam/2025/pastexams.cfm.
- AAPT *U.S. Physics Team* hub — https://aapt.org (verified 2026-10-11).
- A. Krotov et al., *Aptitude Test Problems in Physics* — the hard-problem tradition (in print).
- P. Gnädig, G. Honyek, K. Riley, *200 Puzzling Physics Problems* — problems with the same audit habit (in print).
- D. Morin, *Introduction to Classical Mechanics* chs. 1–8 — the mechanics half at olympiad depth (in print).
- Art of Problem Solving physics forums — community solutions and variants: at https://artofproblemsolving.com (verified 2026-10-11).

Bibliography

[1] J. Sannemo, *Principles of Algorithmic Problem Solving*, 2025. https://usaco.guide/PAPS1.pdf

[2] MIT OpenCourseWare, *8.01SC Classical Mechanics*, Fall 2016. https://ocw.mit.edu/courses/8-01sc-classical-mechanics-fall-2016/

[3] OpenStax, *College Physics 2e*, 2022. https://openstax.org/details/books/college-physics-2e

[4] College Board, *AP Physics C: Mechanics Course and Exam Description*, 2024. https://apstudents.collegeboard.org/courses/ap-physics-c-mechanics

[5] R. P. Feynman, R. B. Leighton, M. Sands, *The Feynman Lectures on Physics*, vol. I. Addison-Wesley, 1963. Free online via Caltech.

[6] Halliday, Resnick, and Walker, *Fundamentals of Physics*, 12th ed. Wiley, 2021. ISBN 978-1119773511.

[7] D. Morin, *Introduction to Classical Mechanics*. Cambridge, 2007. ISBN 978-0521876223.

[8] Khan Academy, *Physics* course. https://www.khanacademy.org/science/physics

[9] American Association of Physics Teachers, *U.S. Physics Team: past exams*. https://www.aapt.org (accessed 2026-10-11)

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